

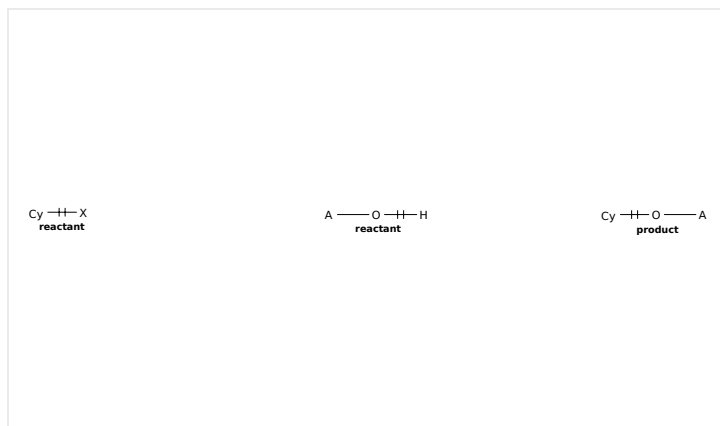
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October 21, 2025, 2:28 PM

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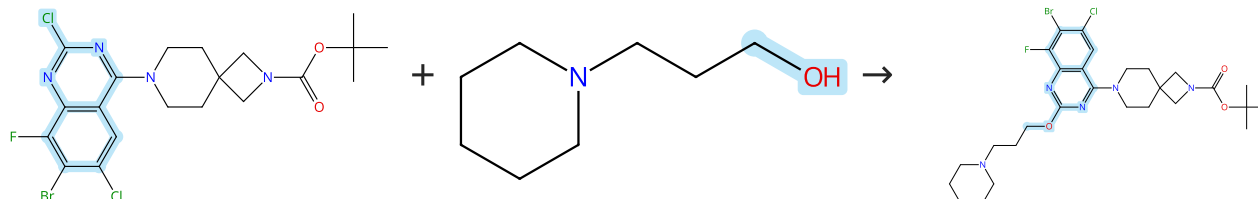
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Reactions (500)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

31-614-CAS-34525920

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Catalysts: Dabco
Solvents: Acetonitrile; 1 h, 95 °C

Experimental Protocols

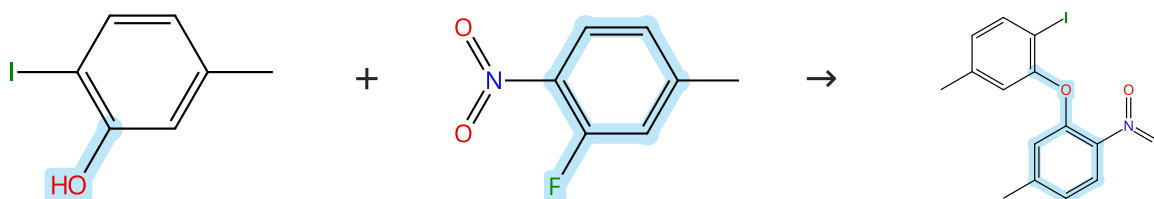
Discovery and biological evaluation of 1-{2,7-diazaspiro[3.5]nonan-2-yl}prop-2-en-1-one derivatives as covalent inhibitors of KRAS G12C with favorable metabolic stability and anti-tumor activity

By: Imaizumi, Tomoyoshi; et al

Bioorganic & Medicinal Chemistry (2022), 71, 116949.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (47)

Suppliers (81)

31-614-CAS-42623270

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; overnight, 100 °C

Experimental Protocols

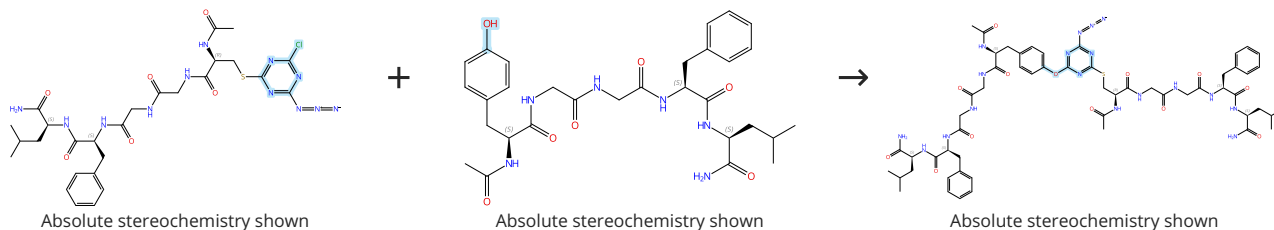
Photostability of Phenoxazine Derivatives

By: Goll, Felix D.; et al

ChemPhysChem (2024), 25(21), e202400506.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%

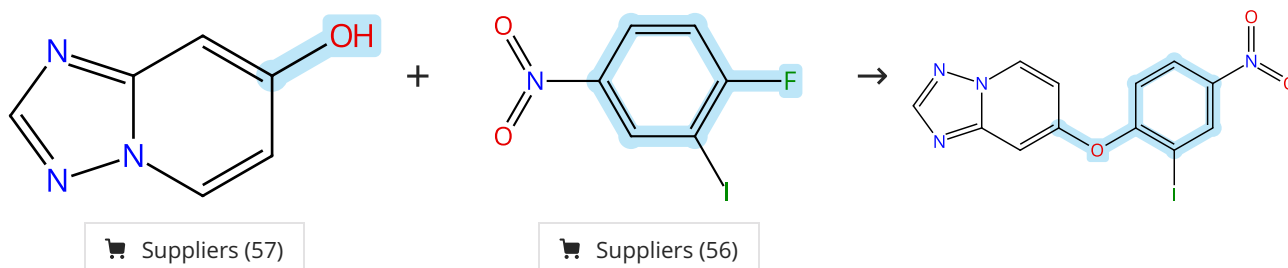


Suppliers (2)

31-614-CAS-34441029	Steps: 1 Yield: 100%	Exploiting azido-dichloro-triazine as a linker for regioselective incorporation of peptides through their N, O, S functional groups
1.1 Reagents: Potassium carbonate; 30 min, > pH 8, 0 °C		By: Sharma, Anamika; et al
Experimental Protocols		Bioorganic Chemistry (2020), 104, 104334.

Scheme 4 (1 Reaction)

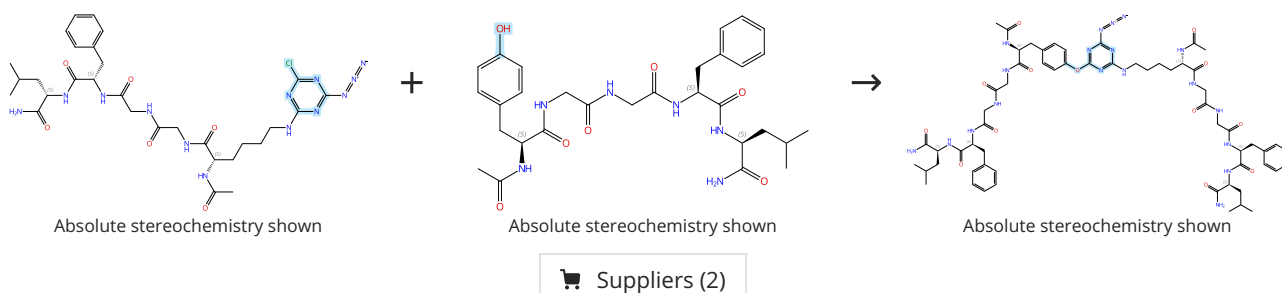
Steps: 1 Yield: 100%



31-614-CAS-41680313	Steps: 1 Yield: 100%	Synthesis and in vivo evaluation of [¹¹ C]tucatinib for HER2-targeted PET imaging
1.1 Reagents: Cesium carbonate Solvents: <i>N</i> -Methyl-2-pyrrolidone; 20 min, 120 °C		By: Muller, Marius; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2023), 80, 129088.

Scheme 5 (1 Reaction)

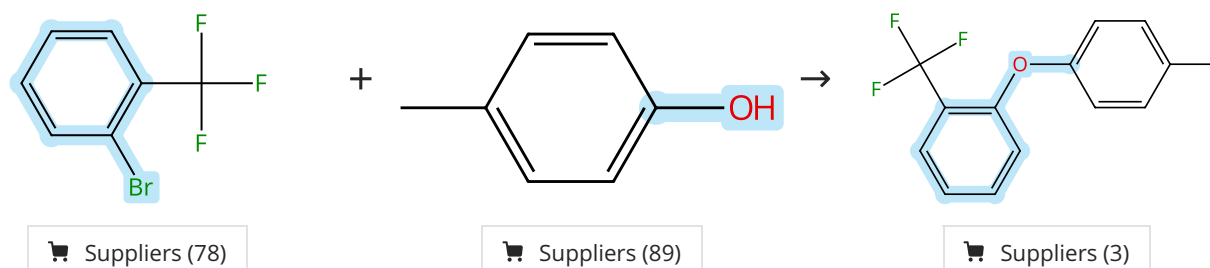
Steps: 1 Yield: 100%



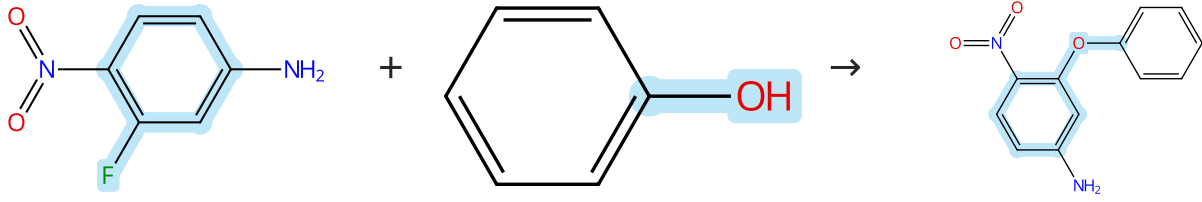
31-614-CAS-34441032	Steps: 1 Yield: 100%	Exploiting azido-dichloro-triazine as a linker for regioselective incorporation of peptides through their N, O, S functional groups
1.1 Reagents: Potassium carbonate; 8 h, rt		By: Sharma, Anamika; et al
Experimental Protocols		Bioorganic Chemistry (2020), 104, 104334.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



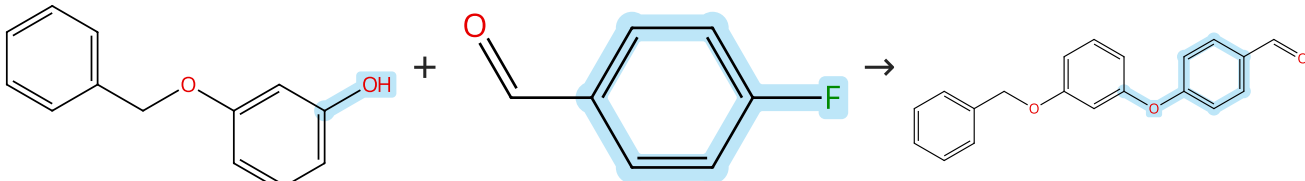
31-614-CAS-36696313	Steps: 1 Yield: 100%	Synthesis of Fluorenones and Xanthenes through Intramolecular C-F Arylation
1.1 Reagents: Potassium carbonate Catalysts: Copper, Potassium iodide Solvents: Pyridine; 24 h, reflux		By: Hamada, Shusuke; et al
Experimental Protocols		Bulletin of the Chemical Society of Japan (2023), 96(5), 401-405.

Scheme 7 (1 Reaction)	Steps: 1 Yield: 100%
	
 Suppliers (89)	 Suppliers (207)

31-614-CAS-31946047	Steps: 1 Yield: 100%	Structure-activity relationships of 1,4-bis(arylsulfonamido)-benzene or naphthalene-N,N'-diacetic acids with varying C2-substituents as inhibitors of Keap1-Nrf2 protein-protein interaction
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, 90 °C		By: Lee, Sumi; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2022), 237, 114380.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Chemical reaction scheme showing the synthesis of 4-(benzyloxy)-4'-formylbiphenyl ether. The reactants are 4-(benzyloxy)phenol and 4-fluorobenzaldehyde. The product is 4-(benzyloxy)-4'-formylbiphenyl ether. The reaction is labeled as Scheme 8 (1 Reaction) and has a yield of 100%.

 Suppliers (69)

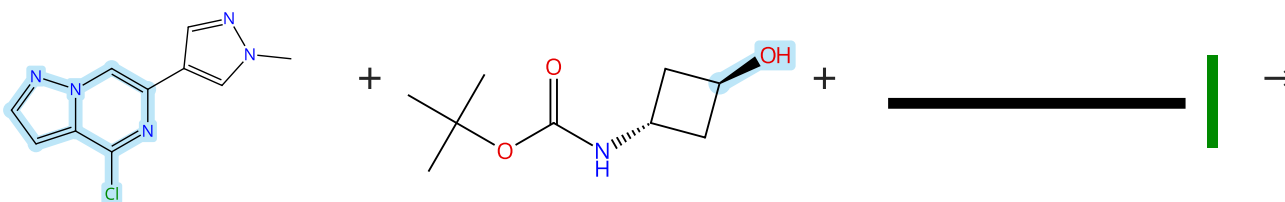
 Suppliers (93)

 Suppliers (15)

31-614-CAS-34652761	Steps: 1 Yield: 100%	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt		By: Chen, Tao; et al
1.2 Reagents: Water; rt		Letters in Drug Design & Discovery (2022), 19(4), 263-268.
Experimental Protocols		

Scheme 9 (1 Reaction)

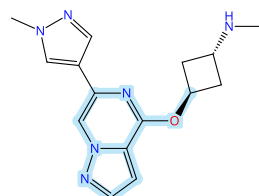
Steps: 1 Yield: 100%



Suppliers (33)

Suppliers (58)

Suppliers (86)



Relative stereochemistry shown

31-614-CAS-40236730

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Sodium hydride
Solvents: Dimethylformamide; 30 min, rt
- 1.2 30 min, rt
- 1.3 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; overnight, rt

Experimental Protocols

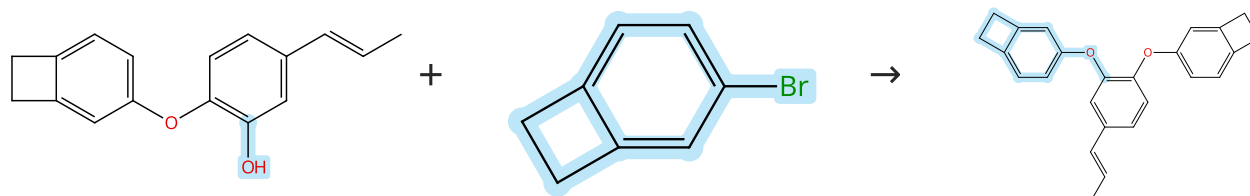
Discovery and Preclinical Characterization of BIIB129, a Covalent, Selective, and Brain-Penetrant BTK Inhibitor for the Treatment of Multiple Sclerosis

By: Himmelbauer, Martin K.; et al

Journal of Medicinal Chemistry (2024), 67(10), 8122-8140.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

31-614-CAS-34581269

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Sodium ascorbate, Cesium carbonate
Catalysts: Cuprous chloride, *N*¹,*N*²-Bis(2-thienylmethyl)ethanediamide
Solvents: 1,4-Dioxane; 7 h, 100 °C

Experimental Protocols

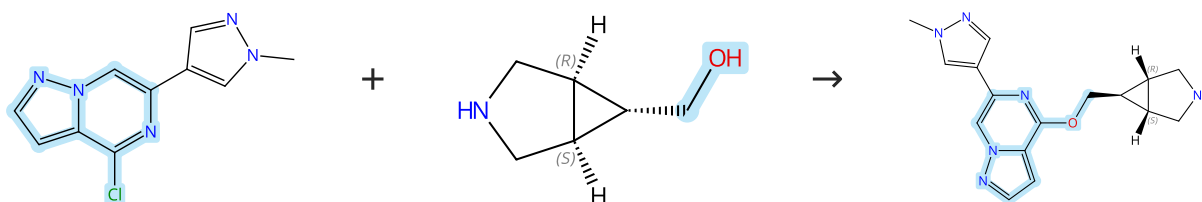
Biobased Low-k Polymers at High Frequency Derived from Isoeugenol

By: Liu, Fengping; et al

ACS Applied Polymer Materials (2022), 4(10), 7173-7181.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (33)

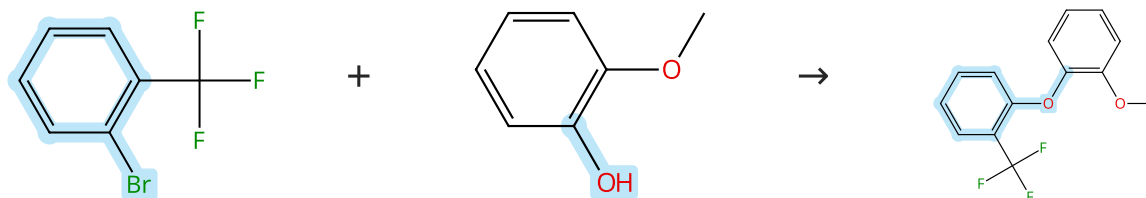
Suppliers (60)

Relative stereochemistry shown

31-614-CAS-40236712	Steps: 1 Yield: 100%	Discovery and Preclinical Characterization of BIIB129, a Covalent, Selective, and Brain-Penetrant BTK Inhibitor for the Treatment of Multiple Sclerosis
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: Tetrahydrofuran; 30 min, rt		By: Himmelbauer, Martin K.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2024), 67(10), 8122-8140.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



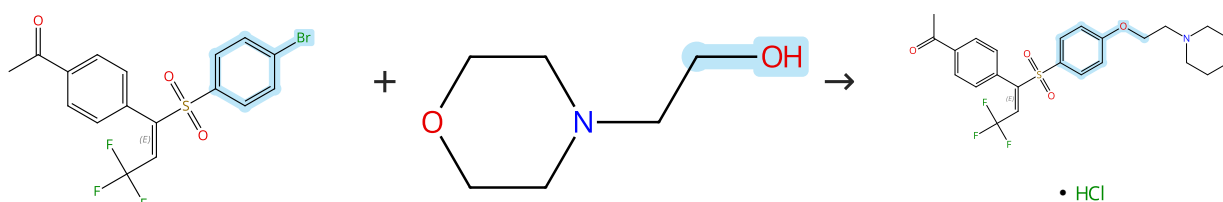
Suppliers (78)

Suppliers (125)

31-614-CAS-36696316	Steps: 1 Yield: 100%	Synthesis of Fluorenones and Xanthenes through Intramolecular C-F Arylation
1.1 Reagents: Potassium carbonate Catalysts: Copper, Potassium iodide Solvents: Pyridine; 24 h, reflux		By: Hamada, Shusuke; et al
Experimental Protocols		Bulletin of the Chemical Society of Japan (2023), 96(5), 401-405.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

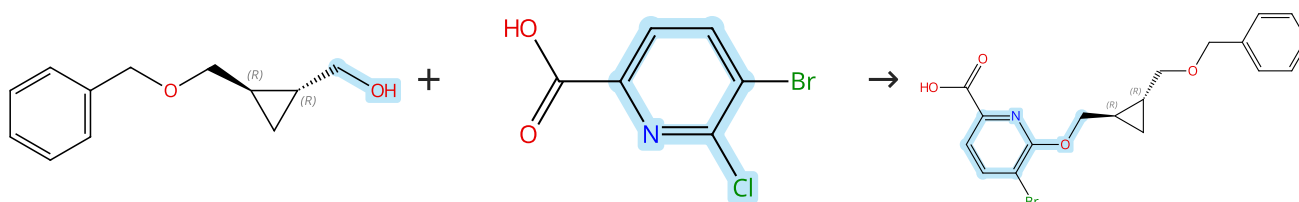
Suppliers (82)

Double bond geometry shown

31-614-CAS-31478498	Steps: 1 Yield: 100%	(E)-β-Trifluoromethyl vinylsulfones as antitumor agents: Synthesis and biological evaluations
1.1 Reagents: Cesium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, Bis(1,1-dimethylethyl)[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine Solvents: Toluene; 2 h, 110 °C		By: Zhang, Jun; et al
1.2 Reagents: Hydrochloric acid Solvents: 1,4-Dioxane, Ethyl acetate		European Journal of Medicinal Chemistry (2022), 232, 114197.
Experimental Protocols		

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Suppliers (65)

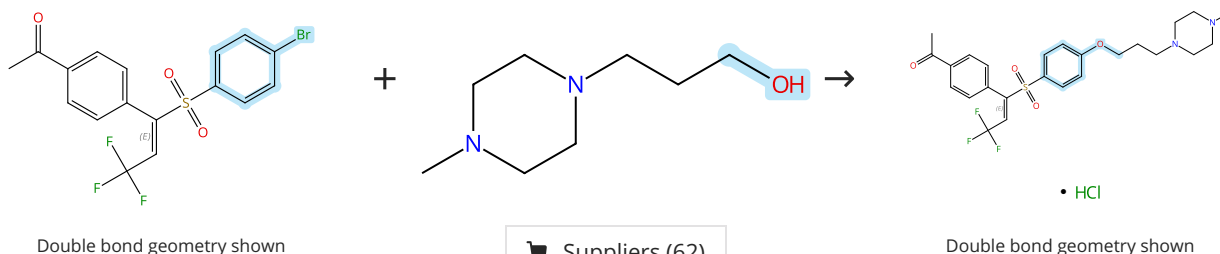
Relative stereochemistry shown

Suppliers (14)

31-170-CAS-22590888 Steps: 1 Yield: 100% 1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 0 °C; 30 min, rt 1.2 3 h, 80 °C	Identification and Preclinical Development of a 2,5,6-Trisubstituted Fluorinated Pyridine Derivative as a Radioligand for the Positron Emission Tomography Imaging of Cannabinoid Type 2 Receptors By: Haider, Achi; et al Journal of Medicinal Chemistry (2020), 63(18), 10287-10306.
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Scheme 15 (1 Reaction)

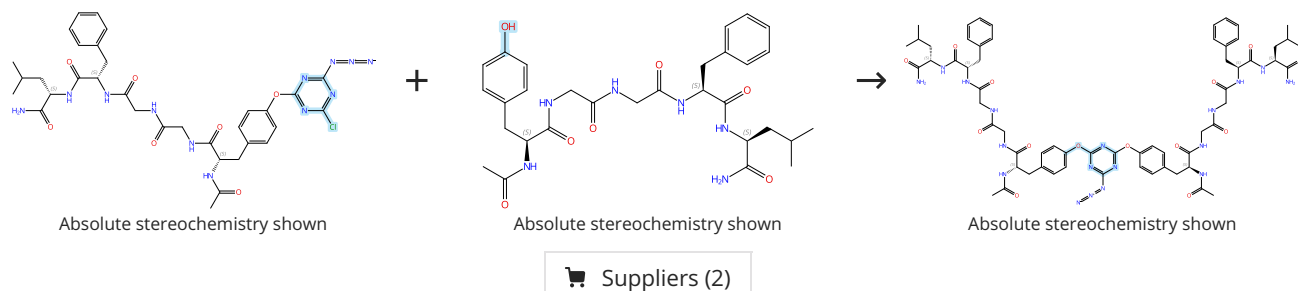
Steps: 1 Yield: 100%



31-614-CAS-31478502 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, Bis(1,1-dimethylethyl)[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl] phosphine Solvents: Toluene; 2 h, 110 °C 1.2 Reagents: Hydrochloric acid Solvents: 1,4-Dioxane, Ethyl acetate Experimental Protocols	(E)-β-Trifluoromethyl vinylsulfones as antitumor agents: Synthesis and biological evaluations By: Zhang, Jun; et al European Journal of Medicinal Chemistry (2022), 232, 114197.
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Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-34441031 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate; 4 h, rt Experimental Protocols	Exploiting azido-dichloro-triazine as a linker for regioselective incorporation of peptides through their N, O, S functional groups By: Sharma, Anamika; et al Bioorganic Chemistry (2020), 104, 104334.
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Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Chemical & Pharmaceutical Bulletin (2024), 72(10), 862-883.



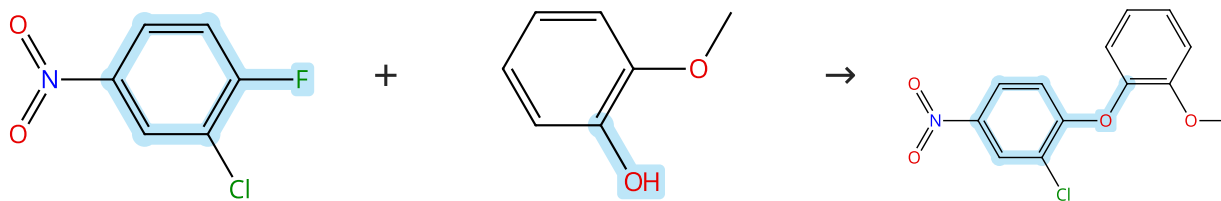
Experimental Protocols



Experimental Protocols

Scheme 20 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (125)

Suppliers (11)

31-614-CAS-31980147

Steps: 1 Yield: 100%

Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; rt

By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

31-170-CAS-23043758

Steps: 1 Yield: 100%

Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 17 h, rt

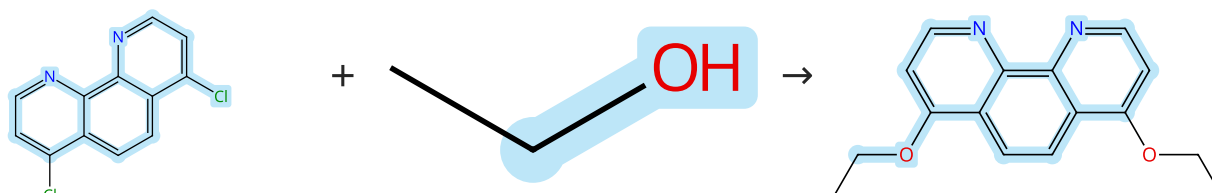
By: Fujii, Shinya; et al

Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (618)

Suppliers (2)

31-170-CAS-22584342

Steps: 1 Yield: 100%

Aryl Ether Syntheses via Aromatic Substitution Proceeding under Mild Conditions

1.1 Reagents: Potassium *tert*-butoxide
Solvents: Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 2 h, 0 °C

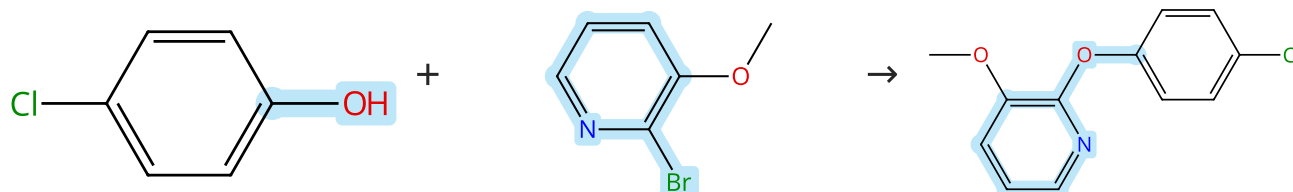
By: Ando, Shin; et al

Journal of Organic Chemistry (2020), 85(17), 11181-11189.

1.2 Reagents: Ammonium chloride
Solvents: Water; 0 °C

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



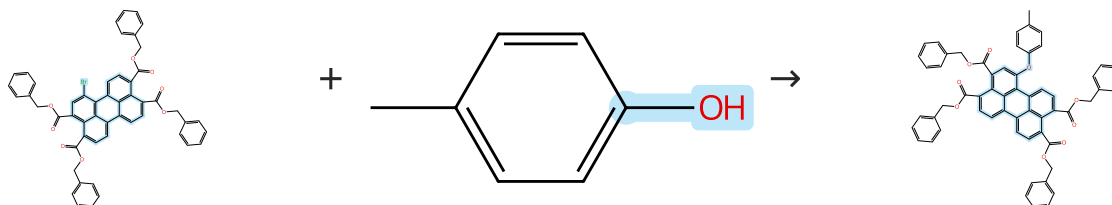
Suppliers (90)

Suppliers (90)

31-170-CAS-22914103 Steps: 1 Yield: 100% 1.1 Reagents: Potassium <i>tert</i>-butoxide Solvents: 1,2-Dimethoxyethane; 20 min, rt 1.2 Catalysts: Copper(1+), tetrakis(acetonitrile)-, (<i>T</i>-4)-, 1,1,1-trifluoromethanesulfonate (1:1) Solvents: 1,2-Dimethoxyethane; overnight, 100 °C Experimental Protocols	¹⁹F- and ¹⁸F-arene deoxyfluorination via organic photoredox-catalysed polarity-reversed nucleophilic aromatic substitution By: Tay, Nicholas E. S.; et al Nature Catalysis (2020), 3(9), 734-742.
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Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%

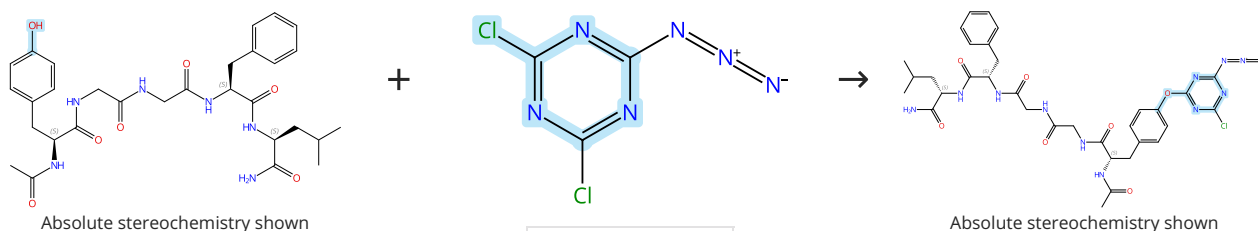


Suppliers (89)

31-170-CAS-23442286 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate, 18-Crown-6 Solvents: Dimethylformamide; rt; 3 h, 50 °C 1.2 Reagents: Water Experimental Protocols	Curved Perylene Diimides Fused with Seven- Membered Rings By: Fujimoto, Keisuke; et al Chemistry - An Asian Journal (2021), 16(6), 690-695.
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Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



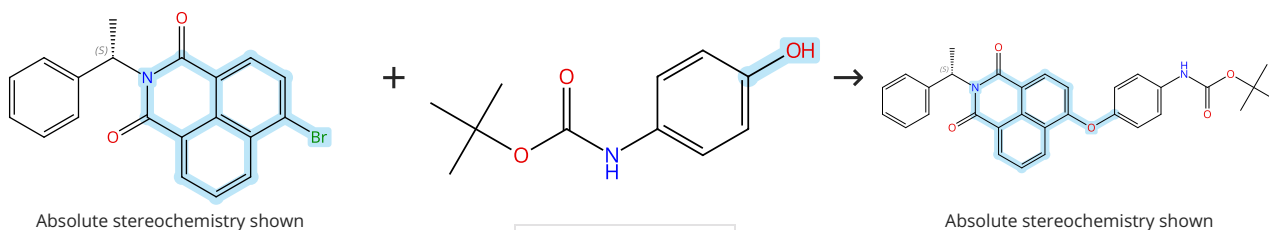
Suppliers (2)

Suppliers (4)

31-614-CAS-34441022 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Acetonitrile, Water; 30 min, > p H 8, 0 °C Experimental Protocols	Exploiting azido-dichloro-triazine as a linker for regioselective incorporation of peptides through their N, O, S functional groups By: Sharma, Anamika; et al Bioorganic Chemistry (2020), 104, 104334.
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Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%

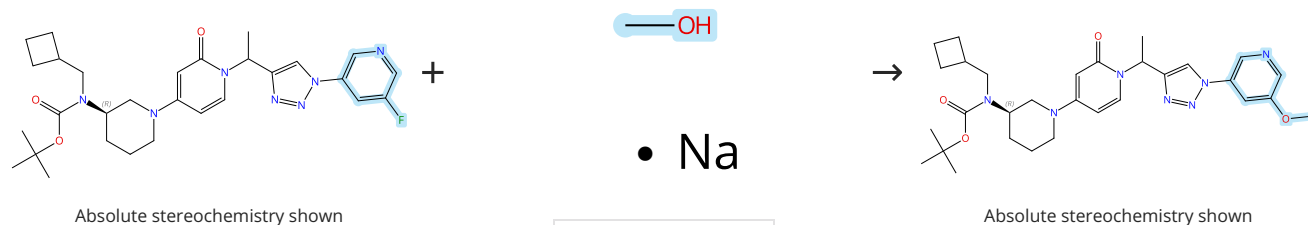


Suppliers (73)

31-614-CAS-35949565	Steps: 1 Yield: 100%	Pathway Complexity in Supramolecular Copolymerization and Blocky Star Copolymers by a Hetero-Seeding Effect
1.1 Solvents: Dimethylformamide; 12 h, 110 °C		By: Khanra, Payel; et al
Experimental Protocols		Journal of the American Chemical Society (2023), 145(9), 5270-5284.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%

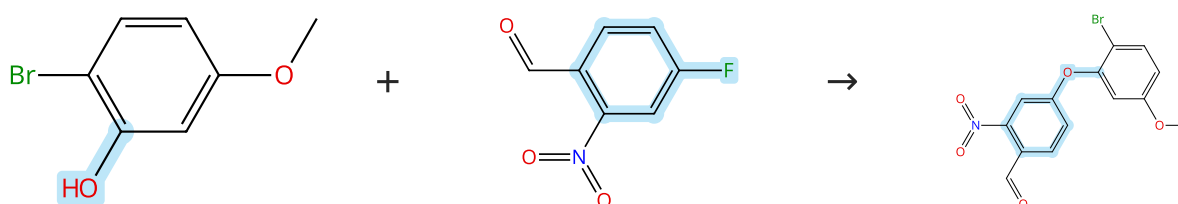


Suppliers (99)

31-614-CAS-44468517	Steps: 1 Yield: 100%	Discovery, Optimization, and Preclinical Pharmacology of E P652, a METTL3 Inhibitor with Efficacy in Liquid and Solid Tumor Models
1.1 Solvents: Methanol; 2.5 h, reflux		By: Dutheuil, Guillaume; et al
Experimental Protocols		Journal of Medicinal Chemistry (2025), 68(3), 2981-3003.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



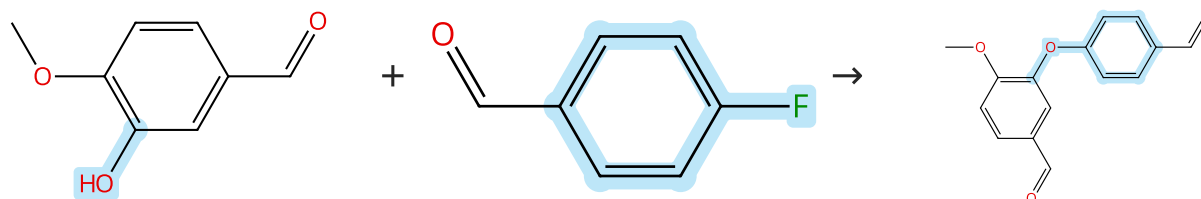
Suppliers (66)

Suppliers (69)

31-170-CAS-21519190	Steps: 1 Yield: 100%	Methoxy-Substituted Nitrodibenzofuran-Based Protecting Group with an Improved Two-Photon Action Cross-Section for Thiol Protection in Solid Phase Peptide Synthesis
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 6 h, rt; 12 h, 50 °C; 50 °C → rt		By: Bader, Taysir K.; et al
1.2 Reagents: Ammonium chloride Solvents: Water; rt		Journal of Organic Chemistry (2020), 85(3), 1614-1625.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (106)

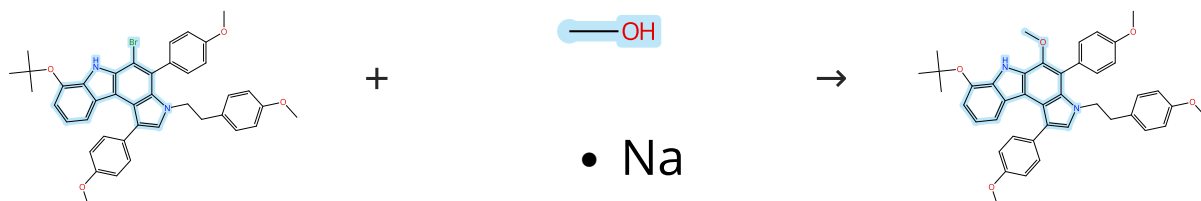
Suppliers (93)

Suppliers (4)

31-170-CAS-22912733	Steps: 1 Yield: 100%	Total synthesis of polymorphatin A, a macrocyclic bisbibenzyl with boat configured arenes
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 18 h, reflux		By: Almalki, Faisal A.; et al
Experimental Protocols		Tetrahedron (2020), 76(42), 131521.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%

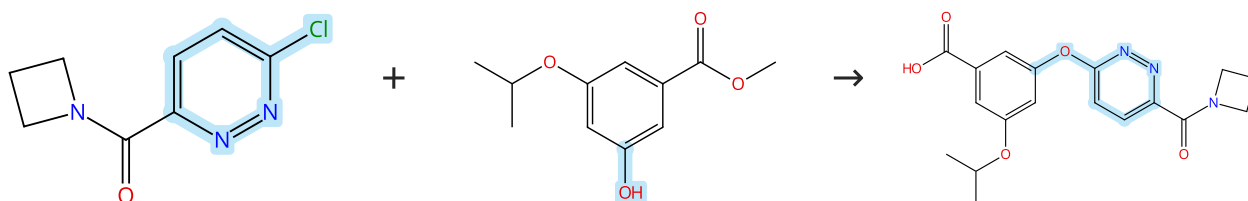


Suppliers (99)

31-170-CAS-22585895	Steps: 1 Yield: 100%	Total Synthesis of Dictyodendrins A-F by the Gold-Catalyzed Cascade Cyclization of Conjugated Diyne with Pyrrole
1.1 Reagents: Sodium Catalysts: Cuprous iodide Solvents: Methanol, Dimethylformamide; rt; 11 h, 80 °C		By: Matsuoka, Junpei; et al
Experimental Protocols		Chemistry - A European Journal (2020), 26(49), 11150-11157.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



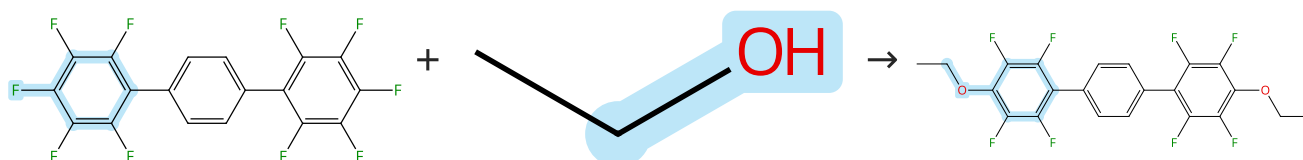
Suppliers (16)

Suppliers (49)

31-614-CAS-40832859	Steps: 1 Yield: 100%	Discovery of a Partial Glucokinase Activator Clinical Candidate: Diethyl ((3-(3-((5-(Azetidine-1-carbonyl)pyrazin-2-yl)oxy)-5-isopropoxybenzamido)-1H-pyrazol-1-yl)methyl)phosphonate (BMS-820132)
1.1 Reagents: Potassium carbonate Solvents: Acetonitrile; 18 h, rt → 80 °C; 80 °C → rt		By: Shi, Yan; et al
1.2 Solvents: Dimethyl sulfoxide; 2 h, 180 °C		Journal of Medicinal Chemistry (2022), 65(5), 4291-4317.
Experimental Protocols		

Scheme 31 (3 Reactions)

Steps: 1 Yield: 98-100%

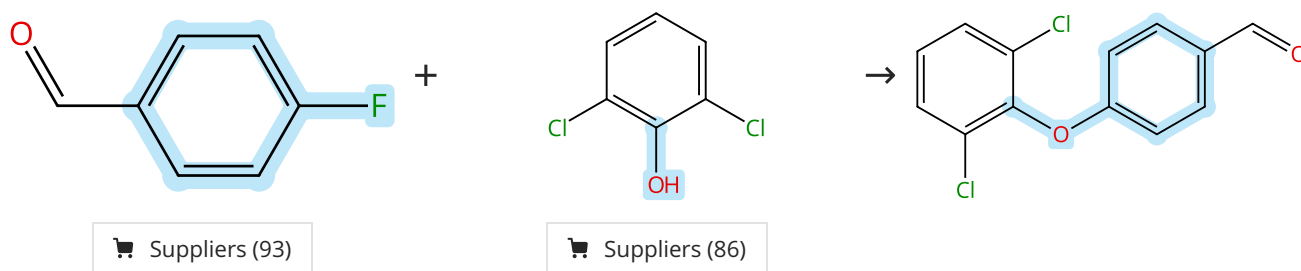


Suppliers (618)

31-614-CAS-24228473 Steps: 1 Yield: 100% 1.1 Reagents: Potassium <i>tert</i>-butoxide Solvents: Tetrahydrofuran Experimental Protocols	Synthesis, characterization, and self-assembly of fluorescent fluorine-containing liquid crystals By: Radwan, Ahmed S.; et al Luminescence (2021), 36(7), 1751-1760.
31-614-CAS-34014719 Steps: 1 Yield: 100% 1.1 Reagents: Potassium <i>tert</i>-butoxide Solvents: Dimethylformamide; 100 °C; 100 °C → 140 °C; cooled Experimental Protocols	Synthesis and characterization of novel fluoroterphenyls: self-assembly of low-molecular-weight fluorescent organogel By: Altoom, Naif Ghazi Luminescence (2021), 36(5), 1285-1299.
31-614-CAS-36920741 Steps: 1 Yield: 98% 1.1 Reagents: Potassium <i>tert</i>-butoxide Solvents: Dimethylformamide; heated Experimental Protocols	Copper(I)-catalysed synthesis of symmetrical perfluoro terphenyl analogues; fluorescence, antioxidant and molecular docking studies By: Alqahtani, Alaa M.; et al Luminescence (2023), 38(8), 1440-1448.

Scheme 32 (1 Reaction)

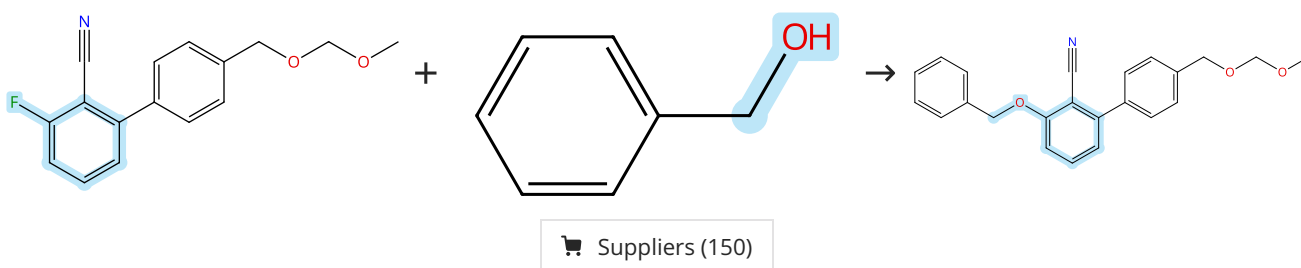
Steps: 1 Yield: 100%



31-614-CAS-34652764 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt 1.2 Reagents: Water; rt Experimental Protocols	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide By: Chen, Tao; et al Letters in Drug Design & Discovery (2022), 19(4), 263-268.
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Scheme 33 (1 Reaction)

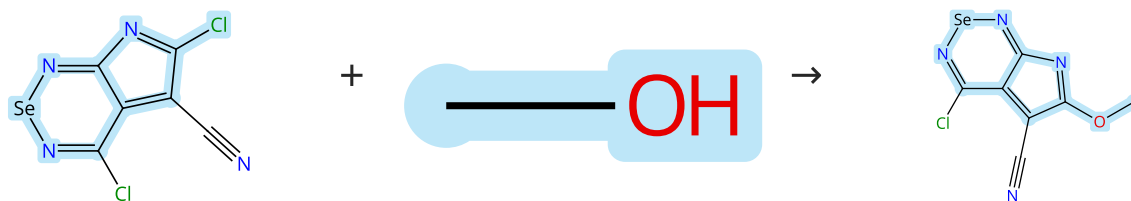
Steps: 1 Yield: 100%



31-614-CAS-40725179 Steps: 1 Yield: 100% 1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 0 °C → rt; 8 h, rt 1.2 Solvents: Ethyl acetate; rt Experimental Protocols	Rapid Evolution of a Fragment-like Molecule to Pan-Metallo-Beta-Lactamase Inhibitors: Initial Leads toward Clinical Candidates By: Mandal, Mihirbaran; et al Journal of Medicinal Chemistry (2022), 65(24), 16234-16251.
--	---

Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (444)

31-614-CAS-41348226

Steps: 1 Yield: 100%

Reaction of tetracyanoethylene with diselenium dichloride: a route to pyrrolo[2,3-c][1,2,6]selenadiazines

1.1 10 min, rt → reflux

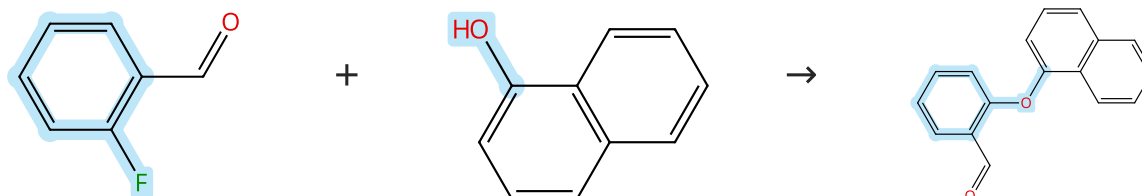
Experimental Protocols

By: Laurent, Cory; et al

New Journal of Chemistry (2024), 48(3), 1206-1211.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (127)

Suppliers (5)

31-614-CAS-36277493

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

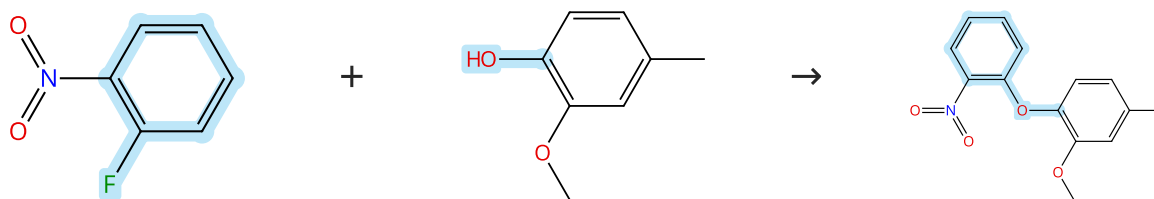
Experimental Protocols

By: Can Uskup, Hacer; et al

ACS Omega (2023), 8(15), 14047-14052.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (103)

Supplier (1)

31-614-CAS-42944372

Steps: 1 Yield: 100%

S_NAr Reactions Using Continuous Plug Flow...in Aqueous Biphasic Media

 1.1 **Reagents:** Potassium hydroxide
Catalysts: Tetrabutylammonium bromide
Solvents: 2-Methyltetrahydrofuran, Water; 10 min, 95 °C

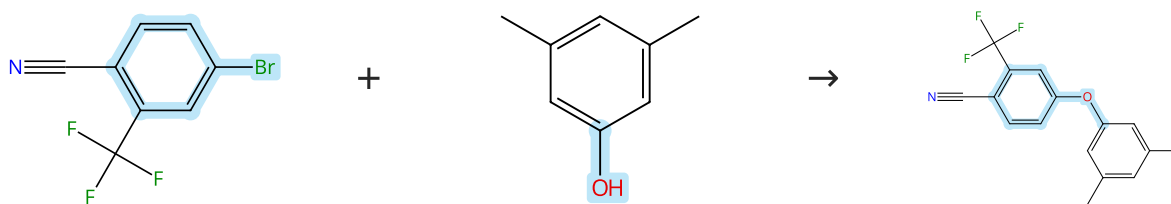
Experimental Protocols

By: Wong, Madison J.; et al

ACS Sustainable Chemistry & Engineering (2024), 12(52), 18725-18734.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (89)

Suppliers (2)

31-614-CAS-43350517

Steps: 1 Yield: 100%

Bidentate NHC-Containing Ligands for Copper Catalyzed Synthesis of Functionalised Diaryl Ethers

By: Richards, Ethan M.; et al

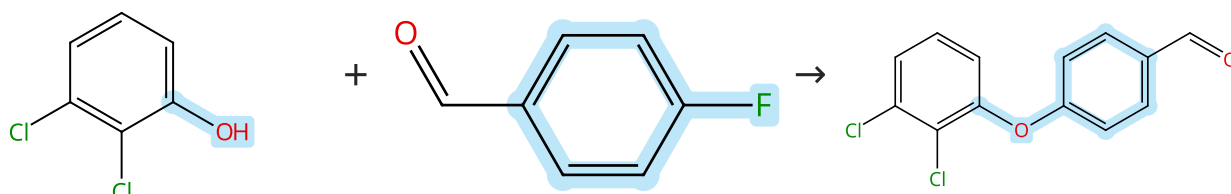
Advanced Synthesis & Catalysis (2025), 367(6), e202400909.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Cuprous iodide, 1-*H*-imidazolium, 3-[4-(trifluoromethyl)-2-pyridinyl]-1-(2,4,6-trimethylphenyl)-, bromide (1:1)
Solvents: Acetonitrile; 20 h, 110 °C

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (59)

Suppliers (93)

Suppliers (7)

31-614-CAS-34652759

Steps: 1 Yield: 100%

Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

By: Chen, Tao; et al

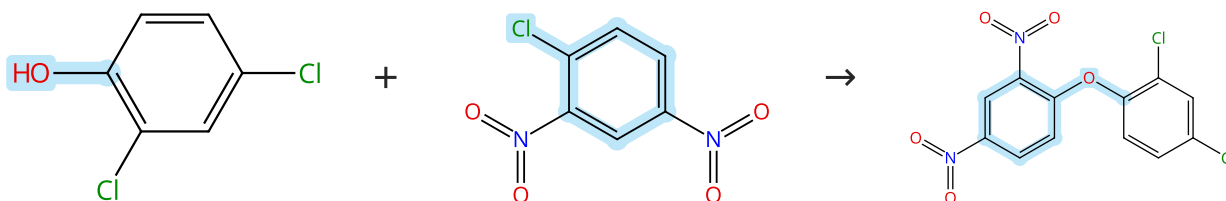
Letters in Drug Design & Discovery (2022), 19(4), 263-268.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt
 1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (102)

Suppliers (75)

Suppliers (4)

31-614-CAS-25139582

Steps: 1 Yield: 100%

Chemical validation of a druggable site on Hsp27/H SPB1 using in silico solvent mapping and biophysical methods

By: Makley, Leah N.; et al

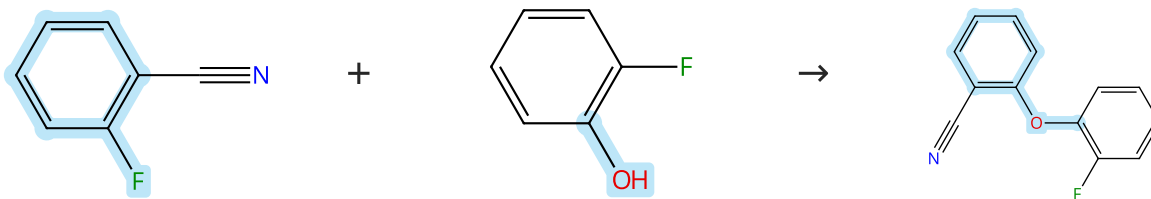
Bioorganic & Medicinal Chemistry (2021), 34, 115990.

- 1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; rt
 1.2 **Reagents:** Hydrochloric acid
Solvents: Water; acidified

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (101)

Suppliers (24)

31-614-CAS-32528351

Steps: 1 Yield: 100%

Access to ortho-Hydroxyphenyl Ketimines via Imine Anion-Mediated Smiles Rearrangement

1.1 Reagents: Potassium hydroxide
Solvents: Dimethyl sulfoxide; 14.5 h, 90 °C; 90 °C → rt

By: Jinno, Shunki; et al

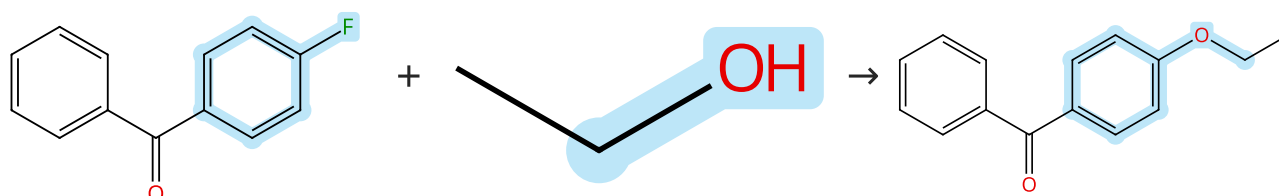
1.2 Reagents: Water; rt

Organic Letters (2022), 24(23), 4140-4144.

Experimental Protocols

Scheme 41 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (71)

Suppliers (618)

Suppliers (47)

31-170-CAS-22584338

Steps: 1 Yield: 100%

Aryl Ether Syntheses via Aromatic Substitution Proceeding under Mild Conditions

1.1 Reagents: Potassium *tert*-butoxide
Solvents: Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 3 h, 0 °C

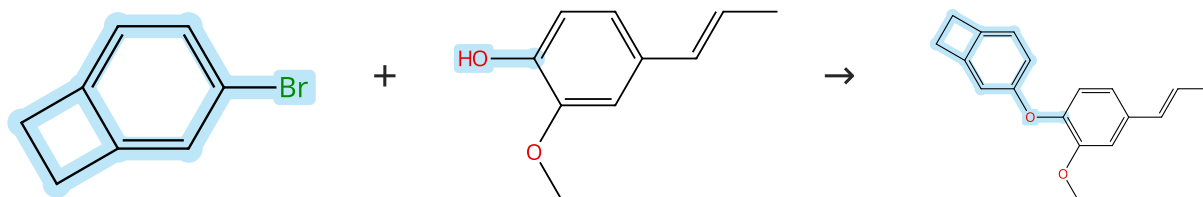
By: Ando, Shin; et al

1.2 Reagents: Ammonium chloride
Solvents: Water; 0 °C

Journal of Organic Chemistry (2020), 85(17), 11181-11189.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

Suppliers (92)

31-614-CAS-34266112

Steps: 1 Yield: 100%

Biobased Low-k Polymers at High Frequency Derived from Isoeugenol

1.1 Reagents: Sodium ascorbate, Cesium carbonate
Catalysts: Cuprous chloride, *N*¹,*N*²-Bis(2-thienylmethyl)ethanediamide
Solvents: 1,4-Dioxane; 6 h, 100 °C

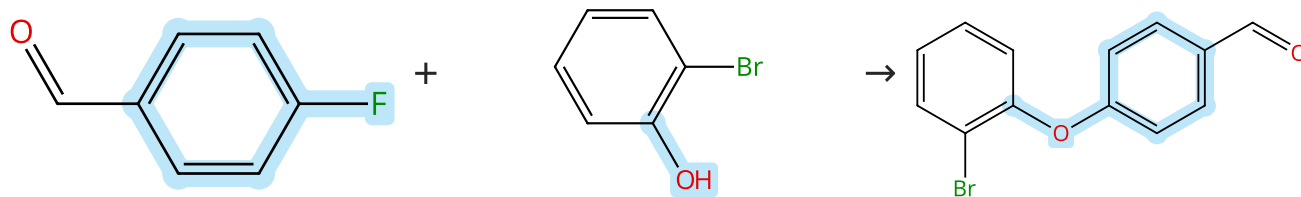
By: Liu, Fengping; et al

ACS Applied Polymer Materials (2022), 4(10), 7173-7181.

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (82)

Suppliers (27)

31-614-CAS-34652755

Steps: 1 Yield: 100%

Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt

By: Chen, Tao; et al

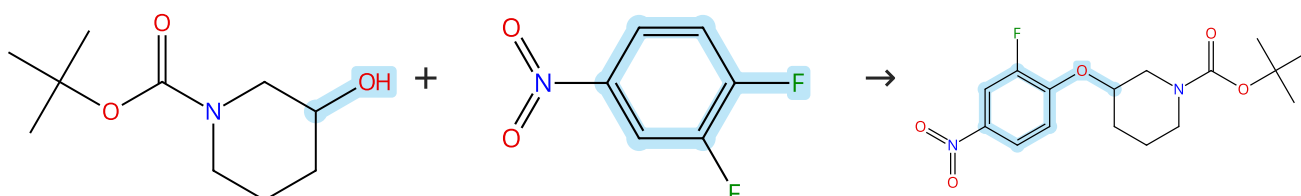
1.2 Reagents: Water; rt

Letters in Drug Design & Discovery (2022), 19(4), 263-268.

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (88)

Supplier (1)

31-614-CAS-40701340

Steps: 1 Yield: 100%

Discovery and Structure-Activity Relationship of Cadazolid: A First-In-Class Quinoxolidinone Antibiotic for the Treatment of Clostridioides difficile Infection

1.1 Reagents: Potassium hydroxide, Sodium hydroxide

Solvents: Dichloromethane, Water; overnight, rt

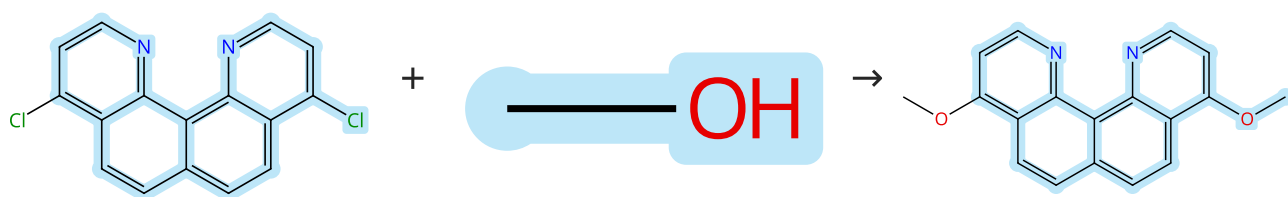
By: Rueedi, Georg; et al

Experimental Protocols

Journal of Medicinal Chemistry (2024), 67(11), 9465-9484.

Scheme 45 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (4)

Suppliers (444)

Supplier (1)

31-170-CAS-22649079

Steps: 1 Yield: 100%

Synthesis and Basicity Studies of Quinolino[7,8-h]quinoline Derivatives

1.1 Reagents: Sodium methoxide

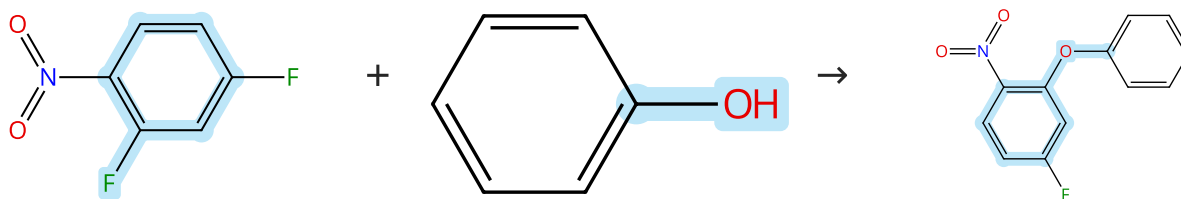
Solvents: Methanol

By: Rowlands, Gareth J.; et al

Journal of Organic Chemistry (2020), 85(17), 11297-11308.

Scheme 46 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (207)

Suppliers (16)

31-170-CAS-22902533

Steps: 1 Yield: 100%

The Effect of the Leaving Group in N- Heterocyclic Carbene-Catalyzed Nucleophilic Aromatic Substitution Reactions

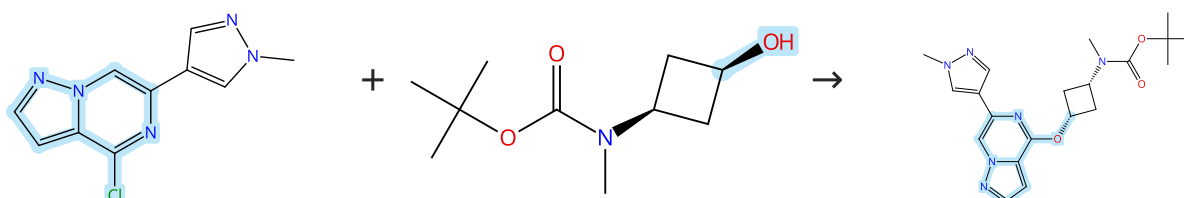
By: Yasui, Kosuke; et al

Bulletin of the Chemical Society of Japan (2020), 93(12), 1424-1429.

1.1 Reagents: Potassium *tert*-butoxide
Solvents: Toluene; 0 °C; 12 h, rt

Scheme 47 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (33)

Relative stereochemistry shown

Suppliers (50)

Relative stereochemistry shown

31-614-CAS-40236732

Steps: 1 Yield: 100%

Discovery and Preclinical Characterization of BIIB129, a Covalent, Selective, and Brain-Penetrant BTK Inhibitor for the Treatment of Multiple Sclerosis

By: Himmelbauer, Martin K.; et al

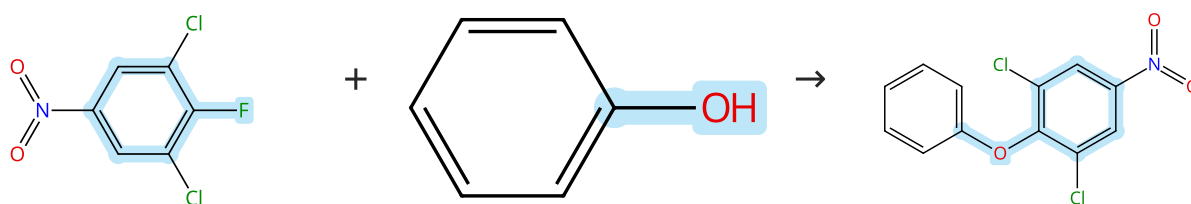
Journal of Medicinal Chemistry (2024), 67(10), 8122-8140.

1.1 Reagents: Potassium *tert*-butoxide
Solvents: Tetrahydrofuran; 5 min, rt

Experimental Protocols

Scheme 48 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (62)

Suppliers (207)

31-170-CAS-23043265

Steps: 1 Yield: 100%

Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling

By: Fujii, Shinya; et al

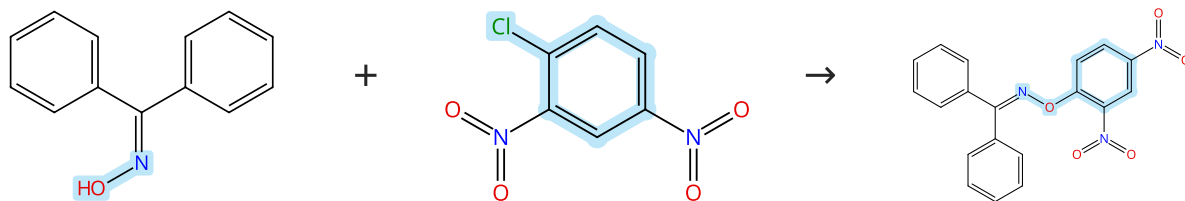
Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 17 h, rt

Experimental Protocols

Scheme 49 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (55)

Suppliers (75)

Suppliers (4)

31-614-CAS-42413493

Steps: 1 Yield: 100%

Photoreductive N-N Homocoupling Catalyzed by a Superphotoreductant

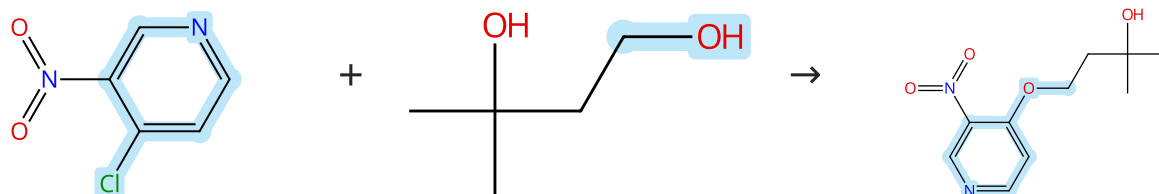
1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 1 h, 30 °C

By: Wang, Yi-Ping; et al

Journal of Organic Chemistry (2024), 89(22), 16804-16808.

Scheme 50 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (69)

31-614-CAS-36012342

Steps: 1 Yield: 100%

Structure-activity relationship (SAR) studies on substituted N-(pyridin-3-yl)-2-amino-isonicotinamides as highly potent and selective glycogen synthase kinase-3 (GSK-3) inhibitors

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 17 h, 75 °C

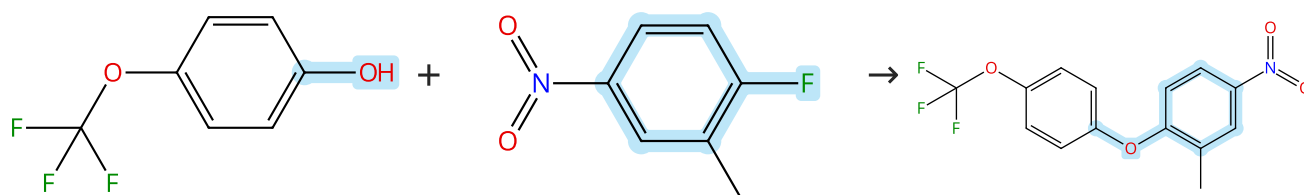
By: Luo, Guanglin; et al

Bioorganic & Medicinal Chemistry Letters (2023), 81, 129143.

Experimental Protocols

Scheme 51 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (84)

Supplier (1)

31-614-CAS-41227168

Steps: 1 Yield: 100%

MPoMA protects against lung epithelial cell injury via p65 degradation

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 22 h, 110 °C; 110 °C → rt

By: Lee, Soheun; et al

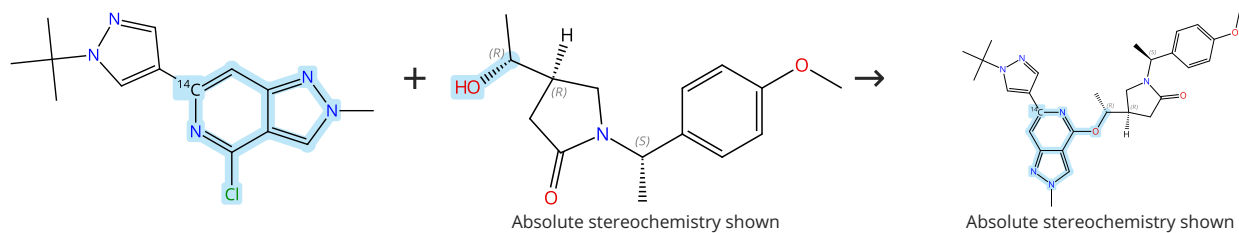
Biomedicine & Pharmacotherapy (2024), 175, 116674.

1.2 **Solvents:** Ethyl acetate, Water; rt

Experimental Protocols

Scheme 52 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (4)

31-614-CAS-36423720

Steps: 1 Yield: 100%

1.1 Reagents: Potassium hydroxide
Solvents: *N*-Methyl-2-pyrrolidone; rt → 70 °C; 36 h, 70 °C

Experimental Protocols

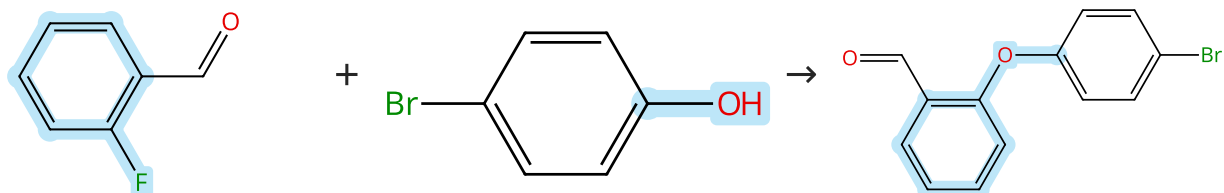
Synthesis of BI 894416 and BI 1342561, two potent and selective spleen tyrosine kinase inhibitors, labeled with carbon 14 and with deuterium

By: Latli, Bachir; et al

Journal of Labelled Compounds and Radiopharmaceuticals (2023), 66(4-6), 155-168.

Scheme 53 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (82)

Suppliers (30)

31-614-CAS-36277495

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

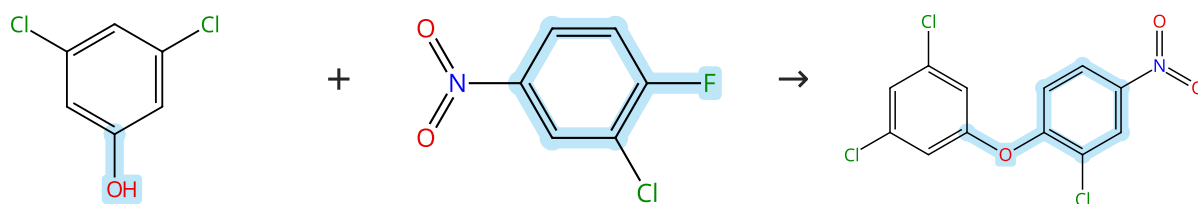
Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

By: Can Uskup, Hacer; et al

ACS Omega (2023), 8(15), 14047-14052.

Scheme 54 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (92)

Suppliers (86)

31-614-CAS-31980142

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; rt

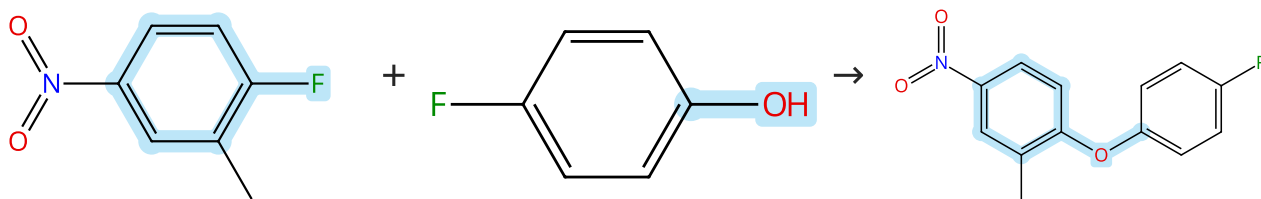
Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

Scheme 55 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

Suppliers (86)

31-614-CAS-41227151

Steps: 1 Yield: 100%

MPoMA protects against lung epithelial cell injury via p65 degradation

1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 19 h, 110 °C; 110 °C → rt

By: Lee, Soheun; et al

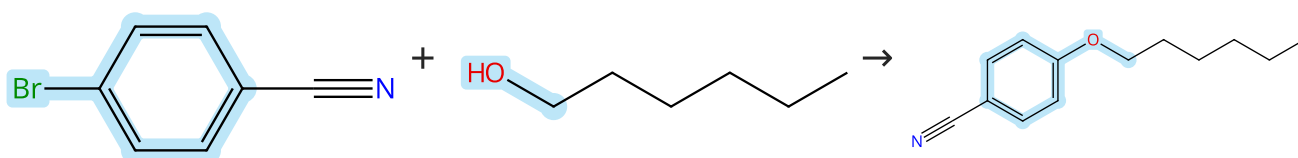
1.2 Solvents: Ethyl acetate, Water; rt

Biomedicine & Pharmacotherapy (2024), 175, 116674.

Experimental Protocols

Scheme 56 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (100)

Suppliers (46)

31-614-CAS-32106643

Steps: 1 Yield: 100%

Discovery and characterization of a novel perylenephore ductant for the activation of aryl halides

1.1 Reagents: Diisopropylethylamine

Catalysts: Nickel sulfate (NiSO₄), 4,4'-Dimethoxy-2,2'-bipyridine, 2767997-30-6

Solvents: Acetonitrile; 24 h, rt

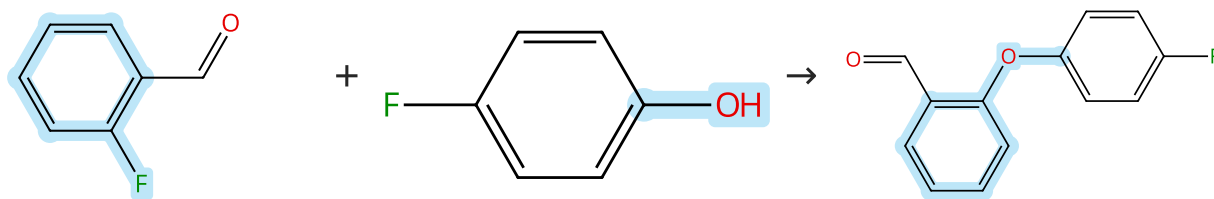
By: Li, Min; et al

Journal of Catalysis (2021), 399, 111-120.

1.2 Reagents: Water; rt

Scheme 57 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (86)

Suppliers (44)

31-614-CAS-36277497

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 2 h, 180 °C

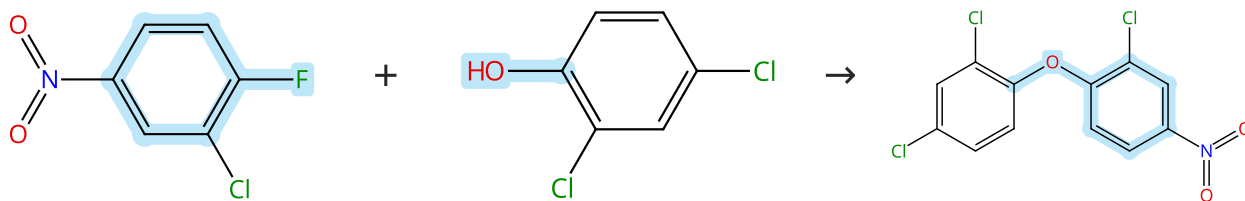
By: Can Uskup, Hacer; et al

Experimental Protocols

ACS Omega (2023), 8(15), 14047-14052.

Scheme 58 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (102)

Suppliers (20)

31-614-CAS-31980141

Steps: 1 Yield: 100%

Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

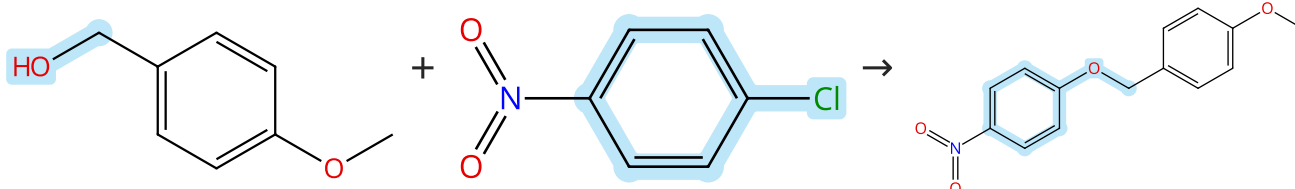
By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt

Scheme 59 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (106)

Suppliers (65)

Suppliers (13)

31-170-CAS-22584349

Steps: 1 Yield: 100%

Aryl Ether Syntheses via Aromatic Substitution Proceeding under Mild Conditions

By: Ando, Shin; et al

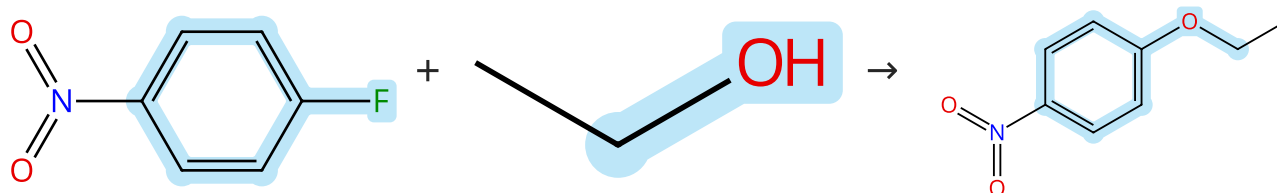
Journal of Organic Chemistry (2020), 85(17), 11181-11189.

1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 2 h, 0 °C

1.2 **Reagents:** Ammonium chloride
Solvents: Water; 0 °C

Scheme 60 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (100)

Suppliers (618)

Suppliers (37)

31-170-CAS-22584331

Steps: 1 Yield: 100%

Aryl Ether Syntheses via Aromatic Substitution Proceeding under Mild Conditions

By: Ando, Shin; et al

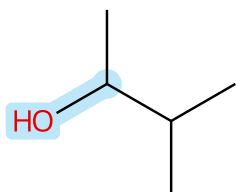
Journal of Organic Chemistry (2020), 85(17), 11181-11189.

1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 2 h, 0 °C

1.2 **Reagents:** Ammonium chloride
Solvents: Water; 0 °C

Scheme 61 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (43)

+

 Multi-component
structure image
available in CAS
SciFinder

→

 Multi-component
structure image
available in CAS
SciFinder

Supplier (1)

31-170-CAS-23461703

Steps: 1 Yield: 100%

1.1 Solvents: Acetonitrile-*d*₃, DMSO-*d*₆; 15 h, 40 °C

Experimental Protocols

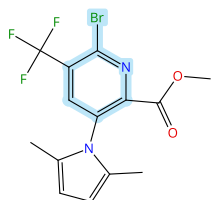
2-Fluoro-5-nitrophenyldiazonium: A Novel Sanger-Type Reagent for the Versatile Functionalization of Alcohols

By: Fischer, Oliver; et al

Chemistry - A European Journal (2021), 27(17), 5417-5421.

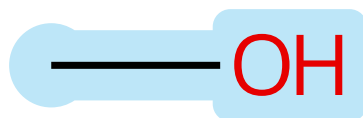
Scheme 62 (1 Reaction)

Steps: 1 Yield: 100%



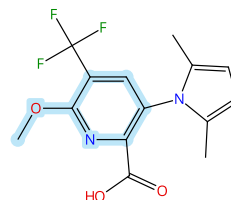
Supplier (1)

+



Suppliers (444)

→



Suppliers (2)

31-614-CAS-32518679

Steps: 1 Yield: 100%

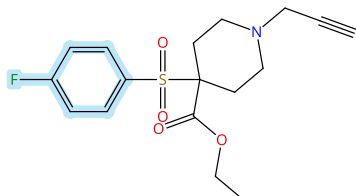
1.1 Reagents: Sodium hydroxide
Solvents: Methanol, Water; 3 h, 90 °C1.2 Reagents: Hydrochloric acid
Solvents: Water; pH 1, rt
Discovery of Icenticaftor (QBW251), a Cystic Fibrosis Transmembrane Conductance Regulator Potentiator with Clinical Efficacy in Cystic Fibrosis and Chronic Obstructive Pulmonary Disease

By: Grand, Darren Le; et al

Journal of Medicinal Chemistry (2021), 64(11), 7241-7260.

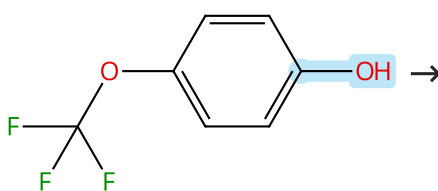
Scheme 63 (1 Reaction)

Steps: 1 Yield: 100%



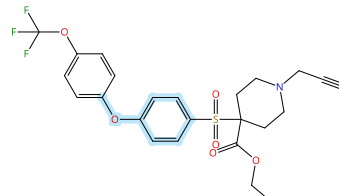
Supplier (1)

+



Suppliers (93)

→



31-614-CAS-40909850

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 19 h, 90 °C

Experimental Protocols

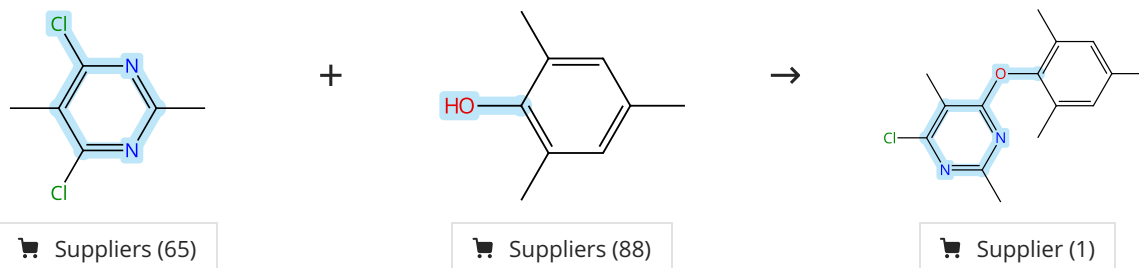
Carborane-Containing Matrix Metalloprotease (MMP) Ligands as Candidates for Boron Neutron-Capture Therapy (BNCT)

By: Lutz, Marlon R. Jr.; et al

ChemMedChem (2020), 15(20), 1897-1908.

Scheme 64 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-42944339

Steps: 1 Yield: 100%

S_NAr Reactions Using Continuous Plug Flow...in Aqueous Biphasic Media

By: Wong, Madison J.; et al

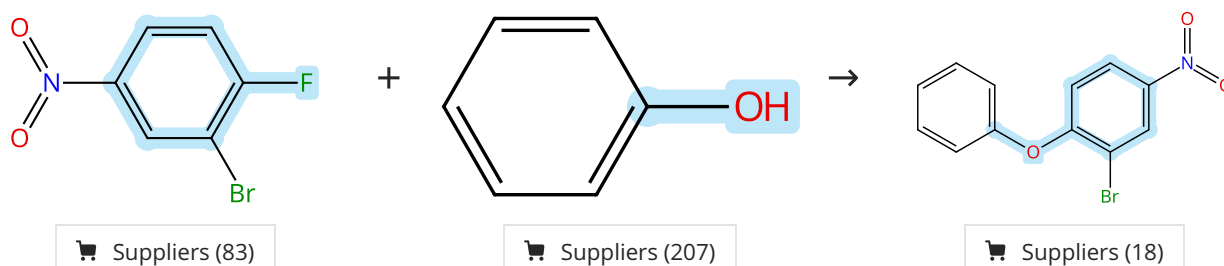
ACS Sustainable Chemistry & Engineering (2024), 12(52), 18725-18734.

1.1 **Reagents:** Potassium hydroxide
Catalysts: Tetrabutylammonium bromide
Solvents: 2-Methyltetrahydrofuran, Water; 10 min, 95 °C

Experimental Protocols

Scheme 65 (2 Reactions)

Steps: 1 Yield: 100%



31-614-CAS-31980156

Steps: 1 Yield: 100%

Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt

31-170-CAS-23043031

Steps: 1 Yield: 100%

Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling

By: Fujii, Shinya; et al

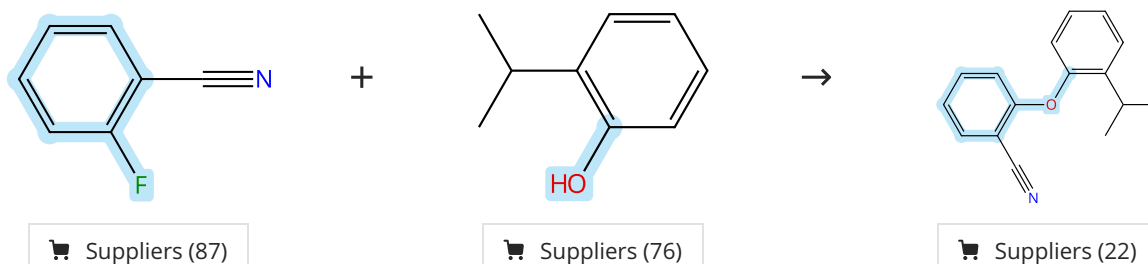
Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 17 h, rt

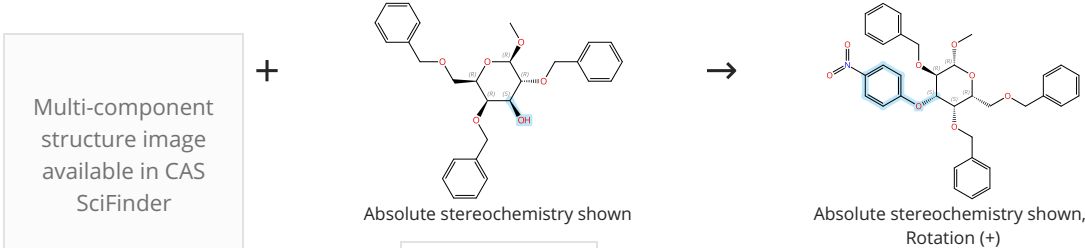
Experimental Protocols

Scheme 66 (1 Reaction)

Steps: 1 Yield: 100%



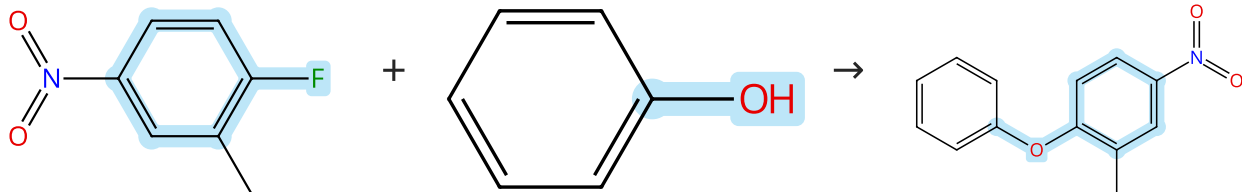
31-614-CAS-32528352	Steps: 1 Yield: 100%	Access to ortho-Hydroxyphenyl Ketimines via Imine Anion-Mediated Smiles Rearrangement
1.1 Reagents: Potassium hydroxide Solvents: Dimethyl sulfoxide; 14.5 h, 90 °C; 90 °C → rt		By: Jinno, Shunki; et al
1.2 Reagents: Water; rt		Organic Letters (2022), 24(23), 4140-4144.
Experimental Protocols		

Scheme 67 (1 Reaction)	Steps: 1 Yield: 100%
 Multi-component structure image available in CAS SciFinder Absolute stereochemistry shown Absolute stereochemistry shown, Rotation (+)	
 Suppliers (42)	 Supplier (1)

31-170-CAS-23463429	Steps: 1 Yield: 100%	Synthesis and Biological Studies of O3-Aryl Galactosides as Galectin Inhibitors
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: Toluene; 1 h, rt; > 2 h, rt		By: Kervefors, Gabriella; et al
Experimental Protocols		Helvetica Chimica Acta (2021), 104(2), e2000220.

Scheme 68 (1 Reaction)

Steps: 1 Yield: 100%



The reaction scheme shows the synthesis of 4-nitrophenyl phenyl ether. The first reactant is 1-fluoro-4-nitrobenzene, a benzene ring with a nitro group (NO₂) at the para position and a fluorine atom (F) at the other para position. The second reactant is phenol, a benzene ring with a hydroxyl group (OH). The product is 4-nitrophenyl phenyl ether, where the phenyl group from phenol is attached to the oxygen atom of the original phenol, and the nitro group remains at the para position.

 Suppliers (84)

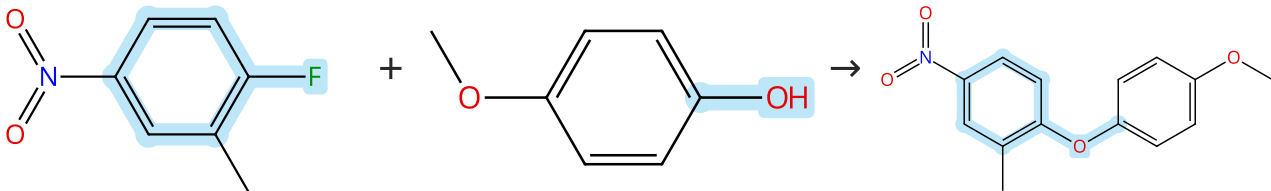
 Suppliers (207)

 Suppliers (17)

31-170-CAS-23043770	Steps: 1 Yield: 100%	Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 17 h, rt		By: Fujii, Shinya; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

Scheme 69 (1 Reaction)

Steps: 1 Yield: 100%



The reaction scheme shows the synthesis of 4-(4-methoxyphenoxy)nitrobenzene. The reactants are 1-fluoro-4-nitrobenzene (a benzene ring with a nitro group at position 1 and a fluorine atom at position 4) and 4-methoxyphenol (a benzene ring with a methoxy group at position 1 and a hydroxyl group at position 4). The product is 4-(4-methoxyphenoxy)nitrobenzene, where the fluorine atom of the first reactant is replaced by the 4-methoxyphenoxy group of the second reactant.

Suppliers (84)

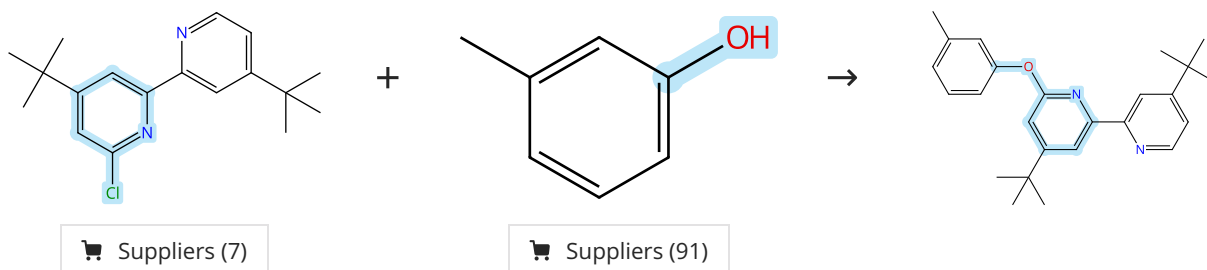
Suppliers (121)

Supplier (1)

31-614-CAS-41227155	Steps: 1 Yield: 100%	MPoMA protects against lung epithelial cell injury via p65 degradation
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide, Ethyl acetate; 22 h, 110 °C; 110 °C → rt		By: Lee, Soheun; et al
1.2 Solvents: Water; rt		Biomedicine & Pharmacotherapy (2024), 175, 116674.
Experimental Protocols		

Scheme 70 (1 Reaction)

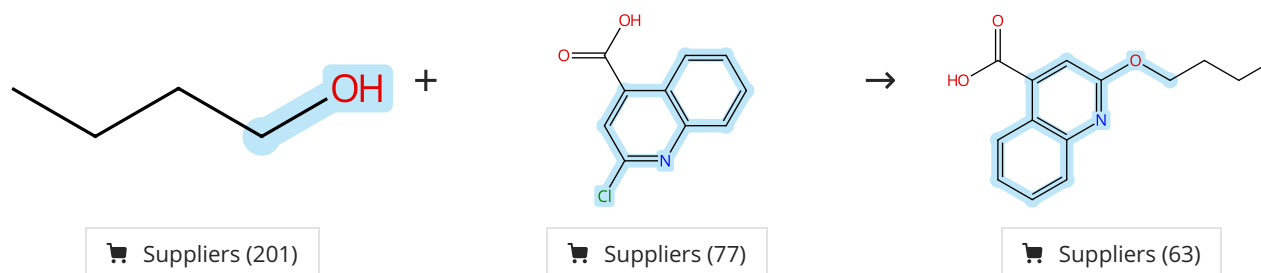
Steps: 1 Yield: 100%



31-170-CAS-23869194	Steps: 1 Yield: 100%	Bipyridine-Type Bidentate Auxiliary-Enabled Copper-Mediated C-H/C-H Biaryl Coupling of Phenols and 1,3-Azoles
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 21 h, 160 °C		By: Kajiwar, Rikuo; et al
1.2 Solvents: Water		Organic Letters (2021), 23(14), 5405-5409.
Experimental Protocols		

Scheme 71 (1 Reaction)

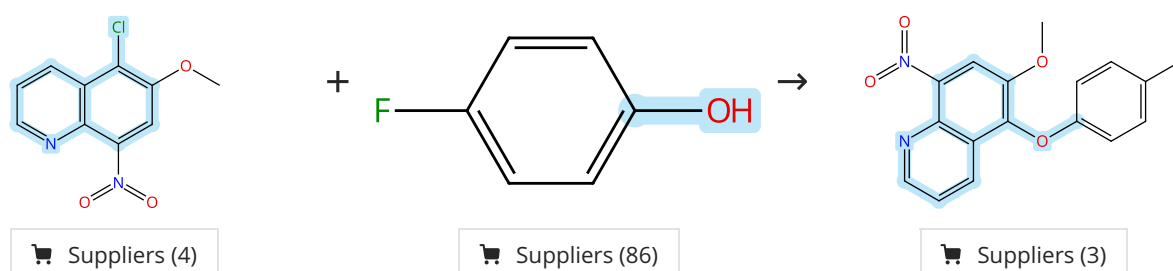
Steps: 1 Yield: 100%



31-614-CAS-41986241	Steps: 1 Yield: 100%	Development of Tailless Homologue Receptor (TLX) Agonist Chemical Tools
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 1 h, rt		By: Hank, Emily C.; et al
1.2 Solvents: Dimethylformamide; rt; overnight, 80 °C; 80 °C → rt		Journal of Medicinal Chemistry (2024), 67(18), 16598-16611.
1.3 Reagents: Hydrochloric acid Solvents: Water; acidified, rt		
Experimental Protocols		

Scheme 72 (1 Reaction)

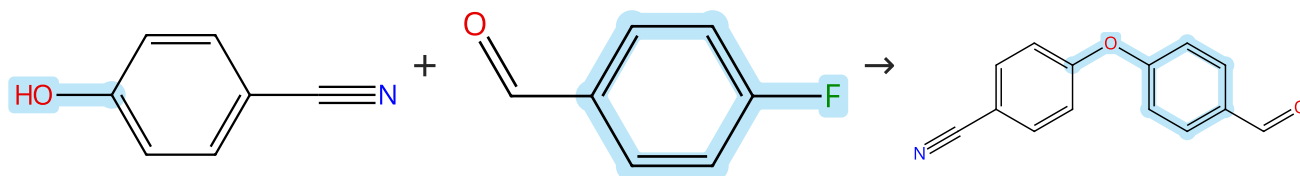
Steps: 1 Yield: 100%



31-614-CAS-24716243	Steps: 1 Yield: 100%	5-Phenoxy Primaquine Analogs and the Tetraoxane Hybrid as Antimalarial Agents
1.1 Reagents: Lithium hydroxide Solvents: Dimethyl sulfoxide; rt; 4 h, 100 °C		By: Jansongsaeng, Somrudee; et al
1.2 Reagents: Water		Molecules (2021), 26(13), 3991.
Experimental Protocols		

Scheme 73 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (90)

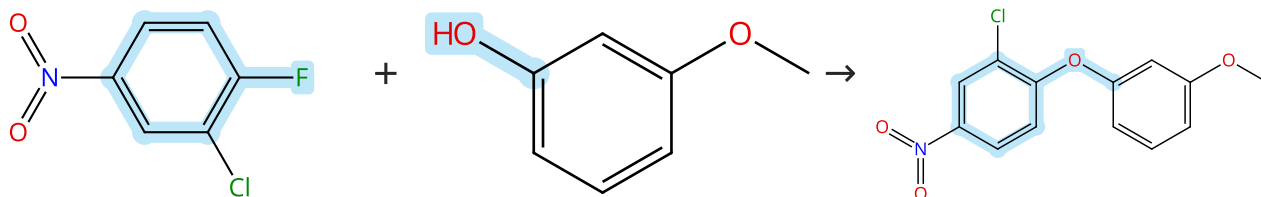
Suppliers (93)

Suppliers (68)

31-614-CAS-34652762	Steps: 1 Yield: 100%	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt		By: Chen, Tao; et al
1.2 Reagents: Water; rt		Letters in Drug Design & Discovery (2022), 19(4), 263-268.
Experimental Protocols		

Scheme 74 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

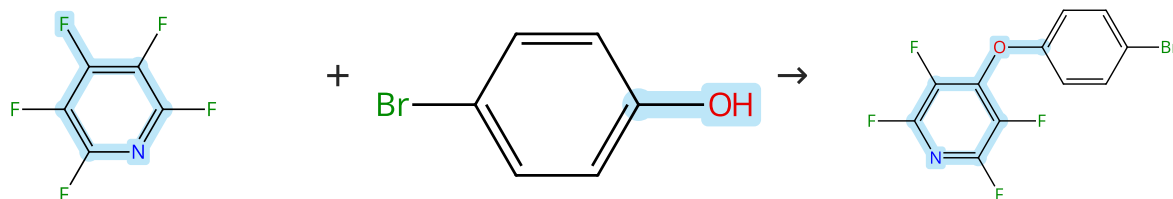
Suppliers (92)

Suppliers (10)

31-170-CAS-23045842	Steps: 1 Yield: 100%	Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 17 h, rt		By: Fujii, Shinya; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

Scheme 75 (2 Reactions)

Steps: 1 Yield: 97-100%



Suppliers (71)

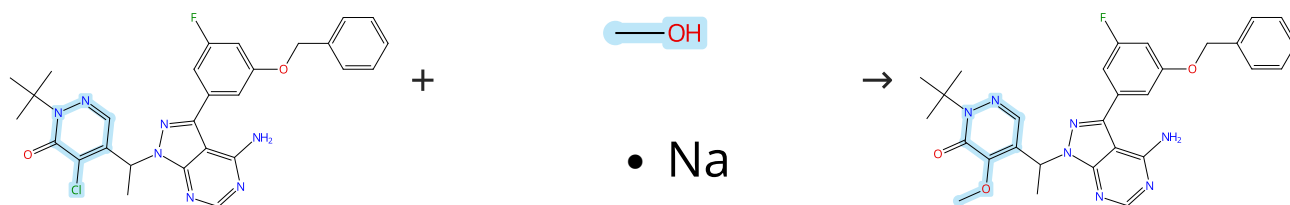
Suppliers (82)

Suppliers (3)

31-614-CAS-39093807 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Acetonitrile; 12 h, 25 °C 1.2 Reagents: Hydrochloric acid Experimental Protocols	Total Synthesis of the Natural Chalcone Lophirone E, Synthetic Studies toward Benzofuran and Indole-Based Analogues, and Investigation of Anti-Leishmanial Activity By: Pozzetti, Luca; et al Molecules (2022), 27(2), 463.
31-614-CAS-40257361 Steps: 1 Yield: 97% 1.1 Reagents: Cesium carbonate Solvents: Acetonitrile; 16 h, rt Experimental Protocols	Mild synthesis of 2,3,5,6-tetrafluoropyridine ethers and their reactivity toward imidazole and pyrazole By: Weeks, Nathan J.; et al Journal of Fluorine Chemistry (2024), 276, 110280.

Scheme 76 (1 Reaction)

Steps: 1 Yield: 100%

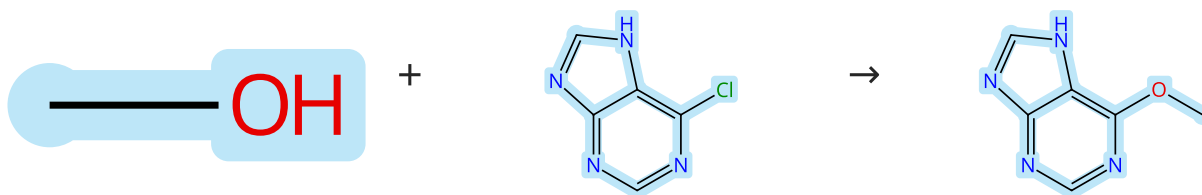


Suppliers (99)

31-614-CAS-41301593 Steps: 1 Yield: 100% 1.1 Solvents: Methanol, 1,4-Dioxane; 1.5 h, rt 1.2 Reagents: Water; rt Experimental Protocols	Discovery and Optimization of Pyridazinones as PI3Kδ Selective Inhibitors for Administration by Inhalation By: Bruno, Paolo; et al Journal of Medicinal Chemistry (2024), 67(13), 11103-11124.
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Scheme 77 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (444)

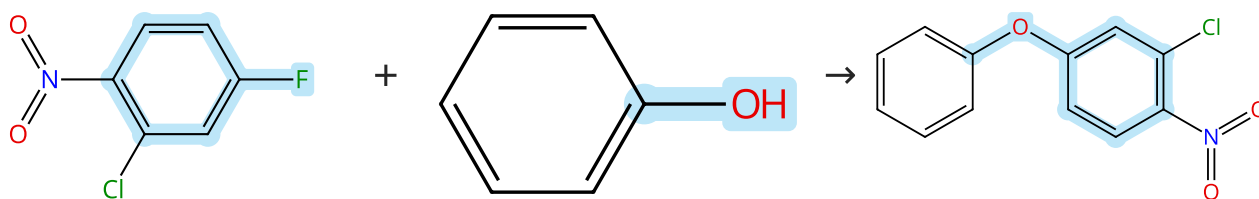
Suppliers (119)

Suppliers (77)

31-170-CAS-23143023 Steps: 1 Yield: 100% 1.1 Reagents: Sodium; 30 min, rt 1.2 rt → 60 °C; 2 h, 60 °C Experimental Protocols	Synthesis and Pharmacological Evaluation of Novel Non-nucleotide Purine Derivatives as P2X7 Antagonists for the Treatment of Neuroinflammation By: Calzaferri, Francesco; et al Journal of Medicinal Chemistry (2021), 64(4), 2272-2290.
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Scheme 78 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (74)

Suppliers (207)

Suppliers (4)

31-614-CAS-31980155

Steps: 1 Yield: 100%

Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

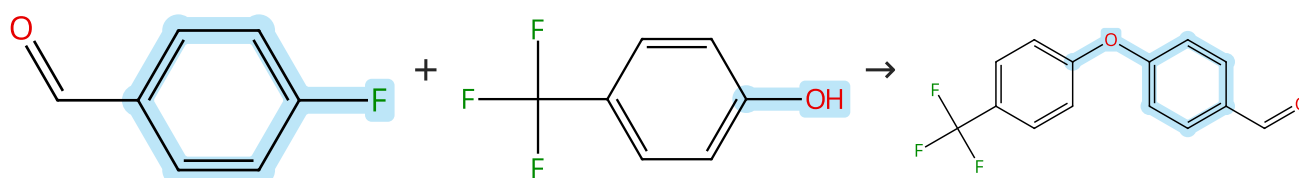
By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt

Scheme 79 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (96)

Suppliers (50)

31-614-CAS-34652760

Steps: 1 Yield: 100%

Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

By: Chen, Tao; et al

Letters in Drug Design & Discovery (2022), 19(4), 263-268.

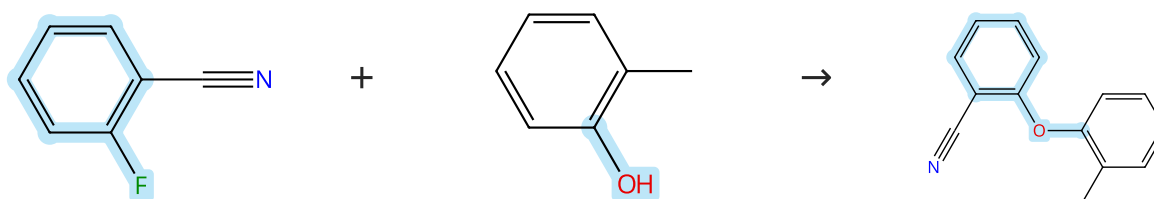
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 80 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (83)

Suppliers (25)

31-614-CAS-32528349

Steps: 1 Yield: 100%

Access to ortho-Hydroxyphenyl Ketimines via Imine Anion-Mediated Smiles Rearrangement

By: Jinno, Shunki; et al

Organic Letters (2022), 24(23), 4140-4144.

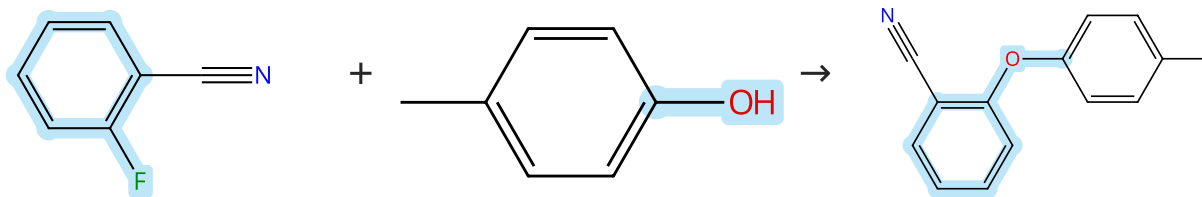
1.1 **Reagents:** Potassium hydroxide
Solvents: Dimethyl sulfoxide; 14.5 h, 90 °C; 90 °C → rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 81 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (89)

Suppliers (20)

31-614-CAS-38942833

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; 15 h, 90 °C; 90 °C → rt

1.2 Reagents: Water; rt

Experimental Protocols

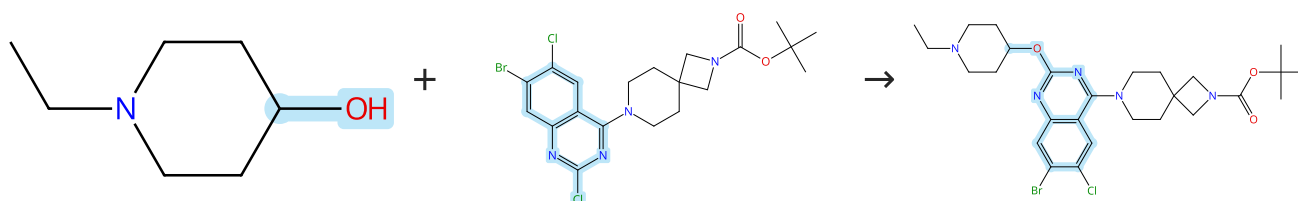
Update of the Imine-Anion-Mediated Smiles Rearrangement: Application to Migration of Electron-Neutral/Rich Aromatic Groups

By: Jinno, Shunki; et al

Synlett (2024), 35(13), 1565-1568.

Scheme 82 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (67)

31-614-CAS-39179678

Steps: 1 Yield: 100%

1.1 Reagents: Sodium hydride
Solvents: Tetrahydrofuran, *N*-Methyl-2-pyrrolidone; 15 min, 0 °C

1.2 2 h, rt

Experimental Protocols

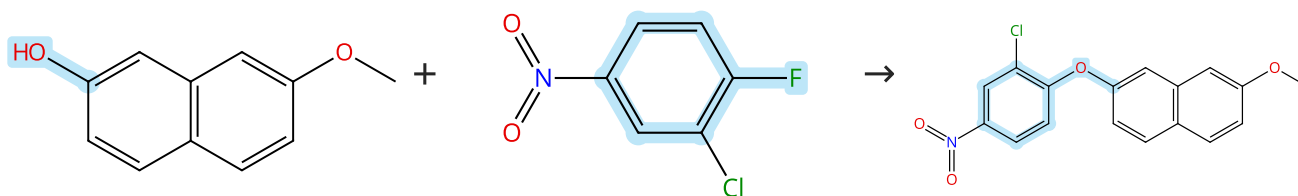
Discovery of ASP6918, a KRAS G12C inhibitor: Synthesis and structure-activity relationships of 1-{2,7-diazaspiro[3.5]non-2-yl}prop-2-en-1-one derivatives as covalent inhibitors with good potency and oral activity for the treatment of solid tumors

By: Imaizumi, Tomoyoshi; et al

Bioorganic & Medicinal Chemistry (2024), 98, 117581.

Scheme 83 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (75)

Suppliers (86)

31-614-CAS-31980145

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; rt

Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

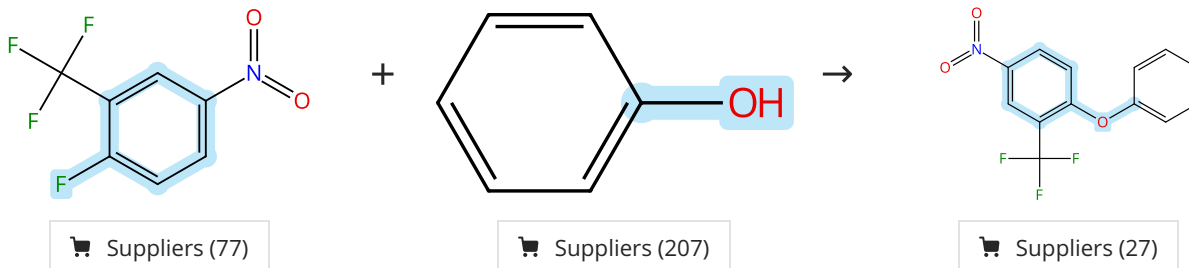
By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

31-170-CAS-23040943	Steps: 1 Yield: 100%	Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling By: Fujii, Shinya; et al Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 17 h, rt		
Experimental Protocols		

Scheme 84 (2 Reactions)

Steps: 1 Yield: 100%

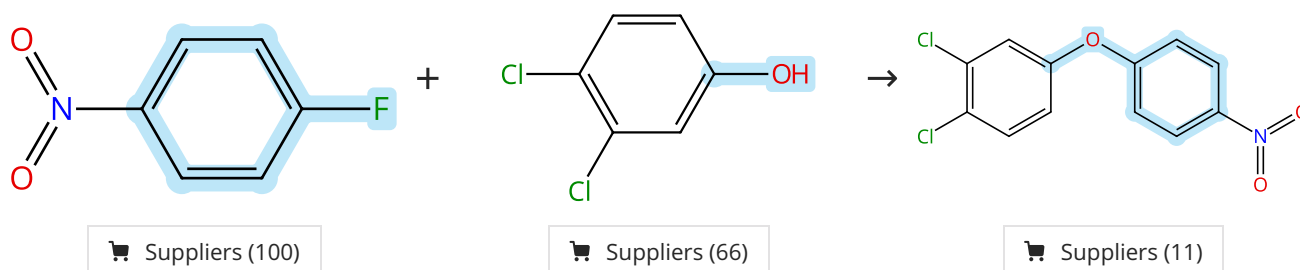


31-614-CAS-31980162	Steps: 1 Yield: 100%	Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents By: Fujii, Shinya; et al ChemMedChem (2021), 16(18), 2817-2822.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; rt		

31-170-CAS-23046247	Steps: 1 Yield: 100%	Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling By: Fujii, Shinya; et al Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 17 h, rt		
Experimental Protocols		

Scheme 85 (1 Reaction)

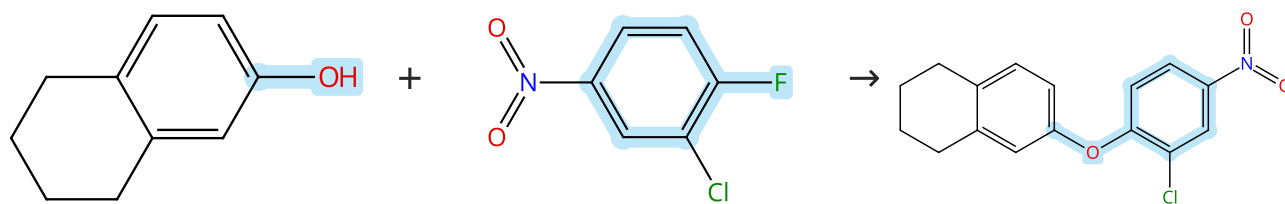
Steps: 1 Yield: 100%



31-614-CAS-37615374	Steps: 1 Yield: 100%	Discovery of N-Arylsulfonyl-Indole-2-Carboxamide Derivatives as Galectin-3 and Galectin-8 C-Terminal Domain Inhibitors By: Zhang, Haoming; et al ACS Medicinal Chemistry Letters (2023), 14(9), 1257-1265.
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 6 h, 50 °C		
Experimental Protocols		

Scheme 86 (2 Reactions)

Steps: 1 Yield: 98-100%



Suppliers (75)

Suppliers (86)

31-614-CAS-31980150

Steps: 1 Yield: 100%

Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

 1.1 Reagents: Potassium carbonate
 Solvents: Dimethylformamide; rt

31-170-CAS-23042849

Steps: 1 Yield: 98%

Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling

By: Fujii, Shinya; et al

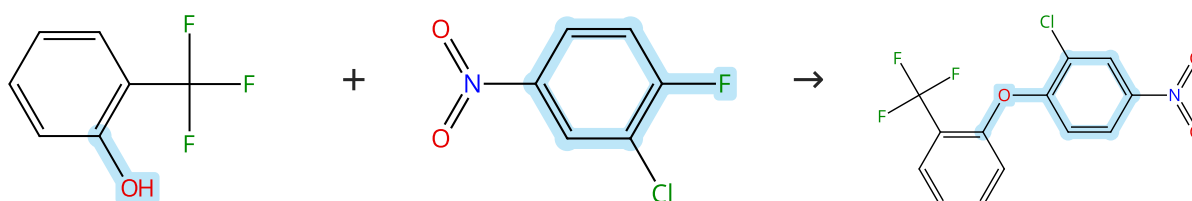
Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

 1.1 Reagents: Potassium carbonate
 Solvents: Dimethylformamide; 17 h, rt

Experimental Protocols

Scheme 87 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

Suppliers (86)

31-170-CAS-23040623

Steps: 1 Yield: 100%

Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling

By: Fujii, Shinya; et al

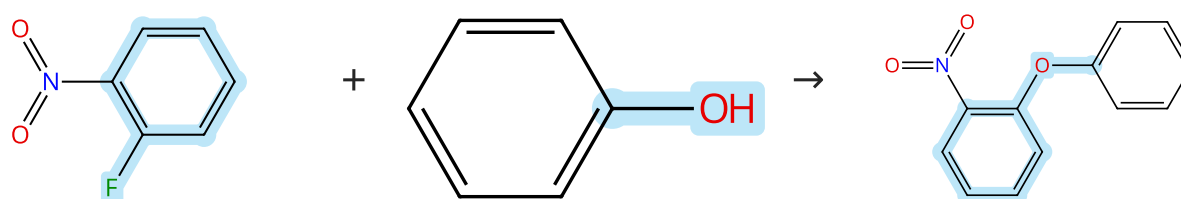
Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

 1.1 Reagents: Potassium carbonate
 Solvents: Dimethylformamide; 17 h, rt

Experimental Protocols

Scheme 88 (2 Reactions)

Steps: 1 Yield: 98-100%



Suppliers (93)

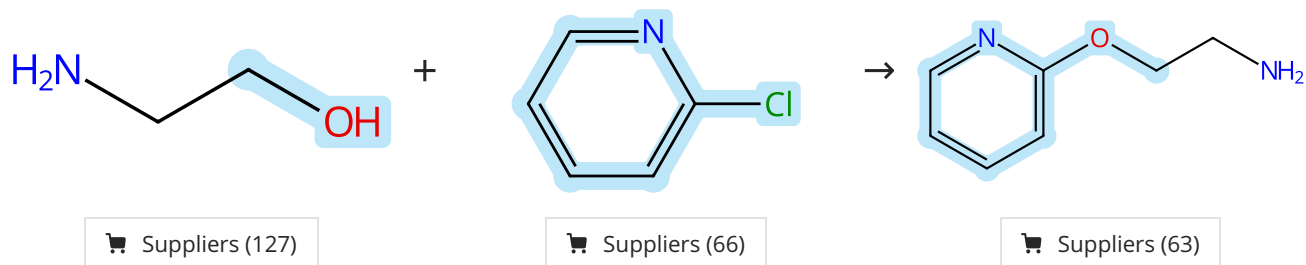
Suppliers (207)

Suppliers (49)

31-614-CAS-42944377 Steps: 1 Yield: 100% 1.1 Reagents: Potassium hydroxide Catalysts: Tetrabutylammonium bromide Solvents: 2-Methyltetrahydrofuran, Water; 10 min, 95 °C Experimental Protocols	S_NAr Reactions Using Continuous Plug Flow...in Aqueous Biphasic Media By: Wong, Madison J.; et al ACS Sustainable Chemistry & Engineering (2024), 12(52), 18725-18734.
31-614-CAS-30040246 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, 100 °C Experimental Protocols	DMF-Assisted Radical Cyclization of o-Isocyanodiaryl Ethers via 1,5-Aryl Migration: Construction of 2-Arylbenzoxazoles By: Yuan, Sitian; et al Journal of Organic Chemistry (2022), 87(2), 1485-1492.

Scheme 89 (1 Reaction)

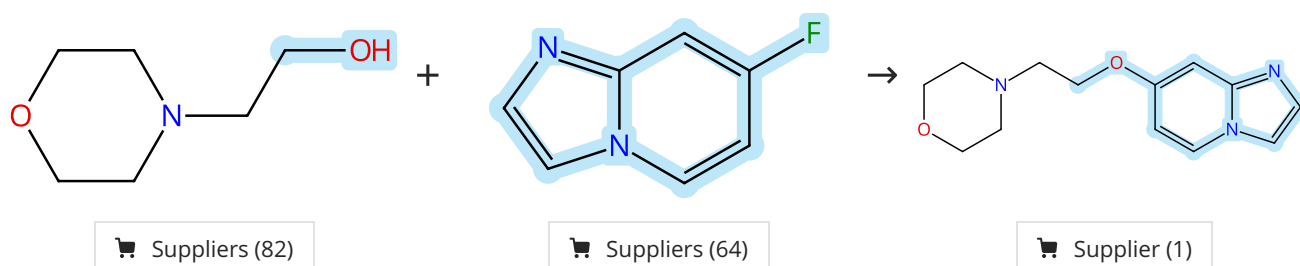
Steps: 1 Yield: 100%



31-614-CAS-33927008 Steps: 1 Yield: 100% 1.1 Reagents: Sodium Solvents: 1,4-Dioxane; 90 min, reflux 1.2 reflux; overnight, reflux 1.3 pH 1 1.4 pH 14 Experimental Protocols	Effect of the replacement of the o-methoxyphenyl moiety with nitrogen-containing aromatic rings within N-phenyl-piperazine and phenoxy-ethylamine-based 1,3-dioxo/oxa thio/dithiolanes as α₁ and 5-HT_{1A} receptor ligands By: Sorbi, Claudia; et al Results in Chemistry (2022), 4, 100425.
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Scheme 90 (1 Reaction)

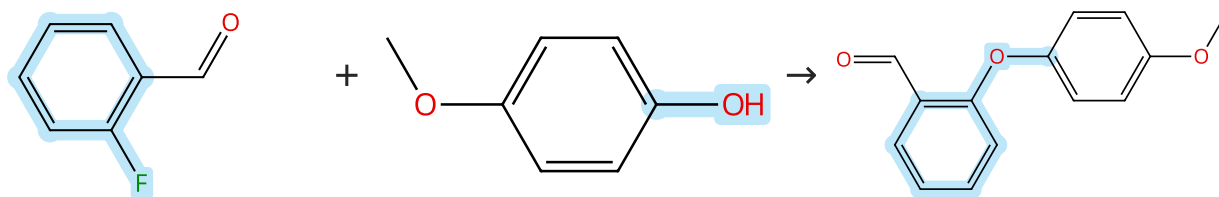
Steps: 1 Yield: 100%



31-614-CAS-39925462 Steps: 1 Yield: 100% 1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; rt; 30 min, rt 1.2 20 h, rt 1.3 Reagents: Sodium bicarbonate Solvents: Ethyl acetate, Water; 15 min, rt Experimental Protocols	Discovery of Clinical Candidate GLPG3970: A Potent and Selective Dual SIK2/SIK3 Inhibitor for the Treatment of Autoimmune and Inflammatory Diseases By: Peixoto, Christophe; et al Journal of Medicinal Chemistry (2024), 67(7), 5233-5258.
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Scheme 91 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (121)

Suppliers (56)

31-614-CAS-36277496

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

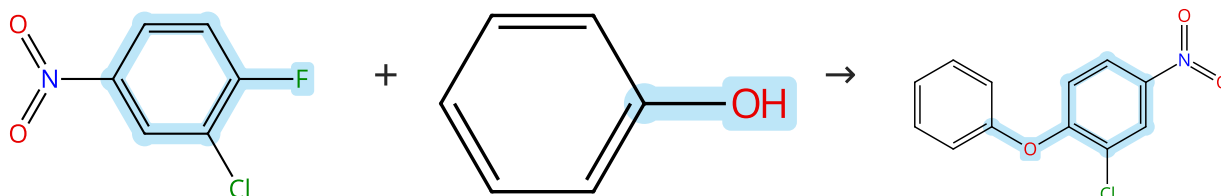
By: Can Uskup, Hacer; et al

Experimental Protocols

ACS Omega (2023), 8(15), 14047-14052.

Scheme 92 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (207)

Suppliers (17)

31-614-CAS-31980138

Steps: 1 Yield: 100%

Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt

By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

31-170-CAS-23039555

Steps: 1 Yield: 100%

Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 17 h, rt

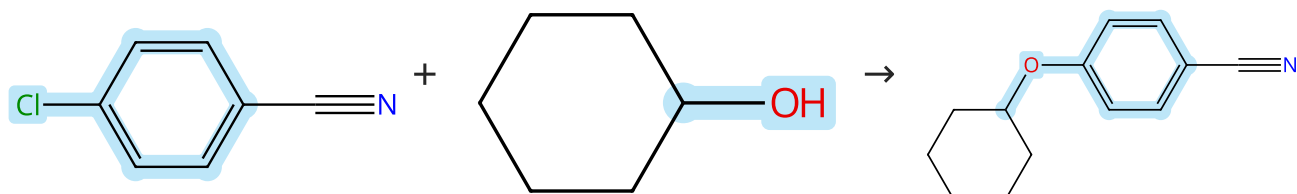
By: Fujii, Shinya; et al

Experimental Protocols

Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

Scheme 93 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (90)

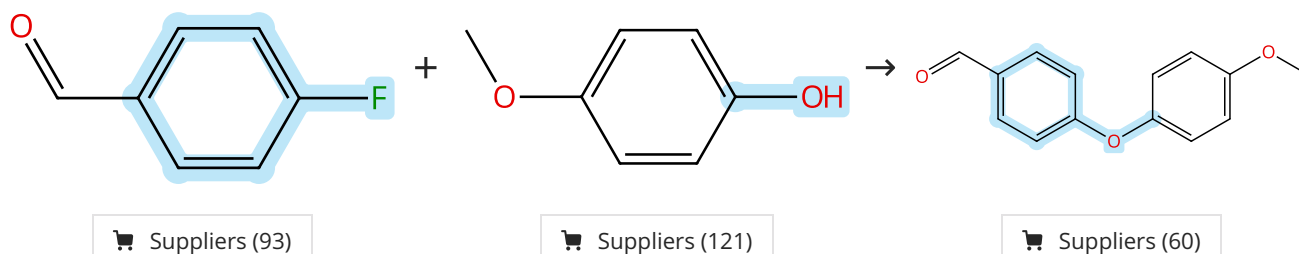
Suppliers (78)

Suppliers (39)

31-614-CAS-32106651 Steps: 1 Yield: 100%	Discovery and characterization of a novel perylenephore ductant for the activation of aryl halides
1.1 Reagents: Diisopropylethylamine Catalysts: Nickel sulfate (NiSO ₄), 4,4'-Dimethoxy-2,2'-bipyridine, 2767997-30-6 Solvents: Acetonitrile; 24 h, rt 1.2 Reagents: Water; rt	By: Li, Min; et al Journal of Catalysis (2021), 399, 111-120.

Scheme 94 (1 Reaction)

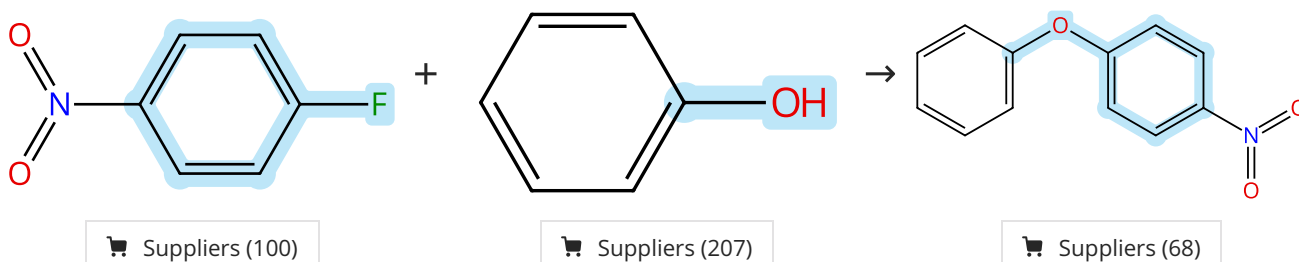
Steps: 1 Yield: 100%



31-614-CAS-41211204 Steps: 1 Yield: 100%	Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 h, 180 °C Experimental Protocols	By: Hasdemir, Belma; et al ChemistrySelect (2024), 9(26), e202401028.

Scheme 95 (3 Reactions)

Steps: 1 Yield: 100%



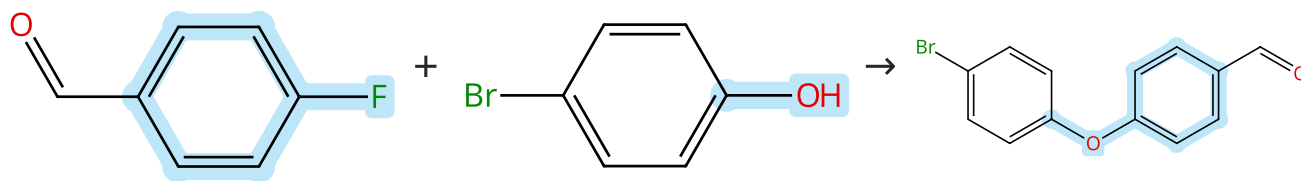
31-614-CAS-42944366 Steps: 1 Yield: 100%	S_NAr Reactions Using Continuous Plug Flow...in Aqueous Biphasic Media
1.1 Reagents: Potassium hydroxide Catalysts: Tetrabutylammonium bromide Solvents: 2-Methyltetrahydrofuran, Water; 10 min, 95 °C Experimental Protocols	By: Wong, Madison J.; et al ACS Sustainable Chemistry & Engineering (2024), 12(52), 18725-18734.

31-614-CAS-31980157 Steps: 1 Yield: 100%	Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; rt	By: Fujii, Shinya; et al ChemMedChem (2021), 16(18), 2817-2822.

31-170-CAS-23044471 Steps: 1 Yield: 100%	Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 17 h, rt Experimental Protocols	By: Fujii, Shinya; et al Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

Scheme 96 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (82)

Suppliers (71)

31-614-CAS-41211201

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Synthesis and Characterization of New Bulky α -Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

31-614-CAS-34652753

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt $\rightarrow$ 100 °C; 100 °C $\rightarrow$ rt

1.2 Reagents: Water; rt

Experimental Protocols

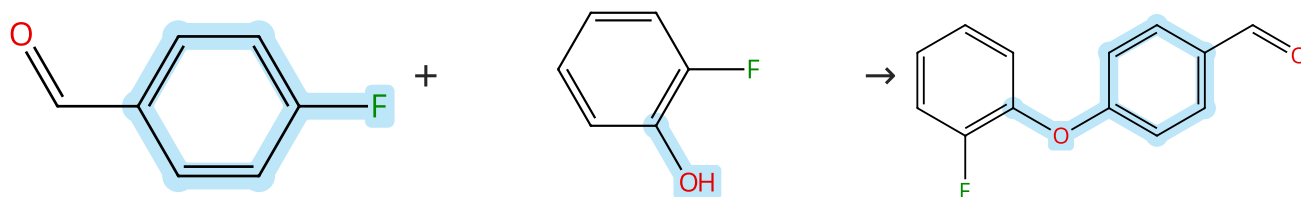
Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

By: Chen, Tao; et al

Letters in Drug Design & Discovery (2022), 19(4), 263-268.

Scheme 97 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (101)

Suppliers (52)

31-614-CAS-34652754

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt $\rightarrow$ 100 °C; 100 °C $\rightarrow$ rt

1.2 Reagents: Water; rt

Experimental Protocols

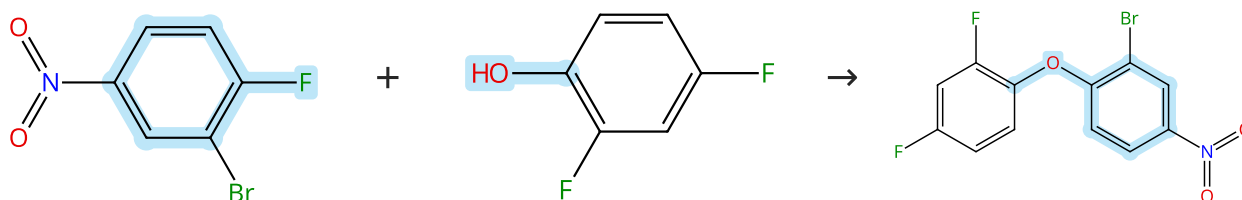
Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

By: Chen, Tao; et al

Letters in Drug Design & Discovery (2022), 19(4), 263-268.

Scheme 98 (3 Reactions)

Steps: 1 Yield: 98-100%



Suppliers (83)

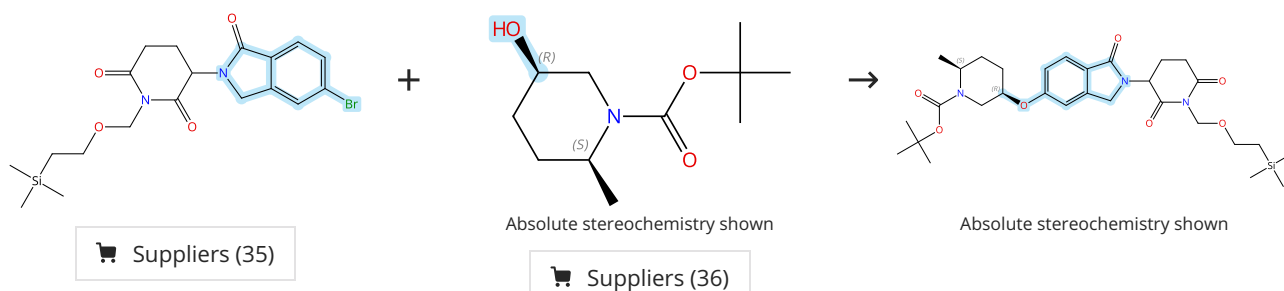
Suppliers (81)

Suppliers (22)

31-614-CAS-32016366 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, rt Experimental Protocols	Structure-guided discovery of novel potent and efficacious proteolysis targeting chimera (PROTAC) degrader of BRD4 By: Xiang, Wang; et al Bioorganic Chemistry (2021), 115, 105238.
31-614-CAS-38998627 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; rt → 110 °C; 2 h, 110 °C Experimental Protocols	Novel Triazolopyridine-Based BRD4 Inhibitors as Potent HIV-1 Latency Reversing Agents By: Wang, Yan-Kai; et al ACS Medicinal Chemistry Letters (2024), 15(1), 60-68.
31-614-CAS-39886652 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 4 h, rt → 80 °C Experimental Protocols	Structure-Based Discovery and Optimization of Furo[3,2-c]pyridin-4(5H)-one Derivatives as Potent and Second Bromodomain (BD2)-Selective Bromo and Extra Terminal Domain (BET) Inhibitors By: Li, Junhua; et al Journal of Medicinal Chemistry (2022), 65(7), 5760-5799.

Scheme 99 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-47306786

Steps: 1 Yield: 100%

- 1.1 Reagents: 2,2,6,6-Tetramethylpiperidine
Catalysts: Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine, Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-κN¹,κN^{1'}]-bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl-κN]phenyl-κC]-, (OC-6-33)-, hexafluorophosphate(1-) (1:1)
Solvents: Acetonitrile; 24 h, rt

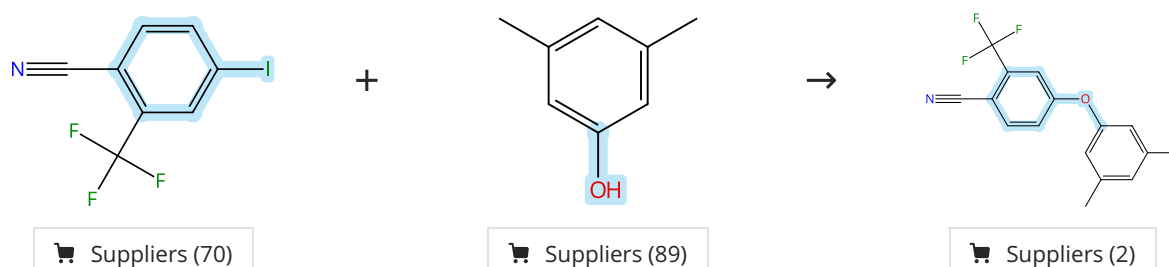
Discovery and Optimization of First-in-Class Molecular Glue Degradors of the WIZ Transcription Factor for Fetal Hemoglobin Induction to Treat Sickle Cell Disease

By: Kerrigan, John Ryan; et al

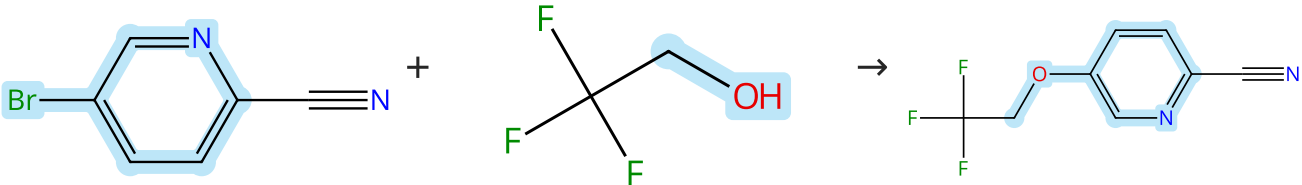
Journal of Medicinal Chemistry (2024), 67(22), 20682-20694.

Scheme 100 (1 Reaction)

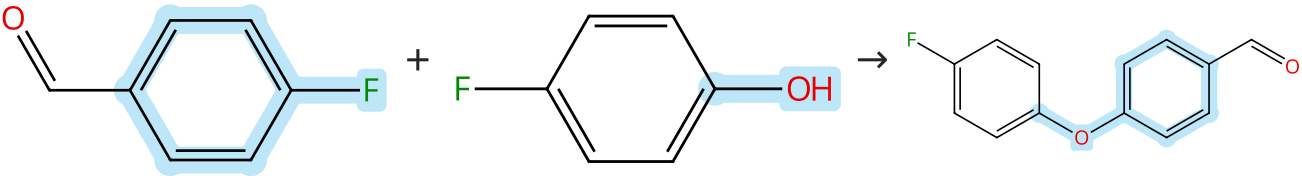
Steps: 1 Yield: 100%



31-614-CAS-43350516 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: Cuprous iodide, 1<i>H</i>-imidazolium, 3-[4-(trifluoromethyl)-2-pyridinyl]-1-(2,4,6-trimethylphenyl)-, bromide (1:1) Solvents: Acetonitrile; 20 h, 110 °C Experimental Protocols	Bidentate NHC-Containing Ligands for Copper Catalyzed Synthesis of Functionalised Diaryl Ethers By: Richards, Ethan M.; et al Advanced Synthesis & Catalysis (2025), 367(6), e202400909.
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Scheme 101 (1 Reaction) Steps: 1 Yield: 100%  Suppliers (103) Suppliers (98) Suppliers (24)		
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31-614-CAS-24212002 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 1 - 2 h, 100 - 105 °C Experimental Protocols	Scalable Asymmetric Synthesis of MK-8998, a T-type Calcium Channel Antagonist By: Zhong, Yong-Li; et al Journal of Organic Chemistry (2022), 87(4), 2120-2128.
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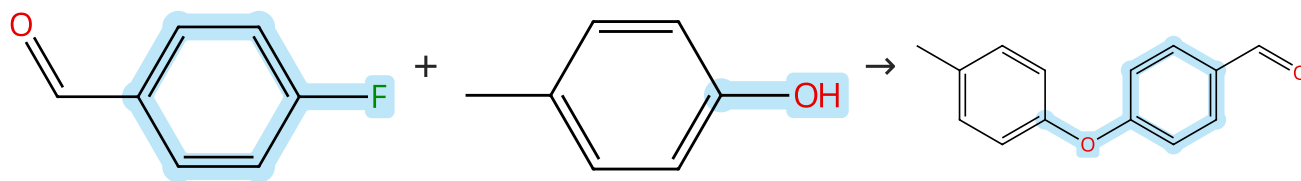
Scheme 102 (2 Reactions) Steps: 1 Yield: 100%  Suppliers (93) Suppliers (86) Suppliers (83)		
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31-614-CAS-41211198 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 h, 180 °C Experimental Protocols	Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation By: Hasdemir, Belma; et al ChemistrySelect (2024), 9(26), e202401028.
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31-614-CAS-34652756 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt 1.2 Reagents: Water; rt Experimental Protocols	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide By: Chen, Tao; et al Letters in Drug Design & Discovery (2022), 19(4), 263-268.
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Scheme 103 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (89)

Suppliers (55)

31-614-CAS-41211196

Steps: 1 Yield: 100%

Synthesis and Characterization of New Bulky α -Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

By: Hasdemir, Belma; et al
ChemistrySelect (2024), 9(26), e202401028.

Experimental Protocols

31-614-CAS-34652758

Steps: 1 Yield: 100%

Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt $\rightarrow$ 100 °C; 100 °C $\rightarrow$ rt

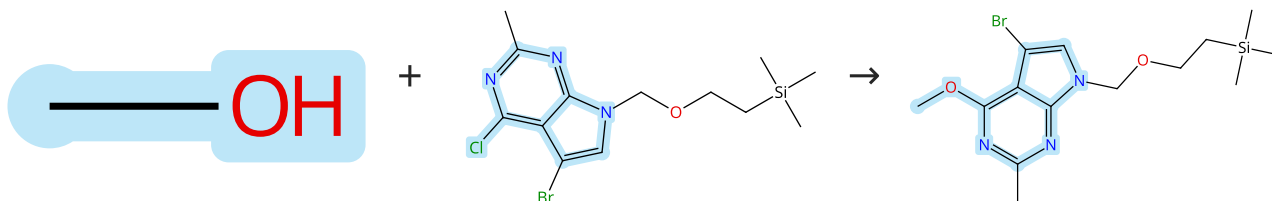
By: Chen, Tao; et al
Letters in Drug Design & Discovery (2022), 19(4), 263-268.

1.2 Reagents: Water; rt

Experimental Protocols

Scheme 104 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (444)

Suppliers (4)

Supplier (1)

31-614-CAS-32679854

Steps: 1 Yield: 100%

Fragment-Derived Selective Inhibitors of Dual-Specificity Kinases DYRK1A and DYRK1B

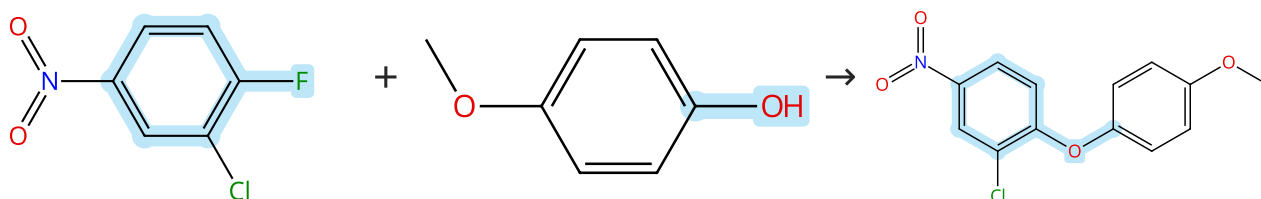
1.1 Reagents: Sodium hydride
Solvents: Tetrahydrofuran; 0 °C; 10 min, 0 °C
1.2 Solvents: Tetrahydrofuran; 30 min, 0 °C; 1 h, 0 °C $\rightarrow$ rt
1.3 Reagents: Ammonium chloride
Solvents: Water; rt

By: Lee Walmsley, David; et al
Journal of Medicinal Chemistry (2021), 64(13), 8971-8991.

Experimental Protocols

Scheme 105 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

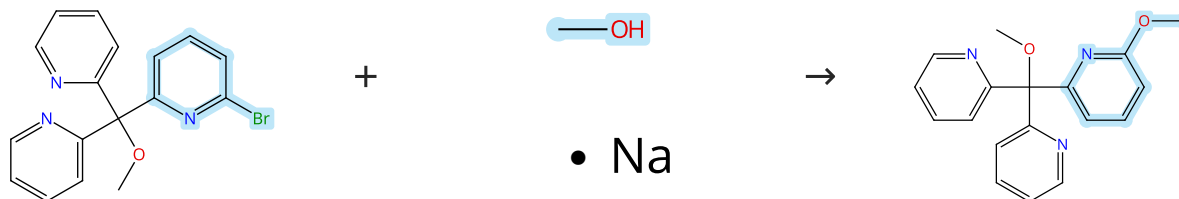
Suppliers (121)

Suppliers (14)

31-614-CAS-31980144 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; rt	Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents By: Fujii, Shinya; et al ChemMedChem (2021), 16(18), 2817-2822.
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Scheme 106 (1 Reaction)

Steps: 1 Yield: 100%

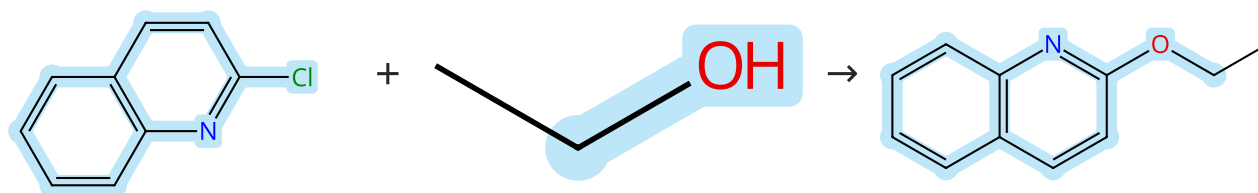


Suppliers (99)

31-614-CAS-47173652 Steps: 1 Yield: 100% 1.1 Solvents: Methanol; 3 d, 80 °C; 80 °C → rt 1.2 Reagents: Hydrochloric acid Solvents: Water; rt 1.3 Reagents: Sodium carbonate Solvents: Water; basified, rt Experimental Protocols	Copper(I) complexes of Methoxytris(2-pyridyl)methane tripod ligands with 6-Alkoxy, 6-Aryloxy or 6-Arylsulfanyl substituent on one Leg. Synthesis and Reactivity studies By: Nagata, Toshii; et al Inorganica Chimica Acta (2025), 576, 122451.
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Scheme 107 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (90)

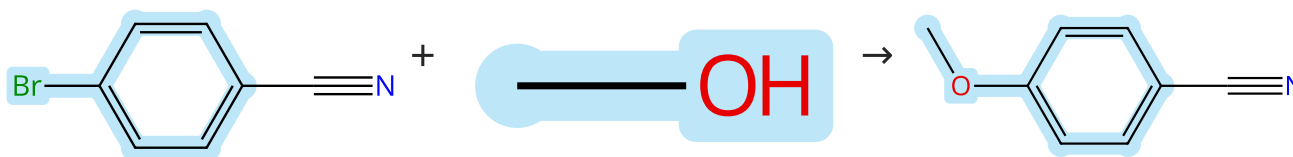
Suppliers (618)

Suppliers (35)

31-170-CAS-22584340 Steps: 1 Yield: 100% 1.1 Reagents: Potassium <i>tert</i>-butoxide Solvents: Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 3 h, 0 °C 1.2 Reagents: Ammonium chloride Solvents: Water; 0 °C	Aryl Ether Syntheses via Aromatic Substitution Proceeding under Mild Conditions By: Ando, Shin; et al Journal of Organic Chemistry (2020), 85(17), 11181-11189.
--	---

Scheme 108 (3 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (93)

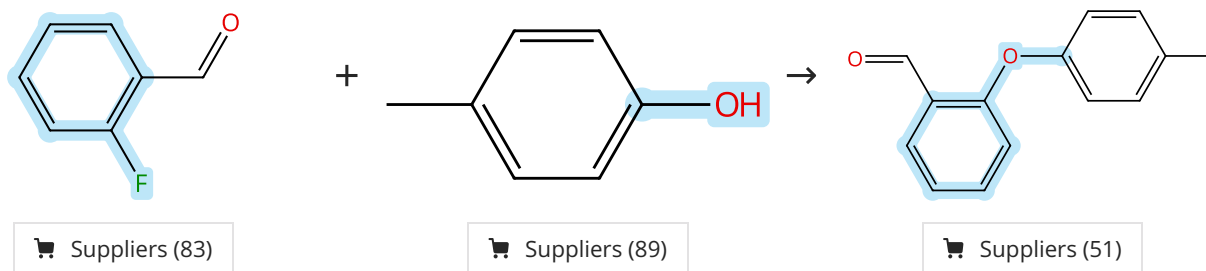
Suppliers (444)

Suppliers (91)

31-614-CAS-35474512 Steps: 1 Yield: 100% 1.1 Reagents: Quinuclidine Catalysts: 2,2'-Bipyridine, Nickel dichloride, [1,1':4',1''-Terphenyl]-4,4''-dicarboxaldehyde, polymer with 5'-(4-aminophenyl)[1,1':3',1''-terphenyl]-4,4''-diamine Solvents: Dimethylacetamide; 5 h, 50 °C Experimental Protocols	Covalent organic frameworks editing for efficient metallaphotoredox catalytic carbon-oxygen cross coupling of aryl halides with alcohols By: Meng, Di; et al Catalysis Science & Technology (2023), 13(5), 1518-1526.
31-614-CAS-34588744 Steps: 1 Yield: 99% 1.1 Reagents: Sodium <i>tert</i>-butoxide Solvents: Dimethyl sulfoxide; 4 h, rt Experimental Protocols	Transition-metal-free arylation of alcohols with aryl bromides at room temperature By: Zhu, Da-Liang; et al Synthesis (2023), 55(4), 637-646.
31-614-CAS-38873230 Steps: 1 Yield: 99% 1.1 Reagents: Quinuclidine Catalysts: Imidazole, Nickel, Carbon nitride (C₃N₄) Solvents: Dimethylacetamide; 6 h, 25 °C Experimental Protocols	Nickel-Coordinated Carbon Nitride as a Metallaphotoredox Platform for the Cross-Coupling of Aryl Halides with Alcohols By: Zhao, Xin; et al ACS Catalysis (2020), 10(24), 15178-15185.

Scheme 109 (1 Reaction)

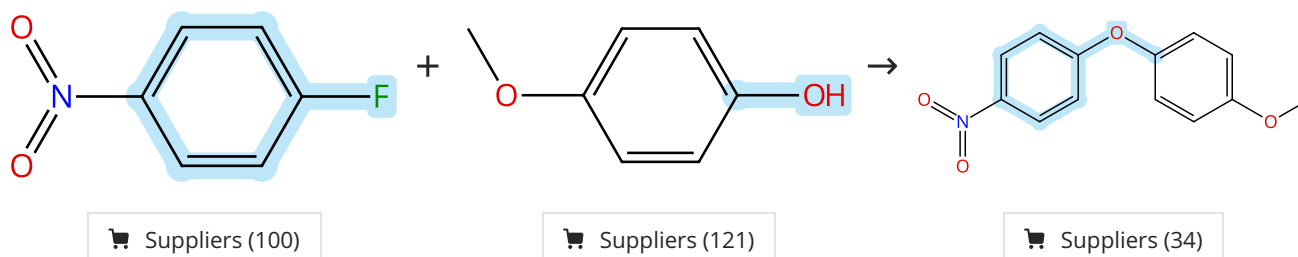
Steps: 1 Yield: 100%



31-614-CAS-36277492 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 h, 180 °C Experimental Protocols	Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization By: Can Uskup, Hacer; et al ACS Omega (2023), 8(15), 14047-14052.
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Scheme 110 (1 Reaction)

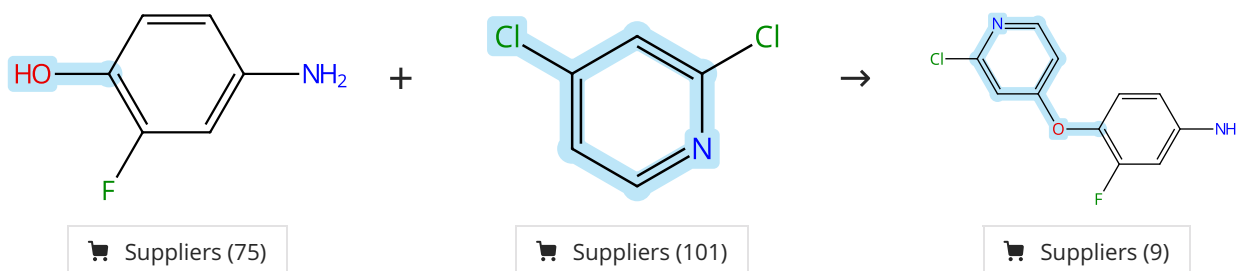
Steps: 1 Yield: 100%



31-614-CAS-42944338 Steps: 1 Yield: 100% 1.1 Reagents: Potassium hydroxide Catalysts: Tetrabutylammonium bromide Solvents: 2-Methyltetrahydrofuran, Water; 10 min, 95 °C Experimental Protocols	S_NAr Reactions Using Continuous Plug Flow...in Aqueous Biphasic Media By: Wong, Madison J.; et al ACS Sustainable Chemistry & Engineering (2024), 12(52), 18725-18734.
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Scheme 111 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-25849612

Steps: 1 Yield: 100%

Discovery of a Pyrimidinedione Derivative as a Potent and Orally Bioavailable Axl Inhibitor

1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Dimethylacetamide; 0.5 h, rt
 1.2 4 h, 85 °C

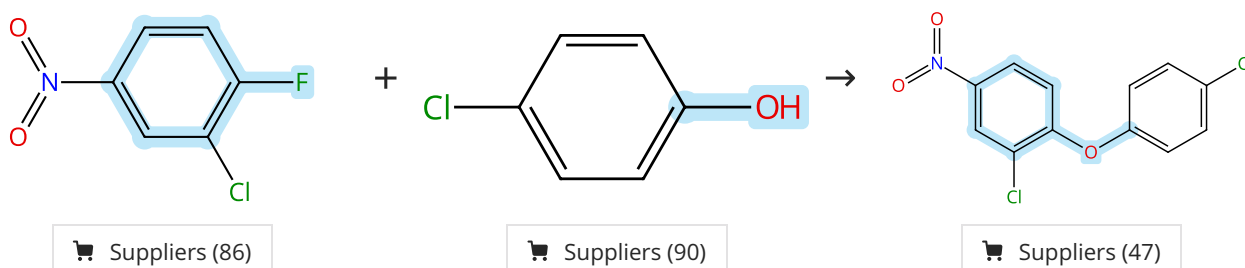
By: Zhang, Hefeng; et al

Journal of Medicinal Chemistry (2021), 64(7), 3956-3975.

Experimental Protocols

Scheme 112 (1 Reaction)

Steps: 1 Yield: 100%



31-170-CAS-23041155

Steps: 1 Yield: 100%

Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 17 h, rt

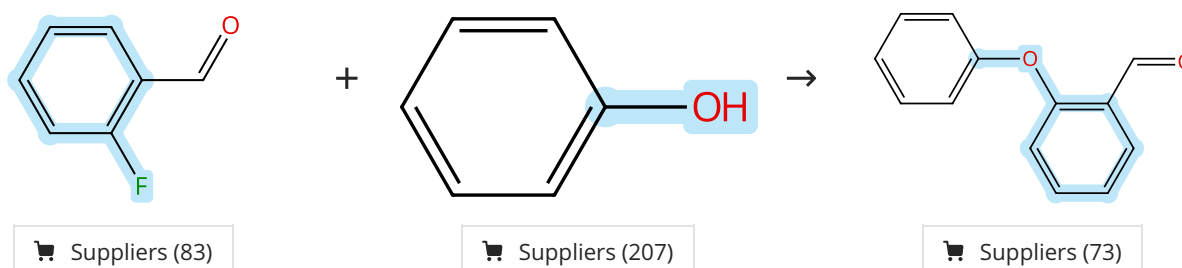
By: Fujii, Shinya; et al

Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

Experimental Protocols

Scheme 113 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-36277494

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

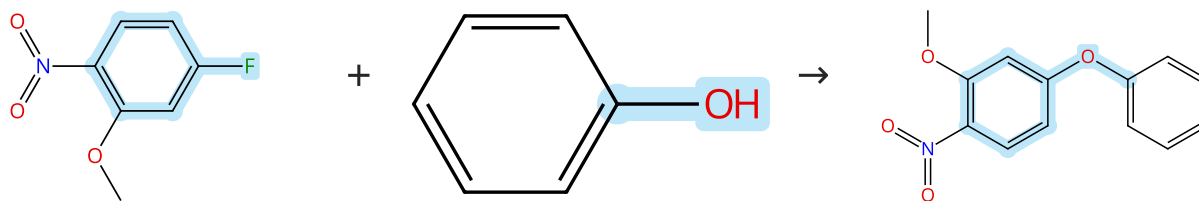
By: Can Uskup, Hacer; et al

ACS Omega (2023), 8(15), 14047-14052.

Experimental Protocols

Scheme 114 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (91)

Suppliers (207)

Suppliers (7)

31-614-CAS-31980159

Steps: 1 Yield: 100%

Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

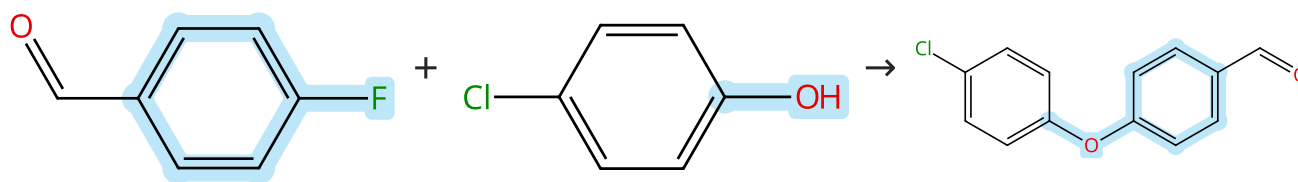
By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt

Scheme 115 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (90)

Suppliers (63)

31-614-CAS-41211200

Steps: 1 Yield: 100%

Synthesis and Characterization of New Bulky α -Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

31-614-CAS-34652751

Steps: 1 Yield: 100%

Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

By: Chen, Tao; et al

Letters in Drug Design & Discovery (2022), 19(4), 263-268.

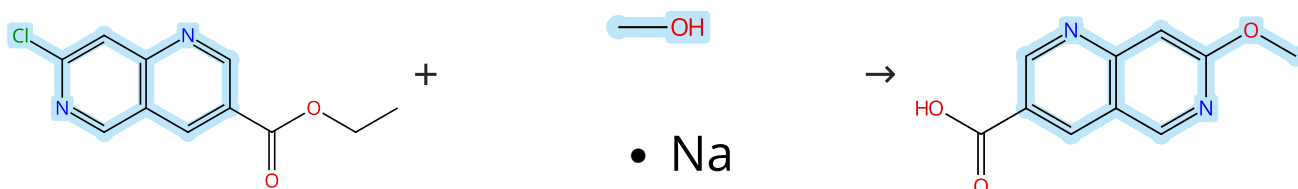
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt $\rightarrow$ 100 °C; 100 °C $\rightarrow$ rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 116 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

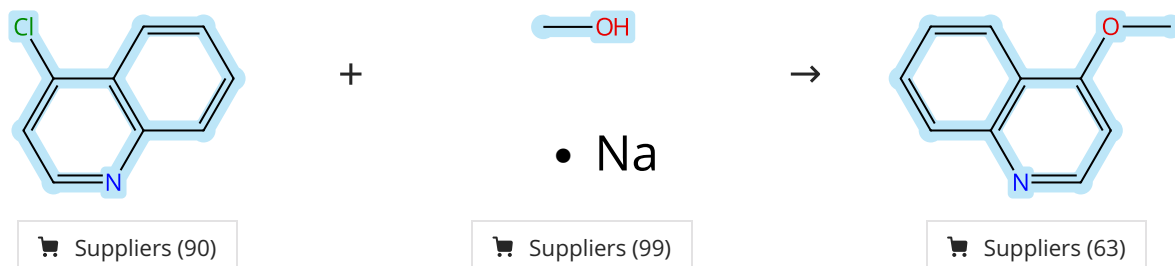
Suppliers (99)

Supplier (1)

31-614-CAS-38094970	Steps: 1 Yield: 100%	The exploration of aza-quinolines as hematopoietic prostaglandin D synthase (H-PGDS) inhibitors with low brain exposure
1.1 Solvents: Methanol; rt; 7 h, rt → 60 °C		
1.2 Reagents: Water; 90 min, 60 °C		
Experimental Protocols		By: Cadilla, Rodolfo; et al Bioorganic & Medicinal Chemistry (2020), 28(23), 115791.

Scheme 117 (1 Reaction)

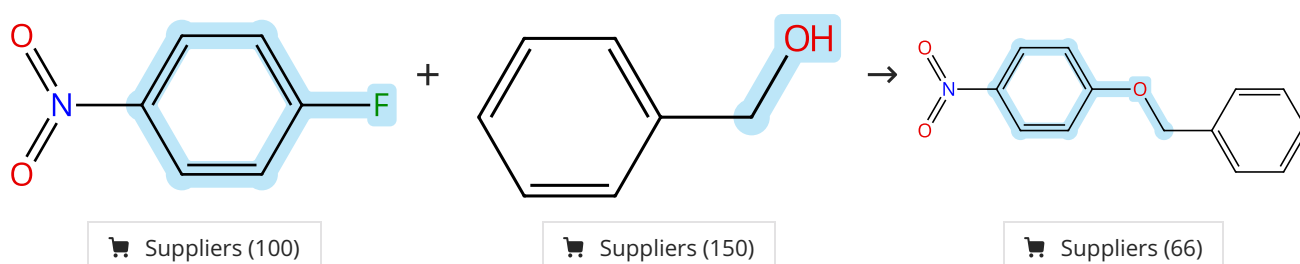
Steps: 1 Yield: 100%



31-614-CAS-37302410	Steps: 1 Yield: 100%	Decarboxylative C-H silylation of N-heteroarenes with silanecarboxylic acids
1.1 Solvents: Methanol; rt; overnight, 70 °C		
Experimental Protocols		By: Zhang, Guodong; et al Chemical Communications (Cambridge, United Kingdom) (2023), 59(17), 2449-2452.

Scheme 118 (1 Reaction)

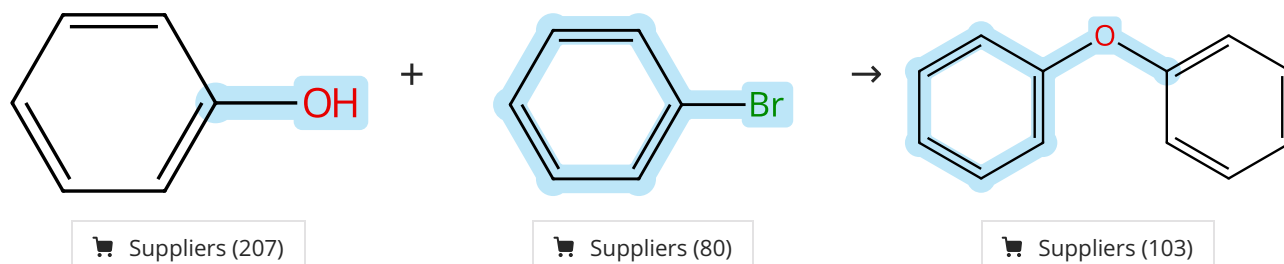
Steps: 1 Yield: 100%



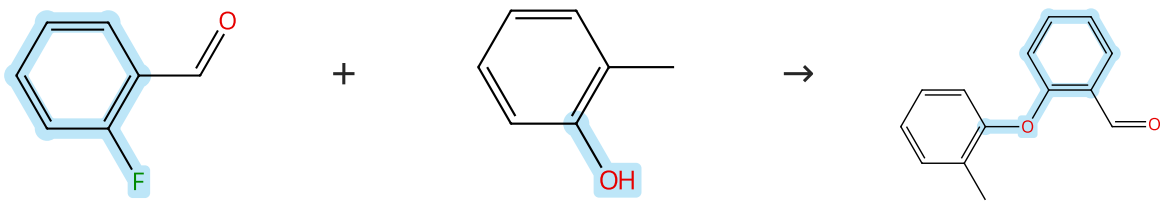
31-614-CAS-42944382	Steps: 1 Yield: 100%	S _N Ar Reactions Using Continuous Plug Flow...in Aqueous Biphasic Media
1.1 Reagents: Potassium hydroxide Catalysts: Tetrabutylammonium bromide Solvents: Water; 10 min, 95 °C		
Experimental Protocols		By: Wong, Madison J.; et al ACS Sustainable Chemistry & Engineering (2024), 12(52), 18725-18734.

Scheme 119 (1 Reaction)

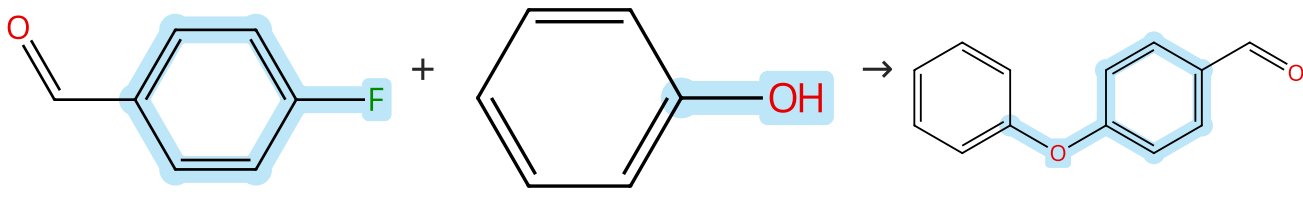
Steps: 1 Yield: 100%



31-614-CAS-35955493 Steps: 1 Yield: 100% 1.1 Reagents: Tripotassium phosphate Catalysts: 2907681-72-3 Solvents: Dimethyl sulfoxide; 15 h, 120 °C Experimental Protocols	Novel 1,2,3-triazolyl phosphine with a pyridyl functionality: synthesis, coinage metal complexes, photophysical studies and Cu(I) catalyzed C-O coupling of phenols with aryl bromides By: Sheokand, Sonu; et al Dalton Transactions (2023), 52(6), 1785-1796.
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Scheme 120 (1 Reaction) Steps: 1 Yield: 100%  Suppliers (83) Suppliers (83) Suppliers (20)		
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31-614-CAS-36277498 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 h, 180 °C Experimental Protocols	Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization By: Can Uskup, Hacer; et al ACS Omega (2023), 8(15), 14047-14052.
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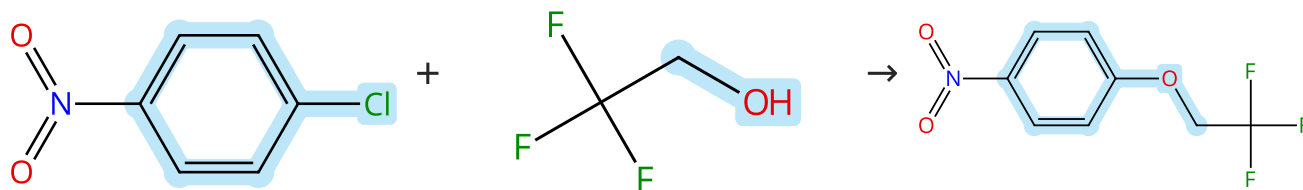
Scheme 121 (2 Reactions) Steps: 1 Yield: 100%  Suppliers (93) Suppliers (207) Suppliers (84)		
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31-614-CAS-41211192 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 h, 180 °C Experimental Protocols	Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation By: Hasdemir, Belma; et al ChemistrySelect (2024), 9(26), e202401028.
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31-614-CAS-34652757 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt 1.2 Reagents: Water; rt Experimental Protocols	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide By: Chen, Tao; et al Letters in Drug Design & Discovery (2022), 19(4), 263-268.
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Scheme 122 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (65)

Suppliers (98)

Suppliers (55)

31-170-CAS-22584352

Steps: 1 Yield: 100%

Aryl Ether Syntheses via Aromatic Substitution Proceeding under Mild Conditions

By: Ando, Shin; et al

Journal of Organic Chemistry (2020), 85(17), 11181-11189.

1.1 Reagents: Potassium *tert*-butoxide

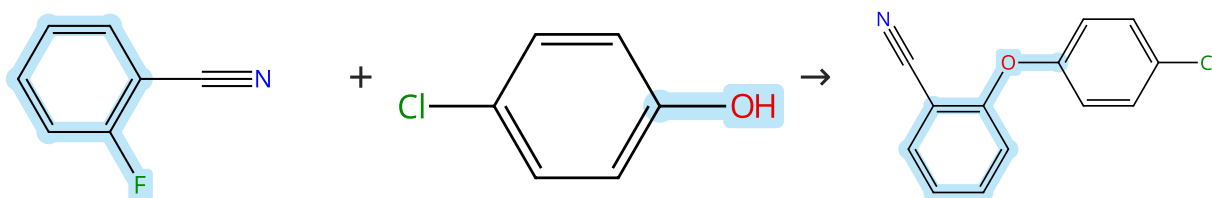
Solvents: Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 1 h, 25 °C

1.2 Reagents: Ammonium chloride

Solvents: Water; 0 °C

Scheme 123 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (90)

Suppliers (17)

31-614-CAS-32528350

Steps: 1 Yield: 100%

Access to ortho-Hydroxyphenyl Ketimines via Imine Anion-Mediated Smiles Rearrangement

By: Jinno, Shunki; et al

Organic Letters (2022), 24(23), 4140-4144.

1.1 Reagents: Potassium hydroxide

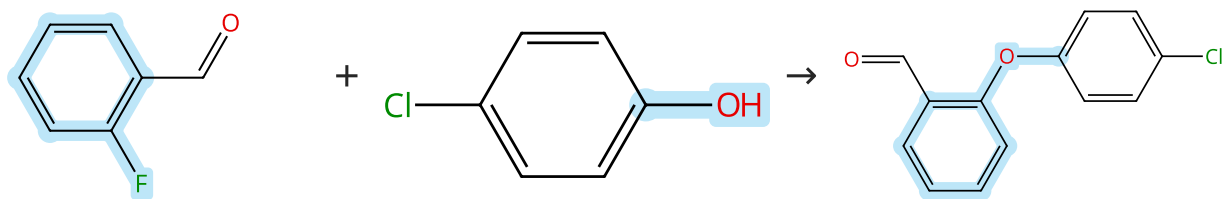
Solvents: Dimethyl sulfoxide; 14.5 h, 90 °C; 90 °C → rt

1.2 Reagents: Water; rt

Experimental Protocols

Scheme 124 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (90)

Suppliers (61)

31-614-CAS-36277490

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

By: Can Uskup, Hacer; et al

ACS Omega (2023), 8(15), 14047-14052.

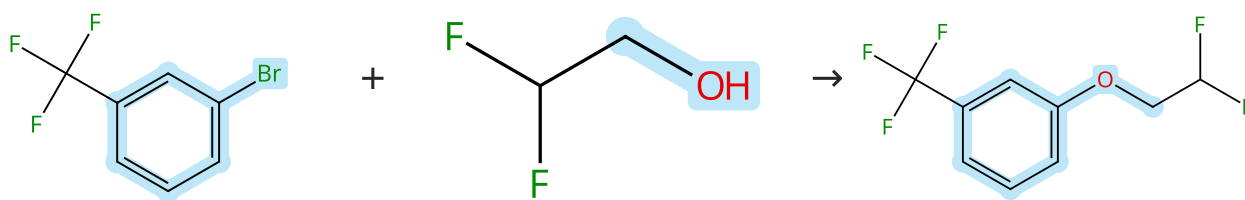
1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Scheme 125 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (73)

Supplier (1)

31-614-CAS-32795091

Steps: 1 Yield: 100%

Copper-catalyzed arylated etherification of 2,2-difluoroethanol and its mechanistic study1.1 Reagents: Potassium *tert*-butoxide

Catalysts: 8-Hydroxyquinoline, Cuprous iodide; 12 h, 100 °C

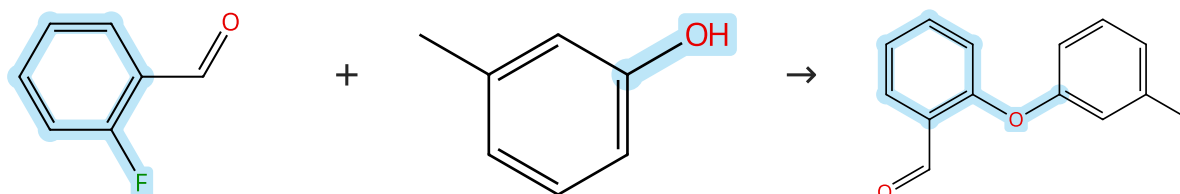
By: Huang, Shuaishuai; et al

Experimental Protocols

Youji Huaxue (2020), 40(7), 2018-2025.

Scheme 126 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (91)

Suppliers (14)

31-614-CAS-36277479

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 2 h, 180 °C

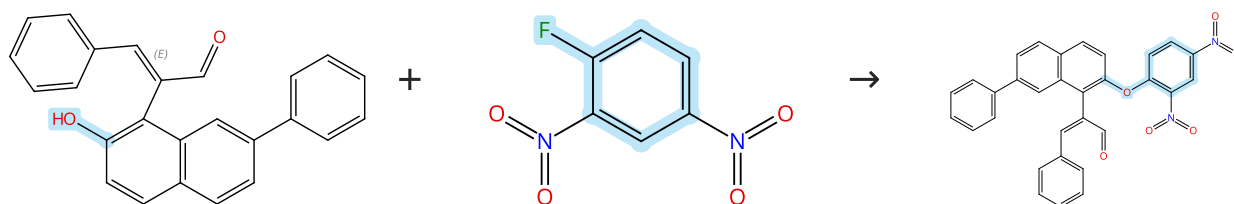
By: Can Uskup, Hacer; et al

Experimental Protocols

ACS Omega (2023), 8(15), 14047-14052.

Scheme 127 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560087

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

1.1 Reagents: Potassium carbonate

Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)

Solvents: Chloroform; 24 h, -10 °C

By: Guo, Fengyuan; et al

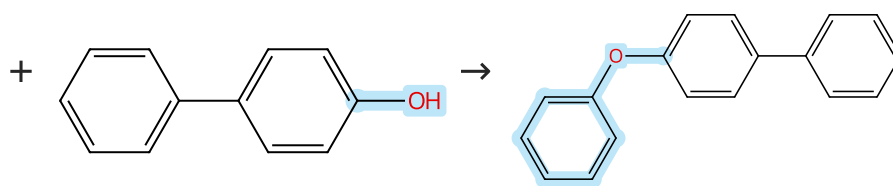
Nature Communications (2023), 14(1), 5050.

Experimental Protocols

Scheme 128 (1 Reaction)

Steps: 1 Yield: 99%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (105)

Suppliers (27)

31-614-CAS-31533998

Steps: 1 Yield: 99%

Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates

By: Kikushima, Kotaro; et al

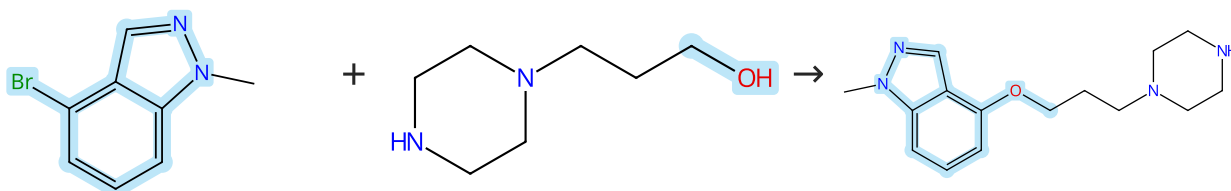
Organic Letters (2022), 24(10), 1924-1928.

- 1.1 **Reagents:** Sodium carbonate
Solvents: Dichloromethane, Water; 18 h, 40 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 129 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (68)

Suppliers (64)

31-614-CAS-40801511

Steps: 1 Yield: 99%

Cu-Catalyzed Amination of Base-Sensitive Aryl Bromides and the Chemoselective N- and O-Arylation of Amino Alcohols

By: Strauss, Michael J.; et al

Journal of the American Chemical Society (2024), 146(27), 18616-18625.

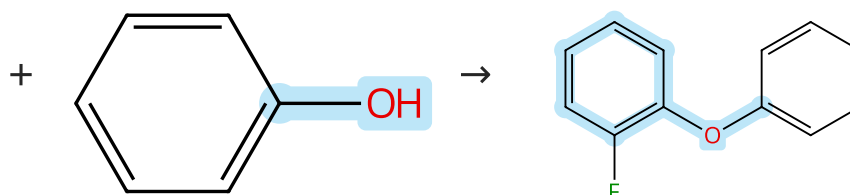
- 1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: Cuprous iodide, 1,2-Benzenediamine, *N*¹,*N*²-bis(2-phenyl-1-naphthalenyl)-
Solvents: Dimethyl sulfoxide; 24 °C; 16 h, 24 °C

Experimental Protocols

Scheme 130 (1 Reaction)

Steps: 1 Yield: 99%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (207)

Suppliers (31)

31-614-CAS-31533997

Steps: 1 Yield: 99%

Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates

By: Kikushima, Kotaro; et al

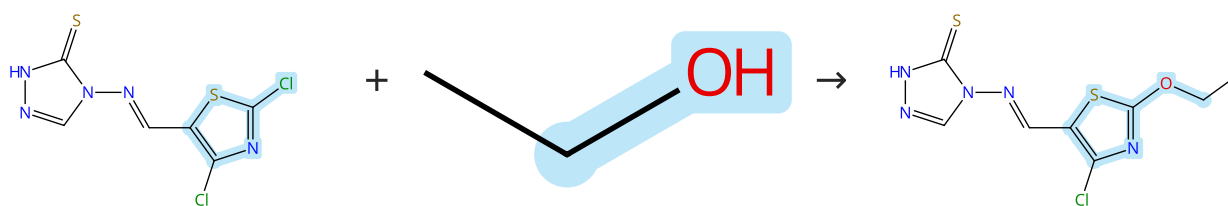
Organic Letters (2022), 24(10), 1924-1928.

- 1.1 **Reagents:** Sodium carbonate
Solvents: Dichloromethane, Water; 18 h, 40 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 131 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (618)

31-170-CAS-23030670

Steps: 1 Yield: 99%

Synthesis, characterization and biological studies of novel series of triazolyl schiff bases bearing thiazole moiety and their s-derivatives

1.1 **Catalysts:** Potassium hydroxide
Solvents: Water; 6 h, rt

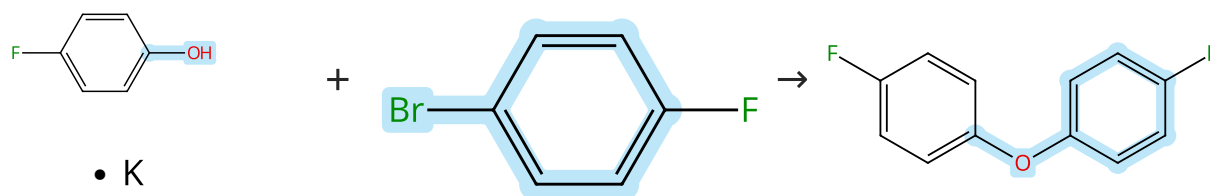
Experimental Protocols

By: Nkurunziza, Jean Baptiste; et al

Journal of Applicable Chemistry (Lumami, India) (2020), 9(4), 556-565.

Scheme 132 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (103)

Suppliers (63)

31-614-CAS-37728583

Steps: 1 Yield: 99%

Cross-coupling by a noncan onical mechanism involving the addition of aryl halide to Cu(II)

1.1 **Reagents:** 1-Fluoronaphthalene
Catalysts: 2983730-57-8
Solvents: Dimethyl sulfoxide; rt → 100 °C; 5 min, 100 °C

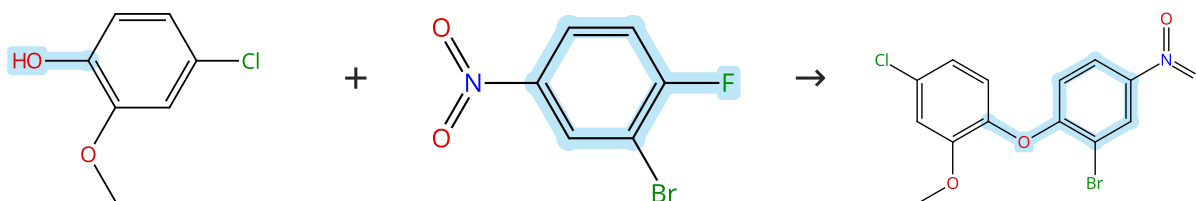
Experimental Protocols

By: Delaney, Connor P.; et al

Science (Washington, DC, United States) (2023), 381(6662), 1079-1085.

Scheme 133 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (90)

Suppliers (83)

31-614-CAS-35312180

Steps: 1 Yield: 99%

Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors

1.1 **Reagents:** Cesium carbonate
Solvents: Dimethyl sulfoxide; 2 h, 110 °C

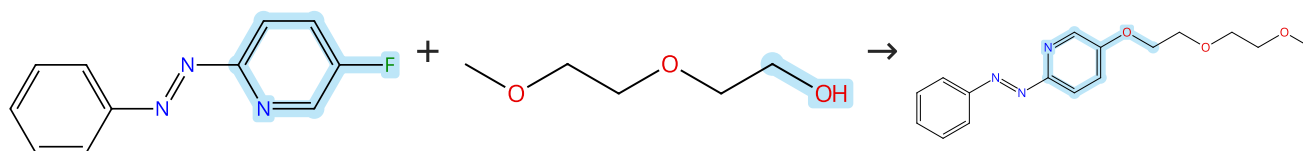
Experimental Protocols

By: Xu, Kai-Yan; et al

Bioorganic & Medicinal Chemistry (2023), 78, 117152.

Scheme 134 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

Suppliers (89)

31-170-CAS-22467505

Steps: 1 Yield: 99%

1.1 **Reagents:** Potassium hydroxide
Solvents: Diethylene glycol monomethyl ether, Water; 18 h, reflux

Experimental Protocols

Structure-activity relationships for osmium(II) arene phenylazopyridine anticancer complexes functionalized with alkoxy and glycolic substituents

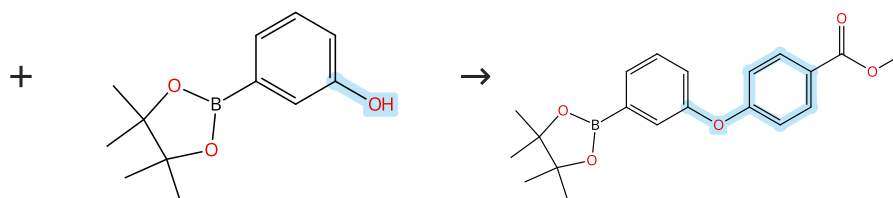
By: Needham, Russell J.; et al

Journal of Inorganic Biochemistry (2020), 210, 111154.

Scheme 135 (1 Reaction)

Steps: 1 Yield: 99%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (90)

31-614-CAS-31534025

Steps: 1 Yield: 99%

1.1 **Reagents:** Sodium carbonate
Solvents: Dichloromethane, Water; 18 h, 40 °C
 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

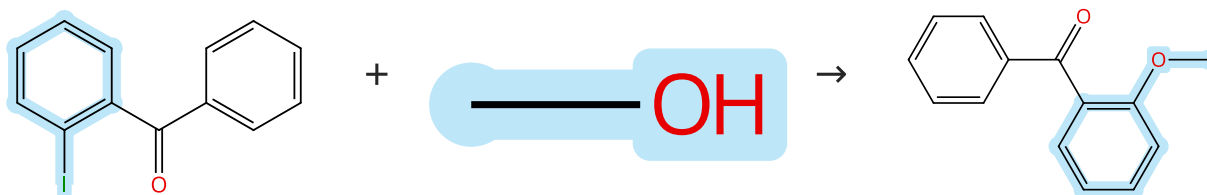
Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates

By: Kikushima, Kotaro; et al

Organic Letters (2022), 24(10), 1924-1928.

Scheme 136 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (60)

Suppliers (444)

Suppliers (59)

31-614-CAS-35849204

Steps: 1 Yield: 99%

1.1 **Reagents:** Tripotassium phosphate
Catalysts: [2-[Bis(tricyclo[3.3.1.1^{3,7}]dec-1-yl)phosphino-κP]-N,N-dimethylbenzenamine]chlorogold
Solvents: 1,2-Dichlorobenzene; 5 min, rt

1.2 **Reagents:** Silver hexafluoroantimonate; 24 h, 80 °C

Experimental Protocols

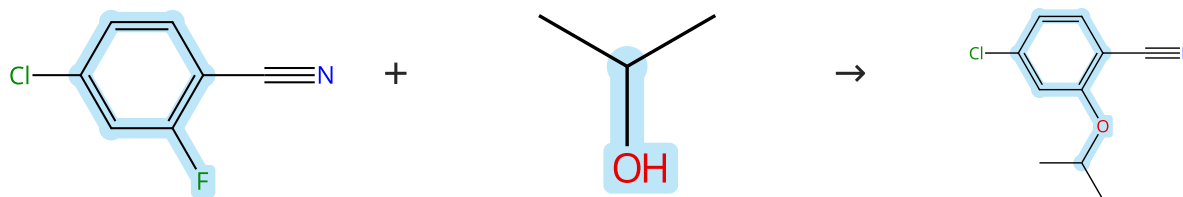
Ligand-Enabled Gold-Catalyzed C(sp²)-O Cross-Coupling Reactions

By: Das, Avishek; et al

ACS Catalysis (2023), 13(6), 3847-3853.

Scheme 137 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (88)

Suppliers (365)

Suppliers (29)

31-614-CAS-38198284

Steps: 1 Yield: 99%

1.1 Reagents: Potassium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; rt; overnight, rt

Experimental Protocols

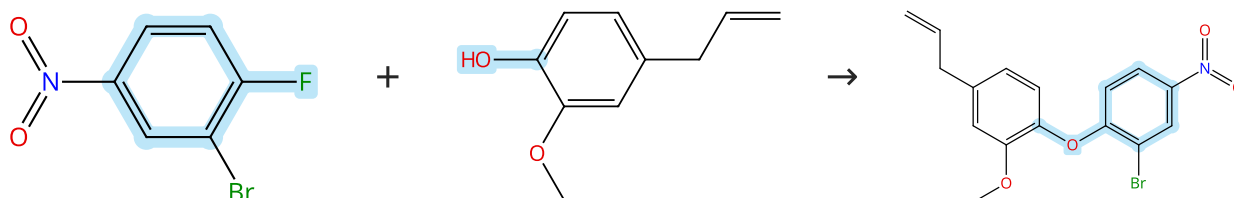
Development of Fluorescent AF64394 Analogues Enables Real-Time Binding Studies for the Orphan Class A GPCR GPR3

By: Bresinsky, Merlin; et al

Journal of Medicinal Chemistry (2023), 66(21), 15025-15041.

Scheme 138 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

Suppliers (120)

31-614-CAS-42944341

Steps: 1 Yield: 99%

1.1 Reagents: Potassium hydroxide
Catalysts: Tetrabutylammonium bromide
Solvents: 2-Methyltetrahydrofuran, Water; 10 min, 95 °C

Experimental Protocols

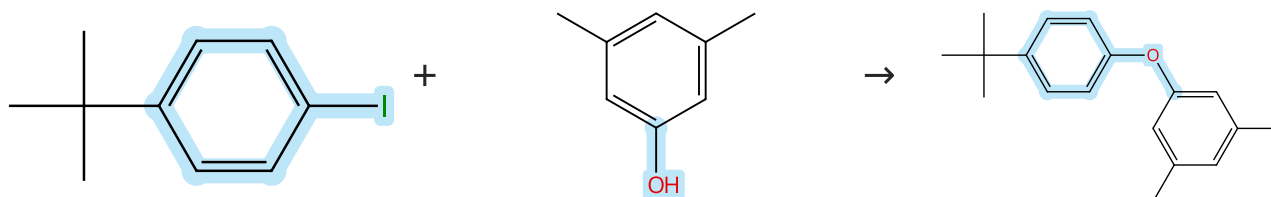
S_NAr Reactions Using Continuous Plug Flow...in Aqueous Biphasic Media

By: Wong, Madison J.; et al

ACS Sustainable Chemistry & Engineering (2024), 12(52), 18725-18734.

Scheme 139 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (68)

Suppliers (89)

31-614-CAS-43350512

Steps: 1 Yield: 99%

1.1 Reagents: Cesium carbonate
Catalysts: Cuprous iodide, 1*H*-imidazolium, 3-[4-(trifluoromethyl)-2-pyridinyl]-1-(2,4,6-trimethylphenyl)-, bromide (1:1)
Solvents: Acetonitrile; 20 h, 110 °C

Experimental Protocols

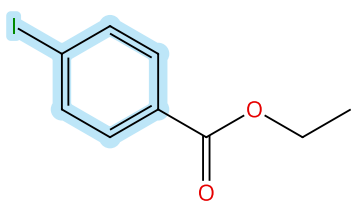
Bidentate NHC-Containing Ligands for Copper Catalyzed Synthesis of Functionalised Diaryl Ethers

By: Richards, Ethan M.; et al

Advanced Synthesis & Catalysis (2025), 367(6), e202400909.

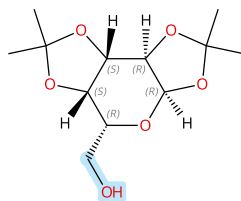
Scheme 140 (1 Reaction)

Steps: 1 Yield: 99%



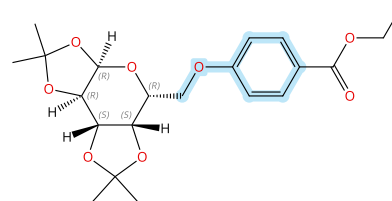
Suppliers (84)

+

Absolute stereochemistry shown,
Rotation (-)

Suppliers (78)

→



Absolute stereochemistry shown

31-614-CAS-35219901

Steps: 1 Yield: 99%

Photoinduced Nickel-Catalyzed Carbon-Heteroatom Coupling

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine**Solvents:** Tetrahydrofuran; 5 min, rt1.2 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene; 36 h, 30 °C1.3 **Solvents:** Water

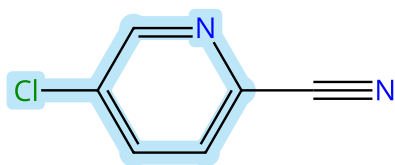
Experimental Protocols

By: Luo, Hang; et al

Chemistry - A European Journal (2023), 29(1), e202202385.

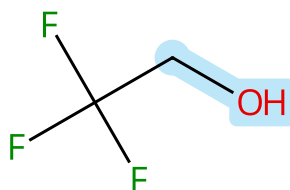
Scheme 141 (1 Reaction)

Steps: 1 Yield: 99%



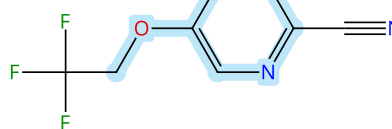
Suppliers (88)

+



Suppliers (98)

→



Suppliers (24)

31-614-CAS-24211999

Steps: 1 Yield: 99%

Scalable Asymmetric Synthesis of MK-8998, a T-type Calcium Channel Antagonist

1.1 **Reagents:** Cesium carbonate**Solvents:** Dimethyl sulfoxide; 1 - 2 h, 100 - 105 °C

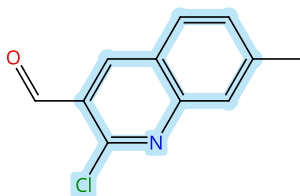
Experimental Protocols

By: Zhong, Yong-Li; et al

Journal of Organic Chemistry (2022), 87(4), 2120-2128.

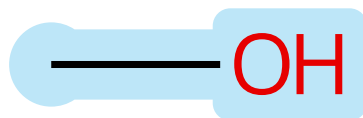
Scheme 142 (1 Reaction)

Steps: 1 Yield: 99%



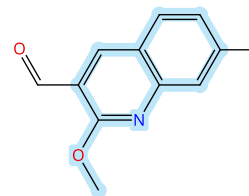
Suppliers (67)

+



Suppliers (444)

→



Suppliers (25)

31-614-CAS-24991585

Steps: 1 Yield: 99%

Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines

1.1 **Reagents:** Phenylboronic acid, Potassium hydroxide**Catalysts:** 2721440-87-3**Solvents:** Methanol; 9 h, reflux

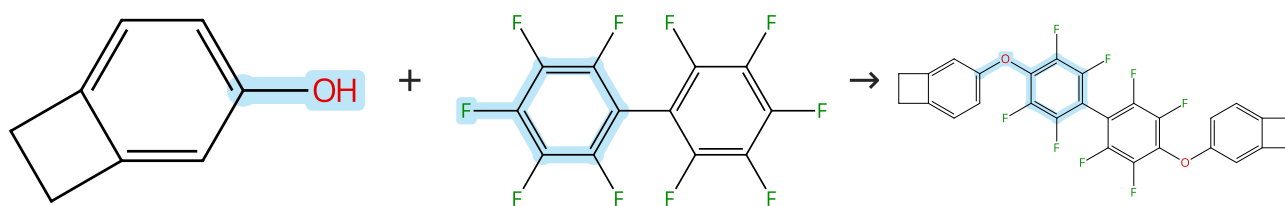
Experimental Protocols

By: Nandhini, Sundar; et al

Applied Organometallic Chemistry (2021), 35(12), e6436.

Scheme 143 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (24)

Suppliers (68)

31-614-CAS-24325633

Steps: 1 Yield: 99%

A fluoropolymer with a low dielectric constant at a high frequency derived from bio-based anethole

1.1 Reagents: Potassium carbonate

Solvents: Dimethyl sulfoxide; rt → 60 °C; 3 h, 60 °C

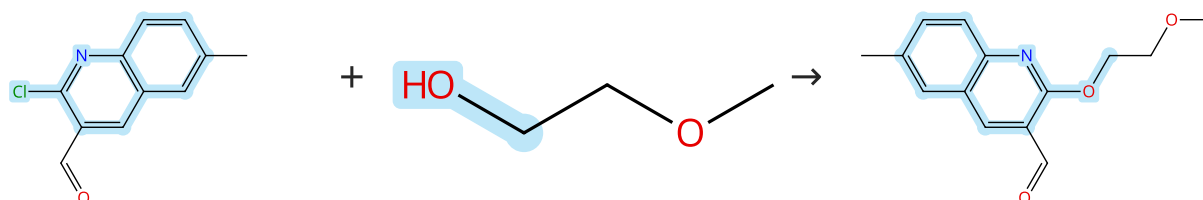
By: Li, Minghui; et al

Experimental Protocols

Polymer Chemistry (2021), 12(31), 4501-4507.

Scheme 144 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (77)

Suppliers (132)

31-614-CAS-24991577

Steps: 1 Yield: 99%

Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines

1.1 Reagents: Phenylboronic acid, Potassium hydroxide

Catalysts: 2721440-87-3

Solvents: 2-Methoxyethanol; 9 h, reflux

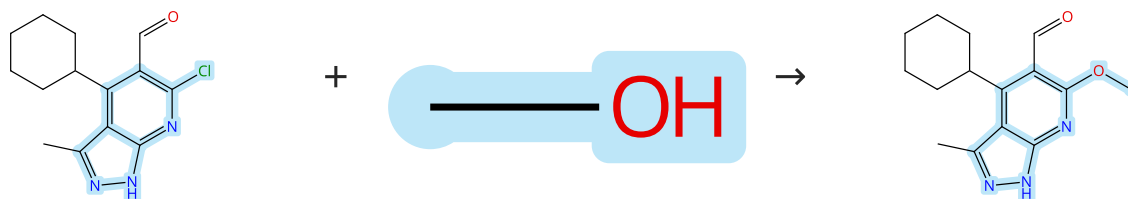
By: Nandhini, Sundar; et al

Experimental Protocols

Applied Organometallic Chemistry (2021), 35(12), e6436.

Scheme 145 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (444)

31-170-CAS-22580547

Steps: 1 Yield: 99%

Fragment-Based Discovery of Pyrazolopyridones as JAK1 Inhibitors with Excellent Subtype Selectivity

1.1 Reagents: Sodium methoxide

Solvents: Methanol; 40 min, 120 °C; 120 °C → rt

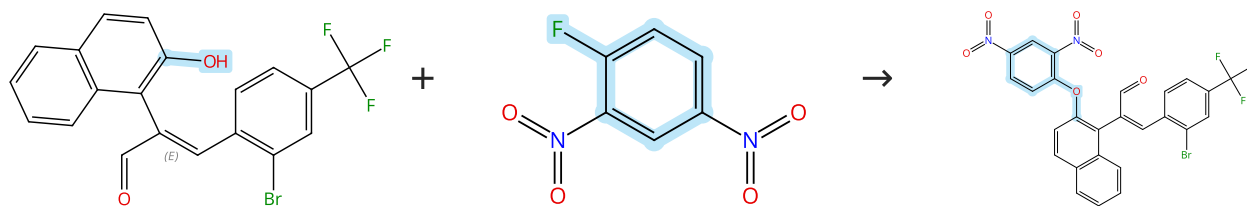
By: Hansen, Bettina Borreschmidt; et al

1.2 Reagents: Acetic acid; 30 min, rt

Journal of Medicinal Chemistry (2020), 63(13), 7008-7032.

Scheme 146 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560064

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

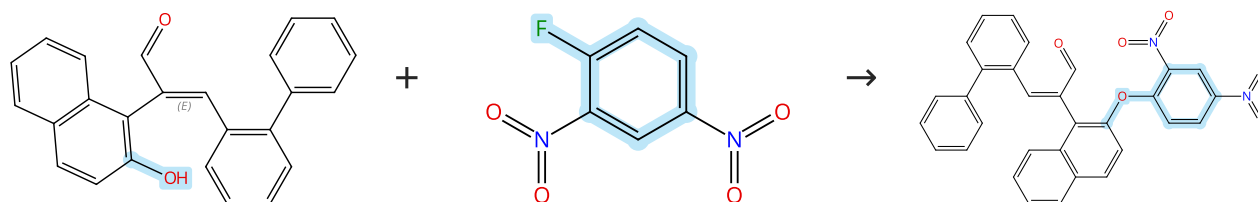
Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

Nature Communications (2023), 14(1), 5050.

Scheme 147 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560038

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

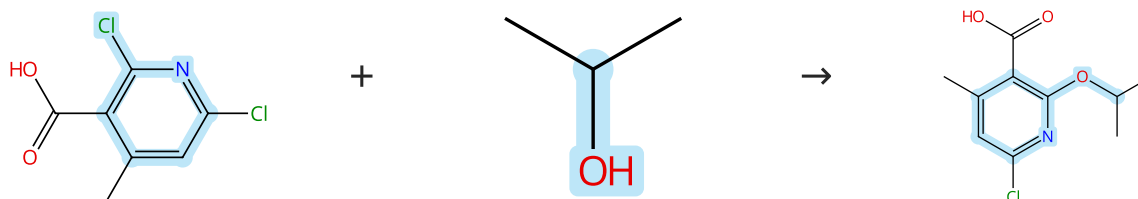
Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

Nature Communications (2023), 14(1), 5050.

Scheme 148 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (69)

Suppliers (365)

Supplier (1)

31-614-CAS-34951950

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran; 0 °C; 3 - 23 h, 70 °C
 1.2 **Reagents:** Sodium hydroxide
Solvents: Water

Experimental Protocols

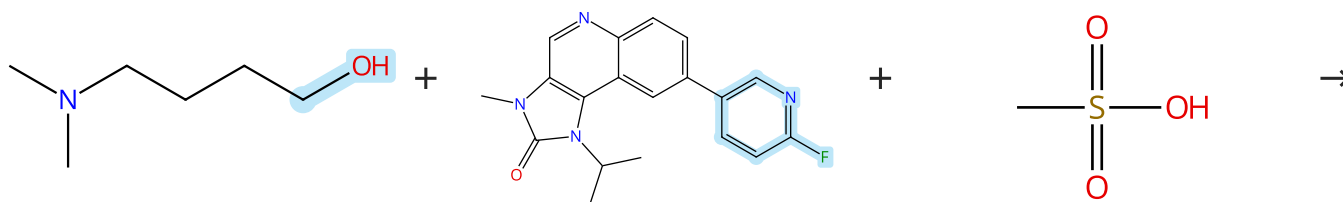
Replacing the oxidation-sensitive triaminoaryl chemotype of problematic $Ky7$ channel openers: Exploration of a nicotini namide scaffold

By: Wurm, Konrad W.; et al

Archiv der Pharmazie (Weinheim, Germany) (2023), 356(2), 2200473.

Scheme 149 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (60)

Suppliers (119)

Multi-component
structure image
available in CAS
SciFinder

31-614-CAS-39405348

Steps: 1 Yield: 99%

Identification of Novel, Selective Ataxia-Telangiectasia
Mutated Kinase Inhibitors with the Ability to Penetrate the
Blood-Brain Barrier: The Discovery of AZD1390

By: Pike, Kurt G.; et al

Journal of Medicinal Chemistry (2024), 67(4), 3090-3111.

1.1 Reagents: Sodium hydride

Solvents: Tetrahydrofuran; 10 - 30 min, 0 °C

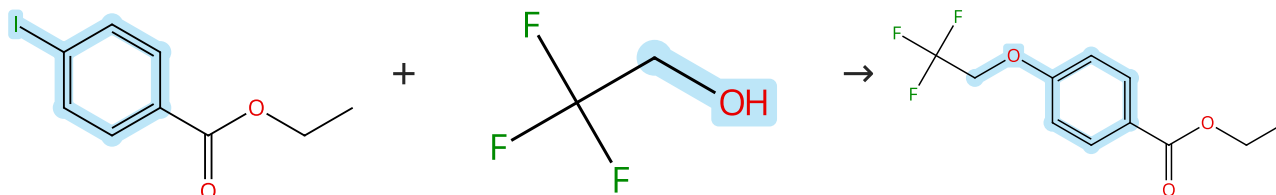
1.2 Solvents: Tetrahydrofuran; 1 - 24 h, rt → 70 °C

1.3 Solvents: Dichloromethane; rt

Experimental Protocols

Scheme 150 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (84)

Suppliers (98)

Suppliers (13)

31-614-CAS-35219840

Steps: 1 Yield: 99%

Photoinduced Nickel-Catalyzed Carbon-Heteroatom Coupling

By: Luo, Hang; et al

Chemistry - A European Journal (2023), 29(1), e202202385.

1.1 Catalysts: Nickel dichloride, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine

Solvents: Tetrahydrofuran; 5 min, rt

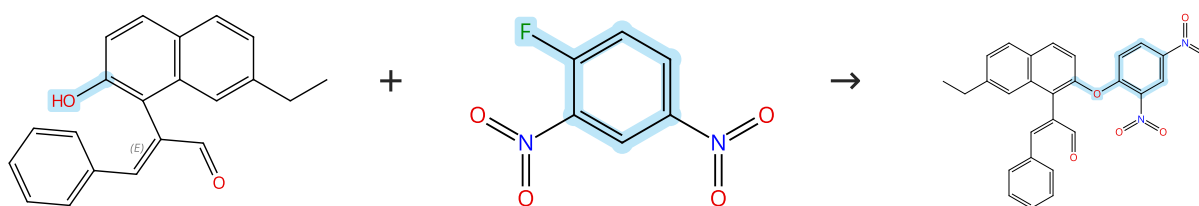
1.2 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene; 18 h, 30 °C

1.3 Solvents: Water

Experimental Protocols

Scheme 151 (1 Reaction)

Steps: 1 Yield: 99%



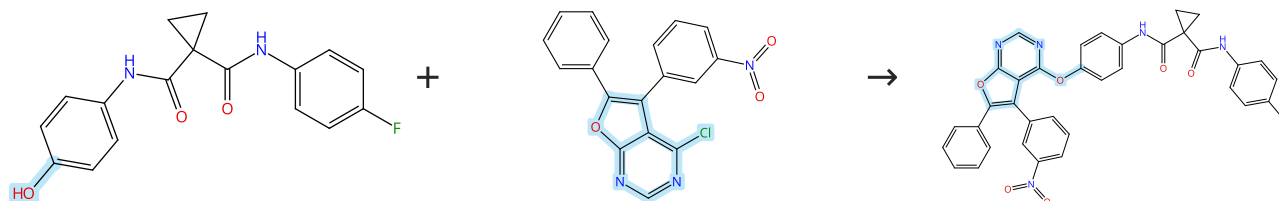
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560085 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
--	--

Scheme 152 (1 Reaction)

Steps: 1 Yield: 99%

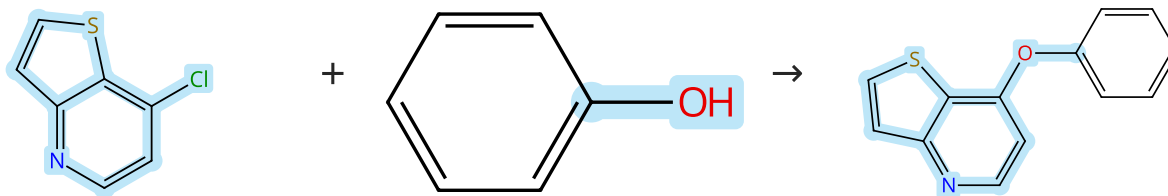


Suppliers (58)

31-614-CAS-41112459 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 0 °C; 20 min, rt; rt → 0 °C 1.2 24 h, rt Experimental Protocols	Discovery of Dual MER/AXL Kinase Inhibitors as Bifunctional Small Molecules for Inhibiting Tumor Growth and Enhancing Tumor Immune Microenvironment By: Li, Mu-Chun; et al Journal of Medicinal Chemistry (2024), 67(13), 10906-10927.
--	---

Scheme 153 (1 Reaction)

Steps: 1 Yield: 99%



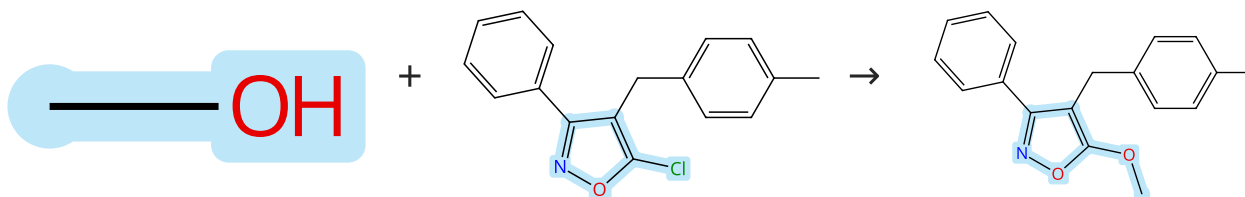
Suppliers (79)

Suppliers (207)

31-614-CAS-24809919 Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 4 h, 135 °C Experimental Protocols	Microwave-Assisted CH Activation of the Thiophene Ring: A Simple Approach to Access Substituted Thienopyridines By: Campos, Joana F.; et al ChemistrySelect (2021), 6(45), 12556-12561.
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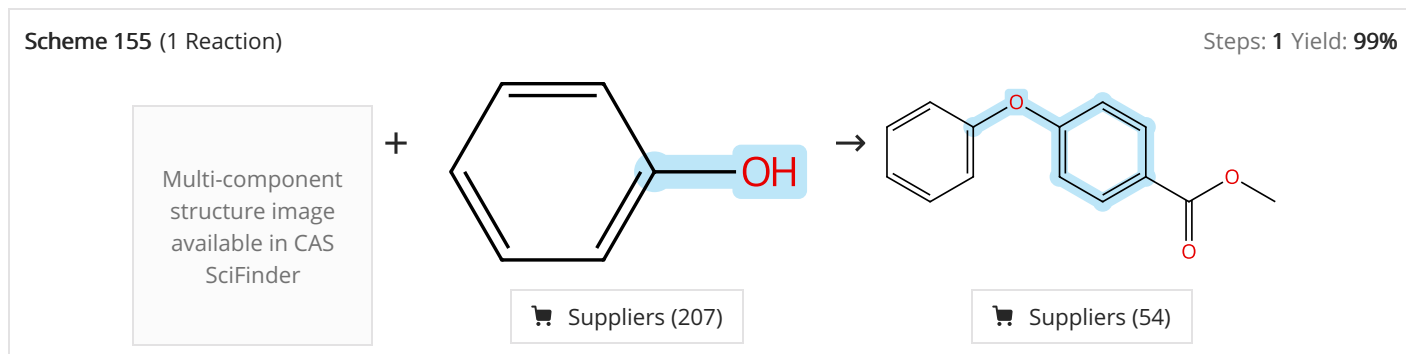
Scheme 154 (1 Reaction)

Steps: 1 Yield: 99%

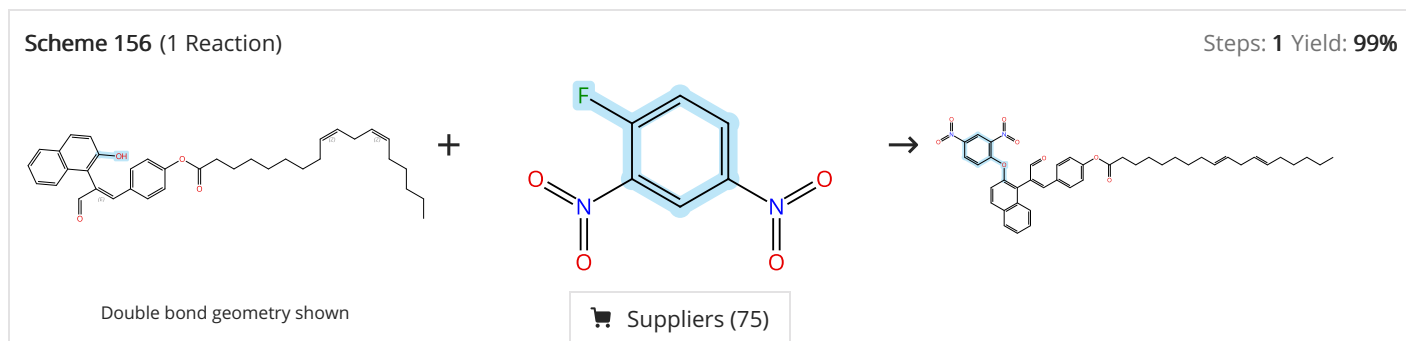


Suppliers (444)

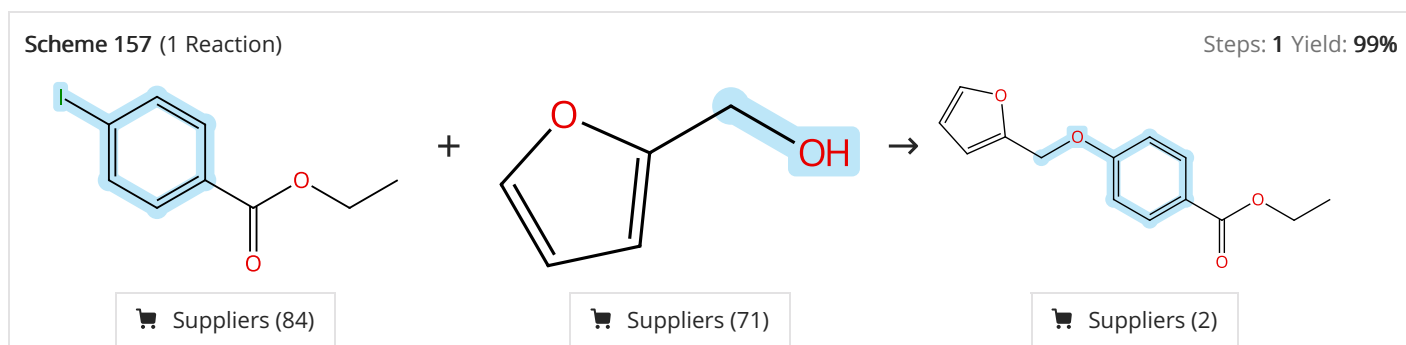
31-614-CAS-32626947	Steps: 1 Yield: 99%	A One-pot Synthesis of Multifunctionalized 1-Pyrrolines from 2-Alkyl-2H-azirines and Diazocarbonyl Compounds
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; 0.5 h, 0 °C		By: Filippov, Ilya P.; et al
1.2 3 - 12 h, reflux; reflux → rt		Journal of Organic Chemistry (2022), 87(13), 8835-8840.
1.3 Reagents: Hydrochloric acid Solvents: Water; pH 3 - 4, rt		
Experimental Protocols		



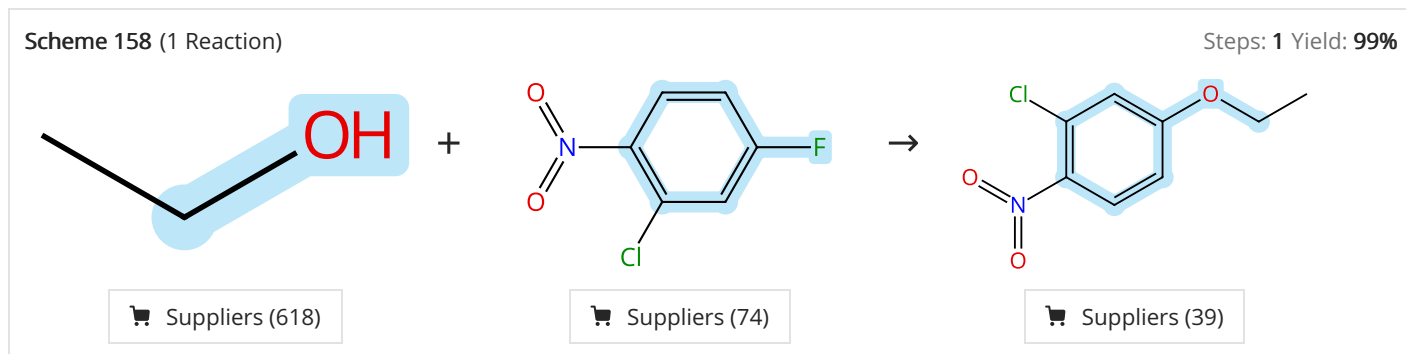
31-614-CAS-31534019	Steps: 1 Yield: 99%	Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates
1.1 Reagents: Sodium carbonate Solvents: Dichloromethane, Water; 18 h, 40 °C		By: Kikushima, Kotaro; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Organic Letters (2022), 24(10), 1924-1928.
Experimental Protocols		



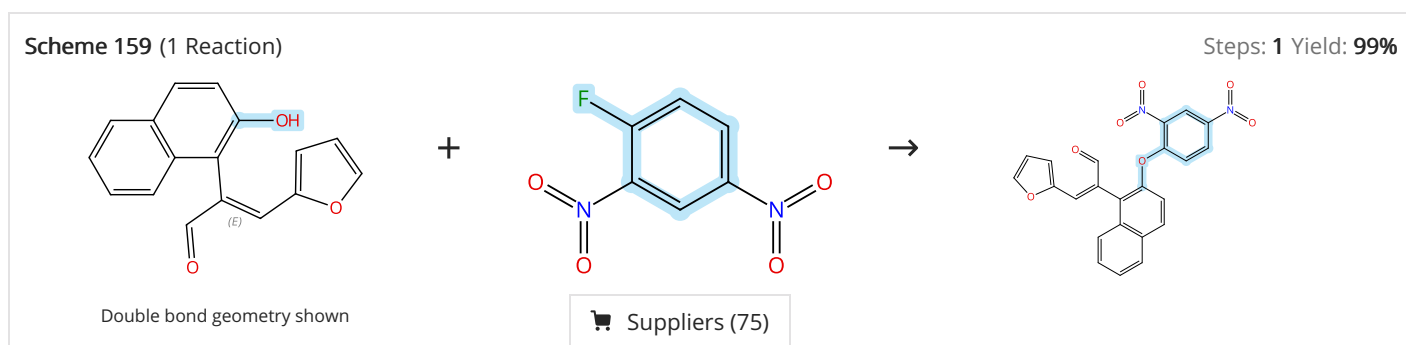
31-614-CAS-37560110	Steps: 1 Yield: 99%	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution
1.1 Reagents: Potassium carbonate Catalysts: 2412758-02-0 Solvents: Chloroform; 24 h, -10 °C		By: Guo, Fengyuan; et al
Experimental Protocols		Nature Communications (2023), 14(1), 5050.



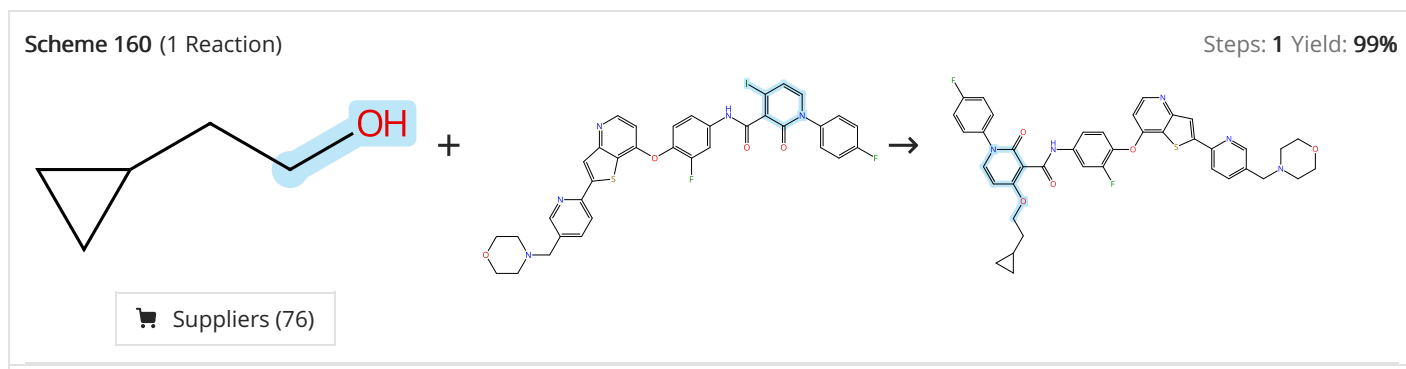
31-614-CAS-35219851	Steps: 1 Yield: 99%	Photoinduced Nickel-Catalyzed Carbon-Heteroatom Coupling
1.1 Catalysts: Nickel dichloride, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Tetrahydrofuran; 5 min, rt		By: Luo, Hang; et al
1.2 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene; 18 h, 30 °C		Chemistry - A European Journal (2023), 29(1), e202202385.
1.3 Solvents: Water		
Experimental Protocols		



31-170-CAS-24143824	Steps: 1 Yield: 99%	Crystallography-guided discovery of carbazole-based retinoic acid-related orphan receptor gamma-t (RORγt) modulators: insights into different protein behaviors with "short" and "long" inverse agonists
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 0 °C; 30 min, rt		By: Yu, Ming-cheng; et al
1.2 rt → 60 °C; 3 h, 60 °C		Acta Pharmacologica Sinica (2021), 42(9), 1524-1534.
Experimental Protocols		



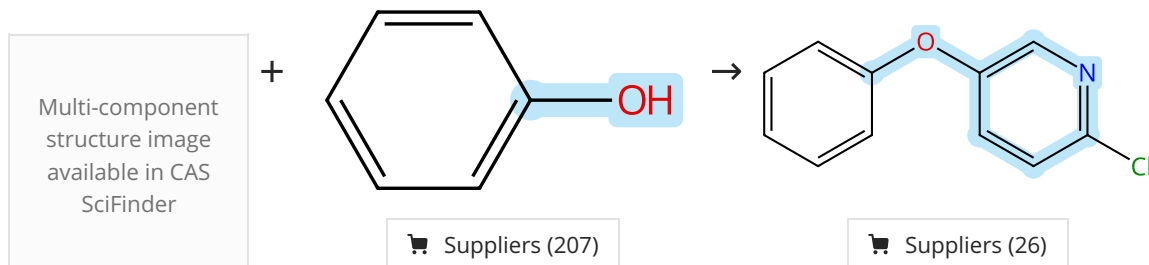
31-614-CAS-37560072	Steps: 1 Yield: 99%	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution
1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2 <i>S</i> ,3 <i>R</i>)-2-[[[(2 <i>R</i> ,3 <i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C		By: Guo, Fengyuan; et al
Experimental Protocols		Nature Communications (2023), 14(1), 5050.



31-614-CAS-37614593	Steps: 1 Yield: 99%	Discovery of Novel, Thienopyridine-Based Tyrosine Kinase Inhibitors Targeting Tumorigenic RON Splice Variants By: Ryu, Hyun; et al ACS Medicinal Chemistry Letters (2023), 14(9), 1198-1207.
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; 10 min, rt		
1.2 Solvents: Tetrahydrofuran; 1 h, rt		

Scheme 161 (1 Reaction)

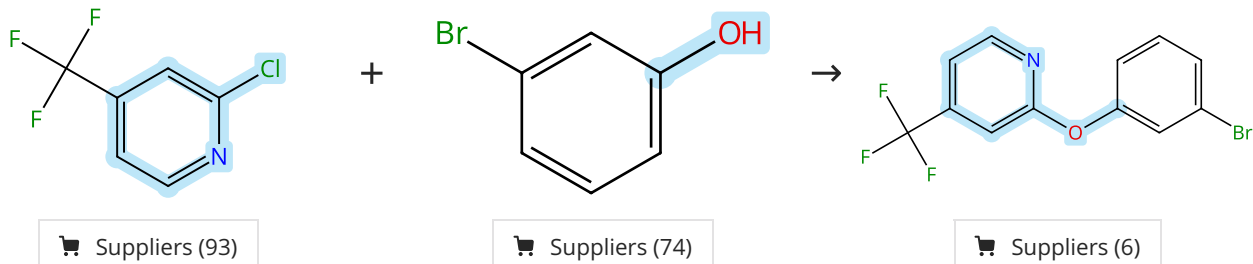
Steps: 1 Yield: 99%



31-614-CAS-31534018	Steps: 1 Yield: 99%	Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates By: Kikushima, Kotaro; et al Organic Letters (2022), 24(10), 1924-1928.
1.1 Reagents: Sodium carbonate Solvents: Dichloromethane, Water; 18 h, 40 °C		
1.2 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 162 (1 Reaction)

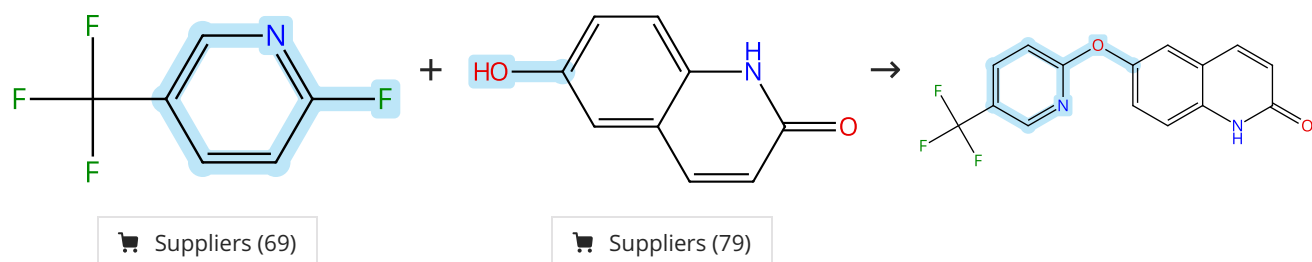
Steps: 1 Yield: 99%



31-614-CAS-39012958	Steps: 1 Yield: 99%	Reversible Monoacylglycerol Lipase Inhibitors: Discovery of a New Class of Benzylpiperidine Derivatives By: Bononi, Giulia; et al Journal of Medicinal Chemistry (2022), 65(10), 7118-7140.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, 110 °C		
Experimental Protocols		

Scheme 163 (1 Reaction)

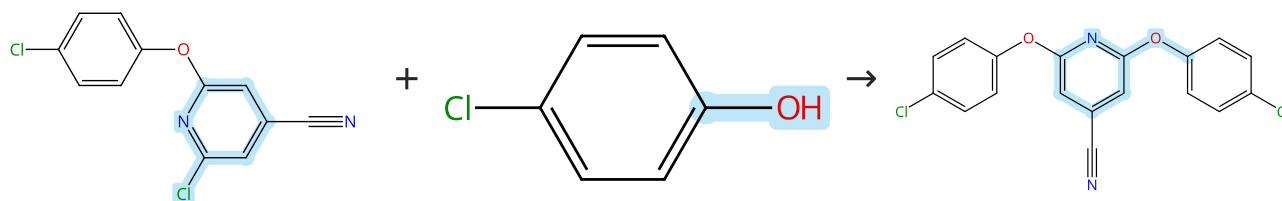
Steps: 1 Yield: 99%



31-614-CAS-32068432	Steps: 1 Yield: 99%	Discovery of (R)-(3-fluoropyrrolidin-1-yl)(6-((5-(trifluoromethyl)pyridin-2-yl)oxy)quinolin-2-yl)methanone (ABBV-318) and analogs as small molecule Na _v 1.7/ Na _v 1.8 blockers for the treatment of pain
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 18 h, 100 °C		By: Patel, Meena V.; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2022), 63, 116743.

Scheme 164 (1 Reaction)

Steps: 1 Yield: 99%



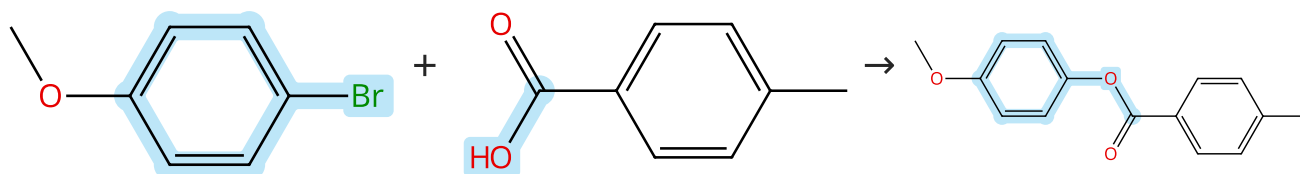
Suppliers (7)

Suppliers (90)

31-170-CAS-21606411	Steps: 1 Yield: 99%	Substrate profiling of the cobalt nitrile hydratase from <i>Rhodococcus rhodochrous</i> ATCC BAA 870
1.1 Catalysts: Tris(dibenzylideneacetone)dipalladium, BINAP Solvents: 1,4-Dioxane; 5 min, 80 °C		By: Mashweu, Adelaide R.; et al
1.2 Reagents: Sodium <i>tert</i> -butoxide; 24 h, 110 - 120 °C		Molecules (2020), 25(1), 238pp..
Experimental Protocols		

Scheme 165 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (71)

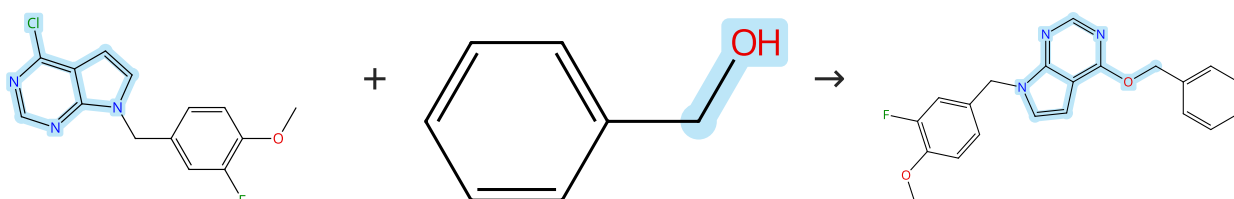
Suppliers (99)

Suppliers (7)

31-614-CAS-44096445	Steps: 1 Yield: 99%	Photosensitized Gold-Catalyzed Cross-Couplings of Aryl Bromides
1.1 Reagents: Sodium carbonate, Silver hexafluoroantimonate Catalysts: Acridinium, 10-methyl-9-(2,4,6-trimethylphenyl)-, perchlorate (1:1), [2-[Bis(tricyclo[3.3.1.1 ^{3,7}]dec-1-yl)phosphino-κP]-N,N-dimethylbenzenamine]chlorogold Solvents: 1,2-Dichloroethane; 18 h, 30 °C		By: Wu, Jiawen; et al
Experimental Protocols		Journal of the American Chemical Society (2025), 147(7), 5839-5850.

Scheme 166 (1 Reaction)

Steps: 1 Yield: 99%

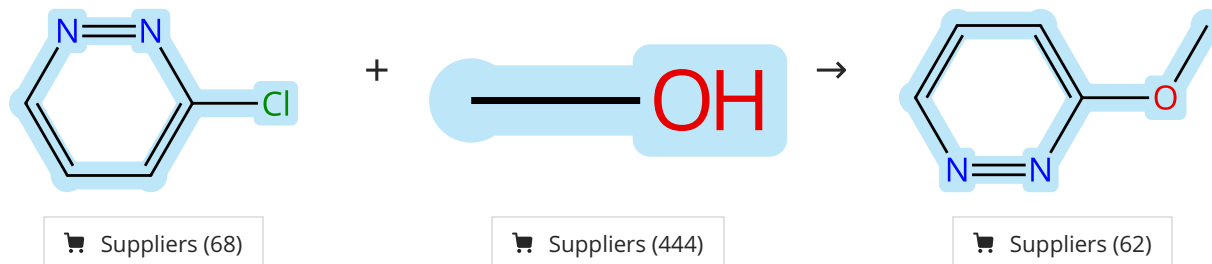


Suppliers (150)

31-614-CAS-24160177	Steps: 1 Yield: 99%	Design, synthesis and in vitro/vivo anticancer activity of 4-substituted 7-(3-fluoro-4-methoxybenzyl)-7H-pyrrolo[2,3-d]pyrimidines
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; 0 °C; 4 h, rt		By: Tsai, Pei-Yi; et al
1.2 Reagents: Water		Journal of the Chinese Chemical Society (Weinheim, Germany) (2021), 68(9), 1761-1770.
Experimental Protocols		

Scheme 167 (1 Reaction)

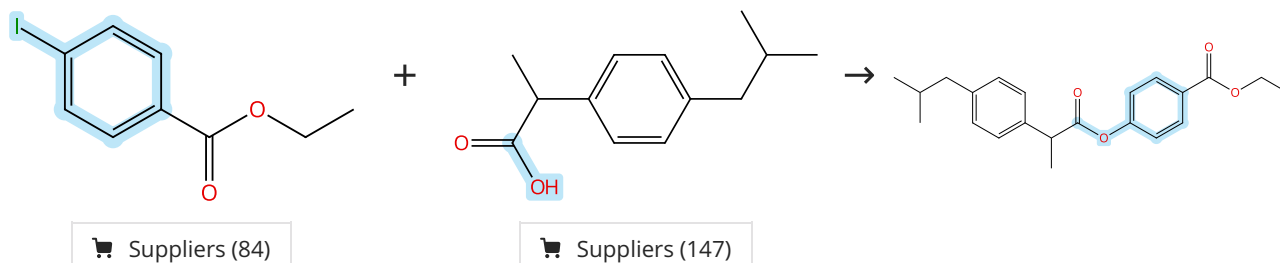
Steps: 1 Yield: 99%



31-170-CAS-21761938	Steps: 1 Yield: 99%	Selectivity and Physicochemical Optimization of Repurposed Pyrazolo[1,5-b]pyridazines for the Treatment of Human African Trypanosomiasis
1.1 Reagents: Sodium <i>tert</i> -butoxide Solvents: Methanol; 16 h, rt → 65 °C		By: Tear, Westley F.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2020), 63(2), 756-783.

Scheme 168 (1 Reaction)

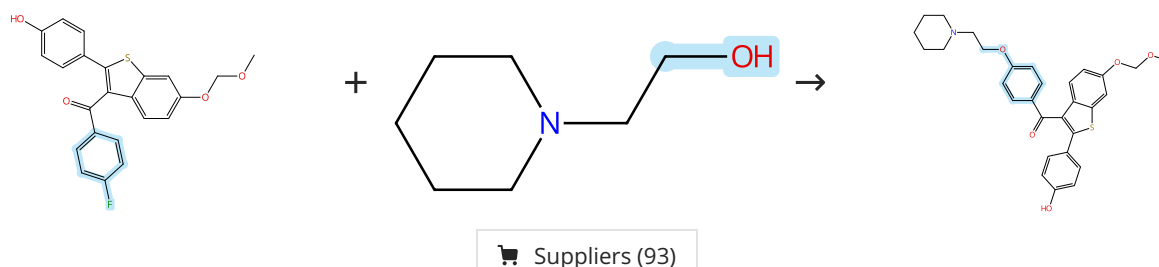
Steps: 1 Yield: 99%



31-614-CAS-35219898	Steps: 1 Yield: 99%	Photoinduced Nickel-Catalyzed Carbon-Heteroatom Coupling
1.1 Catalysts: Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Tetrahydrofuran; 5 min, rt		By: Luo, Hang; et al
1.2 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene; 36 h, 30 °C		Chemistry - A European Journal (2023), 29(1), e202202385.
1.3 Solvents: Water		
Experimental Protocols		

Scheme 169 (1 Reaction)

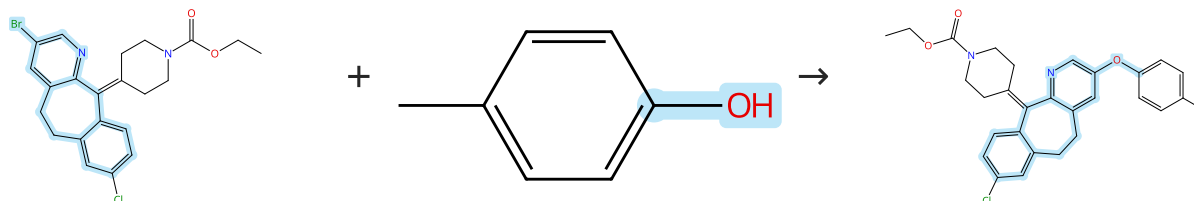
Steps: 1 Yield: 99%



31-614-CAS-41049047	Steps: 1 Yield: 99%	Design, synthesis and antiproliferative activity of raloxifene/histone deacetylase inhibitor hybrids in breast cancer
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 3 h, 50 °C		By: Wang, Yufei; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2024), 274, 116533.

Scheme 170 (1 Reaction)

Steps: 1 Yield: 99%



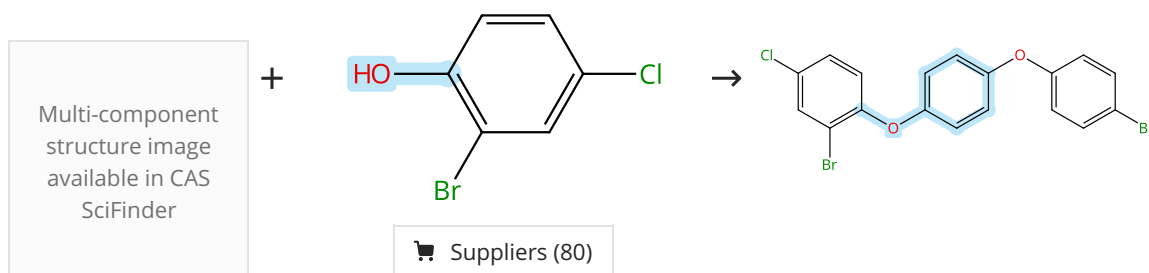
Suppliers (9)

Suppliers (89)

31-170-CAS-23366897	Steps: 1 Yield: 99%	Oxalohydrazide Ligands for Copper-Catalyzed C-O Coupling Reactions with High Turnover Numbers
1.1 Reagents: Tripotassium phosphate Catalysts: Copper bromide (CuBr), <i>N</i> ¹ , <i>N</i> ² -Bis(2,5-dimethyl-1 <i>H</i> -pyrrol-1-yl)ethanediamide Solvents: Dimethyl sulfoxide; 5 - 10 min, rt		By: Ray, Ritwika; et al
1.2 24 h, 100 °C		Angewandte Chemie, International Edition (2021), 60(15), 8203-8211.
Experimental Protocols		

Scheme 171 (1 Reaction)

Steps: 1 Yield: 99%



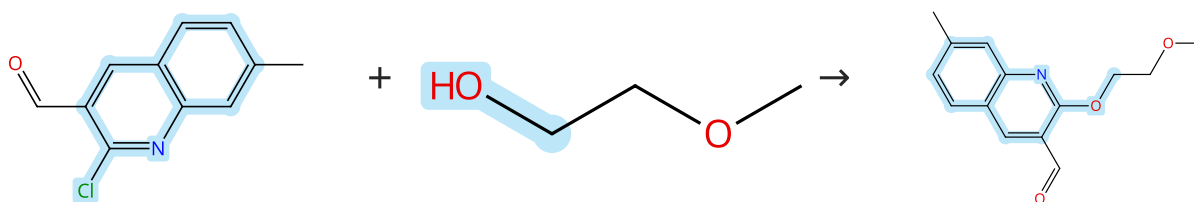
Multi-component
structure image
available in CAS
SciFinder

Suppliers (80)

31-614-CAS-41866199	Steps: 1 Yield: 99%	Synthesis of Diaryl Ethers via Hypervalent Iodine-Mediated C-H Functionalization
1.1 Reagents: Cesium carbonate Solvents: Toluene; 8 h, 80 °C		By: Hu, Chenghu; et al
Experimental Protocols		Organic Letters (2024), 26(37), 7864-7868.

Scheme 172 (1 Reaction)

Steps: 1 Yield: 99%



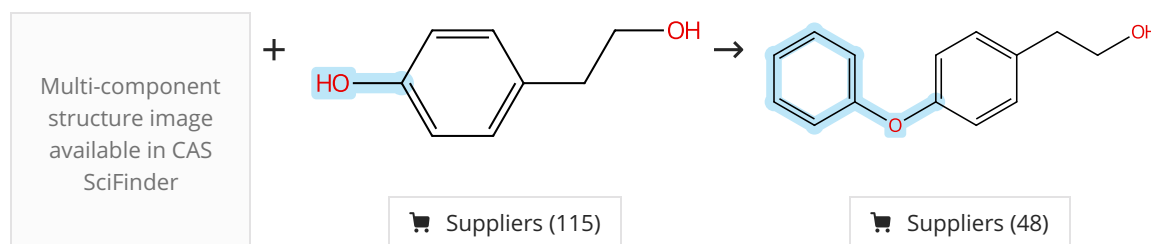
Suppliers (67)

Suppliers (132)

31-614-CAS-24991594	Steps: 1 Yield: 99%	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines By: Nandhini, Sundar; et al Applied Organometallic Chemistry (2021), 35(12), e6436.
1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: 2-Methoxyethanol; 9 h, reflux		
Experimental Protocols		

Scheme 173 (1 Reaction)

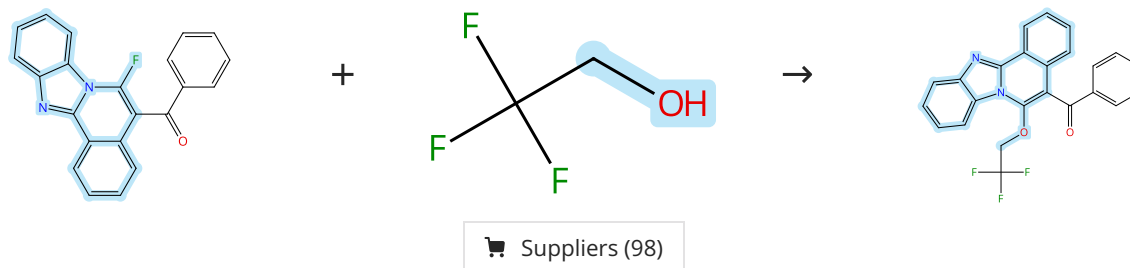
Steps: 1 Yield: 99%



31-614-CAS-31534024	Steps: 1 Yield: 99%	Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates By: Kikushima, Kotaro; et al Organic Letters (2022), 24(10), 1924-1928.
1.1 Reagents: Sodium carbonate Solvents: Dichloromethane, Water; 18 h, 40 °C		
1.2 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 174 (1 Reaction)

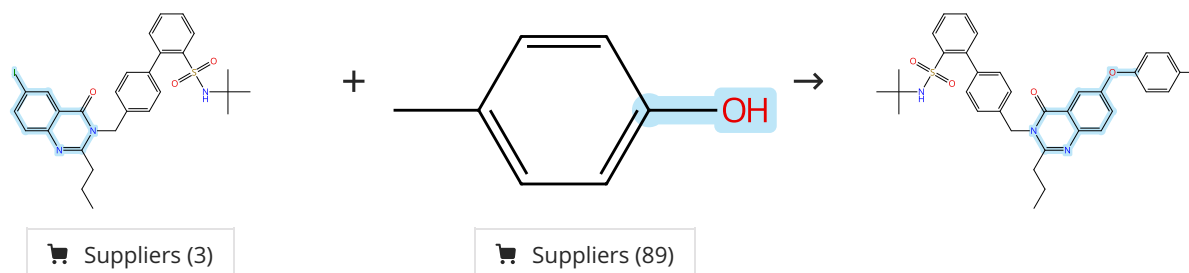
Steps: 1 Yield: 99%



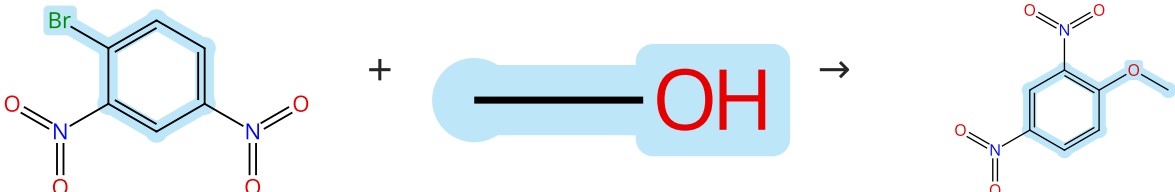
31-614-CAS-36993217	Steps: 1 Yield: 99%	Cascade C-H Activation and Defluorinative Annulation of 2-Arylbenzimidazoles with α-Trifluoromethyl-α-diazoketones: Modular Assembly of 6-Fluorobenzimidazo[2,1-a]isoquinolines By: Dong, Zhongkang; et al Organic Letters (2023), 25(26), 4770-4775.
1.1 Reagents: Potassium carbonate; overnight, 100 °C		
Experimental Protocols		

Scheme 175 (1 Reaction)

Steps: 1 Yield: 99%

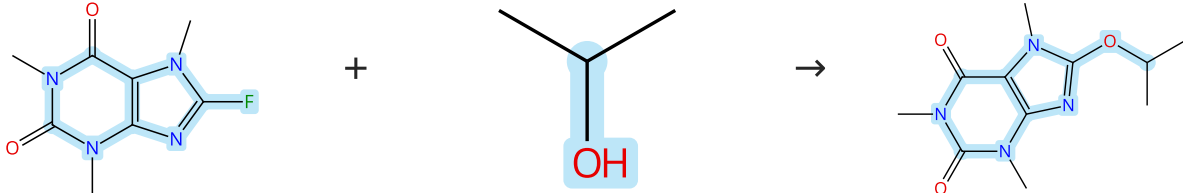


31-170-CAS-23364114	Steps: 1 Yield: 99%	Oxalohydrazide Ligands for Copper-Catalyzed C-O Coupling Reactions with High Turnover Numbers
1.1 Reagents: Tripotassium phosphate Catalysts: Copper bromide (Cu Br), <i>N</i> ¹ , <i>N</i> ² -Bis(2,5-dimethyl-1 <i>H</i> -pyrrol-1-yl)ethanediamide Solvents: Dimethyl sulfoxide; 5 - 10 min, rt	By: Ray, Ritwika; et al	Angewandte Chemie, International Edition (2021), 60(15), 8203-8211.
1.2 24 h, 100 °C		
Experimental Protocols		

Scheme 176 (2 Reactions)		Steps: 1 Yield: 99%
		
 Suppliers (57)	 Suppliers (444)	 Suppliers (28)

31-614-CAS-40947224	Steps: 1 Yield: 99%	Polyetheretherketone fiber-supported polyethylene glycols for phase-transfer catalysis in its surface layer
1.1 Reagents: Sodium hydroxide Catalysts: Polyethylene glycol, Poly(oxy-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene) (chloromethylated) Solvents: Water; 2 h, 65 °C	By: Shi, Xian-Lei; et al	Colloids and Surfaces, A: Physicochemical and Engineering Aspects (2024), 694, 134160.
Experimental Protocols		

31-614-CAS-35190788	Steps: 1 Yield: 99%	Quaternary ammonium salt functionalized polyetheretherketone fiber: A novel fibrous material for phase-transfer catalysis
1.1 Reagents: Sodium hydroxide Catalysts: Chloromethyl methyl ether, Triethylamine, hydrochloride (reaction products with PEEK), Poly(oxy-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene) (reaction products with triethylammonium chloride) Solvents: Methanol, Water; 2 h, 65 °C	By: Liu, Shuangshuang; et al	Reactive & Functional Polymers (2023), 182, 105479.
Experimental Protocols		

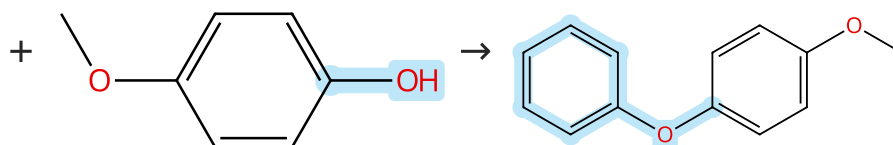
Scheme 177 (1 Reaction)		Steps: 1 Yield: 99%
		
 Suppliers (2)	 Suppliers (365)	 Suppliers (7)

31-614-CAS-39126907	Steps: 1 Yield: 99%	Direct Synthesis of 8-Fluorocaffeine and Its Transformation to 8-Substituted Caffeine
1.1 Reagents: Potassium hydroxide; 1 h, 80 °C	By: Rachmadhaningtiyas, Dyah Ayu; et al	ChemistrySelect (2023), 8(47), e202303074.
Experimental Protocols		

Scheme 178 (1 Reaction)

Steps: 1 Yield: 99%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (121)

Suppliers (52)

31-614-CAS-31534007

Steps: 1 Yield: 99%

Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates

By: Kikushima, Kotaro; et al

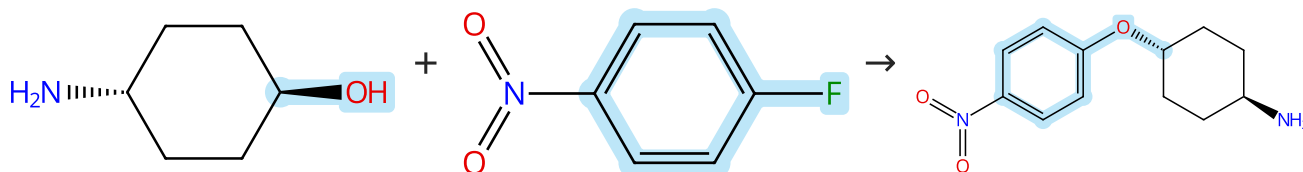
Organic Letters (2022), 24(10), 1924-1928.

- 1.1 **Reagents:** Sodium carbonate
Solvents: Dichloromethane, Water; 18 h, 40 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 179 (1 Reaction)

Steps: 1 Yield: 99%



Relative stereochemistry shown

Suppliers (105)

Suppliers (100)

Relative stereochemistry shown

Supplier (1)

31-614-CAS-34480077

Steps: 1 Yield: 99%

A bioisosteric approach to the discovery of novel N-aryl-N-[4-(aryloxy)cyclohexyl]squaramide-based activators of eukaryotic initiation factor 2 alpha (eIF2α) phosphorylation

By: Kwak, Jinsook; et al

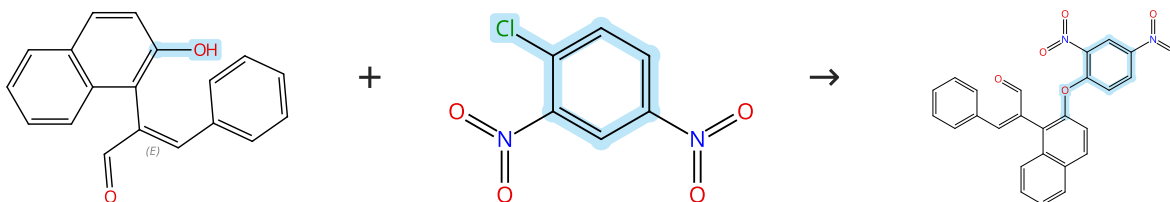
European Journal of Medicinal Chemistry (2022), 239, 114501.

- 1.1 **Reagents:** Sodium hydride
Solvents: Dimethylformamide; 0 °C; 1 h, rt
- 1.2 2 h, 60 °C; 60 °C → rt
- 1.3 **Solvents:** Water; rt

Experimental Protocols

Scheme 180 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560233

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

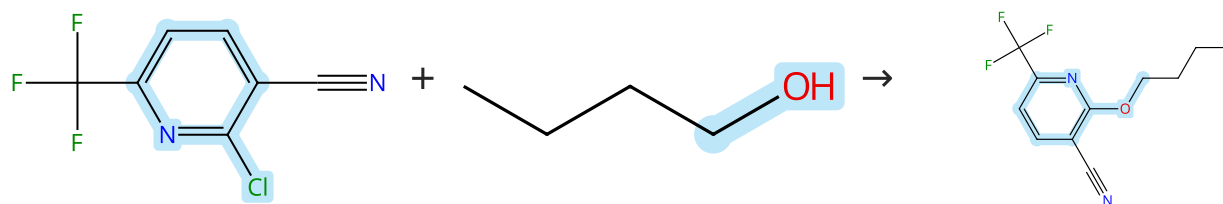
Nature Communications (2023), 14(1), 5050.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

Scheme 181 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (71)

Suppliers (201)

Suppliers (7)

31-614-CAS-40536030

Steps: 1 Yield: 99%

Discovery of N-(1,4-Benzoxazin-3-one) urea analogs as Mode-Selective TRPV1 antagonists

1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 0 °C; overnight, rt

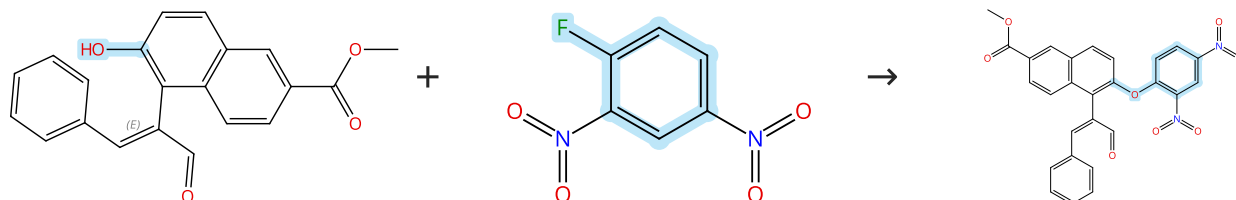
By: Huang, Guocheng; et al

Experimental Protocols

Bioorganic & Medicinal Chemistry Letters (2024), 106, 129735.

Scheme 182 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560103

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

1.1 Reagents: Potassium carbonate

Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)

Solvents: Chloroform; 24 h, -10 °C

By: Guo, Fengyuan; et al

Nature Communications (2023), 14(1), 5050.

Experimental Protocols

Scheme 183 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (106)

Suppliers (82)

Suppliers (7)

31-170-CAS-21785739

Steps: 1 Yield: 99%

A Unified and Practical Method for Carbon-Heteroatom Cross-Coupling using Nickel/Photo Dual Catalysis

1.1 Reagents: Potassium carbonate

Catalysts: 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy- κO)ethane]nickel, *fac*-Tris(2-(2-pyridinyl)phenyl)iridium

Solvents: Dimethylformamide; 5 min, rt

By: Escobar, Randolph A.; et al

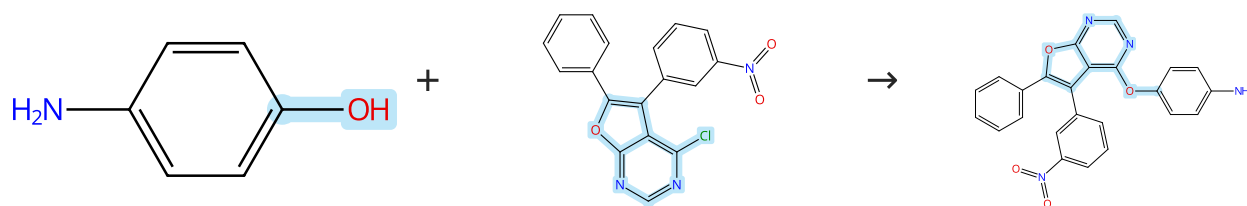
Chemistry - A European Journal (2020), 26(23), 5168-5173.

1.2 18 h, 35 °C

Experimental Protocols

Scheme 184 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (100)

31-614-CAS-41112462

Steps: 1 Yield: 99%

Discovery of Dual MER/AXL Kinase Inhibitors as Bifunctional Small Molecules for Inhibiting Tumor Growth and Enhancing Tumor Immune Microenvironment

By: Li, Mu-Chun; et al

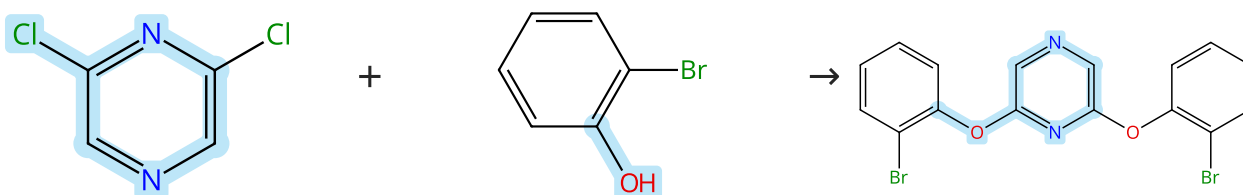
Journal of Medicinal Chemistry (2024), 67(13), 10906-10927.

- 1.1 Reagents: Sodium hydride
Solvents: Dimethylformamide; 30 min, 0 °C
- 1.2 16 h, rt

Experimental Protocols

Scheme 185 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (90)

Suppliers (82)

31-614-CAS-34395870

Steps: 1 Yield: 99%

Effect of Halogens in Bis(haloaryloxy)pyrazine Host Crystals on the Room Temperature Phosphorescence Properties of Bisbenzofuropyrazine Guest Luminophores

By: Tsuboi, Madoka; et al

Chemistry Letters (2022), 51(8), 819-822.

- 1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; 16 h, 160 °C

Experimental Protocols

31-170-CAS-23295464

Steps: 1 Yield: 99%

Room Temperature Phosphorescent Crystals Consisting of Cyclized Guests and Their Uncyclized Mother Host Molecules

By: Nakamura, Shotaro; et al

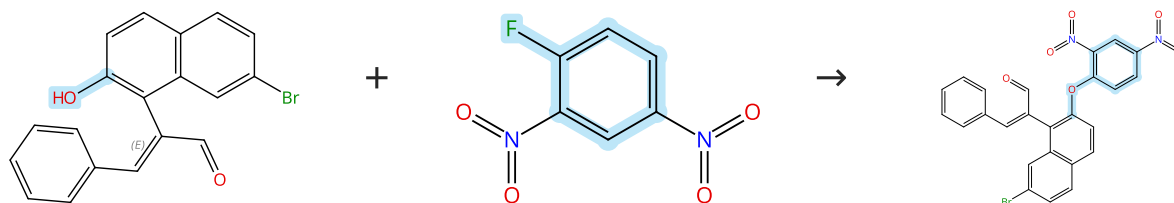
Chemistry Letters (2020), 49(8), 921-924.

- 1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; 16 h, 160 °C

Experimental Protocols

Scheme 186 (1 Reaction)

Steps: 1 Yield: 99%



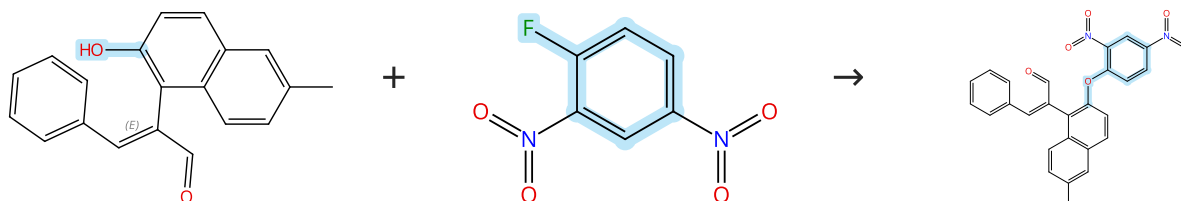
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560095 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
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Scheme 187 (1 Reaction)

Steps: 1 Yield: 99%



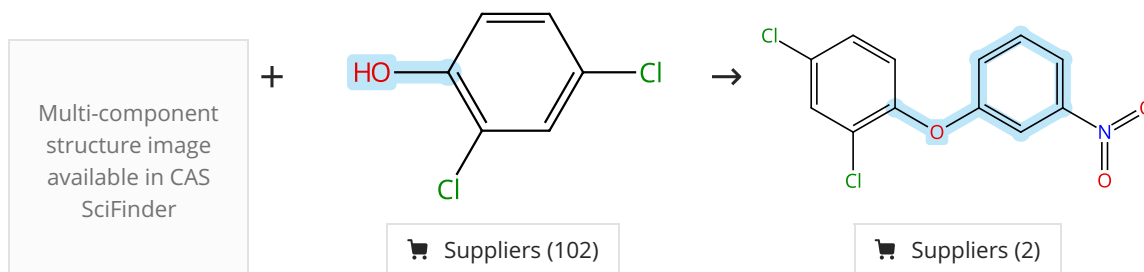
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560097 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
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Scheme 188 (1 Reaction)

Steps: 1 Yield: 99%



Multi-component structure image available in CAS SciFinder

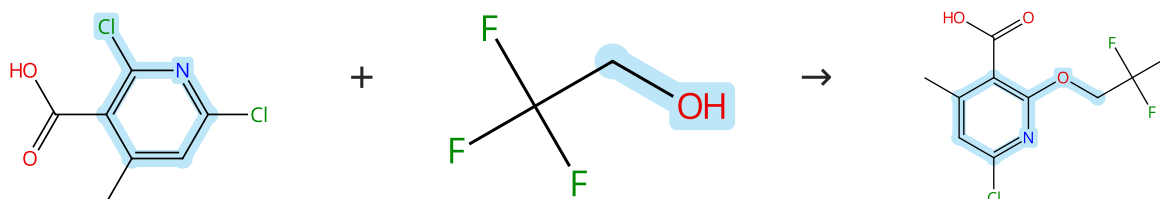
Suppliers (102)

Suppliers (2)

31-614-CAS-41866225 Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate Solvents: Toluene; 10 h, 80 °C Experimental Protocols	Synthesis of Diaryl Ethers via Hypervalent Iodine-Mediated C-H Functionalization By: Hu, Chenghu; et al Organic Letters (2024), 26(37), 7864-7868.
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Scheme 189 (1 Reaction)

Steps: 1 Yield: 99%



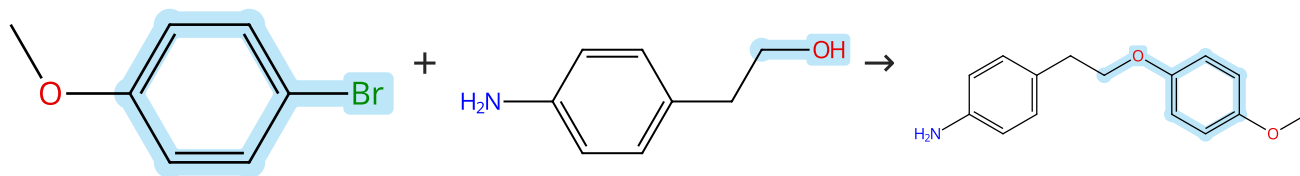
Suppliers (69)

Suppliers (98)

31-614-CAS-34951961	Steps: 1 Yield: 99%	Replacing the oxidation-sensitive triaminoaryl chemotype of problematic Kv7 channel openers: Exploration of a nicotinic amide scaffold
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; 0 °C; 3 - 23 h, 70 °C		By: Wurm, Konrad W.; et al
1.2 Reagents: Sodium hydroxide Solvents: Water		Archiv der Pharmazie (Weinheim, Germany) (2023), 356(2), 2200473.
Experimental Protocols		

Scheme 190 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (71)

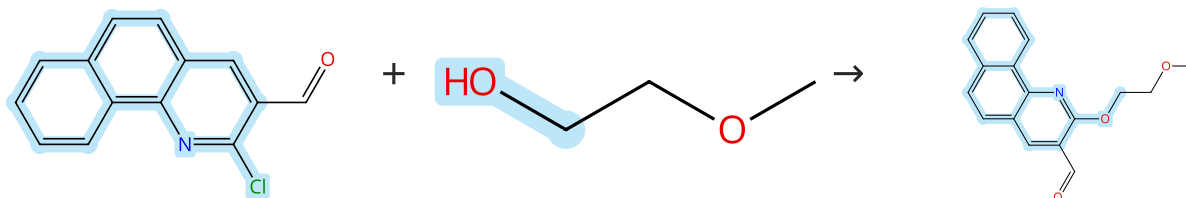
Suppliers (88)

Suppliers (2)

31-614-CAS-40801510	Steps: 1 Yield: 99%	Cu-Catalyzed Amination of Base-Sensitive Aryl Bromides and the Chemoselective N- and O-Arylation of Amino Alcohols
1.1 Reagents: Sodium <i>tert</i> -butoxide Catalysts: Cuprous iodide, 1,2-Benzenediamine, <i>N</i> ¹ , <i>N</i> ² -bis(2-phenyl-1-naphthalenyl)- Solvents: Dimethyl sulfoxide; 24 °C; 16 h, 24 °C		By: Strauss, Michael J.; et al
Experimental Protocols		Journal of the American Chemical Society (2024), 146(27), 18616-18625.

Scheme 191 (1 Reaction)

Steps: 1 Yield: 99%



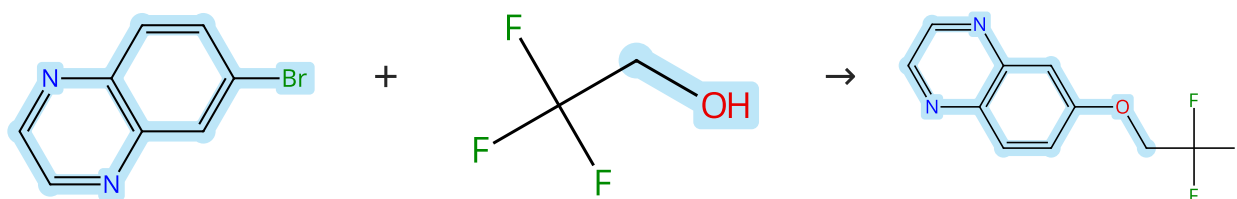
Suppliers (9)

Suppliers (132)

31-614-CAS-24991591	Steps: 1 Yield: 99%	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines
1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: 2-Methoxyethanol; 11 h, reflux		By: Nandhini, Sundar; et al
Experimental Protocols		Applied Organometallic Chemistry (2021), 35(12), e6436.

Scheme 192 (1 Reaction)

Steps: 1 Yield: 99%



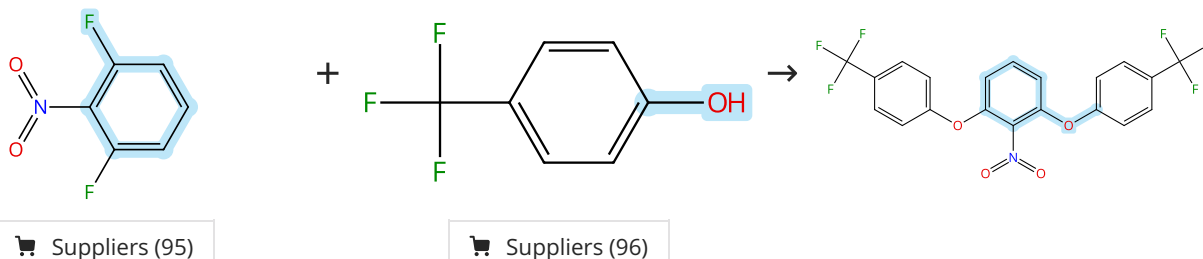
Suppliers (82)

Suppliers (98)

31-170-CAS-22515599 Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][[3,6-dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]bis(1,1-dimethylethyl)phosphine-κP](methanesulfonato-κO) palladium Solvents: Toluene; 2 h, 80 °C	Synthesis of Fluorinated Alkyl Aryl Ethers by Palladium-Catalyzed C-O Cross-Coupling By: Szpera, Robert; et al Organic Letters (2020), 22(16), 6573-6577.
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Scheme 193 (1 Reaction)

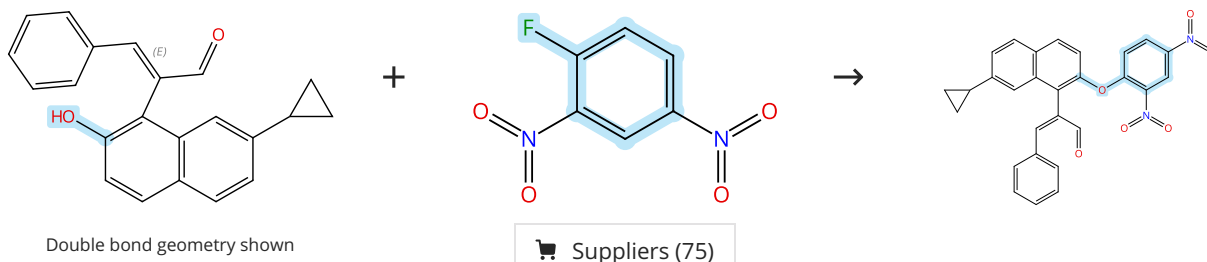
Steps: 1 Yield: 99%



31-614-CAS-40274057 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 4 h, 100 °C; 100 °C → rt 1.2 Reagents: Sodium hydroxide Solvents: Water; rt Experimental Protocols	A Strategy for Improving the Efficiency of Boronic Acid Catalysis in the Synthesis of Amides By: Almetwali, Fatima; et al European Journal of Organic Chemistry (2023), 26(45), e202300720.
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Scheme 194 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-37560084 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
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Scheme 195 (1 Reaction)

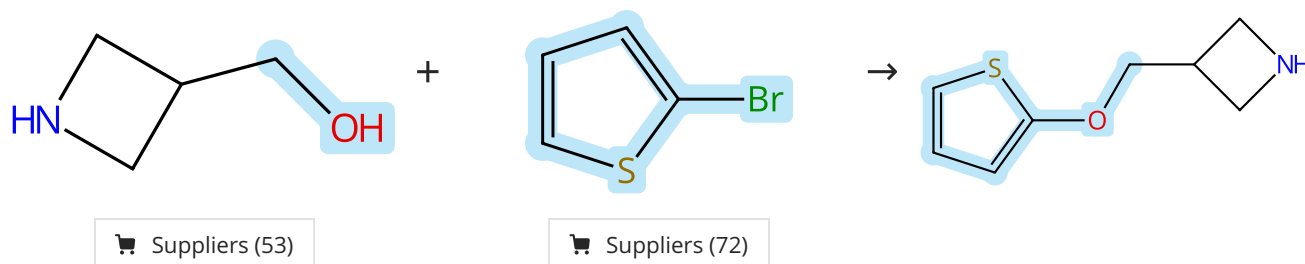
Steps: 1 Yield: 99%



31-614-CAS-24991575	Steps: 1 Yield: 99%	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines By: Nandhini, Sundar; et al Applied Organometallic Chemistry (2021), 35(12), e6436.
1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: Methanol; 11 h, reflux		
Experimental Protocols		

Scheme 196 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-40801500	Steps: 1 Yield: 99%	Cu-Catalyzed Amination of Base-Sensitive Aryl Bromides and the Chemoselective N- and O-Arylation of Amino Alcohols By: Strauss, Michael J.; et al Journal of the American Chemical Society (2024), 146(27), 18616-18625.
1.1 Reagents: Sodium trimethylsilanolate Catalysts: Cuprous iodide, 1,2-Benzenediamine, <i>N</i> ¹ , <i>N</i> ² -bis(2-phenyl-1-naphthalenyl)- Solvents: Dimethyl sulfoxide; 24 °C; 16 h, 24 °C		
Experimental Protocols		

Scheme 197 (1 Reaction)

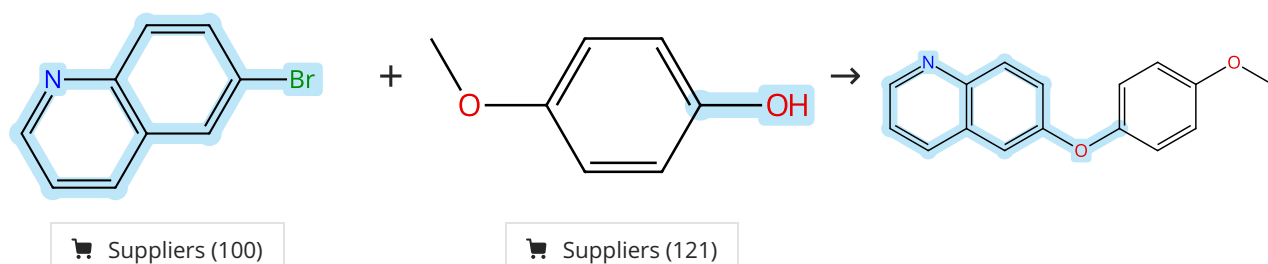
Steps: 1 Yield: 99%



31-614-CAS-24991571	Steps: 1 Yield: 99%	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines By: Nandhini, Sundar; et al Applied Organometallic Chemistry (2021), 35(12), e6436.
1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: Ethanol; 9 h, reflux		
Experimental Protocols		

Scheme 198 (1 Reaction)

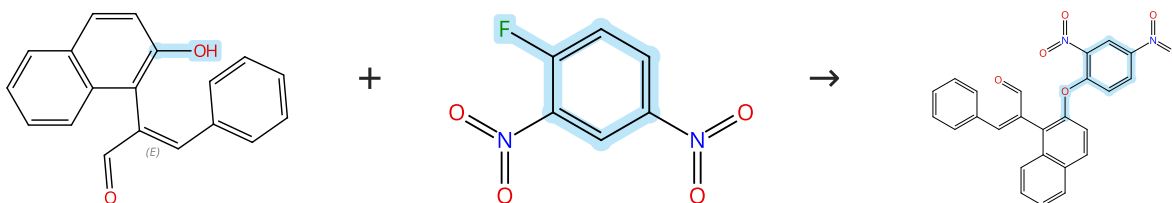
Steps: 1 Yield: 99%



31-614-CAS-43873527	Steps: 1 Yield: 99%	Electromagnetic milling promoted mechanochemical copper-catalyzed C-O coupling reaction for diaryl ether formation
1.1 Reagents: Cesium carbonate Catalysts: Copper(II) acetylacetonate, <i>N</i> ¹ , <i>N</i> ² -Di-1-naphthale nylethanediamide Solvents: Dimethylformamide; 3 h, 79 °C		By: Zeng, Zhaoxuan; et al Organic Chemistry Frontiers (2025), 12(5), 1528-1536.
Experimental Protocols		

Scheme 199 (1 Reaction)

Steps: 1 Yield: 99%



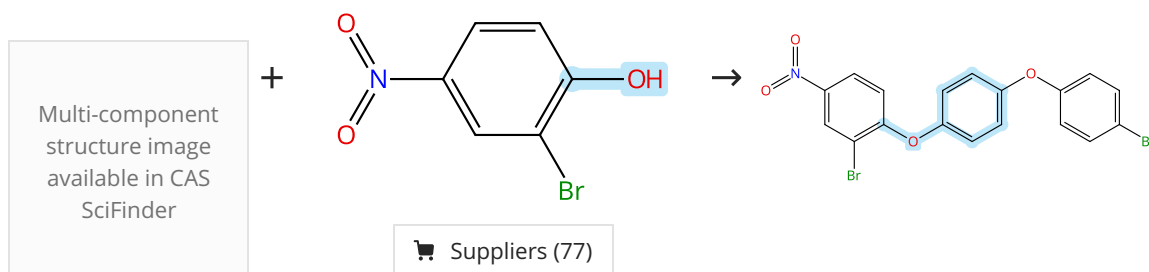
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560035	Steps: 1 Yield: 99%	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution
1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl] methyl][[(2 <i>S</i> ,3 <i>R</i>)-2-[[[(2 <i>R</i> ,3 <i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl] amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl] amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C		By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
Experimental Protocols		

Scheme 200 (1 Reaction)

Steps: 1 Yield: 99%



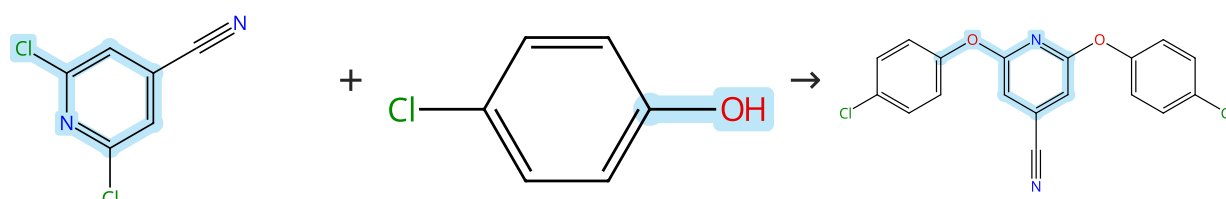
Multi-component structure image available in CAS SciFinder

Suppliers (77)

31-614-CAS-41866186	Steps: 1 Yield: 99%	Synthesis of Diaryl Ethers via Hypervalent Iodine-Mediated C-H Functionalization
1.1 Reagents: Cesium carbonate Solvents: Toluene; 8 h, 80 °C		By: Hu, Chenghu; et al Organic Letters (2024), 26(37), 7864-7868.
Experimental Protocols		

Scheme 201 (1 Reaction)

Steps: 1 Yield: 99%



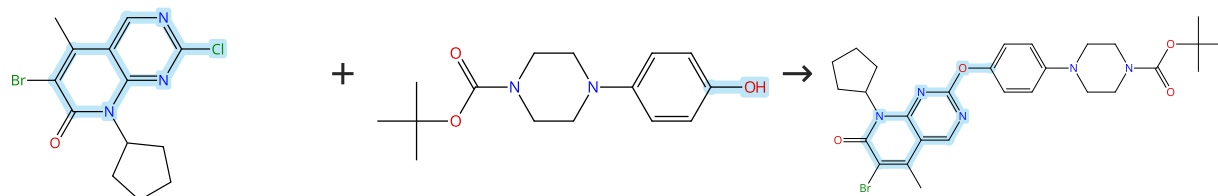
Suppliers (73)

Suppliers (90)

31-170-CAS-23898523 Steps: 1 Yield: 99% 1.1 Reagents: Sodium <i>tert</i>-butoxide Catalysts: Tris(dibenzylideneacetone)dipalladium, BINAP Solvents: 1,4-Dioxane; 24 h, 110 - 120 °C Experimental Protocols	Synthesis of novel pyridine and pyrimidine derivatives as potential inhibitors of HIV-1 reverse transcriptase using palladium-catalysed C-N cross-coupling and nucleophilic aromatic substitution reactions By: Changunda, Charles R. K.; et al ARKIVOC (Gainesville, FL, United States) (2020), (3), 152-170.
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Scheme 202 (1 Reaction)

Steps: 1 Yield: 99%



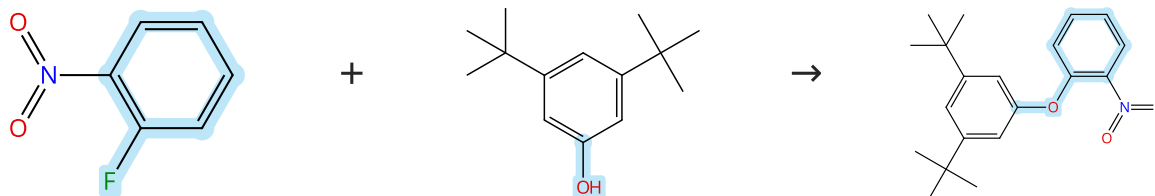
Suppliers (70)

Suppliers (61)

31-614-CAS-39447462 Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate Solvents: Tetrahydrofuran; 30 min, 100 °C Experimental Protocols	Bridged Proteolysis Targeting Chimera (PROTAC) Enables Degradation of Undruggable Targets By: Xiong, Yan; et al Journal of the American Chemical Society (2022), 144(49), 22622-22632.
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Scheme 203 (1 Reaction)

Steps: 1 Yield: 99%



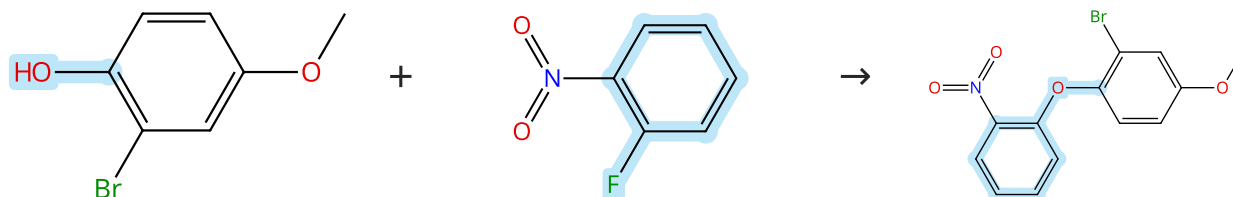
Suppliers (93)

Suppliers (83)

31-170-CAS-22969183 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt → 95 °C; 24 h, 95 °C Experimental Protocols	Synthesis of Cycloheptatrienes, Oxepines, Thiepinines, and Silepinines: A Comparison between Brønsted Acid and Au-Catalysis By: Sprenger, Kristin; et al European Journal of Organic Chemistry (2020), 2020(39), 6245-6254.
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Scheme 204 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (75)

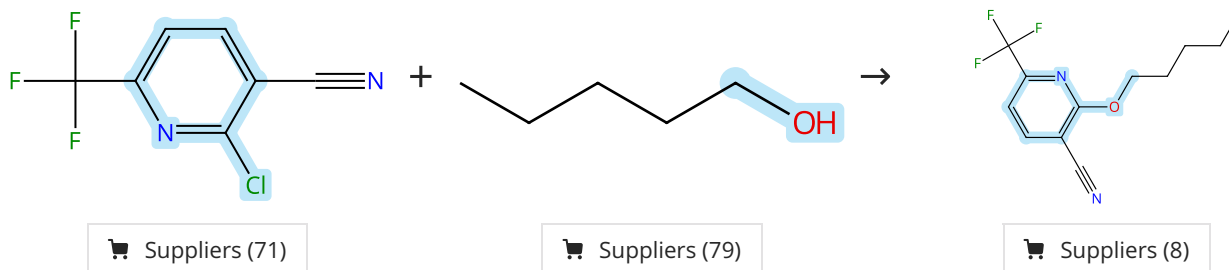
Suppliers (93)

Suppliers (2)

31-614-CAS-31558912	Steps: 1 Yield: 99%	Synthesis and Bond Activation Chemistry of Palladium (II) Pincer Complexes with a Weakly Coordinating Side Arm
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 24 h, 100 °C		By: Prakasham, A. P.; et al
Experimental Protocols		Organometallics (2022), 41(5), 634-641.

Scheme 205 (1 Reaction)

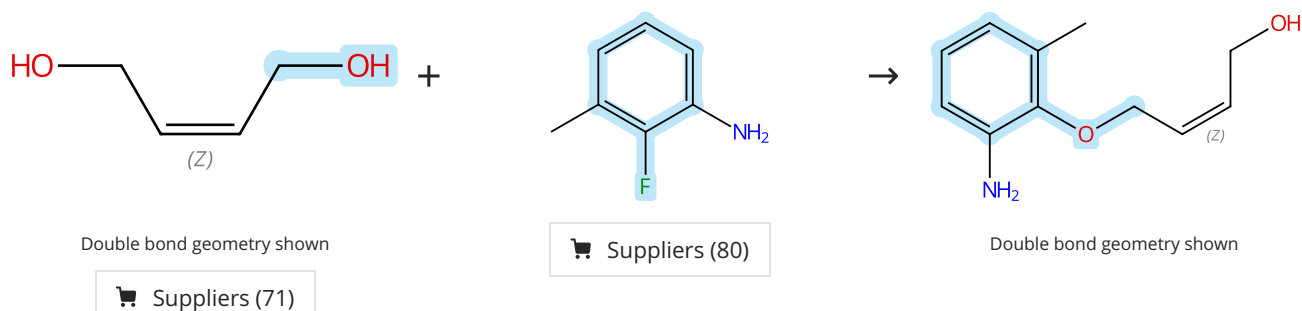
Steps: 1 Yield: 99%



31-614-CAS-40536023	Steps: 1 Yield: 99%	Discovery of N-(1,4-Benzoxazin-3-one) urea analogs as Mode-Selective TRPV1 antagonists
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 0 °C; overnight, rt		By: Huang, Guocheng; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2024), 106, 129735.

Scheme 206 (1 Reaction)

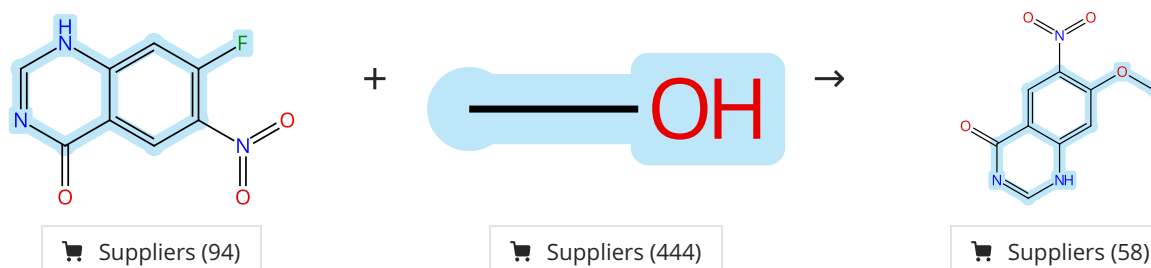
Steps: 1 Yield: 99%



31-614-CAS-41266141	Steps: 1 Yield: 99%	Enantioselective Synthesis of Dihydrobenzofurans, Dihydrobenzosulfones, and Dihydroindoles by Merging One-pot Intramolecular Heck-Matsuda Reactions from Anilines with Redox-Relay Process
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; 0 °C; 30 min, 0 °C; 0 °C → rt; 30 min, rt; rt → 0 °C		By: Scarpa de Souza, Edson Leonardo; et al
1.2 0 °C; 30 min, 0 °C; overnight, rt		Chemistry - A European Journal (2024), 30(44), e202401115.
1.3 Reagents: Water; rt		
1.4 Reagents: Acetic acid, Iron Solvents: Acetone, Water; overnight, rt → 60 °C		
Experimental Protocols		

Scheme 207 (1 Reaction)

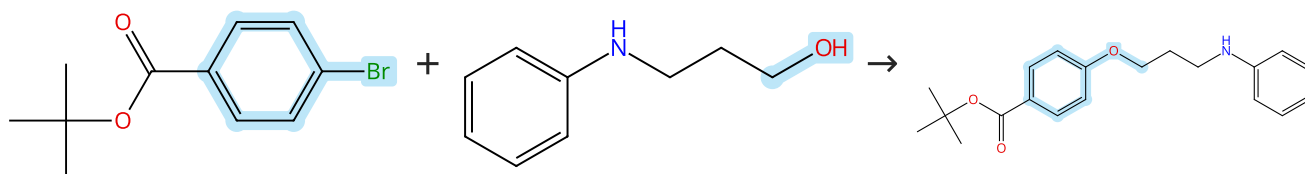
Steps: 1 Yield: 99%



31-614-CAS-45933962	Steps: 1 Yield: 99%	Synthesis and antitumor activity of 4- aminoquinazoline derivatives containing morpholine moieties
1.1 Reagents: Potassium hydroxide, Tetrabutylammonium bromide; 6 h, 80 °C; 80 °C → rt		By: Zhou, Xinxin; et al
1.2 Reagents: Acetic acid; pH 6		Journal of the Indian Chemical Society (2025), 102(7), 101770.
Experimental Protocols		

Scheme 208 (1 Reaction)

Steps: 1 Yield: 99%



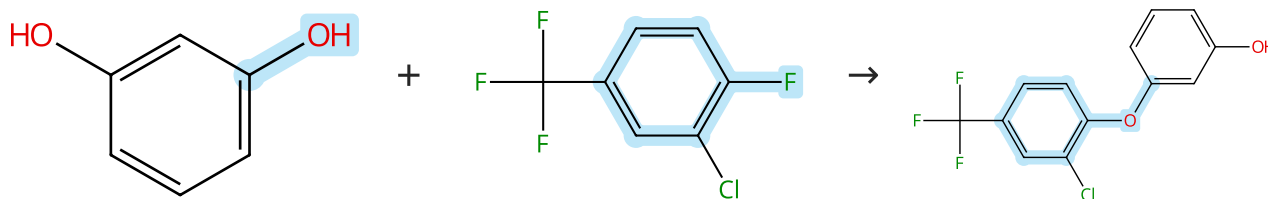
Suppliers (72)

Suppliers (46)

31-614-CAS-40801501	Steps: 1 Yield: 99%	Cu-Catalyzed Amination of Base-Sensitive Aryl Bromides and the Chemoselective N- and O-Arylation of Amino Alcohols
1.1 Reagents: Sodium trimethylsilanolate Catalysts: Cuprous iodide, 1,2-Benzenediamine, <i>N</i> ¹ , <i>N</i> ² -bis(2-phenyl-1-naphthalenyl)- Solvents: Dimethyl sulfoxide; 24 °C; 16 h, 24 °C		By: Strauss, Michael J.; et al
Experimental Protocols		Journal of the American Chemical Society (2024), 146(27), 18616-18625.

Scheme 209 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (109)

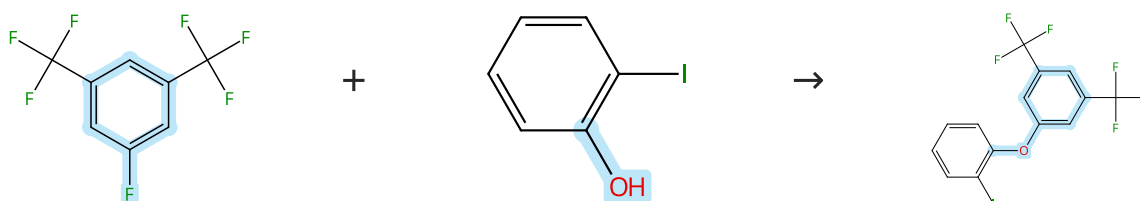
Suppliers (68)

Suppliers (6)

31-614-CAS-32517468	Steps: 1 Yield: 99%	Highly Enantioselective Rhodium-Catalyzed Transfer Hydrogenation of Tetrasubstituted Olefins: Application toward the Synthesis of GPR40 Agonist MK-2305
1.1 Reagents: Potassium hydroxide Solvents: Dimethyl sulfoxide; 20 - 30 °C		By: Zhong, Yong-Li; et al
1.2 3 h, 50 °C; 24 h, 130 °C; 130 °C → 50 °C		Organic Letters (2022), 24(17), 3254-3258.
1.3 Reagents: Hydrochloric acid Solvents: Water		
Experimental Protocols		

Scheme 210 (1 Reaction)

Steps: 1 Yield: 99%



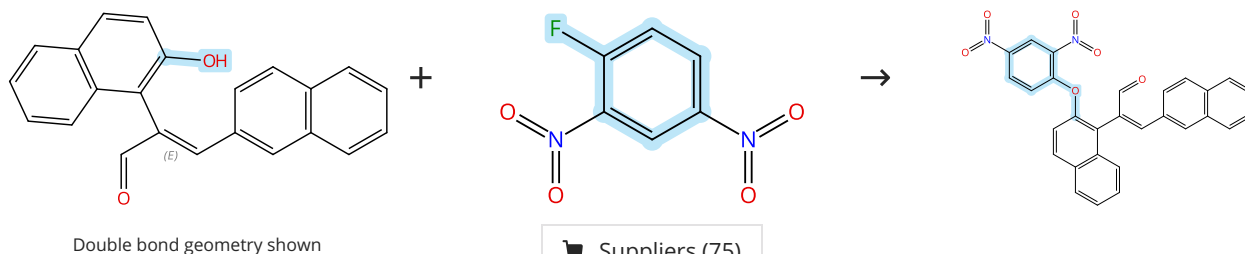
Suppliers (45)

Suppliers (85)

31-614-CAS-40274042	Steps: 1 Yield: 99%	A Strategy for Improving the Efficiency of Boronic Acid Catalysis in the Synthesis of Amides
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 100 °C; 100 °C → 25 °C		By: Almetwali, Fatima; et al
1.2 Reagents: Sodium hydroxide Solvents: Water; 25 °C		European Journal of Organic Chemistry (2023), 26(45), e202300720.
Experimental Protocols		

Scheme 211 (1 Reaction)

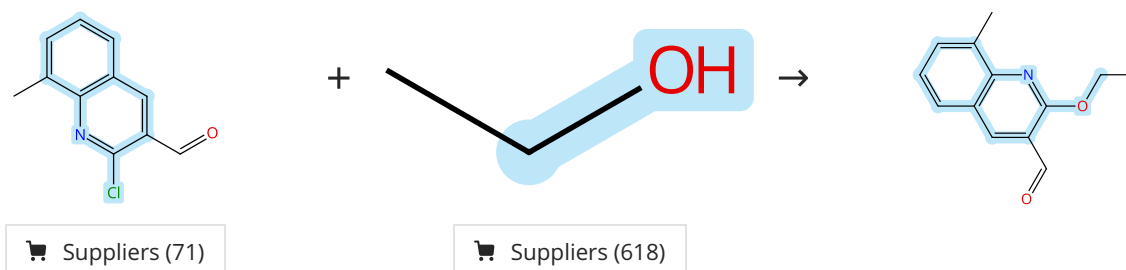
Steps: 1 Yield: 99%



31-614-CAS-37560065	Steps: 1 Yield: 99%	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution
1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][(2 <i>S</i> ,3 <i>R</i>)-2-[[[(2 <i>R</i> ,3 <i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C		By: Guo, Fengyuan; et al
Experimental Protocols		Nature Communications (2023), 14(1), 5050.

Scheme 212 (1 Reaction)

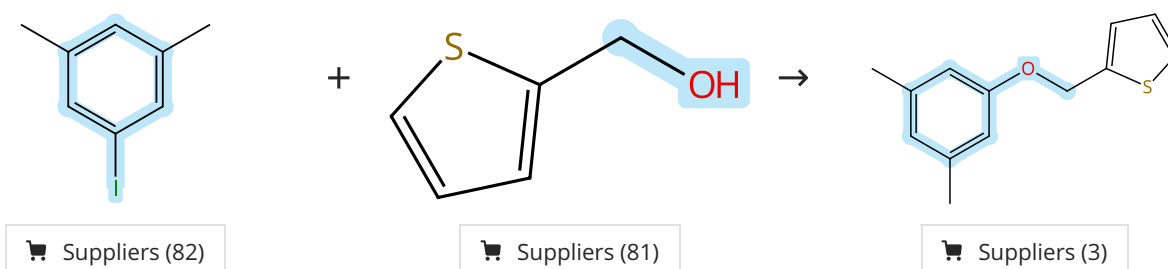
Steps: 1 Yield: 99%



31-614-CAS-24991573	Steps: 1 Yield: 99%	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines
1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: Ethanol; 9 h, reflux		By: Nandhini, Sundar; et al
Experimental Protocols		Applied Organometallic Chemistry (2021), 35(12), e6436.

Scheme 213 (1 Reaction)

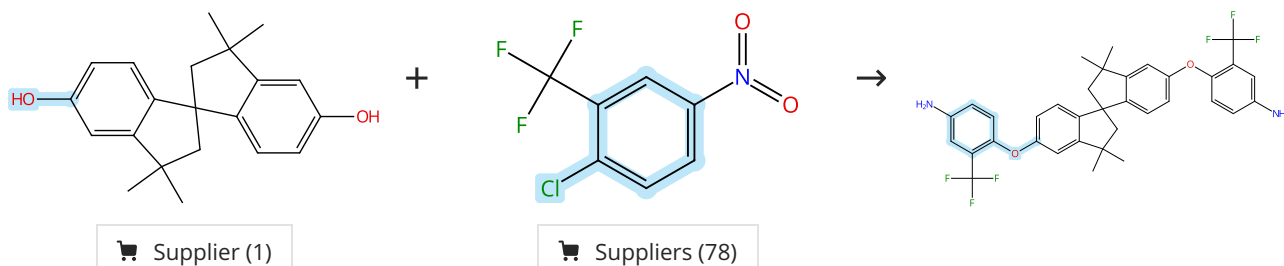
Steps: 1 Yield: 99%



31-614-CAS-35219875 Steps: 1 Yield: 99% 1.1 Catalysts: Nickel dichloride, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Tetrahydrofuran; 5 min, rt 1.2 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene; 18 h, 30 °C 1.3 Solvents: Water Experimental Protocols	Photoinduced Nickel-Catalyzed Carbon-Heteroatom Coupling By: Luo, Hang; et al Chemistry - A European Journal (2023), 29(1), e202202385.
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Scheme 214 (1 Reaction)

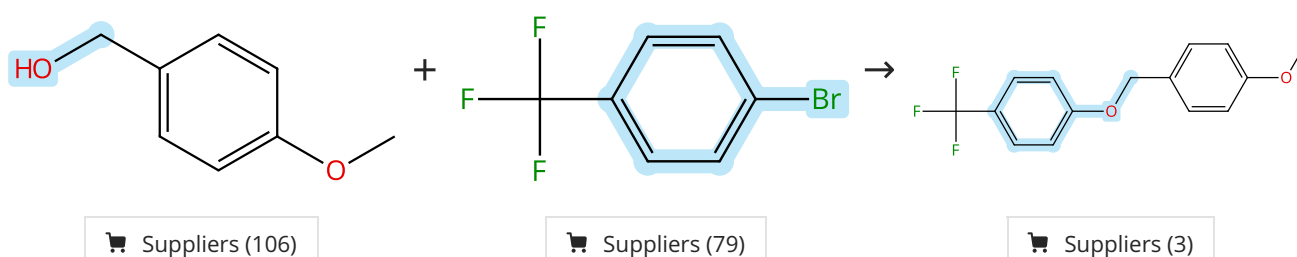
Steps: 1 Yield: 99%



31-170-CAS-21732533 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate; 120 °C 1.2 Reagents: Hydrazine hydrate (1:1) Catalysts: Palladium Experimental Protocols	Gas separation properties of aromatic polyimides with bulky groups. Comparison of experimental and simulated results By: Alvarez, Cristina; et al Journal of Membrane Science (2020), 602, 117959.
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Scheme 215 (1 Reaction)

Steps: 1 Yield: 99%



31-170-CAS-21785683 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy-κO)ethane]nickel, <i>fac</i>-Tris(2-(2-pyridinyl)phenyl)iridium Solvents: Dimethylformamide; 5 min, rt 1.2 18 h, 35 °C Experimental Protocols	A Unified and Practical Method for Carbon-Heteroatom Cross-Coupling using Nickel/Photo Dual Catalysis By: Escobar, Randolph A.; et al Chemistry - A European Journal (2020), 26(23), 5168-5173.
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Scheme 216 (1 Reaction)

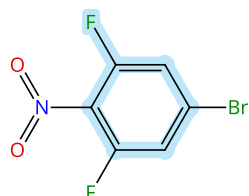
Steps: 1 Yield: 99%



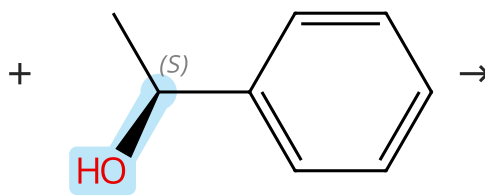
31-614-CAS-24991579	Steps: 1 Yield: 99%	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines
1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: Ethanol; 8 h, reflux		
Experimental Protocols		By: Nandhini, Sundar; et al Applied Organometallic Chemistry (2021), 35(12), e6436.

Scheme 217 (1 Reaction)

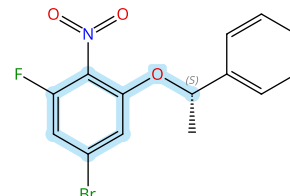
Steps: 1 Yield: 99%



Suppliers (75)

Absolute stereochemistry shown,
Rotation (-)

Suppliers (89)

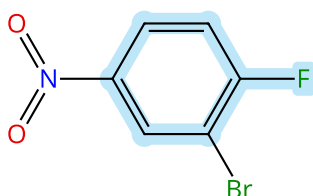


Absolute stereochemistry shown

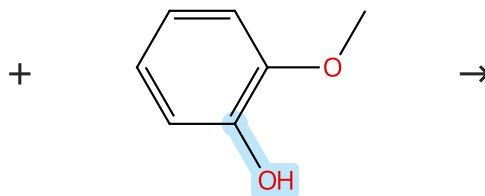
31-170-CAS-22860756	Steps: 1 Yield: 99%	Design and Synthesis of a Highly Selective and In Vivo- Capable Inhibitor of the Second Bromodomain of the Bromodomain and Extra Terminal Domain Family of Proteins
1.1 Reagents: Lithium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; rt; 1 h, 100 °C		
1.2 Solvents: Water		By: Preston, Alex; et al Journal of Medicinal Chemistry (2020), 63(17), 9070-9092.

Scheme 218 (1 Reaction)

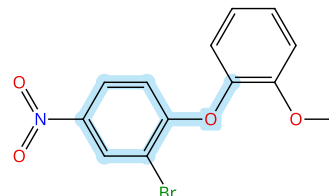
Steps: 1 Yield: 99%



Suppliers (83)



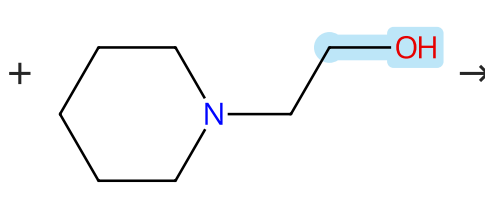
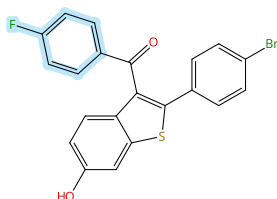
Suppliers (125)



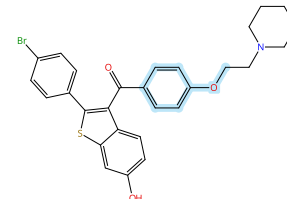
31-614-CAS-35312185	Steps: 1 Yield: 99%	Design, synthesis, and biological evaluation of quinoxaline derivatives as potent BRD4 inhibitors
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 2 h, 110 °C		
Experimental Protocols		By: Xu, Kai-Yan; et al Bioorganic & Medicinal Chemistry (2023), 78, 117152.

Scheme 219 (1 Reaction)

Steps: 1 Yield: 99%



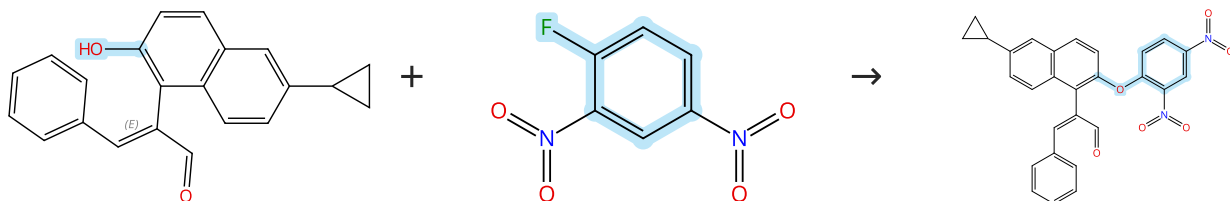
Suppliers (93)



31-614-CAS-41049023	Steps: 1 Yield: 99%	Design, synthesis and antiproliferative activity of raloxifen e/histone deacetylase inhibitor hybrids in breast cancer
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 3 h, 50 °C		By: Wang, Yufei; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2024), 274, 116533.

Scheme 220 (1 Reaction)

Steps: 1 Yield: 99%



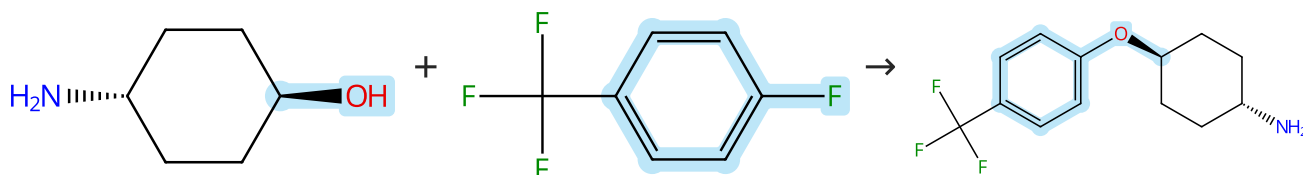
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560092	Steps: 1 Yield: 99%	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution
1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2 <i>S</i> ,3 <i>R</i>)-2-[[[(2 <i>R</i> ,3 <i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C		By: Guo, Fengyuan; et al
Experimental Protocols		Nature Communications (2023), 14(1), 5050.

Scheme 221 (1 Reaction)

Steps: 1 Yield: 99%



Relative stereochemistry shown

Suppliers (105)

Suppliers (69)

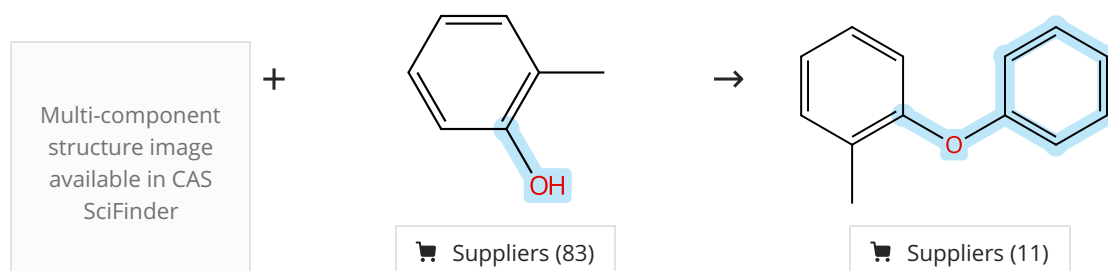
Relative stereochemistry shown

Suppliers (4)

31-614-CAS-34480076	Steps: 1 Yield: 99%	A bioisosteric approach to the discovery of novel N-aryl-N-[4-(aryloxy)cyclohexyl]squaramide-based activators of eukaryotic initiation factor 2 alpha (eIF2α) phosphorylation
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 0 °C; 1 h, rt		By: Kwak, Jinsook; et al
1.2 2 h, 60 °C; 60 °C → rt		
1.3 Solvents: Water; rt		
Experimental Protocols		European Journal of Medicinal Chemistry (2022), 239, 114501.

Scheme 222 (1 Reaction)

Steps: 1 Yield: 99%



Multi-component structure image available in CAS SciFinder

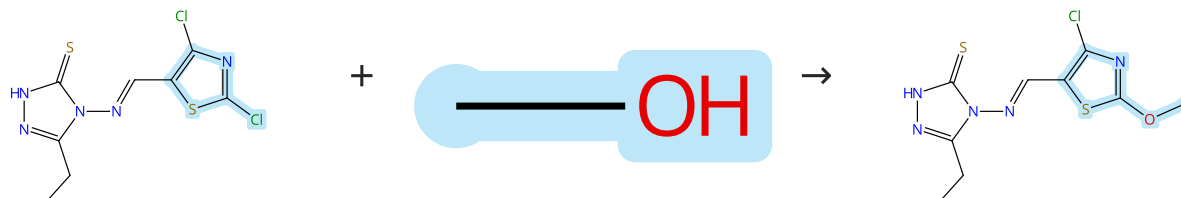
Suppliers (83)

Suppliers (11)

31-614-CAS-31534011	Steps: 1 Yield: 99%	Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates
1.1 Reagents: Sodium carbonate Solvents: Dichloromethane, Water; 18 h, 40 °C		By: Kikushima, Kotaro; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Organic Letters (2022), 24(10), 1924-1928.
Experimental Protocols		

Scheme 223 (1 Reaction)

Steps: 1 Yield: 99%

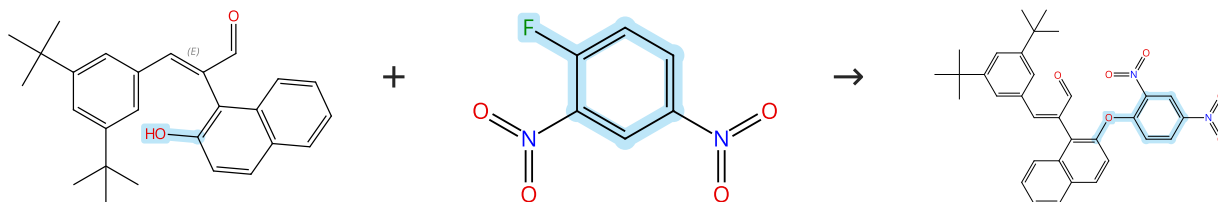


Suppliers (444)

31-170-CAS-23034423	Steps: 1 Yield: 99%	Synthesis, characterization and biological studies of novel series of triazolyl schiff bases bearing thiazole moiety and their s-derivatives
1.1 Catalysts: Potassium hydroxide Solvents: Water; 6 h, rt		By: Nkurunziza, Jean Baptiste; et al
Experimental Protocols		Journal of Applicable Chemistry (Lumami, India) (2020), 9(4), 556-565.

Scheme 224 (1 Reaction)

Steps: 1 Yield: 99%



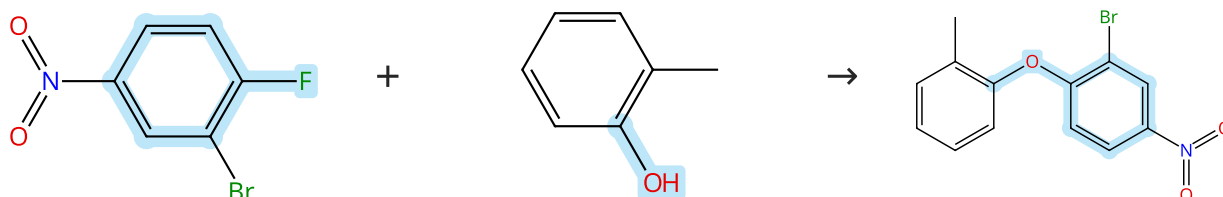
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560070	Steps: 1 Yield: 99%	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution
1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2 <i>S</i> ,3 <i>R</i>)-2-[[[(2 <i>R</i> ,3 <i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C		By: Guo, Fengyuan; et al
Experimental Protocols		Nature Communications (2023), 14(1), 5050.

Scheme 225 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

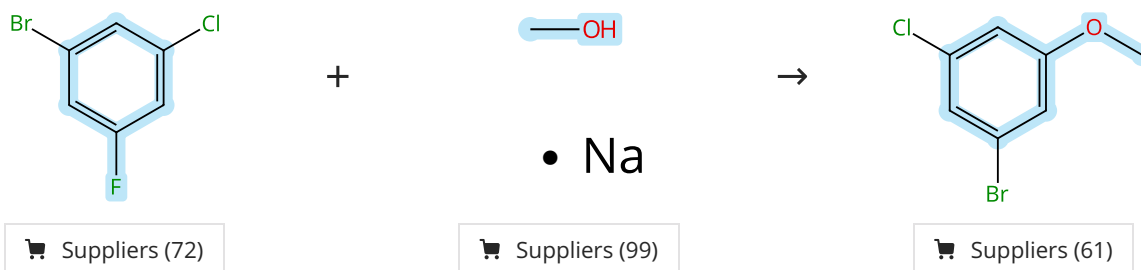
Suppliers (83)

Suppliers (3)

31-614-CAS-35312184	Steps: 1 Yield: 99%	Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 2 h, 110 °C		By: Xu, Kai-Yan; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2023), 78, 117152.

Scheme 226 (1 Reaction)

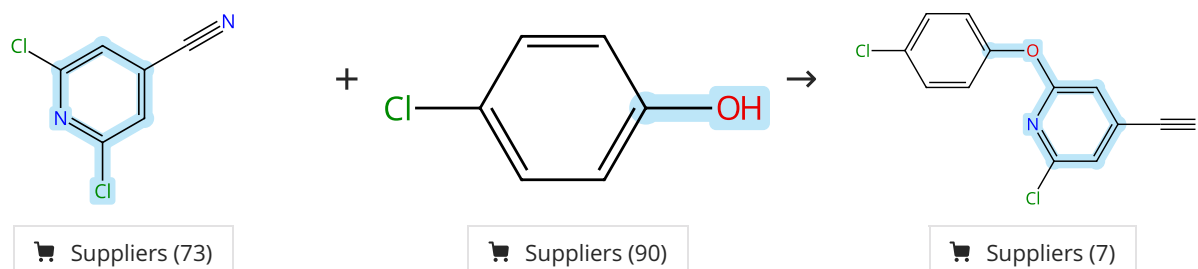
Steps: 1 Yield: 99%



31-170-CAS-21657487	Steps: 1 Yield: 99%	Synthesis of mono- and dimethoxylated polychlorinated biphenyl derivatives starting from fluoroarene derivatives
1.1 Solvents: Dimethylformamide; 3 h, 110 °C		By: Dhakal, Ram; et al
1.2 Reagents: Water; cooled		Environmental Science and Pollution Research (2020), 27(9), 8905-8925.
Experimental Protocols		

Scheme 227 (1 Reaction)

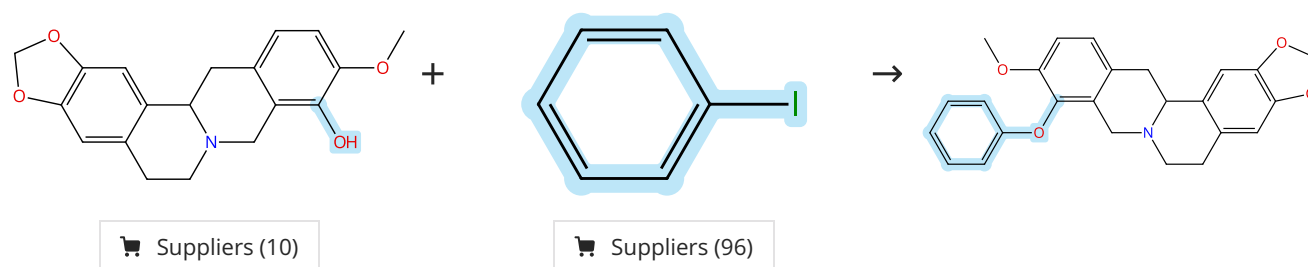
Steps: 1 Yield: 99%



31-170-CAS-23899510	Steps: 1 Yield: 99%	Synthesis of novel pyridine and pyrimidine derivatives as potential inhibitors of HIV-1 reverse transcriptase using palladium-catalysed C-N cross-coupling and nucleophilic aromatic substitution reactions
1.1 Reagents: Sodium <i>tert</i> -butoxide; 24 h, 110 - 120 °C		By: Changunda, Charles R. K.; et al
Experimental Protocols		ARKIVOC (Gainesville, FL, United States) (2020), (3), 152-170.

Scheme 228 (1 Reaction)

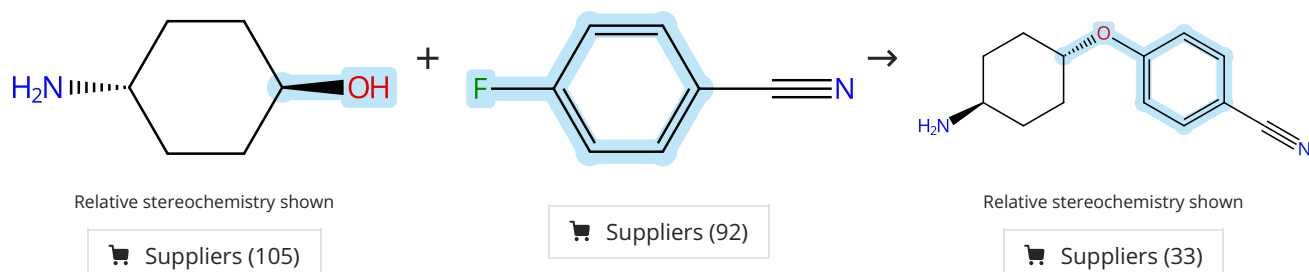
Steps: 1 Yield: 99%



31-170-CAS-23784116	Steps: 1 Yield: 99%	Synthesis of 9-O-Arylated Berberine with a Polystyrene Resin Supported Copper(II) Catalyst
1.1 Reagents: Potassium carbonate Catalysts: Cu ²⁺ (supported on polyethyleneimine-polystyrene resin) Solvents: Dimethyl sulfoxide; 48 h, 100 °C		By: Meng, Qi; et al ChemistrySelect (2021), 6(16), 4011-4014.
Experimental Protocols		

Scheme 229 (1 Reaction)

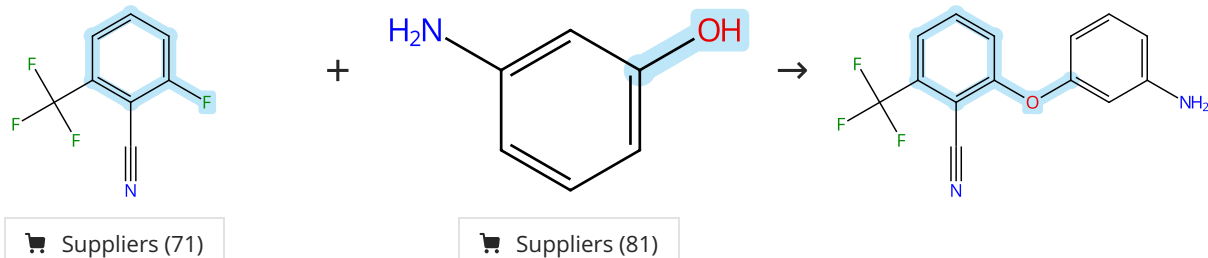
Steps: 1 Yield: 99%



31-614-CAS-34480081	Steps: 1 Yield: 99%	A bioisosteric approach to the discovery of novel N-aryl-N-[4-(aryloxy)cyclohexyl]squaramide-based activators of eukaryotic initiation factor 2 alpha (eIF2α) phosphorylation
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 0 °C; 1 h, rt 1.2 2 h, 60 °C; 60 °C → rt 1.3 Solvents: Water; rt		By: Kwak, Jinsook; et al European Journal of Medicinal Chemistry (2022), 239, 114501.
Experimental Protocols		

Scheme 230 (1 Reaction)

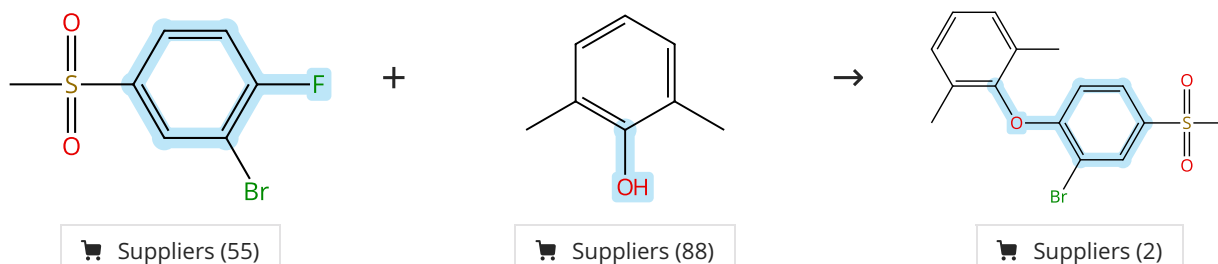
Steps: 1 Yield: 99%



31-614-CAS-34902390	Steps: 1 Yield: 99%	Structure-Activity Relationship Development Efforts towards Peripherally Selective Analogs of the Cannabinoid Receptor Partial Agonist BAY 59-3074
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 15 min, rt; 4 h, 80 °C		By: Amato, George; et al Molecules (2022), 27(17), 5672.
Experimental Protocols		

Scheme 231 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-24358140 Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 18 h, 110 °C Experimental Protocols	Discovery of N-Ethyl-4-[2-(4-fluoro-2,6-dimethyl-phenoxy)-5-(1-hydroxy-1-methyl-ethyl)phenyl]-6-methyl-7-oxo-1H-pyrrolo [2,3-c]pyridine-2-carboxamide (ABBV-744), a BET Bromodomain Inhibitor with Selectivity for the Second Bromodomain By: Sheppard, George S.; et al Journal of Medicinal Chemistry (2020), 63(10), 5585-5623.
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Scheme 232 (1 Reaction) Suppliers (77) Suppliers (444) Suppliers (11)	Steps: 1 Yield: 99%
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31-614-CAS-24991576 Steps: 1 Yield: 99% 1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: Methanol; 9 h, reflux Experimental Protocols	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines By: Nandhini, Sundar; et al Applied Organometallic Chemistry (2021), 35(12), e6436.
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Scheme 233 (1 Reaction) Suppliers (79) Suppliers (444) Suppliers (69)	Steps: 1 Yield: 99%
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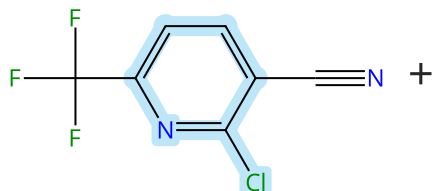
31-170-CAS-22250438 Steps: 1 Yield: 99% 1.1 Reagents: Quinuclidine Catalysts: Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis (1,1-dimethylethyl)-2,2'-bipyridine Solvents: Acetonitrile; 24 h, rt Experimental Protocols	Solar-driven tandem photoredox nickel-catalysed cross-coupling using modified carbon nitride By: Qin, Yangzhong; et al Chemical Science (2020), 11(28), 7456-7461.
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Scheme 234 (1 Reaction) Suppliers (65) Suppliers (10)	Steps: 1 Yield: 99% Absolute stereochemistry shown, Rotation (+) Absolute stereochemistry shown
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31-170-CAS-22590870	Steps: 1 Yield: 99%	Identification and Preclinical Development of a 2,5,6-Trisubstituted Fluorinated Pyridine Derivative as a Radioligand for the Positron Emission Tomography Imaging of Cannabinoid Type 2 Receptors
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 20 min, 0 °C; 3 h, 80 °C		By: Haider, Achi; et al Journal of Medicinal Chemistry (2020), 63(18), 10287-10306.

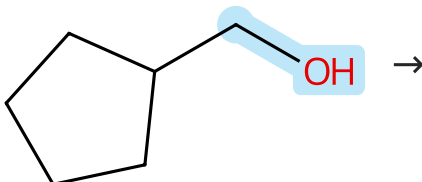
Scheme 235 (1 Reaction)

Steps: 1 Yield: 99%

N#Cc1cc(C(F)(F)F)c(F)c(F)n1Cl



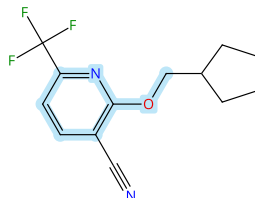
Suppliers (71)

OC1CCCC1



Suppliers (97)



N#Cc1cc(C(F)(F)F)c(F)c(OC2CCCC2)n1

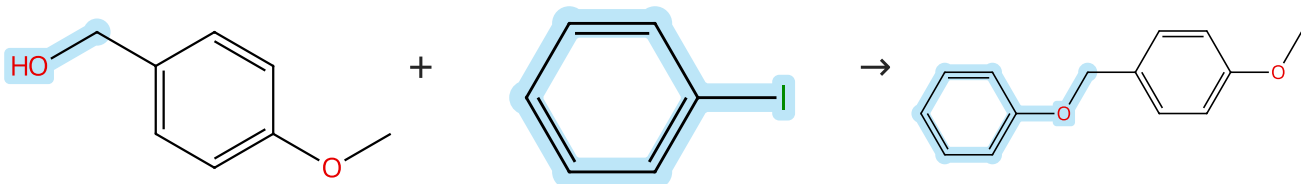


Suppliers (2)

31-614-CAS-40536025	Steps: 1 Yield: 99%	Discovery of N-(1,4-Benzoxazin-3-one) urea analogs as Mode-Selective TRPV1 antagonists
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 0 °C; overnight, rt		By: Huang, Guocheng; et al Bioorganic & Medicinal Chemistry Letters (2024), 106, 129735.
Experimental Protocols		

Scheme 236 (1 Reaction)

Steps: 1 Yield: 99%



The reaction scheme shows the synthesis of 4-(benzyloxy)benzaldehyde from 4-(methoxymethyl)benzaldehyde and iodobenzene. The reactants are 4-(methoxymethyl)benzaldehyde (a benzene ring with a -CH₂OH group at position 1 and a -OCH₃ group at position 4) and iodobenzene (a benzene ring with an -I atom). The product is 4-(benzyloxy)benzaldehyde (a benzene ring with a -CH₂OCH₂Ph group at position 1 and a -OCH₃ group at position 4). The reaction is indicated by a plus sign between the reactants and a right-pointing arrow leading to the product.

Suppliers (106)

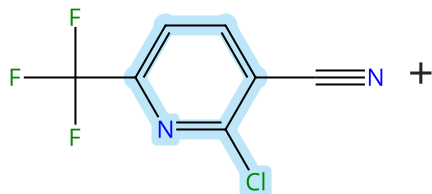
Suppliers (96)

Suppliers (13)

31-170-CAS-21785736	Steps: 1 Yield: 99%	A Unified and Practical Method for Carbon-Heteroatom Cross-Coupling using Nickel/Photo Dual Catalysis
1.1 Reagents: Potassium carbonate Catalysts: 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy-κO)ethane]nickel, fac-Tris(2-(2-pyridinyl)phenyl)iridium Solvents: Dimethylformamide; 5 min, rt		By: Escobar, Randolph A.; et al Chemistry - A European Journal (2020), 26(23), 5168-5173.
1.2 18 h, 35 °C		
Experimental Protocols		

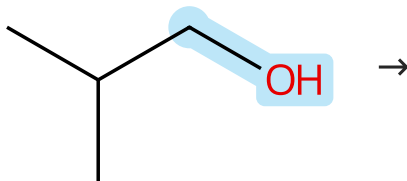
Scheme 237 (1 Reaction)

Steps: 1 Yield: 99%

Fc1cc(C#N)c(Cl)c(F)c1C(F)(F)F

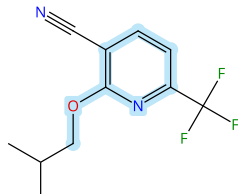


Suppliers (71)

CC(C)CO



Suppliers (122)

CC(C)COc1cc(C#N)c(C(F)(F)F)c(F)c1

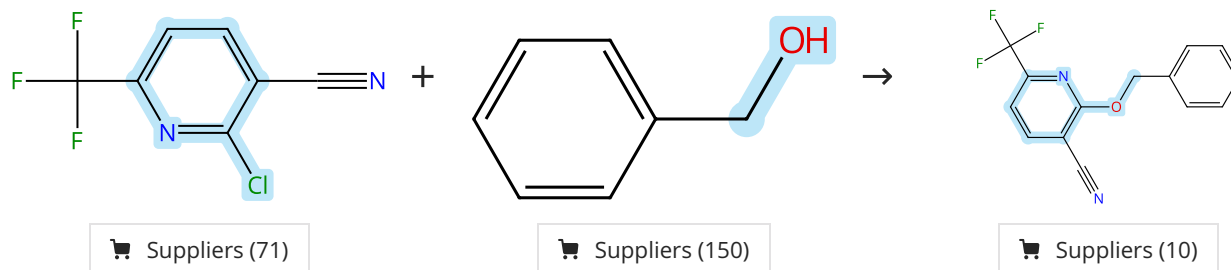


Suppliers (7)

31-614-CAS-40536036	Steps: 1 Yield: 99%	Discovery of N-(1,4-Benzoxazin-3-one) urea analogs as Mode-Selective TRPV1 antagonists
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 0 °C; overnight, rt		By: Huang, Guocheng; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2024), 106, 129735.

Scheme 238 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-40536037	Steps: 1 Yield: 99%	Discovery of N-(1,4-Benzoxazin-3-one) urea analogs as Mode-Selective TRPV1 antagonists
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 0 °C; overnight, rt		By: Huang, Guocheng; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2024), 106, 129735.

Scheme 239 (1 Reaction)

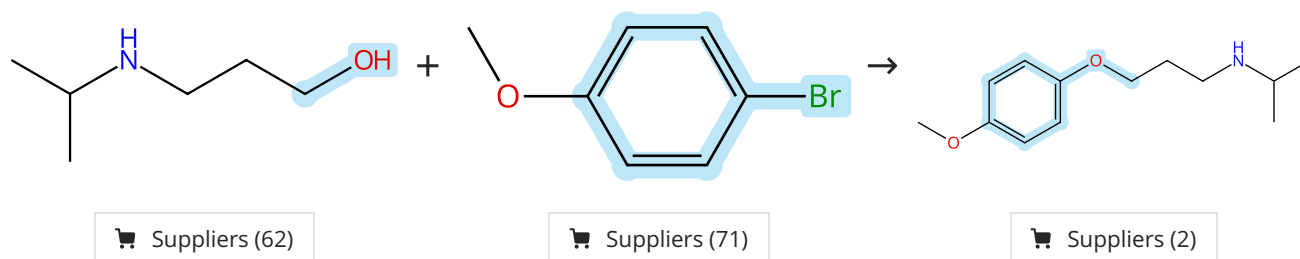
Steps: 1 Yield: 99%



31-614-CAS-24991569	Steps: 1 Yield: 99%	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines
1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: Methanol; 9 h, reflux		By: Nandhini, Sundar; et al
Experimental Protocols		Applied Organometallic Chemistry (2021), 35(12), e6436.

Scheme 240 (1 Reaction)

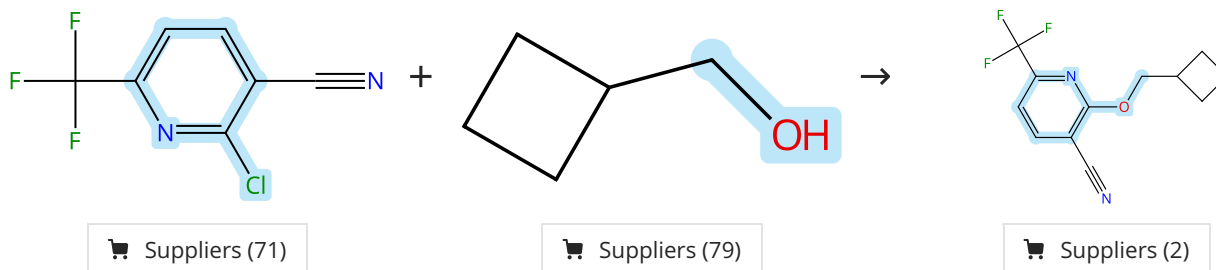
Steps: 1 Yield: 99%



31-614-CAS-40801502	Steps: 1 Yield: 99%	Cu-Catalyzed Amination of Base-Sensitive Aryl Bromides and the Chemoselective N- and O-Arylation of Amino Alcohols
1.1 Reagents: Sodium trimethylsilanolate Catalysts: Cuprous iodide, 1,2-Benzenediamine, <i>N</i> ¹ , <i>N</i> ² -bis(2-phenyl-1-naphthalenyl)- Solvents: Dimethyl sulfoxide; 24 °C; 16 h, 24 °C		By: Strauss, Michael J.; et al Journal of the American Chemical Society (2024), 146(27), 18616-18625.
Experimental Protocols		

Scheme 241 (1 Reaction)

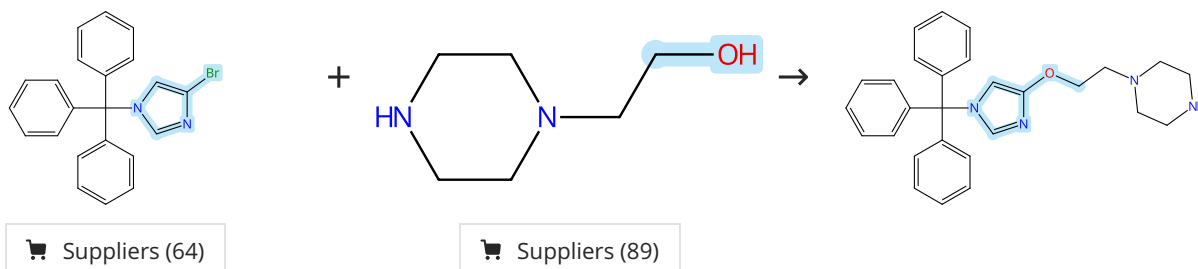
Steps: 1 Yield: 99%



31-614-CAS-40536035	Steps: 1 Yield: 99%	Discovery of N-(1,4-Benzoxazin-3-one) urea analogs as Mode-Selective TRPV1 antagonists
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 0 °C; overnight, rt		By: Huang, Guocheng; et al Bioorganic & Medicinal Chemistry Letters (2024), 106, 129735.
Experimental Protocols		

Scheme 242 (1 Reaction)

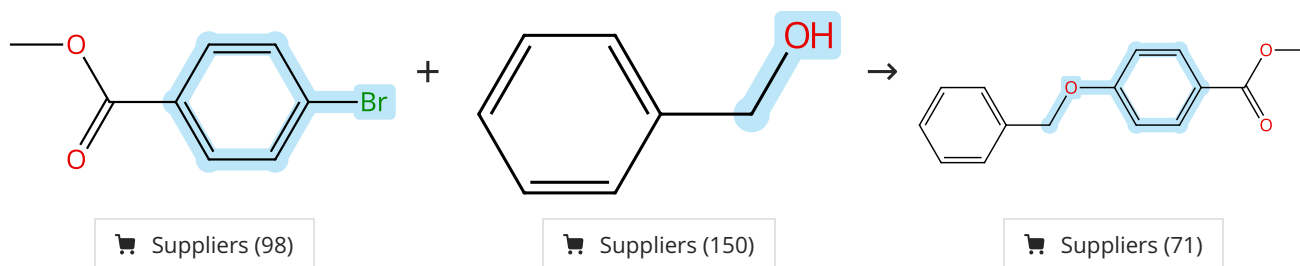
Steps: 1 Yield: 99%



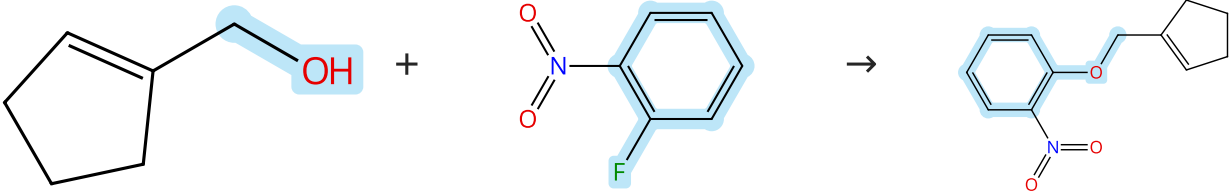
31-614-CAS-40801507	Steps: 1 Yield: 99%	Cu-Catalyzed Amination of Base-Sensitive Aryl Bromides and the Chemoselective N- and O-Arylation of Amino Alcohols
1.1 Reagents: Sodium trimethylsilanolate Catalysts: Cuprous iodide, 1,2-Benzenediamine, <i>N</i> ¹ , <i>N</i> ² -bis(2-phenyl-1-naphthalenyl)- Solvents: Dimethyl sulfoxide; 24 °C; 16 h, 24 °C		By: Strauss, Michael J.; et al Journal of the American Chemical Society (2024), 146(27), 18616-18625.
Experimental Protocols		

Scheme 243 (1 Reaction)

Steps: 1 Yield: 99%

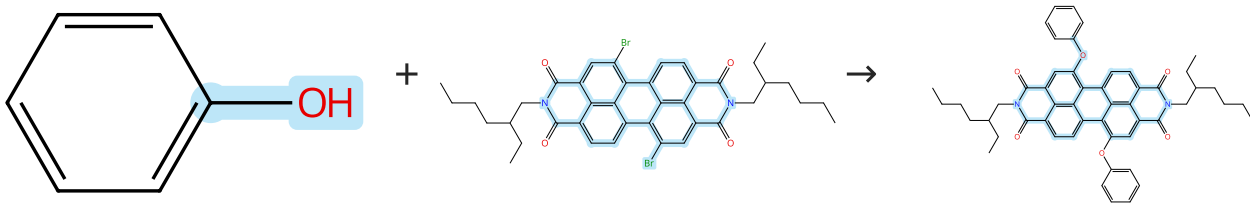


31-614-CAS-35219891	Steps: 1 Yield: 99%	Photoinduced Nickel-Catalyzed Carbon-Heteroatom Coupling
1.1 Catalysts: Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: 2-Methyltetrahydrofuran; 5 min, rt		By: Luo, Hang; et al
1.2 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene; 36 h, 30 °C		Chemistry - A European Journal (2023), 29(1), e202202385.
1.3 Solvents: Water		
Experimental Protocols		

Scheme 244 (2 Reactions)		Steps: 1 Yield: 99%
		
 Suppliers (35)	 Suppliers (93)	 Suppliers (4)

31-614-CAS-46452992	Steps: 1 Yield: 99%	Intramolecular Heck-Matsuda reaction for the asymmetric synthesis of a spirobenzofuran using chiral N,N-Ligands: Experimental and theoretical insights into the catalytic cycle and enantioselectivity
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; 0 °C; 30 min, 0 °C; 0 °C → rt; 30 min, rt; rt → 0 °C		By: da Silva, Vitor H. Menezes; et al
1.2 0 °C; 30 min, 0 °C; 16 h, rt		Journal of Molecular Structure (2025), 1346, 143188.
1.3 Reagents: Water; rt		
Experimental Protocols		

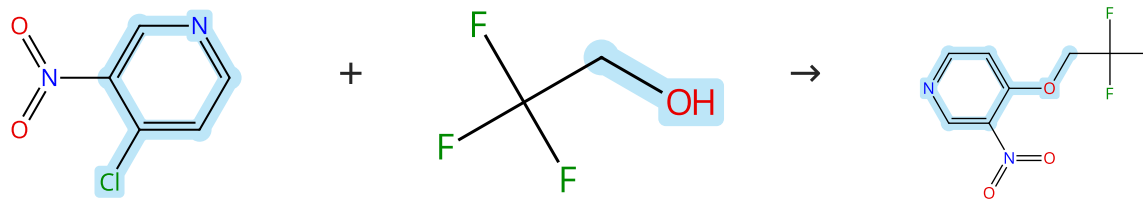
31-614-CAS-35372475	Steps: 1 Yield: 99%	In Tandem Enantioselective Intramolecular Heck-Matsuda Reactions directly from Anilines
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; 0 °C; 30 min, 0 °C; 0 °C → rt; 30 min, rt; rt → 0 °C		By: Chorro, Tomaz Henrique Duarte; et al
1.2 0 °C; 30 min, 0 °C; overnight, rt		Advanced Synthesis & Catalysis (2023), 365(2), 211-223.
1.3 Solvents: Water; rt		
Experimental Protocols		

Scheme 245 (1 Reaction)		Steps: 1 Yield: 99%
		
 Suppliers (207)	 Suppliers (46)	

31-170-CAS-22390965	Steps: 1 Yield: 99%	Fine-Tuning the Electronic Properties of Azo Chromophore-Incorporated Perylene Bisimide Dyads
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 0.5 h, rt → 60 °C		By: Turel, Tankut; et al
1.2 12 h, 120 °C		Journal of Organic Chemistry (2020), 85(16), 10593-10602.

Scheme 246 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (86)

Suppliers (98)

Suppliers (2)

31-614-CAS-36012346

Steps: 1 Yield: 99%

Structure-activity relationship (SAR) studies on substituted N-(pyridin-3-yl)-2-amino-isonicotinamides as highly potent and selective glycogen synthase kinase-3 (GSK-3) inhibitors

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 17 h, 75 °C

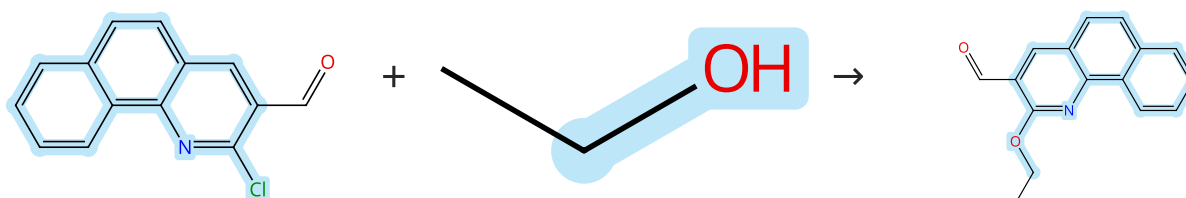
Experimental Protocols

By: Luo, Guanglin; et al

Bioorganic & Medicinal Chemistry Letters (2023), 81, 129143.

Scheme 247 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (9)

Suppliers (618)

31-614-CAS-24991587

Steps: 1 Yield: 99%

Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinoline and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines

1.1 Reagents: Phenylboronic acid, Potassium hydroxide
Catalysts: 2721440-87-3
Solvents: Ethanol; 11 h, reflux

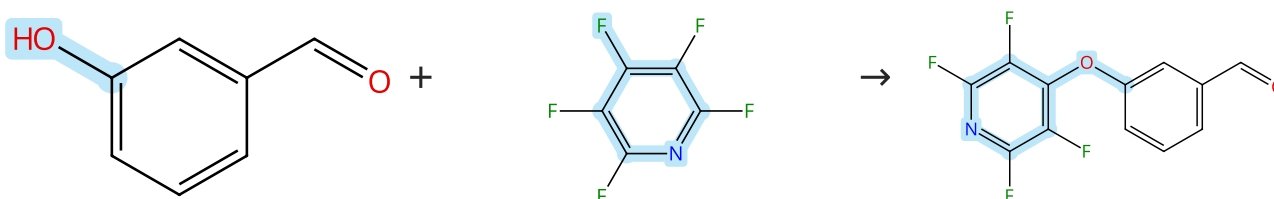
Experimental Protocols

By: Nandhini, Sundar; et al

Applied Organometallic Chemistry (2021), 35(12), e6436.

Scheme 248 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (97)

Suppliers (71)

31-614-CAS-36601620

Steps: 1 Yield: 99%

Site-selective nucleophilic substitution reactions of pentafluoropyridine with hydroxybenzaldehydes: synthesis of triarylmethanes comprising perfluoropyridine moieties

1.1 Reagents: Potassium carbonate
Solvents: Acetonitrile; 30 min, rt
1.2 40 min, rt

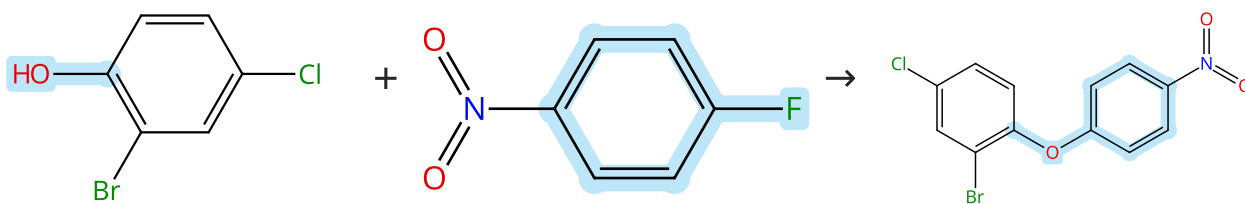
Experimental Protocols

By: Vari, Zahra; et al

New Journal of Chemistry (2023), 47(22), 10645-10658.

Scheme 249 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (80)

Suppliers (100)

Suppliers (6)

31-614-CAS-37615387

Steps: 1 Yield: 99%

Discovery of N-Arylsulfonyl-Indole-2-Carboxamide Derivatives as Galectin-3 and Galectin-8 C-Terminal Domain Inhibitors

By: Zhang, Haoming; et al

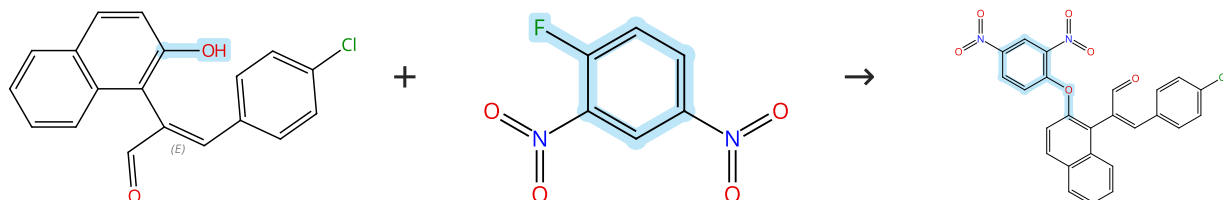
ACS Medicinal Chemistry Letters (2023), 14(9), 1257-1265.

1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 6 h, 50 °C

Experimental Protocols

Scheme 250 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560057

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

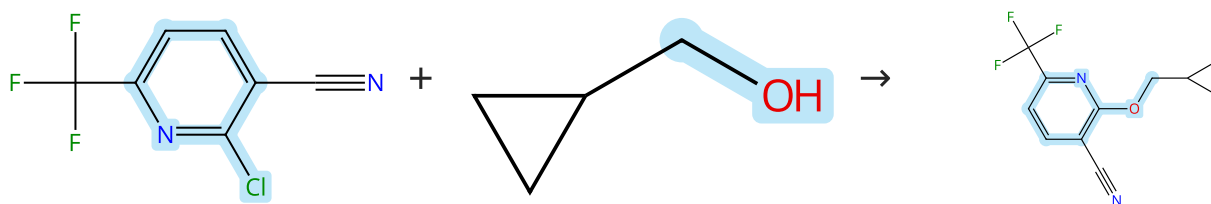
Nature Communications (2023), 14(1), 5050.

1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

Scheme 251 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (71)

Suppliers (66)

Suppliers (8)

31-614-CAS-40536034

Steps: 1 Yield: 99%

Discovery of N-(1,4-Benzoxazin-3-one) urea analogs as Mode-Selective TRPV1 antagonists

By: Huang, Guocheng; et al

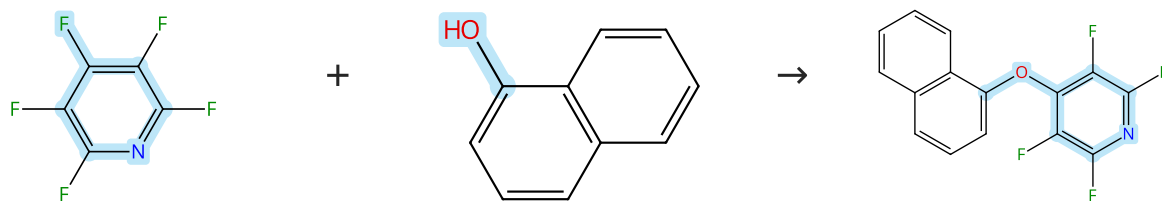
Bioorganic & Medicinal Chemistry Letters (2024), 106, 129735.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 0 °C; overnight, rt

Experimental Protocols

Scheme 252 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (71)

Suppliers (127)

Suppliers (2)

31-170-CAS-21983427

Steps: 1 Yield: 99%

Ionic hydrogenation of naphthyl tetrafluoropyridin-4-yl ethers as a new route to 5,6,7,8-tetrahydronaphthols

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 1 h, 20 °C

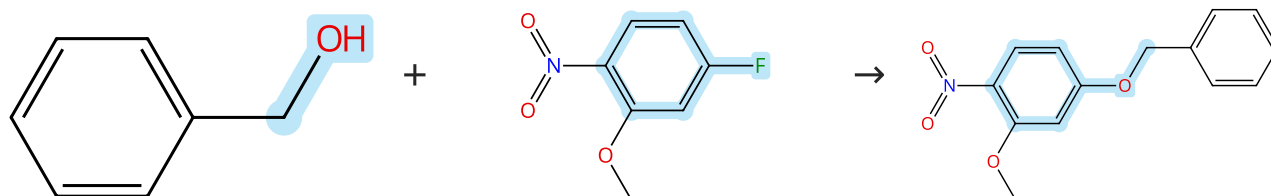
By: Zhu, Zhongwei; et al

Experimental Protocols

Mendelev Communications (2020), 30(2), 190-191.

Scheme 253 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (150)

Suppliers (91)

Suppliers (17)

31-170-CAS-22809318

Steps: 1 Yield: 99%

Discovery of IACS-9439, a Potent, Exquisitely Selective, and Orally Bioavailable Inhibitor of CSF1R

1.1 **Reagents:** Sodium hydride
Solvents: Dimethylformamide; 0.5 h, 0 °C

By: Czako, Barbara; et al

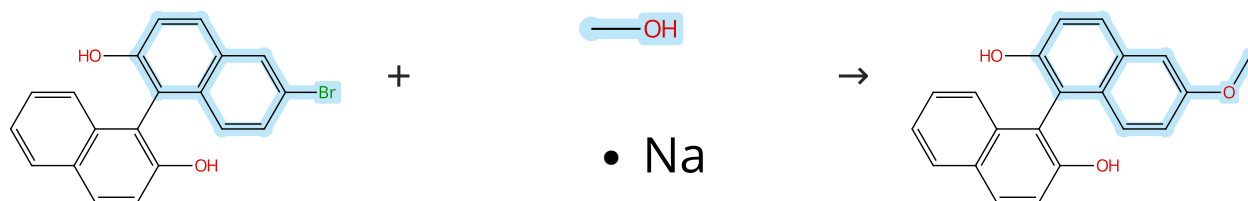
1.2 1 h, rt

Journal of Medicinal Chemistry (2020), 63(17), 9888-9911.

1.3 **Reagents:** Water

Scheme 254 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

Suppliers (99)

31-614-CAS-40024442

Steps: 1 Yield: 99%

General, Modular Access toward Immobilized Chiral Phosphoric Acid Catalysts and Their Application in Flow Chemistry

1.1 **Reagents:** Cuprous chloride
Solvents: Methanol, Dimethylformamide; 16 h, 90 °C; 90 °C → rt

By: Laue, Michael; et al

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; neutralized, 0 °C

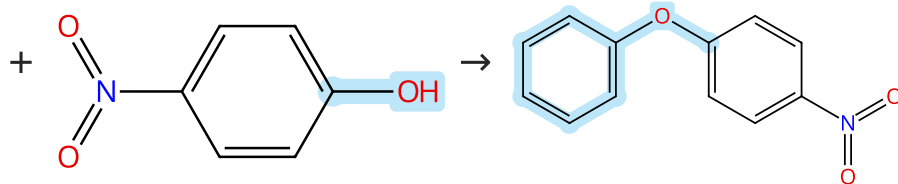
ACS Catalysis (2024), 14(8), 5550-5559.

Experimental Protocols

Scheme 255 (1 Reaction)

Steps: 1 Yield: 99%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (133)

Suppliers (68)

31-614-CAS-31534001

Steps: 1 Yield: 99%

Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates

By: Kikushima, Kotaro; et al

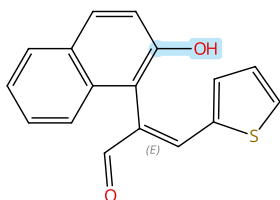
Organic Letters (2022), 24(10), 1924-1928.

- 1.1 **Reagents:** Sodium carbonate
Solvents: Dichloromethane, Water; 18 h, 40 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

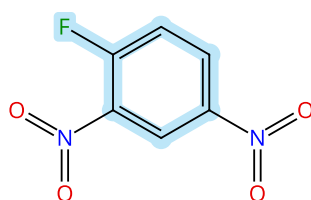
Scheme 256 (1 Reaction)

Steps: 1 Yield: 99%

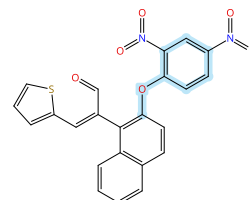


Double bond geometry shown

+



→



Suppliers (75)

31-614-CAS-37560076

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

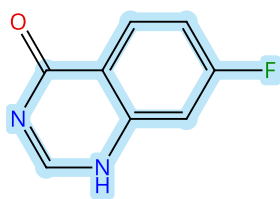
Nature Communications (2023), 14(1), 5050.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)]
- Solvents:** Chloroform; 24 h, -10 °C

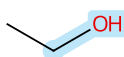
Experimental Protocols

Scheme 257 (1 Reaction)

Steps: 1 Yield: 99%

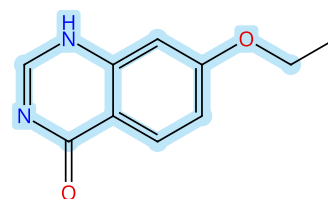


+



• Na

→



Suppliers (96)

Suppliers (94)

Suppliers (3)

31-614-CAS-38180250

Steps: 1 Yield: 99%

Identification of Small Molecule Inhibitors and Ligand Directed Degradors of Calcium/Calmodulin Dependent Protein Kinase Kinase 1 and 2 (CaMKK1/2)

By: Chen, Young; et al

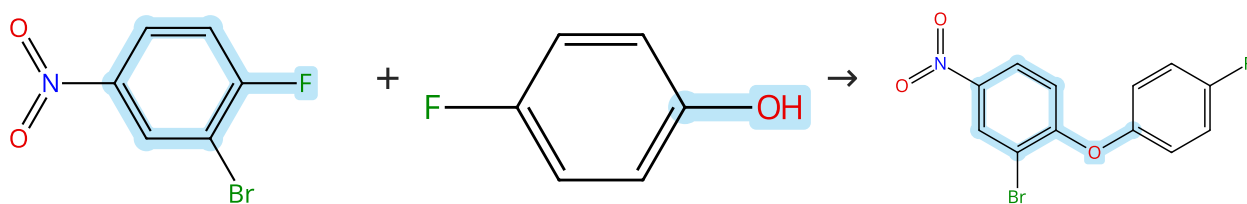
Journal of Medicinal Chemistry (2023), 66(23), 15750-15760.

- 1.1 **Solvents:** Dimethylformamide; rt; 1 h, 110 °C
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water; pH 5 - 6, 0 °C

Experimental Protocols

Scheme 258 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

Suppliers (86)

Suppliers (28)

31-614-CAS-35312181

Steps: 1 Yield: 99%

Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors

By: Xu, Kai-Yan; et al

Bioorganic & Medicinal Chemistry (2023), 78, 117152.

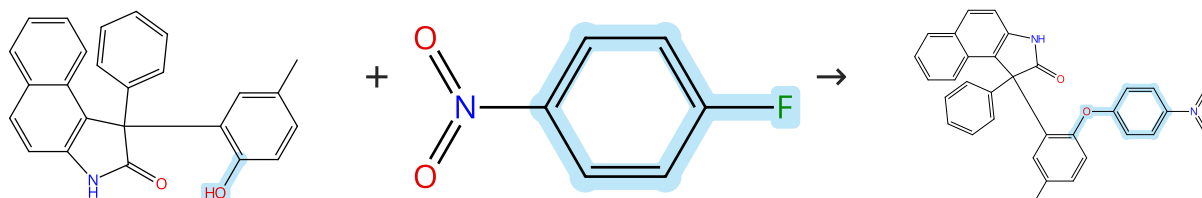
1.1 Reagents: Cesium carbonate

Solvents: Dimethyl sulfoxide; 2 h, 110 °C

Experimental Protocols

Scheme 259 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (100)

31-170-CAS-23191824

Steps: 1 Yield: 99%

I₂-Mediated Cross-Dehydrogenative Coupling and Amidation of 3-Aryl Benzofuranones with Aryl Amines for the Synthesis of 3,3-Diaryl Indolin-2-ones

By: Tang, Zhi; et al

Journal of Organic Chemistry (2021), 86(3), 2965-2973.

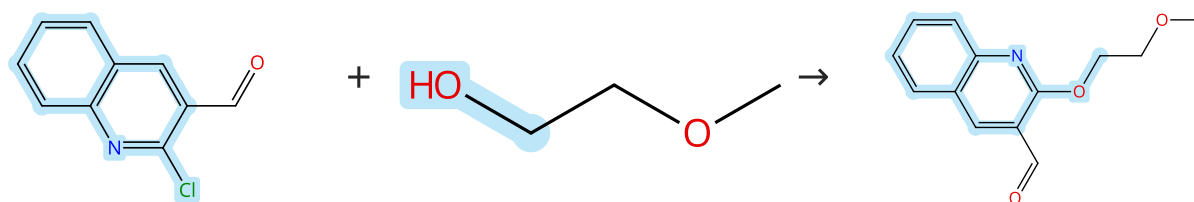
1.1 Reagents: Tempo

Solvents: Dimethyl sulfoxide; 4 h, 40 °C

Experimental Protocols

Scheme 260 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (97)

Suppliers (132)

Suppliers (6)

31-614-CAS-24991572

Steps: 1 Yield: 99%

Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinoline and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines

By: Nandhini, Sundar; et al

Applied Organometallic Chemistry (2021), 35(12), e6436.

1.1 Reagents: Phenylboronic acid, Potassium hydroxide

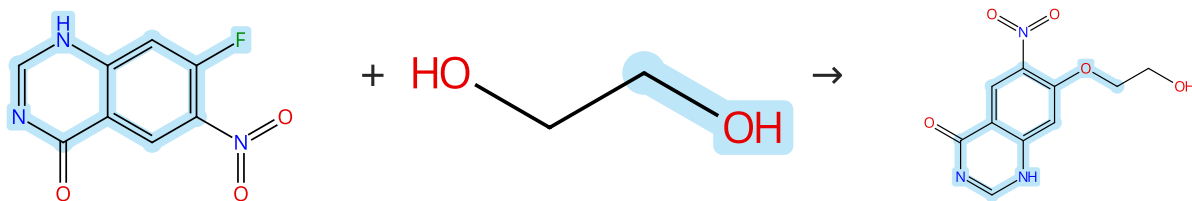
Catalysts: 2721440-87-3

Solvents: 2-Methoxyethanol; 8 h, reflux

Experimental Protocols

Scheme 261 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (94)

Suppliers (166)

Supplier (1)

31-170-CAS-21768067

Steps: 1 Yield: 99%

Design and synthesis of a novel class E GFR/HER2 dual inhibitors containing tricyclic oxazine fused quinazolines scaffold

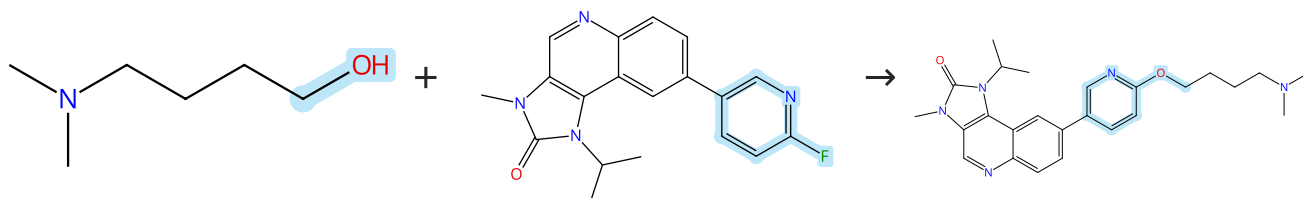
By: Sun, Min; et al

Bioorganic & Medicinal Chemistry Letters (2020), 30(9), 127045.

1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran; 20 h, 75 °C

Scheme 262 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (60)

31-614-CAS-39405346

Steps: 1 Yield: 99%

Identification of Novel, Selective Ataxia-Telangiectasia Mutated Kinase Inhibitors with the Ability to Penetrate the Blood-Brain Barrier: The Discovery of AZD1390

By: Pike, Kurt G.; et al

Journal of Medicinal Chemistry (2024), 67(4), 3090-3111.

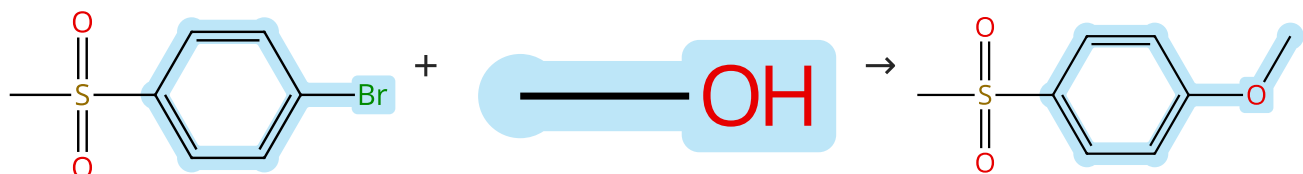
1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran; 10 - 30 min, 0 °C

1.2 **Solvents:** Tetrahydrofuran; 1 - 24 h, rt → 70 °C

Experimental Protocols

Scheme 263 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (74)

Suppliers (444)

Suppliers (63)

31-614-CAS-43604757

Steps: 1 Yield: 99%

Modulating π -bridge in donor- π -acceptor covalent organic frameworks for low-energy-light-driven photocatalytic reaction

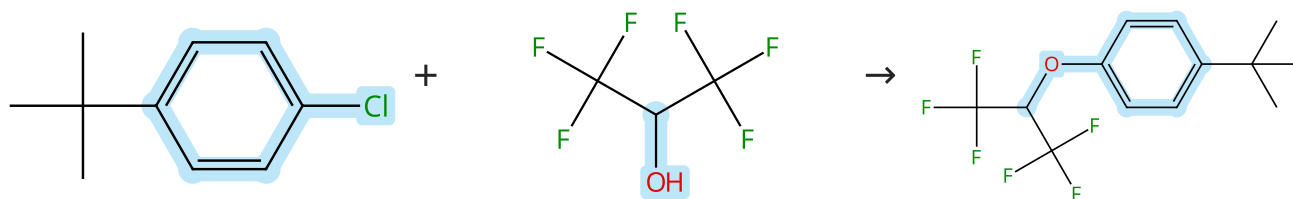
By: Yang, Jiahuan; et al

Journal of Colloid and Interface Science (2025), 683(Part_2), 612-621.

1.1 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene
Catalysts: Nickel bromide, 1,3,5-Benzenetricarboxaldehyde, 2, 4,6-trihydroxy-, polymer with 4,4'-(1,2-ethynediyl)bis[benzen amine]; 24 h

Scheme 264 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (56)

Suppliers (154)

31-614-CAS-36552412

Steps: 1 Yield: 99%

Catalytic S_NAr Hexafluoroisopropoxylation of Aryl Chlorides and Bromides

By: Su, Jiang; et al

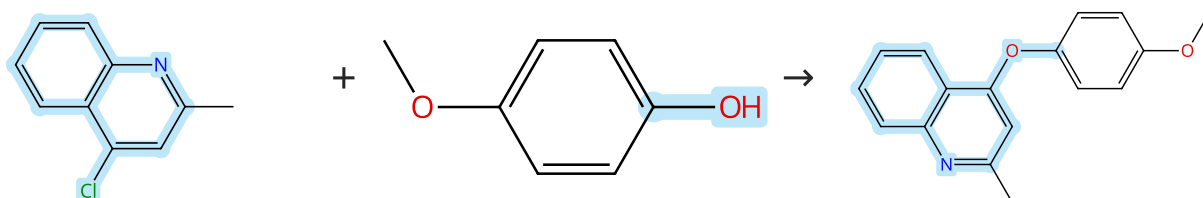
Angewandte Chemie, International Edition (2023), 62(24), e202302908.

1.1 **Reagents:** Silver carbonate
Catalysts: Bis[dichloro[η^5 -(pentamethylcyclopentadienyl)]rhodium], [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl- κO]methanesulfonamidato- κO]silver; 15 h, 80 °C

Experimental Protocols

Scheme 265 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (75)

Suppliers (121)

Suppliers (3)

31-614-CAS-38967451

Steps: 1 Yield: 99%

Ligand-Enabled Nickel Catalysis for the O-Arylation of Alcohols and Phenols with (Hetero)aryl Chlorides Using a Soluble Organic Base

By: Morrison, Kathleen M.; et al

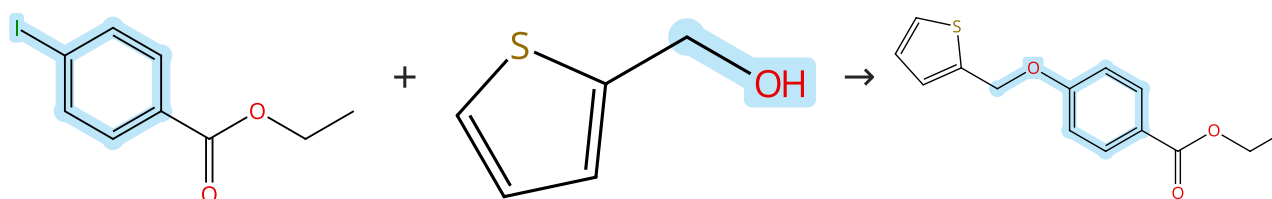
ACS Catalysis (2024), 14(1), 566-573.

1.1 **Reagents:** Sodium trifluoroacetate, 2-Methyl-*N*-(tri-1-pyrrolidinylphosphoranylidene)-2-propanamine
Catalysts: Bis(1,5-cyclooctadiene)nickel, 3026232-46-9
Solvents: Toluene; 18 h, 110 °C

Experimental Protocols

Scheme 266 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (84)

Suppliers (81)

Suppliers (2)

31-614-CAS-35219845

Steps: 1 Yield: 99%

Photoinduced Nickel-Catalyzed Carbon-Heteroatom Coupling

By: Luo, Hang; et al

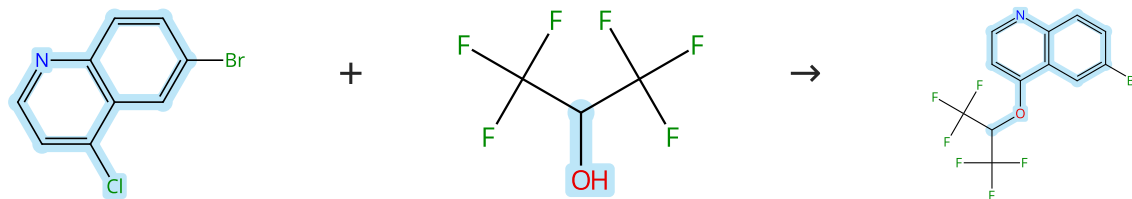
Chemistry - A European Journal (2023), 29(1), e202202385.

1.1 **Catalysts:** Nickel dichloride, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Tetrahydrofuran; 5 min, rt
 1.2 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene; 18 h, 30 °C
 1.3 **Solvents:** Water

Experimental Protocols

Scheme 267 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (88)

Suppliers (154)

31-614-CAS-24281686

Steps: 1 Yield: 99%

Catalyst- and Additive-Free Synthesis of Fluoroalkoxyquinolines1.1 **Solvents:** 1,1,1,3,3,3-Hexafluoro-2-propanol; 24 h, 120 °C; 120 °C → rt1.2 **Reagents:** Sodium bicarbonate
Solvents: Water; neutralized, rt

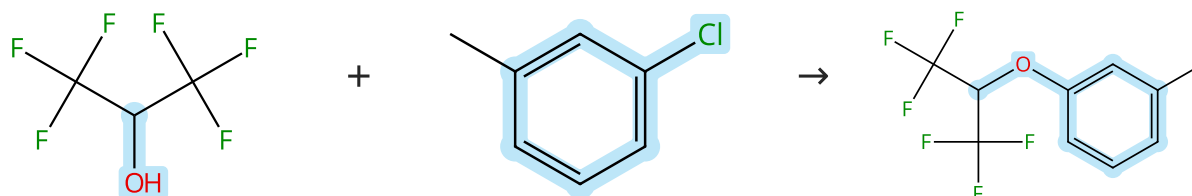
By: Dhiman, Ankit Kumar; et al

Synthesis (2021), 53(21), 4124-4130.

Experimental Protocols

Scheme 268 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (154)

Suppliers (62)

31-614-CAS-36552445

Steps: 1 Yield: 99%

Catalytic S_NAr Hexafluoroisopropoxylation of Aryl Chlorides and Bromides1.1 **Reagents:** Silver carbonate
Catalysts: Bis[dichloro[η^5 -(pentamethylcyclopentadienyl)]rhodium], [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl- κO]methanesulfonamido- κO]silver; 12 h, 80 °C

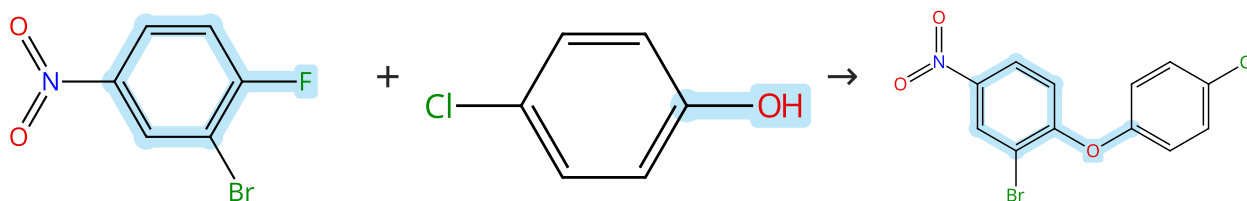
By: Su, Jiang; et al

Angewandte Chemie, International Edition (2023), 62(24), e202302908.

Experimental Protocols

Scheme 269 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

Suppliers (90)

Suppliers (3)

31-614-CAS-35312187

Steps: 1 Yield: 99%

Design, synthesis, and biological evaluation of quinoxaline derivatives as potent BRD4 inhibitors1.1 **Reagents:** Cesium carbonate
Solvents: Dimethyl sulfoxide; 2 h, 110 °C

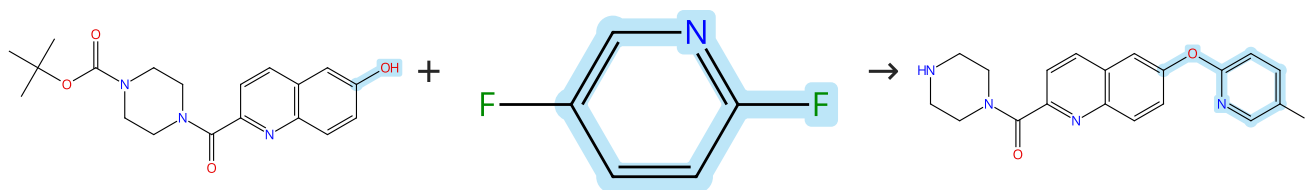
By: Xu, Kai-Yan; et al

Bioorganic & Medicinal Chemistry (2023), 78, 117152.

Experimental Protocols

Scheme 270 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

Suppliers (67)

Suppliers (2)

31-614-CAS-32068470

Steps: 1 Yield: 99%

Discovery of (R)-(3-fluoropyrrolidin-1-yl)(6-((5-(trifluoromethyl)pyridin-2-yl)oxy)quinolin-2-yl)methanone (ABBV-318) and analogs as small molecule Na_v1.7/ Na_v1.8 blockers for the treatment of pain

By: Patel, Meena V.; et al

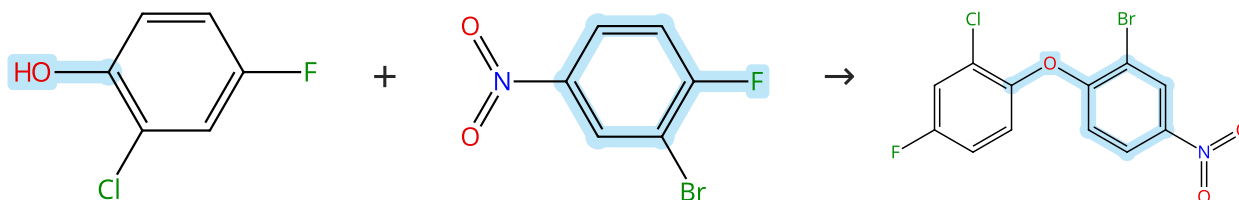
Bioorganic & Medicinal Chemistry (2022), 63, 116743.

- 1.1 **Reagents:** Sodium hydride
Solvents: Dimethylformamide; 10 min
- 1.2 18 h, 65 °C
- 1.3 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; 2 h, rt

Experimental Protocols

Scheme 271 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (78)

Suppliers (83)

31-614-CAS-35312183

Steps: 1 Yield: 99%

Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors

By: Xu, Kai-Yan; et al

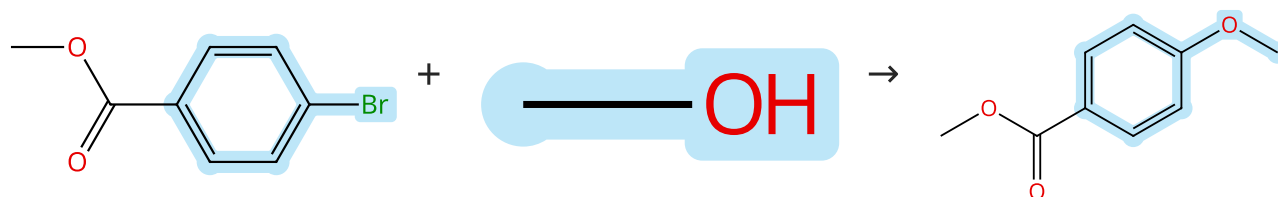
Bioorganic & Medicinal Chemistry (2023), 78, 117152.

- 1.1 **Reagents:** Cesium carbonate
Solvents: Dimethyl sulfoxide; 2 h, 110 °C

Experimental Protocols

Scheme 272 (2 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (98)

Suppliers (444)

Suppliers (92)

31-614-CAS-43604741

Steps: 1 Yield: 99%

Modulating π -bridge in donor- π -acceptor covalent organic frameworks for low-energy-light-driven photocatalytic reaction

By: Yang, Jiahuan; et al

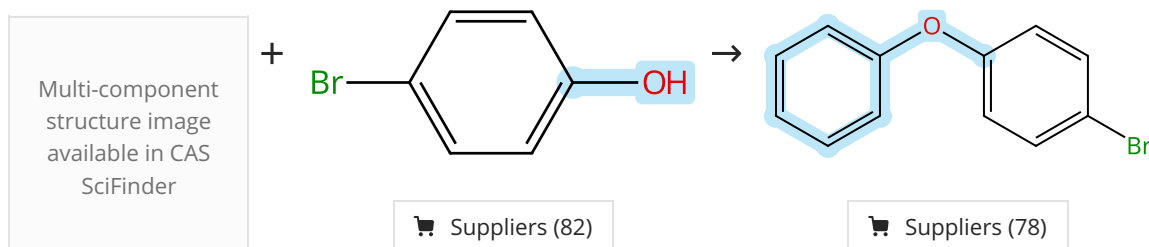
Journal of Colloid and Interface Science (2025), 683(Part_2), 612-621.

- 1.1 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene
Catalysts: Nickel bromide, 1,3,5-Benzenetricarboxaldehyde, 2, 4,6-trihydroxy-, polymer with 4,4'-(1,2-ethynediyl)bis[benzen amine]
Solvents: Methanol; 24 h

31-614-CAS-38944049	Steps: 1 Yield: 98%	Bottom-Up Construction of Ni(II)-Incorporated Covalent Organic Framework for Metallaphotoredox Catalysis
1.1 Reagents: Triethylamine Catalysts: 3026433-57-5 Solvents: Methanol, Acetonitrile; 24 h, rt		By: Hu, Hai-Chao; et al
Experimental Protocols		Chemistry - A European Journal (2024), 30(10), e202303476.

Scheme 273 (1 Reaction)

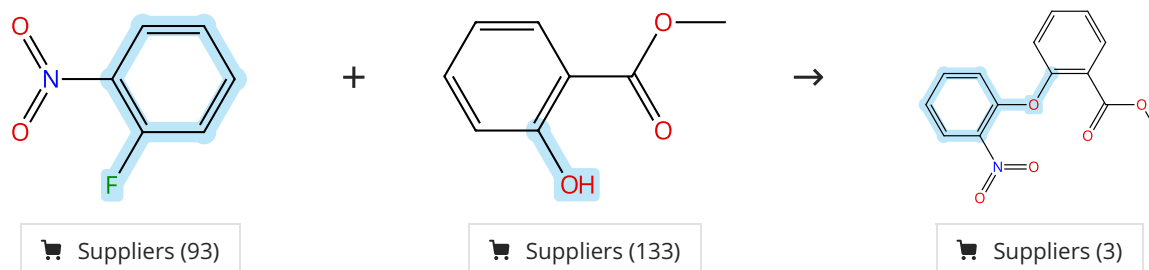
Steps: 1 Yield: 99%



31-614-CAS-31534029	Steps: 1 Yield: 99%	Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates
1.1 Reagents: Sodium carbonate Solvents: Dichloromethane, Water; 18 h, 40 °C		By: Kikushima, Kotaro; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Organic Letters (2022), 24(10), 1924-1928.
Experimental Protocols		

Scheme 274 (1 Reaction)

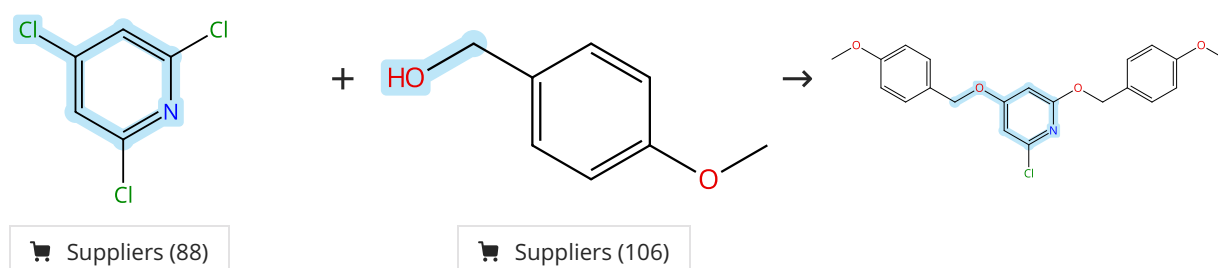
Steps: 1 Yield: 99%



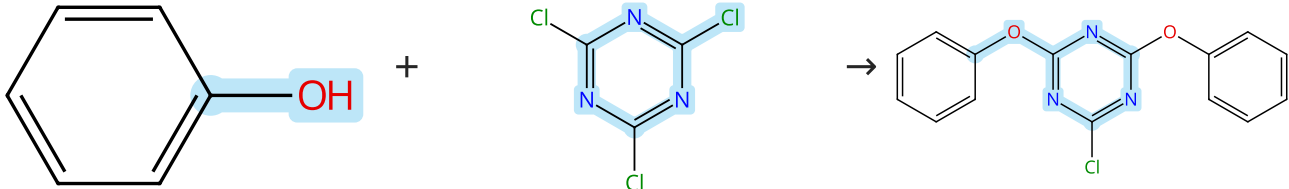
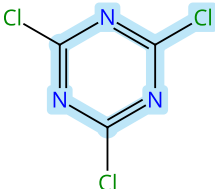
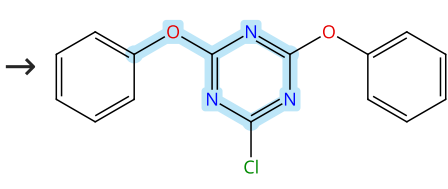
31-614-CAS-38294812	Steps: 1 Yield: 99%	Herringbone Helical Foldamers from Aromatic Ether Derived ε-Amino Acid Peptides
1.1 Reagents: Potassium carbonate Solvents: DMSO- <i>d</i> ₆ ; overnight, rt		By: Yang, Yong; et al
Experimental Protocols		Chemistry - A European Journal (2023), 29(66), e202301832.

Scheme 275 (1 Reaction)

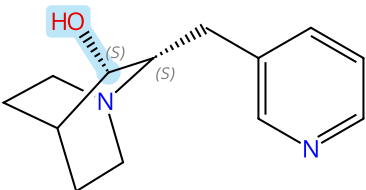
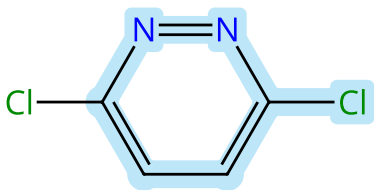
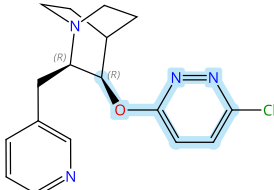
Steps: 1 Yield: 99%



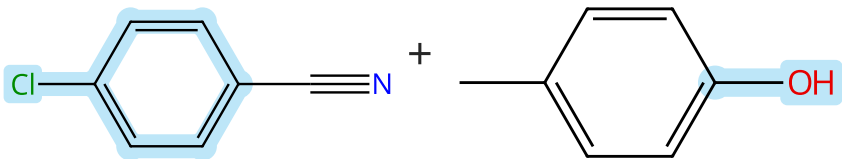
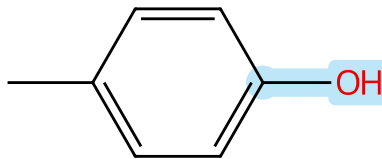
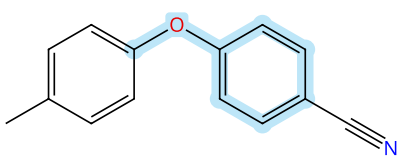
31-614-CAS-39486828	Structure-Activity Relationship Studies and Optimization of 4-Hydroxypyridones as GPR84 Agonists
Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 0 °C; 1 h, 0 °C; 0 °C → rt; 48 h, rt 1.2 Reagents: Ammonium chloride Solvents: Water; rt	By: Ieremias, Loukas; et al Journal of Medicinal Chemistry (2024), 67(5), 3542-3570.
Experimental Protocols	

Scheme 276 (1 Reaction)			Steps: 1 Yield: 99%
			 Suppliers (207)  Suppliers (71)  Suppliers (9)

31-614-CAS-35047948	Molecular duplexes featuring NH...N, CH...O and CH...π interactions in solid-state self-assembly of triazine based compounds
Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Tetrahydrofuran; 72 h, neutralized, rt	
Experimental Protocols	By: Asghar, Shazia; et al Royal Society Open Science (2022), 9(11), 220603.

Scheme 277 (1 Reaction)			Steps: 1 Yield: 99%
 Relative stereochemistry shown		 Relative stereochemistry shown	 Suppliers (87)

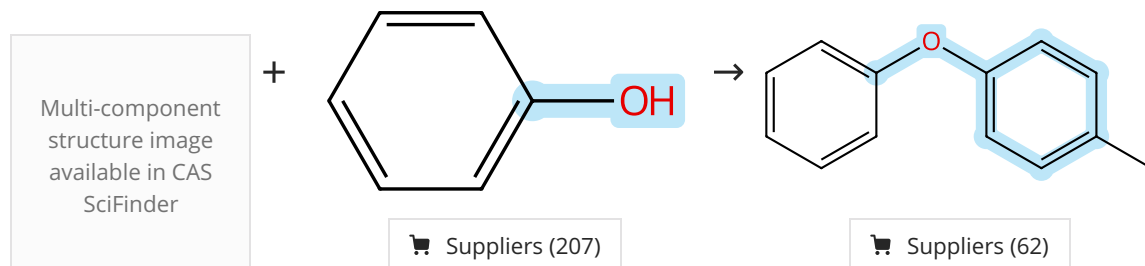
31-614-CAS-35597167	Novel α7 nicotinic acetylcholine receptor modulators as potential antitussive agents
Steps: 1 Yield: 99% 1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: Tetrahydrofuran; 1 h, 25 °C 1.2 1 h, 25 °C	By: Mazurov, Anatoly; et al Bioorganic & Medicinal Chemistry Letters (2023), 80, 129067.
Experimental Protocols	

Scheme 278 (1 Reaction)			Steps: 1 Yield: 99%
			 Suppliers (90)  Suppliers (89)  Suppliers (56)

31-170-CAS-23365413	Steps: 1 Yield: 99%	Oxalohydrazide Ligands for Copper-Catalyzed C-O Coupling Reactions with High Turnover Numbers
1.1 Reagents: Tripotassium phosphate Catalysts: Copper bromide (Cu Br), <i>N</i> ¹ , <i>N</i> ² -Bis(2,5-dimethyl-1 <i>H</i> -pyrrol-1-yl)ethanediamide Solvents: Dimethyl sulfoxide; 5 - 10 min, rt		By: Ray, Ritwika; et al Angewandte Chemie, International Edition (2021), 60(15), 8203-8211.
1.2 24 h, 120 °C; 120 °C → rt		
1.3 Reagents: Sodium hydroxide Solvents: Water; rt		
Experimental Protocols		

Scheme 279 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-31534004	Steps: 1 Yield: 99%	Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates
1.1 Reagents: Sodium carbonate Solvents: Dichloromethane, Water; 18 h, 40 °C		By: Kikushima, Kotaro; et al Organic Letters (2022), 24(10), 1924-1928.
1.2 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 280 (1 Reaction)

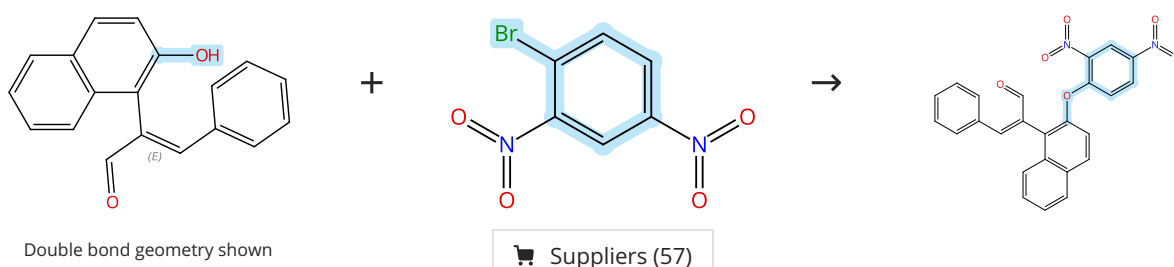
Steps: 1 Yield: 99%



31-614-CAS-24991581	Steps: 1 Yield: 99%	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines
1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: Methanol; 10 h, reflux		By: Nandhini, Sundar; et al Applied Organometallic Chemistry (2021), 35(12), e6436.
Experimental Protocols		

Scheme 281 (1 Reaction)

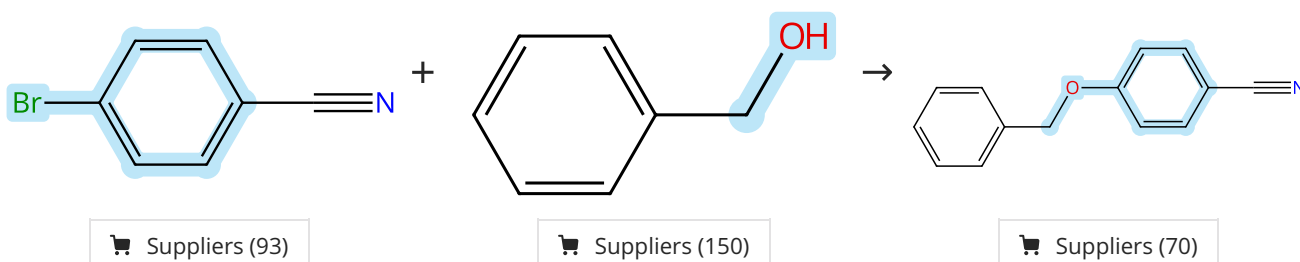
Steps: 1 Yield: 99%



31-614-CAS-37560235 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
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Scheme 282 (1 Reaction)

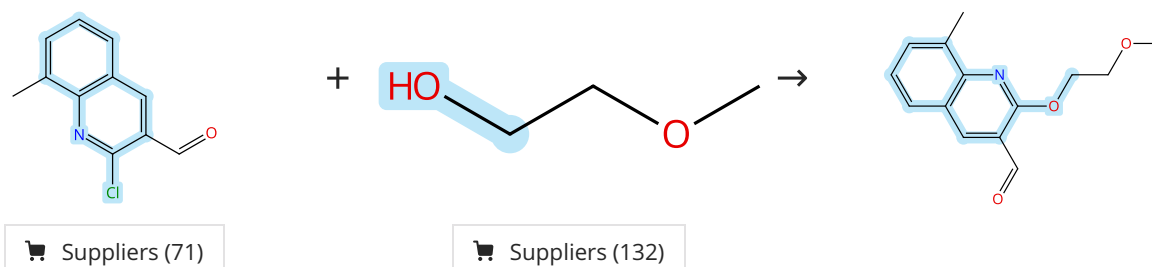
Steps: 1 Yield: 99%



31-170-CAS-23363638 Steps: 1 Yield: 99% 1.1 Reagents: Tripotassium phosphate Catalysts: Copper bromide (CuBr), <i>N</i>¹,<i>N</i>²-Bis(2,5-dimethyl-1<i>H</i>-pyrrol-1-yl)ethanediamide Solvents: Dimethyl sulfoxide; 5 - 10 min, rt 1.2 24 h, 120 °C Experimental Protocols	Oxalohydrazide Ligands for Copper-Catalyzed C-O Coupling Reactions with High Turnover Numbers By: Ray, Ritwika; et al Angewandte Chemie, International Edition (2021), 60(15), 8203-8211.
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Scheme 283 (1 Reaction)

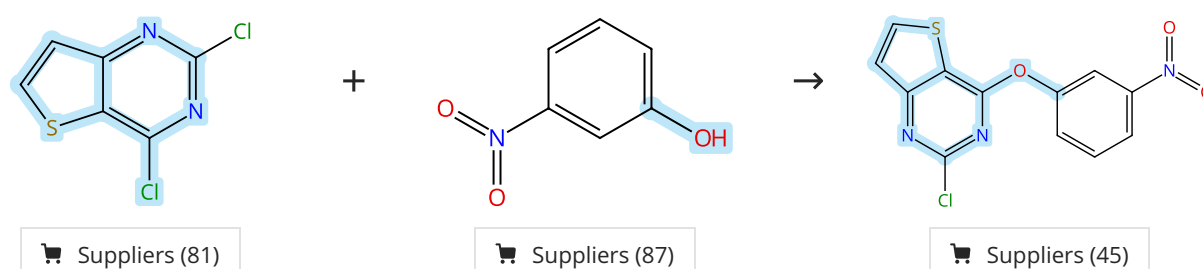
Steps: 1 Yield: 99%



31-614-CAS-24991584 Steps: 1 Yield: 99% 1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: 2-Methoxyethanol; 9 h, reflux Experimental Protocols	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines By: Nandhini, Sundar; et al Applied Organometallic Chemistry (2021), 35(12), e6436.
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Scheme 284 (1 Reaction)

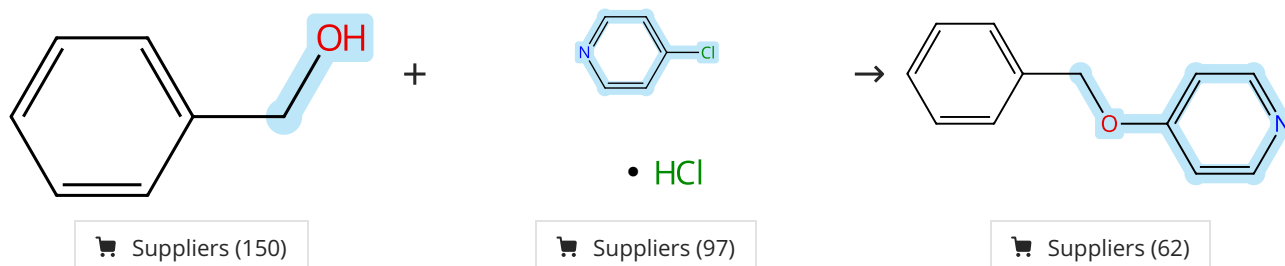
Steps: 1 Yield: 99%



31-170-CAS-22330757	Steps: 1 Yield: 99%	Design, synthesis and antitumor activity of novel thiophene-pyrimidine derivatives as EGFR inhibitors overcoming T790M and L858R/T790M mutations
1.1 Catalysts: Cesium carbonate Solvents: 1,4-Dioxane; 4 h, rt		By: Xiao, Zhen; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2020), 203, 112511.

Scheme 285 (1 Reaction)

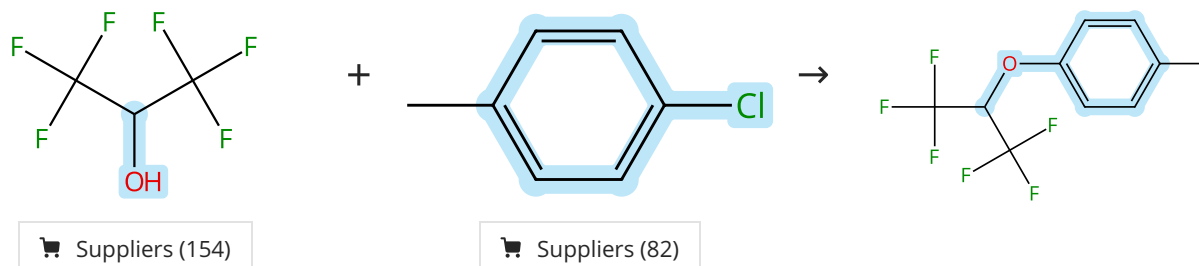
Steps: 1 Yield: 99%



31-614-CAS-40869842	Steps: 1 Yield: 99%	Discovery of a Promising CBP/p300 Degrader XYD129 for the Treatment of Acute Myeloid Leukemia
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 15 min, 0 °C		By: Wu, Tianbang; et al
1.2 5 min, 0 °C; 10 min, 0 °C; 0 °C → 60 °C; 2 h, 60 °C		Journal of Medicinal Chemistry (2024), 67(11), 9194-9213.
Experimental Protocols		

Scheme 286 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-36552413	Steps: 1 Yield: 99%	Catalytic S _N Ar Hexafluoroisopropoxylation of Aryl Chlorides and Bromides
1.1 Reagents: Silver carbonate Catalysts: Bis[dichloro[η ⁵ -(pentamethylcyclopentadienyl)]rhodium], [1,1,1-Trifluoro- <i>N</i> -[(trifluoromethyl)sulfonyl-κ <i>O</i>]methanesulfonamido-κ <i>O</i>]silver Solvents: 1,2-Dichloroethane; 12 h, 80 °C		By: Su, Jiang; et al
Experimental Protocols		Angewandte Chemie, International Edition (2023), 62(24), e202302908.

Scheme 287 (1 Reaction)

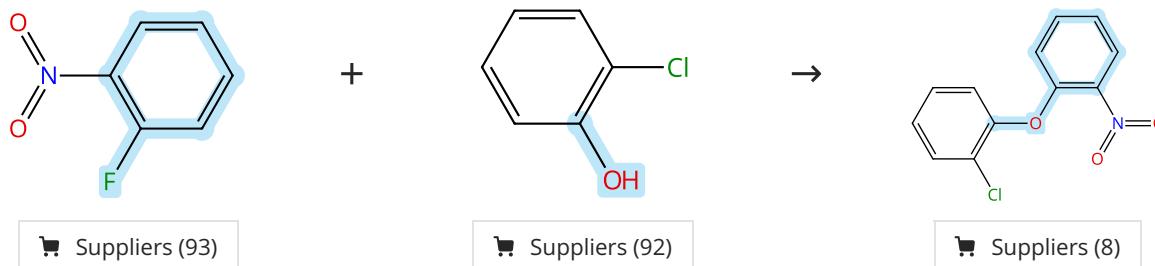
Steps: 1 Yield: 99%



31-170-CAS-23364768	Steps: 1 Yield: 99%	Oxalohydrazide Ligands for Copper-Catalyzed C-O Coupling Reactions with High Turnover Numbers
1.1 Reagents: Tripotassium phosphate Catalysts: Copper bromide (Cu Br), <i>N</i> ¹ , <i>N</i> ² -Bis(2,5-dimethyl-1 <i>H</i> -pyrrol-1-yl)ethanediamide Solvents: Dimethyl sulfoxide; 5 - 10 min, rt		By: Ray, Ritwika; et al
1.2 24 h, 120 °C		Angewandte Chemie, International Edition (2021), 60(15), 8203-8211.
Experimental Protocols		

Scheme 288 (1 Reaction)

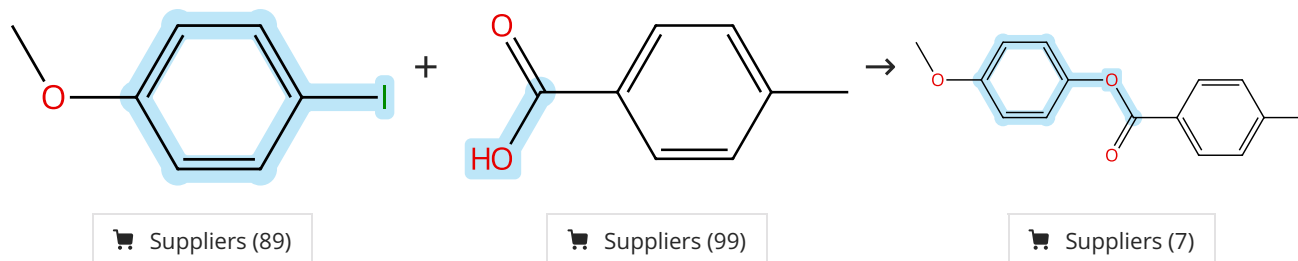
Steps: 1 Yield: 99%



31-170-CAS-22969850	Steps: 1 Yield: 99%	Synthesis of Cycloheptatrienes, Oxepines, Thiepinines, and Silepinines: A Comparison between Broensted Acid and Au-Catalysis
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt → 95 °C; 24 h, 95 °C		By: Sprenger, Kristin; et al
Experimental Protocols		European Journal of Organic Chemistry (2020), 2020(39), 6245-6254.

Scheme 289 (1 Reaction)

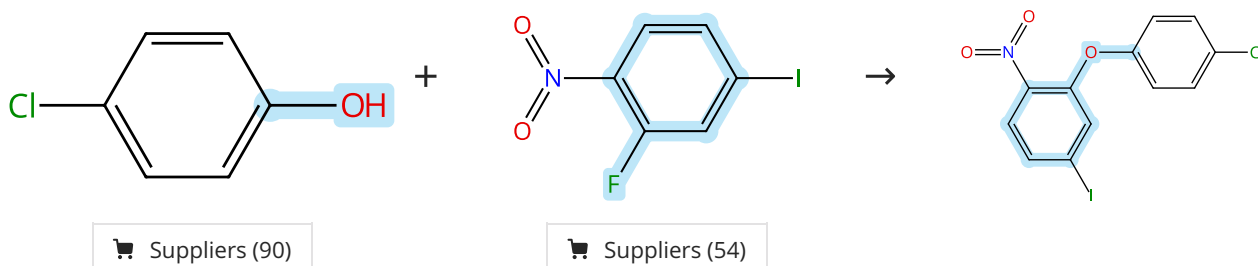
Steps: 1 Yield: 99%



31-614-CAS-41256823	Steps: 1 Yield: 99%	Photosensitized Reductive Elimination of Gold(III) to Enable Esterification of Aryl Iodides with Carboxylic Acids
1.1 Reagents: Sodium carbonate Catalysts: Benzophenone, Silver hexafluoroantimonate, [2-[Bis(tricyclo[3.3.1.1 ^{3,7}]dec-1-yl)phosphino-κ ²]- <i>N,N</i> -dimethylbenzenamine]chlorogold Solvents: 1,2-Dichloroethane; 18 h, 30 °C		By: Wu, Jiawen; et al
Experimental Protocols		JACS Au (2024), 4(8), 3084-3093.

Scheme 290 (1 Reaction)

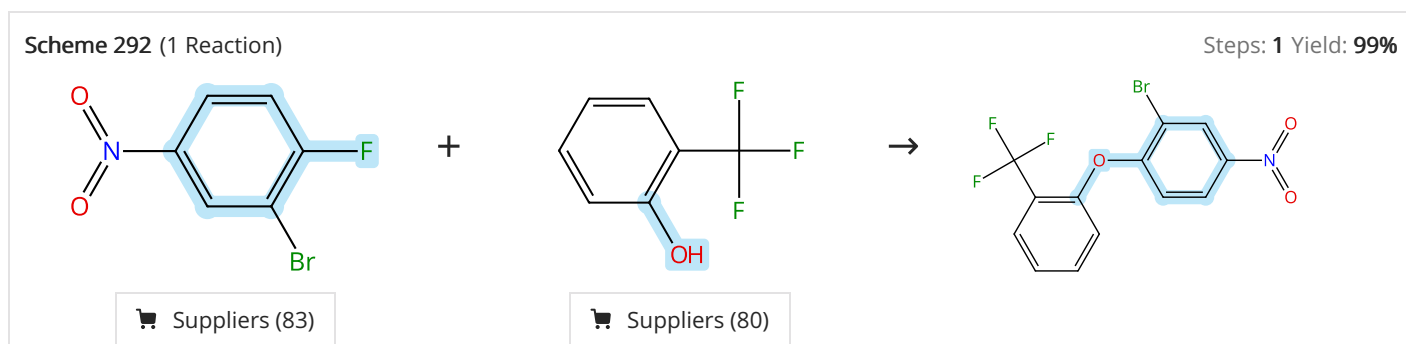
Steps: 1 Yield: 99%



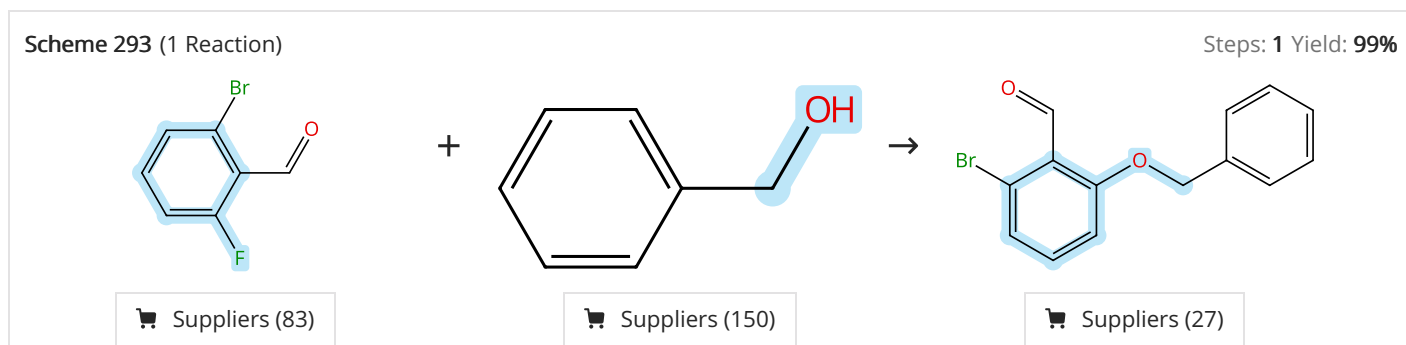
31-614-CAS-38471053	Steps: 1 Yield: 99%	Exploring the Translational Gap of a Novel Class of Escherichia coli IspE Inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 20 min, rt		By: Ropponen, Henni-Karoliina; et al
1.2 3 h, 90 °C		ChemMedChem (2023), 18(19), e202300346.
Experimental Protocols		



31-614-CAS-40536029	Steps: 1 Yield: 99%	Discovery of N-(1,4-Benzoxazin-3-one) urea analogs as Mode-Selective TRPV1 antagonists
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 0 °C; overnight, rt		By: Huang, Guocheng; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2024), 106, 129735.



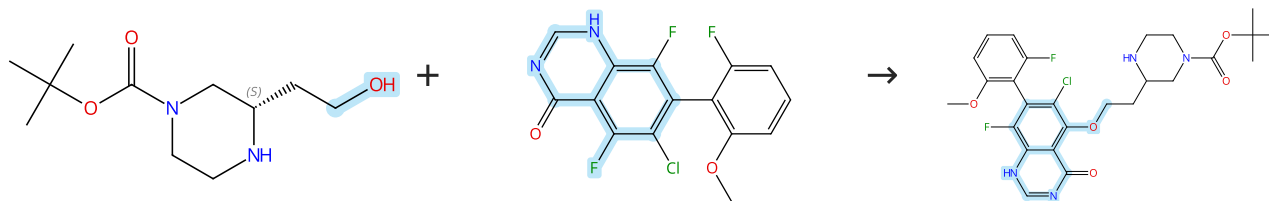
31-614-CAS-35312186	Steps: 1 Yield: 99%	Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 2 h, 110 °C		By: Xu, Kai-Yan; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2023), 78, 117152.



31-614-CAS-34555895	Steps: 1 Yield: 99%	The First Class of Small Molecules Potently Disrupting the Y A P-TEAD Interaction by Direct Competition
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; overnight, 80 °C		By: Furet, Pascal; et al
Experimental Protocols		ChemMedChem (2022), 17(19), e202200303.

Scheme 294 (1 Reaction)

Steps: 1 Yield: 99%



Absolute stereochemistry shown

Suppliers (56)

31-614-CAS-40188184

Steps: 1 Yield: 99%

Discovery of AZD4625, a Covalent Allosteric Inhibitor of the Mutant GTPase KRAS^{G12C}

By: Kettle, Jason G.; et al

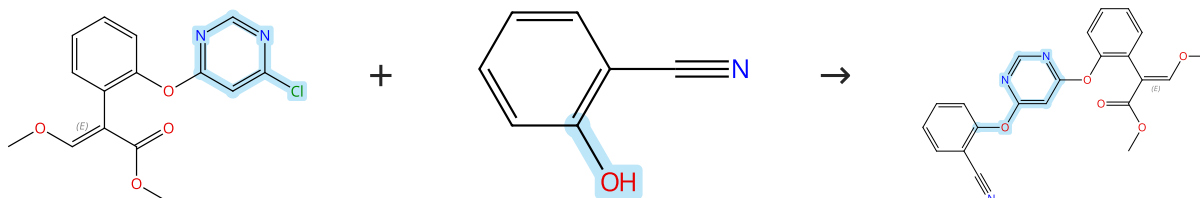
Journal of Medicinal Chemistry (2022), 65(9), 6940-6952.

- 1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran; 15 min, rt
- 1.2 3.5 h, 60 °C; 60 °C → rt
- 1.3 **Reagents:** Ammonium chloride
Solvents: Water; rt

Experimental Protocols

Scheme 295 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (46)

Suppliers (98)

Double bond geometry shown

Suppliers (78)

31-614-CAS-40217024

Steps: 1 Yield: 99%

Development of a Trimethylamine-Catalyzed Novel Synthesis of Azoxystrobin

By: Yang, Binglian; et al

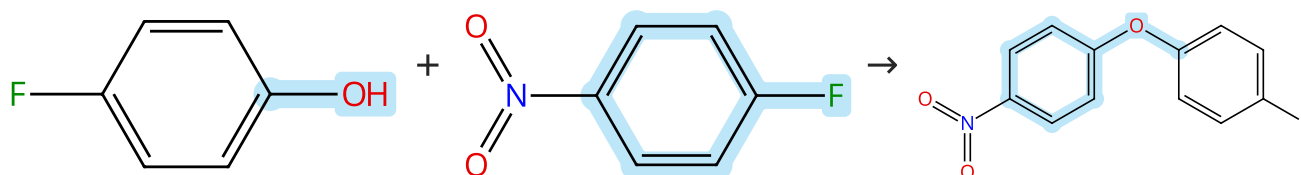
Organic Process Research & Development (2023), 27(7), 1276-1282.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Trimethylamine
Solvents: Toluene, Water; 30 - 60 min, rt → 80 °C; 8 h, 80 °C

Experimental Protocols

Scheme 296 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (86)

Suppliers (100)

Suppliers (42)

31-614-CAS-37615360

Steps: 1 Yield: 99%

Discovery of N-Arylsulfonyl-Indole-2-Carboxamide Derivatives as Galectin-3 and Galectin-8 C-Terminal Domain Inhibitors

By: Zhang, Haoming; et al

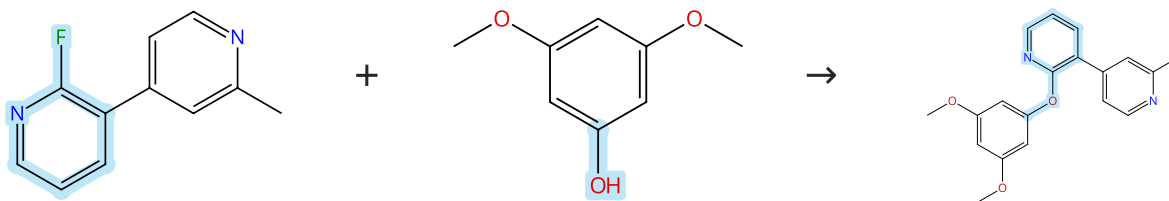
ACS Medicinal Chemistry Letters (2023), 14(9), 1257-1265.

- 1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 6 h, 50 °C

Experimental Protocols

Scheme 297 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (15)

Suppliers (92)

31-614-CAS-35040234

Steps: 1 Yield: 99%

2-Phenoxy-3, 4'-bipyridine derivatives inhibit AURKB-dependent mitotic processes by disrupting its localization

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; rt; 8 h, 115 °C

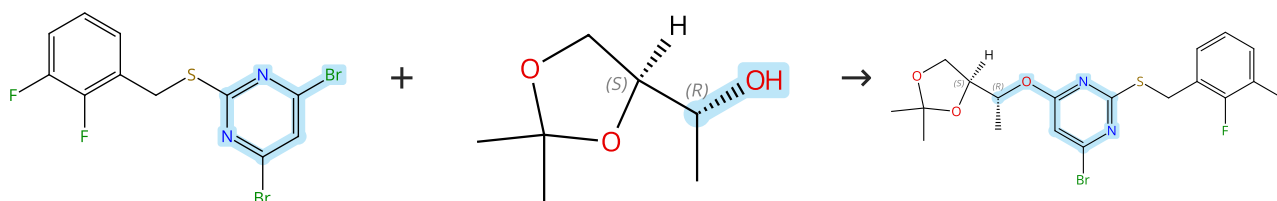
By: Lv, Gang; et al

Experimental Protocols

European Journal of Medicinal Chemistry (2023), 245(Part_1), 114904.

Scheme 298 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (4)

Suppliers (22)

31-614-CAS-32734949

Steps: 1 Yield: 99%

Development and Scale-Up of a Copper-Catalyzed Sulfamidation Coupling Reaction

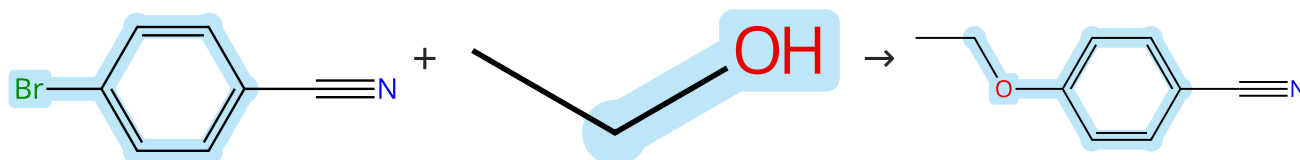
1.1 Reagents: Potassium *tert*-amylate
Solvents: 2-Methyltetrahydrofuran; 0 °C

By: Adlington, Neil K.; et al

Organic Process Research & Development (2022), 26(8), 2337-2350.

Scheme 299 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (93)

Suppliers (618)

Suppliers (75)

31-614-CAS-34588746

Steps: 1 Yield: 99%

Transition-metal-free arylation of alcohols with aryl bromides at room temperature

1.1 Reagents: Sodium *tert*-butoxide
Solvents: Dimethyl sulfoxide; 4 h, rt

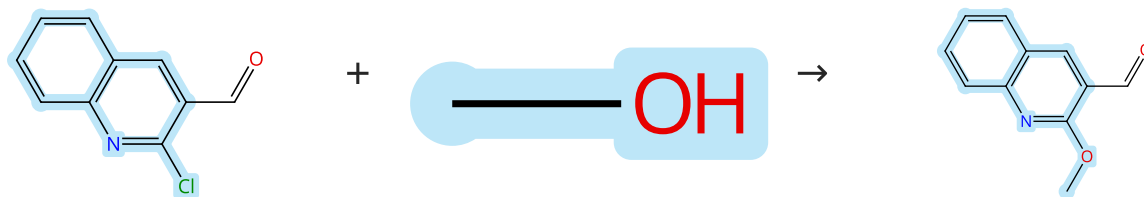
By: Zhu, Da-Liang; et al

Experimental Protocols

Synthesis (2023), 55(4), 637-646.

Scheme 300 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (97)

Suppliers (444)

Suppliers (63)

31-614-CAS-24991574

Steps: 1 Yield: 99%

Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines

By: Nandhini, Sundar; et al

Applied Organometallic Chemistry (2021), 35(12), e6436.

1.1 Reagents: Phenylboronic acid, Potassium hydroxide

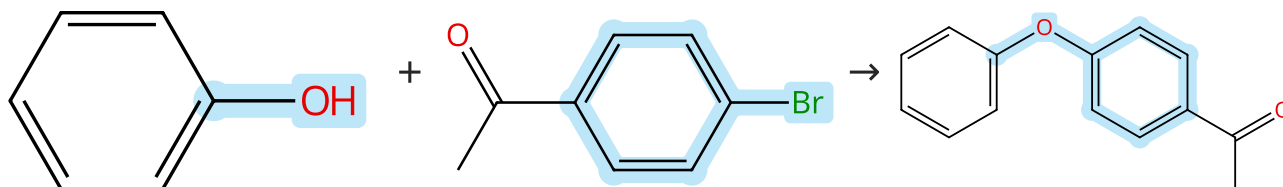
Catalysts: 2721440-87-3

Solvents: Methanol; 8 h, reflux

Experimental Protocols

Scheme 301 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (207)

Suppliers (85)

Suppliers (86)

31-614-CAS-43873506

Steps: 1 Yield: 99%

Electromagnetic milling promoted mechanochemical copper-catalyzed C-O coupling reaction for diaryl ether formation

By: Zeng, Zhaoxuan; et al

Organic Chemistry Frontiers (2025), 12(5), 1528-1536.

1.1 Reagents: Cesium carbonate

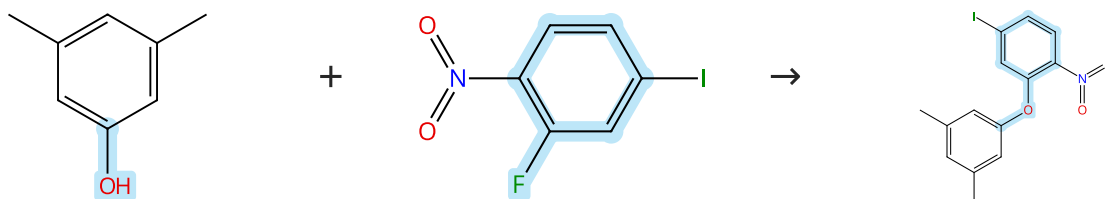
Catalysts: Copper(II) acetylacetonate, *N*¹,*N*²-Di-1-naphthale nylethanediamide

Solvents: Dimethylformamide; 3 h, 79 °C

Experimental Protocols

Scheme 302 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (89)

Suppliers (54)

31-614-CAS-38471052

Steps: 1 Yield: 99%

Exploring the Translational Gap of a Novel Class of Escherichia coli IspE Inhibitors

By: Ropponen, Henni-Karoliina; et al

ChemMedChem (2023), 18(19), e202300346.

1.1 Reagents: Potassium carbonate

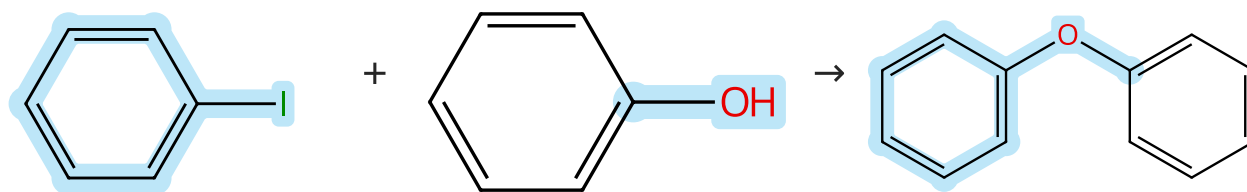
Solvents: Dimethyl sulfoxide; 20 min, rt

1.2 3 h, 90 °C

Experimental Protocols

Scheme 303 (9 Reactions)

Steps: 1 Yield: 97-99%



Suppliers (96)

Suppliers (207)

Suppliers (103)

31-170-CAS-22063626

Steps: 1 Yield: 99%

Pd based on 2-Aminopyrimidine and 1 H-benzo[d]imidazol-2-amine functionalized Fe₃O₄ nanoparticles as novel recyclable magnetic nanocatalysts for Ullmann coupling reaction

By: Chen, Jin; et al

Applied Organometallic Chemistry (2020), 34(8), e5708.

1.1 **Reagents:** Potassium carbonate
Catalysts: *N*-[3-(Trimethoxysilyl)propyl]-2-pyrimidinamine (Fe₃O₄ magnetic nanoparticles; post-treated with Pd(OAc)₂)
Solvents: Dimethylformamide; 1.5 h, 70 °C

Experimental Protocols

31-614-CAS-32504410

Steps: 1 Yield: 98%

Cu/cellulose-modified magnetite nanocomposites as a highly active and selective catalyst for ultrasound-promoted aqueous O-arylation Ullmann and sp-sp² Sonogashira cross-coupling reactions

By: Ghamari Kargar, Pouya; et al

Sustainable Chemistry and Pharmacy (2022), 27, 100672.

1.1 **Reagents:** Potassium hydroxide
Catalysts: Magnetite (Fe₃O₄), Copper, Cellulose
Solvents: Water; 10 min, 40 °C

Experimental Protocols

31-614-CAS-24177522

Steps: 1 Yield: 98%

Magnetization of graphene oxide nanosheets using nickel magnetic nanoparticles as a novel support for the fabrication of copper as a practical, selective, and reusable nanocatalyst in C-C and C-O coupling reactions

By: Moradi, Parisa; et al

RSC Advances (2021), 11(42), 25867-25879.

1.1 **Reagents:** Potassium hydroxide
Catalysts: Nickel
Solvents: Dimethyl sulfoxide; 2 h, 130 °C

Experimental Protocols

31-614-CAS-31921350

Steps: 1 Yield: 98%

Fe-MIL-101 modified by isatin-Schiff-base-Co: a heterogeneous catalyst for C-C, C-O, C-N, and C-P cross coupling reactions

By: Rouzifar, Majid; et al

New Journal of Chemistry (2021), 45(42), 19963-19976.

1.1 **Reagents:** Potassium carbonate
Catalysts: 2765562-72-7
Solvents: Dimethylformamide; 4 h, 80 °C

Experimental Protocols

31-614-CAS-44202637

Steps: 1 Yield: 97%

Green approach to Ullmann and Sonogashira cross-coupling reactions in water: Co-MOF as a robust and recyclable nanostructure

By: Algethami, Norah; et al

Journal of Organometallic Chemistry (2025), 1029, 123563.

1.1 **Reagents:** Potassium hydroxide
Catalysts: 4,4'-[1,4-Phenylenebis[(*E*)-methylidynenitrilo]]bis[3-hydroxybenzoic acid] (Co MOF)
Solvents: Water; 45 min, 50 °C

Experimental Protocols

31-170-CAS-23698435

Steps: 1 Yield: 97%

An efficient, cost-effective, and magnetically recoverable copper catalyst for O-arylation of phenols with aryl halides in choline chloride-based deep eutectic solvents

By: Niakan, Mahsa; et al

Colloids and Surfaces, A: Physicochemical and Engineering Aspects (2021), 620, 126603.

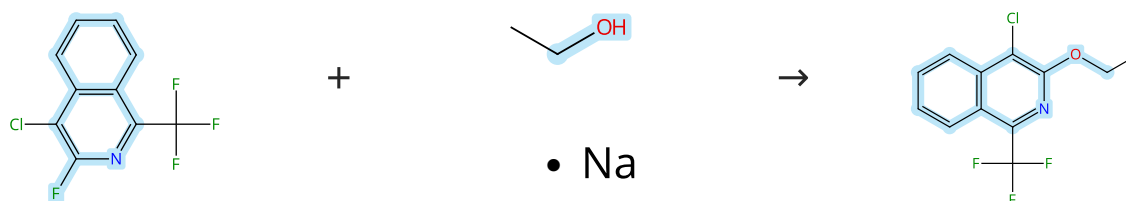
1.1 **Reagents:** Potassium carbonate
Catalysts: 2642302-91-6 (supported on Fe₃O₄-SiO₂)
Solvents: Urea, Choline chloride; 6 h, 100 °C

Experimental Protocols

31-170-CAS-23916243 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Gold (starch-coated iron-oxide supported) Solvents: Water; 12 h, 100 °C Experimental Protocols	On-Water and Efficient Ullmann-Type O-Arylation Cross Coupling Reaction of Phenols and Aryl Tosylates in the Presence of Fe₃O₄@Starch-Au as Nanocatalyst By: Mousavi-Mashhadi, Seyed Ali; et al ChemistrySelect (2021), 6(16), 3941-3951.
31-170-CAS-22063638 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: <i>N</i>-[3-(Trimethoxysilyl)propyl]-1<i>H</i>-benzimidazol-2-amine (Fe₃O₄ magnetic nanoparticles; post-treated with Pd (OAc)₂) Solvents: Dimethylformamide; 1.5 h, 70 °C Experimental Protocols	Pd based on 2-Aminopyrimidine and 1<i>H</i>-benzo[d]imidazol-2-amine functionalized Fe₃O₄ nanoparticles as novel recyclable magnetic nanocatalysts for Ullmann coupling reaction By: Chen, Jin; et al Applied Organometallic Chemistry (2020), 34(8), e5708.
31-170-CAS-21502631 Steps: 1 Yield: 97% 1.1 Reagents: Potassium hydroxide Catalysts: Copper Solvents: Water; 120 min, reflux Experimental Protocols	Glycerol Cu(II) Complex Supported on Fe₃O₄ Magnetic Nanoparticles: A New and Highly Efficient Reusable Catalyst for the Formation of Aryl-Sulfur and Aryl-Oxygen Bonds By: Ashraf, Muhammad Aqeel; et al Catalysis Letters (2020), 150(4), 1128-1141.

Scheme 304 (1 Reaction)

Steps: 1 Yield: 99%

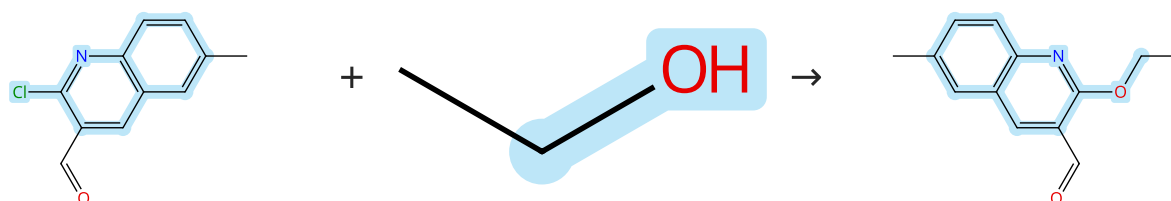


Suppliers (94)

31-614-CAS-41167327 Steps: 1 Yield: 99% 1.1 Solvents: Ethanol; 155 °C Experimental Protocols	One-pot multistep synthesis of 1-fluoroalkylisoquinolines and fused fluoroalkylpyridines from <i>N</i>-fluoroalkyl-1,2,3-triazoles By: Kubickova, Anna; et al Organic Chemistry Frontiers (2024), 11(16), 4442-4448.
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Scheme 305 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (77)

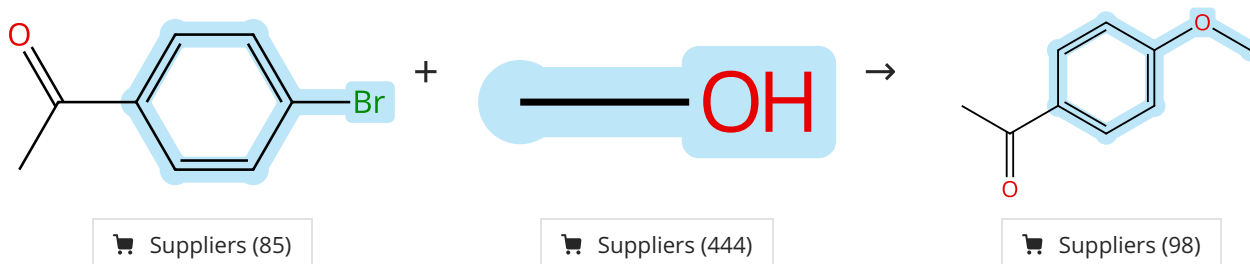
Suppliers (618)

Supplier (1)

31-614-CAS-24991570	Steps: 1 Yield: 99%	Synthesis of tetranuclear complex of Pd(II) with thiosemicarbazone ligands derived from 2-quinolone and its catalytic evaluation in Suzuki-Miyaura-type coupling reactions and alkoxylation of chloroquinolines
1.1 Reagents: Phenylboronic acid, Potassium hydroxide Catalysts: 2721440-87-3 Solvents: Ethanol; 9 h, reflux		By: Nandhini, Sundar; et al
Experimental Protocols		Applied Organometallic Chemistry (2021), 35(12), e6436.

Scheme 306 (1 Reaction)

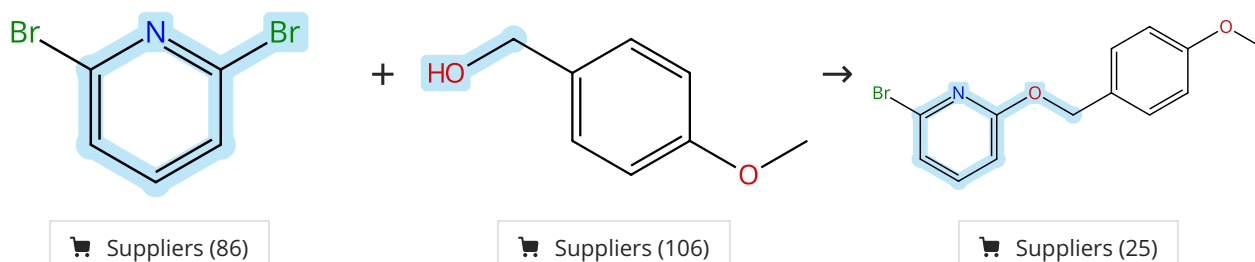
Steps: 1 Yield: 99%



31-170-CAS-22250439	Steps: 1 Yield: 99%	Solar-driven tandem photoredox nickel-catalysed cross-coupling using modified carbon nitride
1.1 Reagents: Quinuclidine Catalysts: Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Acetonitrile; 20 h, rt		By: Qin, Yangzhong; et al
		Chemical Science (2020), 11(28), 7456-7461.

Scheme 307 (1 Reaction)

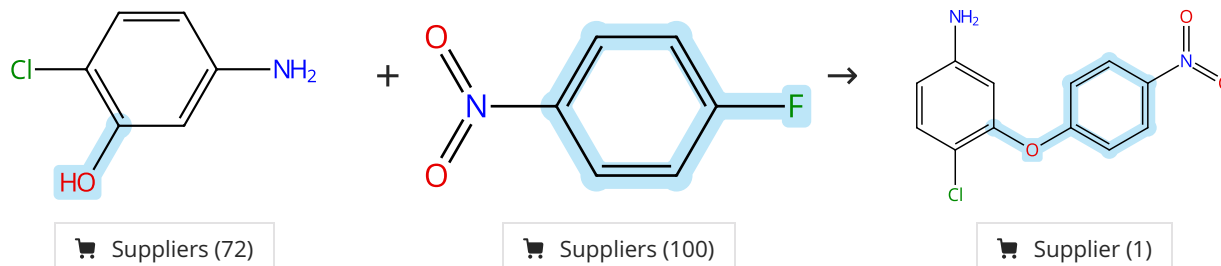
Steps: 1 Yield: 99%



31-614-CAS-36010276	Steps: 1 Yield: 99%	Ligand-Enabled Palladium(II)-Catalyzed Enantioselective β -C(sp ³)-H Arylation of Aliphatic Tertiary Amides
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: 1,4-Dioxane; overnight, 100 °C		By: Yuan, Chen-Hui; et al
Experimental Protocols		Angewandte Chemie, International Edition (2023), 62(17), e202300854.

Scheme 308 (1 Reaction)

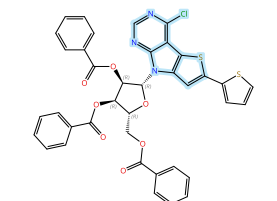
Steps: 1 Yield: 98%



31-614-CAS-40338184	Steps: 1 Yield: 98%	Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C		By: Xu, Yue; et al
Experimental Protocols		Bioorganic Chemistry (2023), 131, 106339.

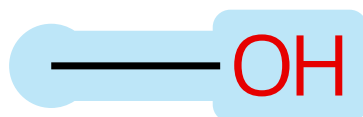
Scheme 309 (1 Reaction)

Steps: 1 Yield: 98%

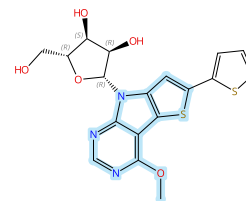


Absolute stereochemistry shown

+



→



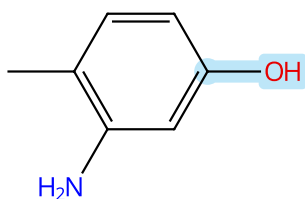
Absolute stereochemistry shown

Suppliers (444)

31-614-CAS-39014196	Steps: 1 Yield: 98%	Synthesis of Polycyclic Hetero-Fused 7-Deazapurine Heterocycles and Nucleosides through C-H Dibenzothiophenation and Negishi Coupling
1.1 Reagents: Sodium methoxide Solvents: Methanol, 1,4-Dioxane; overnight, 65 °C		By: Yang, Chao; et al
Experimental Protocols		Journal of the American Chemical Society (2022), 144(42), 19437-19446.

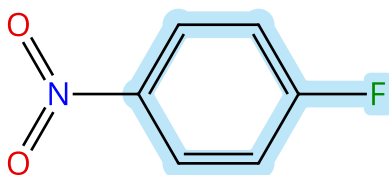
Scheme 310 (1 Reaction)

Steps: 1 Yield: 98%

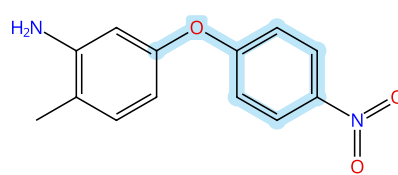


Suppliers (72)

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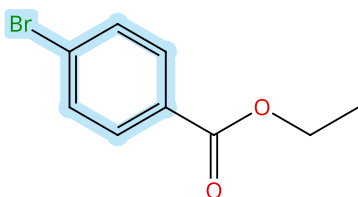


Suppliers (100)

31-614-CAS-40338182	Steps: 1 Yield: 98%	Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C		By: Xu, Yue; et al
Experimental Protocols		Bioorganic Chemistry (2023), 131, 106339.

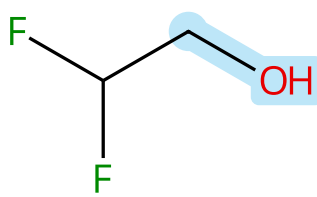
Scheme 311 (1 Reaction)

Steps: 1 Yield: 98%

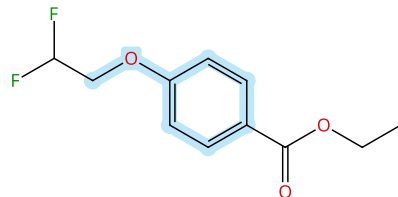


Suppliers (83)

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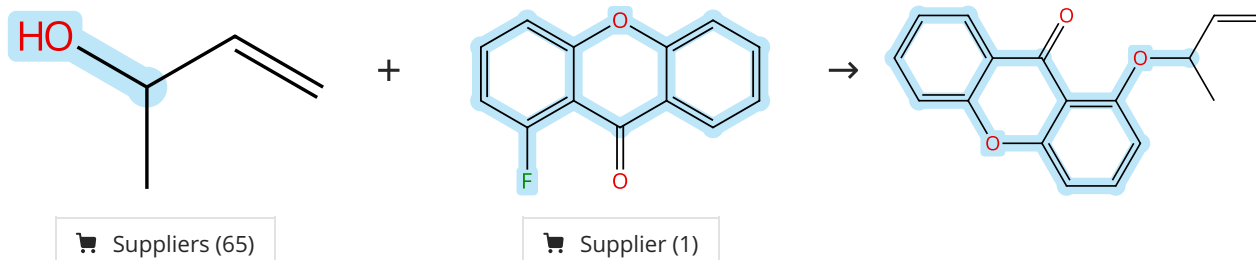
Suppliers (73)

Suppliers (5)

31-170-CAS-22515587 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][[3,6-dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]bis(1,1-dimethylethyl)phosphine-κP](methanesulfonato-κO) palladium Solvents: Toluene; 1 h, 80 °C	Synthesis of Fluorinated Alkyl Aryl Ethers by Palladium-Catalyzed C-O Cross-Coupling By: Szpera, Robert; et al Organic Letters (2020), 22(16), 6573-6577.
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Scheme 312 (1 Reaction)

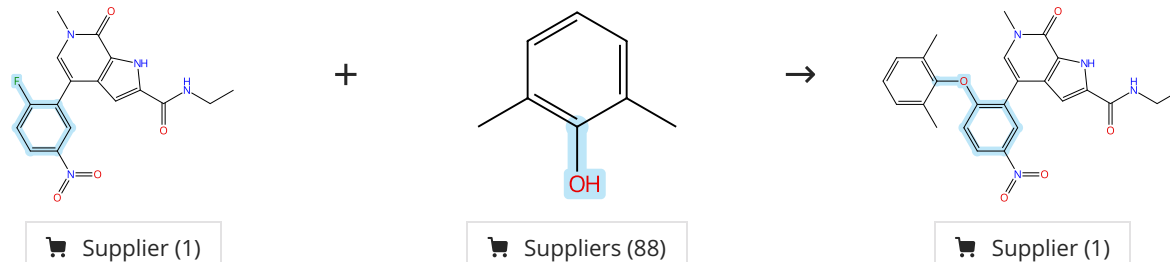
Steps: 1 Yield: 98%



31-170-CAS-22503807 Steps: 1 Yield: 98% 1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 0 °C; 1 h, 25 °C 1.2 Solvents: Dimethylformamide; 0 °C; 20 min, 25 °C 1.3 Solvents: Water; pH 7 Experimental Protocols	An Efficient Two-Step Protocol for the Isoprenylation of Xanthone at the C2 Position Starting from 1-Fluoroxanthone Derivative By: Fujimoto, Yuuki; et al Synlett (2020), 31(14), 1423-1429.
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Scheme 313 (1 Reaction)

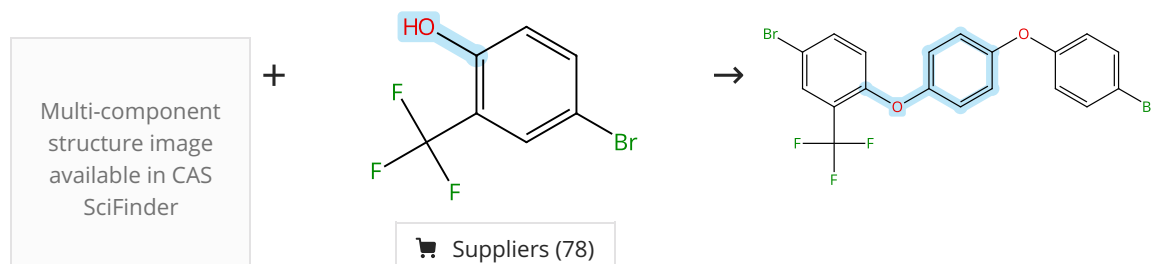
Steps: 1 Yield: 98%



31-614-CAS-24358170 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 90 min, 50 °C Experimental Protocols	Discovery of N-Ethyl-4-[2-(4-fluoro-2,6-dimethyl-phenoxy)-5-(1-hydroxy-1-methyl-ethyl)phenyl]-6-methyl-7-oxo-1H-pyrrolo[2,3-c]pyridine-2-carboxamide (ABBV-744), a BET Bromodomain Inhibitor with Selectivity for the Second Bromodomain By: Sheppard, George S.; et al Journal of Medicinal Chemistry (2020), 63(10), 5585-5623.
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Scheme 314 (1 Reaction)

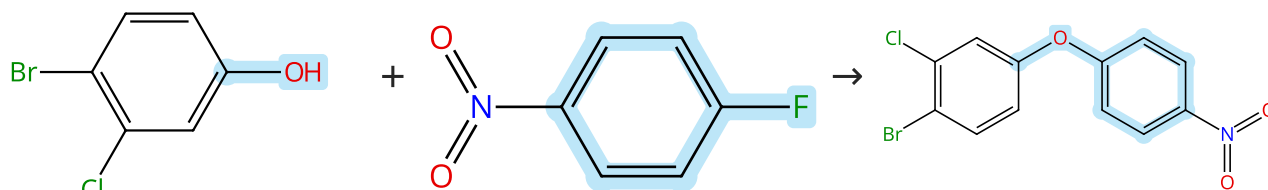
Steps: 1 Yield: 98%



31-614-CAS-41866187	Steps: 1 Yield: 98%	Synthesis of Diaryl Ethers via Hypervalent Iodine-Mediated C-H Functionalization
1.1 Reagents: Cesium carbonate Solvents: Toluene; 8 h, 80 °C		By: Hu, Chenghu; et al
Experimental Protocols		Organic Letters (2024), 26(37), 7864-7868.

Scheme 315 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

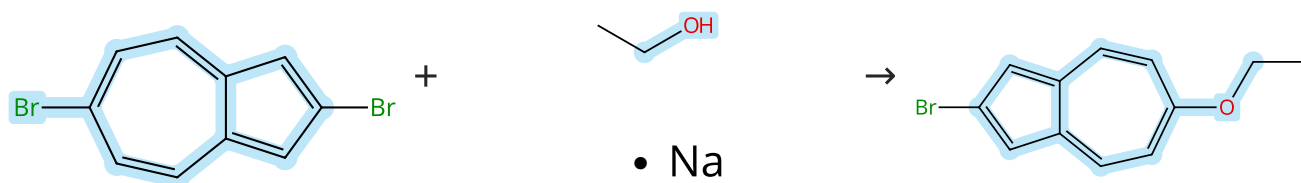
Suppliers (100)

Supplier (1)

31-614-CAS-37615376	Steps: 1 Yield: 98%	Discovery of N-Arylsulfonyl-Indole-2-Carboxamide Derivatives as Galectin-3 and Galectin-8 C-Terminal Domain Inhibitors
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 6 h, 50 °C		By: Zhang, Haoming; et al
Experimental Protocols		ACS Medicinal Chemistry Letters (2023), 14(9), 1257-1265.

Scheme 316 (1 Reaction)

Steps: 1 Yield: 98%



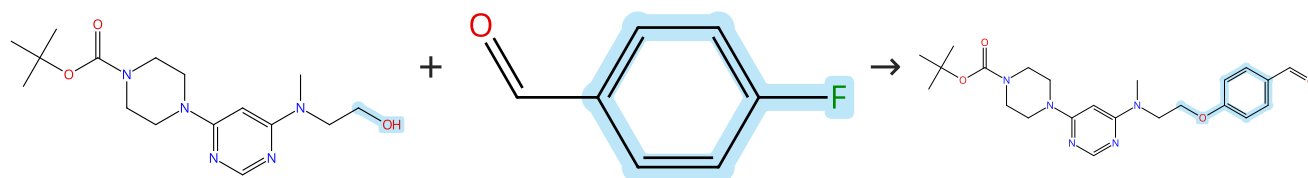
Suppliers (3)

Suppliers (94)

31-614-CAS-41031852	Steps: 1 Yield: 98%	Poly(Vinyl Azulene)s: An Emerging Class of Poly(Vinyl Arene)s
1.1 21 h, 80 °C		By: Xiang, Junjun; et al
Experimental Protocols		Macromolecules (Washington, DC, United States) (2024), 57(15), 7516-7523.

Scheme 317 (1 Reaction)

Steps: 1 Yield: 98%

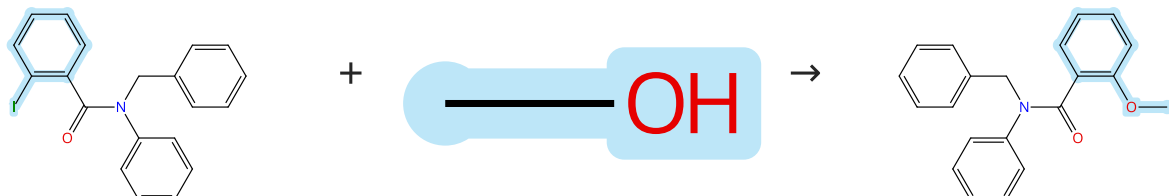


Suppliers (93)

31-170-CAS-22215759	Steps: 1 Yield: 98%	An efficient and scalable approach for the synthesis of piperazine based glitazone and its derivatives
1.1 Reagents: Potassium hydroxide Solvents: Dimethylformamide; 10 min, rt		By: Sharma, Vijay Kumar; et al
1.2 rt; 16 h, rt; rt → 0 °C		Synthetic Communications (2020), 50(14), 2181-2187.
1.3 Solvents: Water; 0 °C; 15 min, 0 °C		
Experimental Protocols		

Scheme 318 (1 Reaction)

Steps: 1 Yield: 98%



The reaction scheme shows the conversion of N-(2-chlorophenyl)-N-(diphenylmethyl)benzamide to N-(2-methoxyphenyl)-N-(diphenylmethyl)benzamide. The starting material is a benzamide derivative with a 2-chlorophenyl group. It reacts with a hydroxyl group (OH), represented by a blue pill-shaped icon. The product is the corresponding 2-methoxyphenyl derivative, where the chlorine atom has been replaced by a methoxy group (OCH₃).

 Suppliers (3)

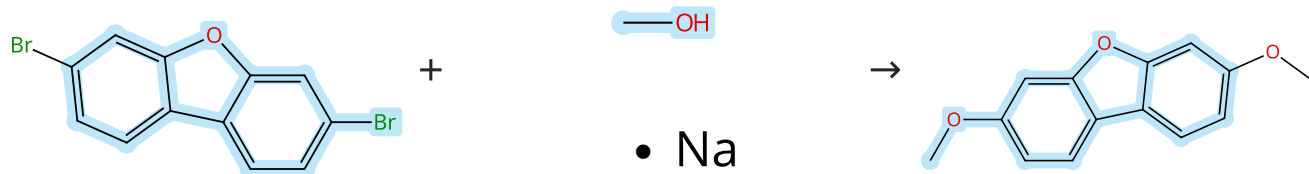
 Suppliers (444)

 Suppliers (10)

31-614-CAS-35849243	Steps: 1 Yield: 98%	Ligand-Enabled Gold-Catalyzed C(sp ²)-O Cross-Coupling Reactions
1.1 Reagents: Tripotassium phosphate Catalysts: [2-[Bis(tricyclo[3.3.1.1 ^{3,7}]dec-1-yl)phosphino-κP]-N,N-dimethylbenzenamine]chlorogold Solvents: 1,2-Dichlorobenzene; 5 min, rt		By: Das, Avishek; et al
1.2 Reagents: Silver hexafluoroantimonate; 24 h, 80 °C		ACS Catalysis (2023), 13(6), 3847-3853.
Experimental Protocols		

Scheme 319 (1 Reaction)

Steps: 1 Yield: 98%



Chemical reaction scheme showing the conversion of 2,6-dibromo-3,4-dihydro-1H-benzo[b][1,4]dioxole (a benzodioxole derivative with bromine atoms at positions 2 and 6) reacting with a hydroxyl group (OH) and sodium (Na) to form 2,6-dimethoxy-3,4-dihydro-1H-benzo[b][1,4]dioxole (a benzodioxole derivative with methoxy groups at positions 2 and 6).

 Suppliers (47)

 Suppliers (99)

 Suppliers (5)

31-614-CAS-40396298	Steps: 1 Yield: 98%	Synthesis and properties of an O-doped aromatic belt
1.1 Catalysts: Copper bromide (Cu Br) Solvents: Methyl acetate, Toluene; 5 min, rt		By: Han, Ying; et al
1.2 Solvents: Methanol; 72 h, 110 °C; 110 °C → rt		Chemical Communications (Cambridge, United Kingdom) (2024), 60(44), 5719-5722.
1.3 Reagents: Water		
Experimental Protocols		

Scheme 320 (1 Reaction)

Steps: **1** Yield: **98%**

Chemical reaction scheme showing the synthesis of a fluorinated ether compound. The reactants are 4-cyanobenzyl alcohol and hexafluorobenzene. The product is 4-(4-cyanophenyl)-2-(4-cyanophenyl)-1,2,3,4,5,6-hexafluorobenzene.

Supplier (1)

Suppliers (78)

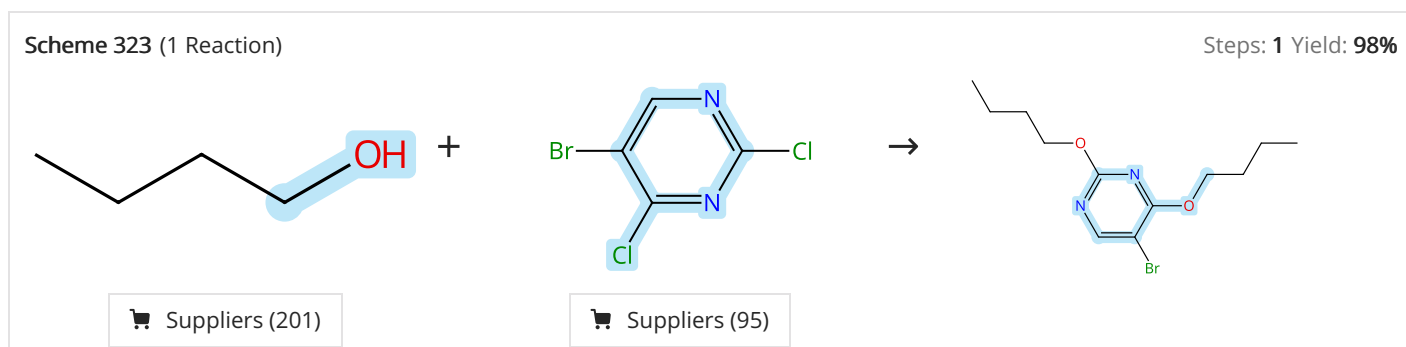
31-614-CAS-25098807	Steps: 1 Yield: 98%	Influence of multifluorophenoxy terminus on the mesomorphism of the alkoxy and alkyl cyanobiphenyl compounds in search of new ambient nematic liquid crystals and mixtures
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, 40 - 50 °C		By: Wang, Kunlun; et al
Experimental Protocols		Liquid Crystals (2021), 48(5), 672-688.



31-170-CAS-23920986	Steps: 1 Yield: 98%	Enantioselective Construction of Pyrimidine-Fused Diazepinone Derivatives Bearing a Tertiary Stereogenic Center Enabled by Iridium-Catalysed Intramolecular Allylic Substitution
1.1 Solvents: Methanol; 1 h, 70 °C		By: Jiang, Xiaoding; et al
Experimental Protocols		Advanced Synthesis & Catalysis (2021), 363(13), 3227-3232.



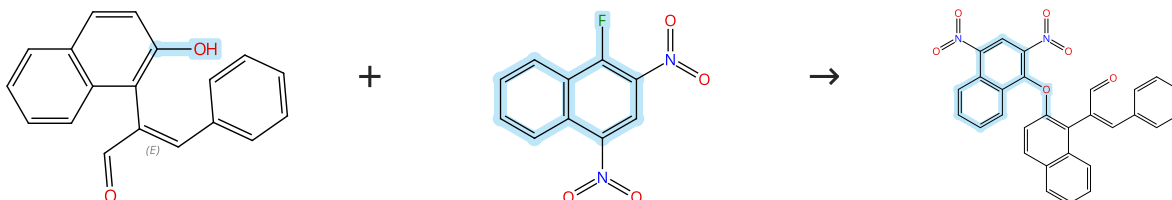
31-614-CAS-38944042	Steps: 1 Yield: 98%	Bottom-Up Construction of Ni(II)-Incorporated Covalent Organic Framework for Metallaphotoredox Catalysis
1.1 Reagents: Triethylamine Catalysts: 3026433-57-5 Solvents: Methanol, Acetonitrile; 24 h, rt		By: Hu, Hai-Chao; et al
Experimental Protocols		Chemistry - A European Journal (2024), 30(10), e202303476.



31-614-CAS-39342701 Steps: 1 Yield: 98% 1.1 Reagents: Sodium; 0 °C; 20 min, 0 °C 1.2 overnight, rt Experimental Protocols	Development and evaluation of 2,4-disubstituted-5-aryl pyrimidine derivatives as antibacterial agents By: Khalifa, Hend; et al Archiv der Pharmazie (Weinheim, Germany) (2024), 357(4), 2300656.
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Scheme 324 (1 Reaction)

Steps: 1 Yield: 98%



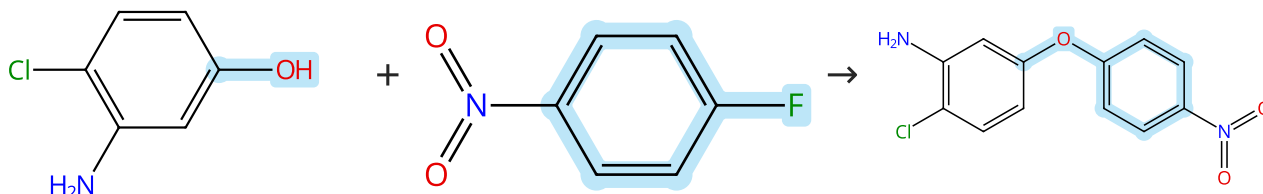
Double bond geometry shown

Suppliers (2)

31-614-CAS-37560123 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
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Scheme 325 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (79)

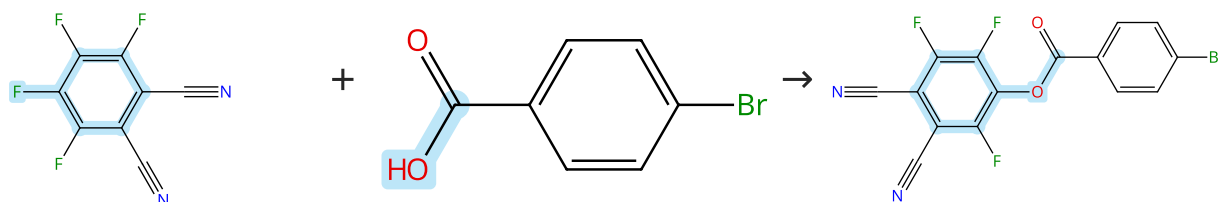
Suppliers (100)

Suppliers (2)

31-614-CAS-40338192 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C Experimental Protocols	Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors By: Xu, Yue; et al Bioorganic Chemistry (2023), 131, 106339.
---	---

Scheme 326 (1 Reaction)

Steps: 1 Yield: 98%



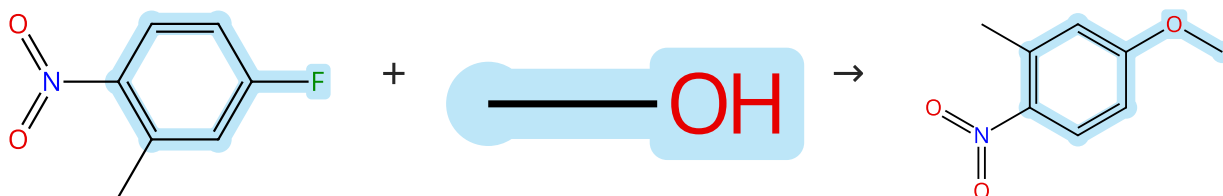
Suppliers (76)

Suppliers (100)

31-614-CAS-35473867	Steps: 1 Yield: 98%	The Electron Deficient Fluoroarene Mediated Synthesis of Trifluoromethyl Ketones from Carboxylic Acids
1.1 Reagents: Triethylamine Solvents: Acetonitrile; 2 h, rt		By: Said, Madhukar S.; et al
Experimental Protocols		Organic Letters (2023), 25(7), 1094-1098.

Scheme 327 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (85)

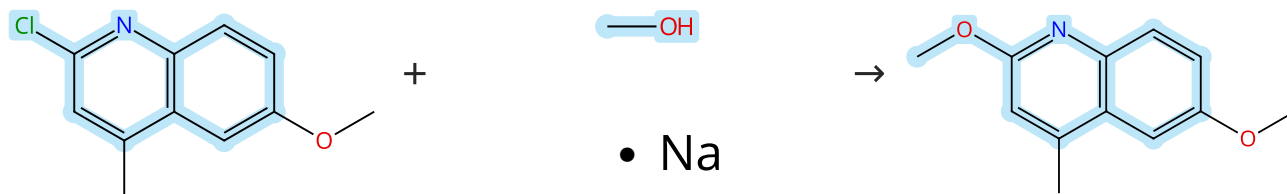
Suppliers (444)

Suppliers (80)

31-614-CAS-39884298	Steps: 1 Yield: 98%	Discovery of Novel Polycyclic Heterocyclic Derivatives from Evodiamine for the Potential Treatment of Triple-Negative Breast Cancer
1.1 Reagents: Sodium methoxide Solvents: Methanol; 0 °C; 0 °C → rt; 3 h, rt		By: Xu, Shengtao; et al
1.2 Solvents: Water; rt		Journal of Medicinal Chemistry (2021), 64(23), 17346-17365.
Experimental Protocols		

Scheme 328 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (61)

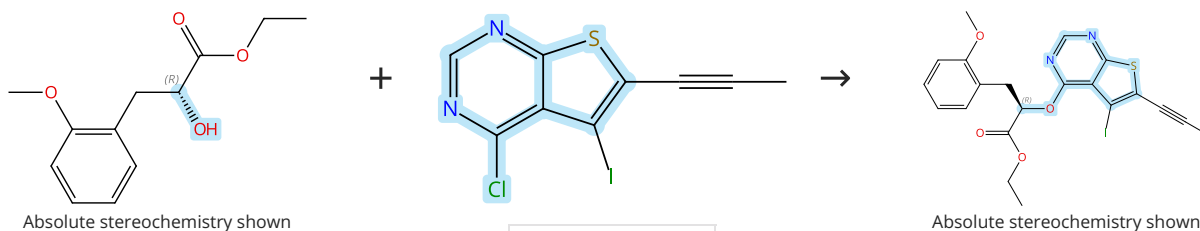
Suppliers (99)

Suppliers (9)

31-614-CAS-34978638	Steps: 1 Yield: 98%	Efficient and Sustainable Synthesis of the Antimalarial Drug Tafenoquine
1.1 Solvents: Methanol; 48 h, 70 °C		By: Kavthe, Rahul D.; et al
Experimental Protocols		ACS Sustainable Chemistry & Engineering (2022), 10(50), 16896-16902.

Scheme 329 (1 Reaction)

Steps: 1 Yield: 98%



Absolute stereochemistry shown

Absolute stereochemistry shown

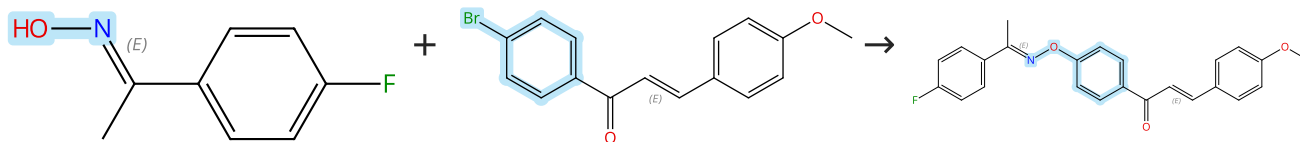
Supplier (1)

Suppliers (8)

31-170-CAS-23904223	Steps: 1 Yield: 98%	Discovery of S64315, a Potent and Selective Mcl-1 Inhibitor
1.1 Reagents: Cesium carbonate Solvents: <i>tert</i> -Butanol; 2 h, 50 °C		By: Szlavik, Zoltan; et al
Experimental Protocols		Journal of Medicinal Chemistry (2020), 63(22), 13762-13795.

Scheme 330 (1 Reaction)

Steps: 1 Yield: 98%



Double bond geometry shown

Double bond geometry shown

Double bond geometry shown

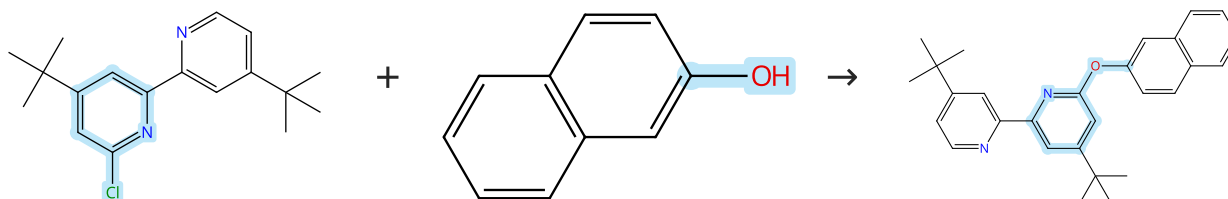
Suppliers (11)

Suppliers (15)

31-170-CAS-22571083	Steps: 1 Yield: 98%	An Easy Access to Oxime Ethers by Pd- Catalyzed C-O Cross-Coupling of Activated Aryl Bromides with Ketoximes and Chalcone Oximes
1.1 Reagents: Cesium carbonate Catalysts: Di-μ-chlorobis(η ³ -2-propenyl)dipalladium, Bis(1,1-dimethylethyl)[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl] phosphine Solvents: Toluene; 3 h, 75 °C		By: Reeta; et al
Experimental Protocols		Chinese Journal of Chemistry (2020), 38(8), 830-836.

Scheme 331 (2 Reactions)

Steps: 1 Yield: 98%



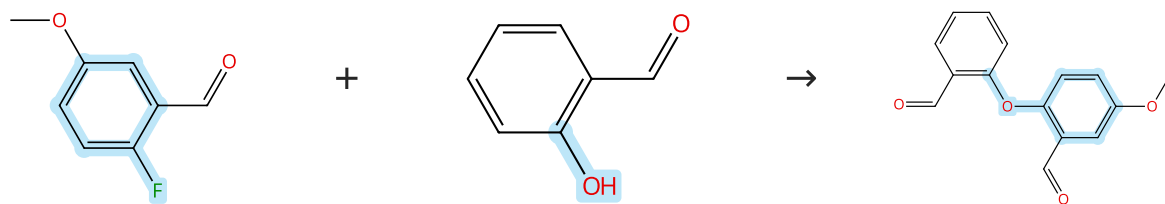
Suppliers (7)

Suppliers (117)

31-170-CAS-23868690	Steps: 1 Yield: 98%	Bipyridine-Type Bidentate Auxiliary-Enabled Copper-Mediated C-H/C-H Biaryl Coupling of Phenols and 1,3-Azoles
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 21 h, 160 °C		By: Kajiware, Rikuo; et al
1.2 Solvents: Water		Organic Letters (2021), 23(14), 5405-5409.
Experimental Protocols		
31-614-CAS-24403444	Steps: 1 Yield: 98%	Copper-mediated Regioselective C-H Cyanation of Phenols with Assistance of Bipyridine-type Bidentate Auxiliary
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 21 h, 160 °C		By: Kajiware, Rikuo; et al
1.2 Reagents: Water		Chemistry Letters (2021), 50(10), 1814-1817.
Experimental Protocols		

Scheme 332 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (84)

Suppliers (95)

31-170-CAS-23309288

Steps: 1 Yield: 98%

N-Heterocyclic carbene (NHC)-catalyzed intramolecular benzoin condensation-oxidation

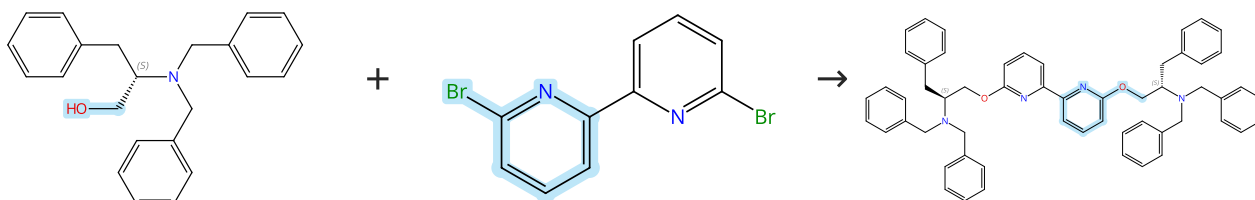
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylacetamide; 4 h, reflux

By: Satyam, Killari; et al

Organic & Biomolecular Chemistry (2021), 19(7), 1488-1492.

Scheme 333 (1 Reaction)

Steps: 1 Yield: 98%



Absolute stereochemistry shown,
Rotation (+)

Suppliers (70)

Suppliers (31)

31-614-CAS-31203570

Steps: 1 Yield: 98%

Chiral Bipyridine Ligand with Flexible Molecular Recognition Site: Development and Application to Copper-Catalyzed Asymmetric Borylation of α,β -Unsaturated Ketones

1.1 **Reagents:** Sodium hydride
Solvents: Dimethylformamide, Tetrahydrofuran; rt; 1 h, rt

By: Tsutsumi, Ryosuke; et al

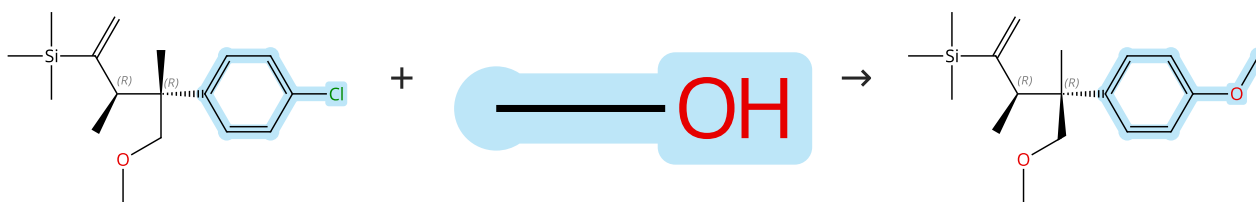
1.2 24 h, 75 °C

ChemCatChem (2022), 14(2), e202101278.

Experimental Protocols

Scheme 334 (1 Reaction)

Steps: 1 Yield: 98%



Absolute stereochemistry shown,
Rotation (-)

Suppliers (444)

31-614-CAS-39189327

Steps: 1 Yield: 98%

Catalytic enantioselective reductive Eschenmoser-Claisen rearrangements

1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: [2'-(Amino- κ M)[1,1'-biphenyl]-2-yl- κ C][[3,6-dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]bis(1,1-dimethylethyl)phosphine- κ P](methanesulfonato- κ O) palladium
Solvents: 1,4-Dioxane; 18 h, 80 °C

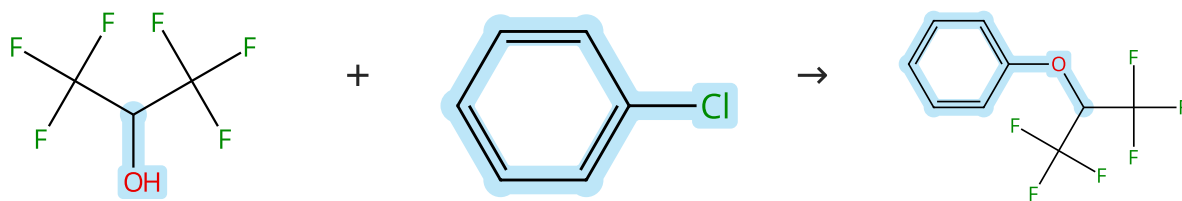
By: Zhang, Guoting; et al

Science (Washington, DC, United States) (2024), 383(6681), 395-401.

Experimental Protocols

Scheme 335 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (154)

Suppliers (127)

Suppliers (6)

31-614-CAS-36552416

Steps: 1 Yield: 98%

Catalytic S_NAr Hexafluoroisopropoxylation of Aryl Chlorides and Bromides

1.1 Reagents: Silver carbonate

By: Su, Jiang; et al

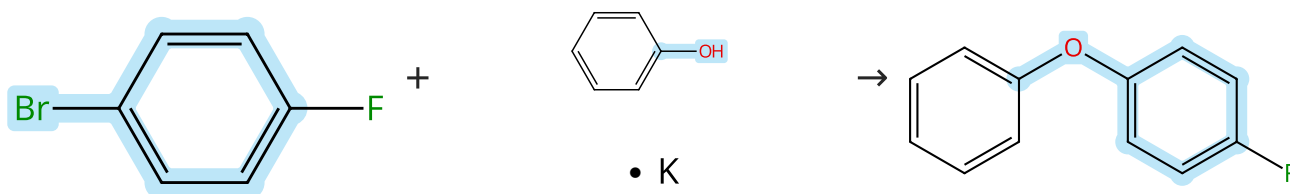
Catalysts: Bis[dichloro[η^5 -(pentamethylcyclopentadienyl)]rhodium], [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl- κO]methanesulfonamido- κO]silver; 15 h, 80 °C

Angewandte Chemie, International Edition (2023), 62(24), e202302908.

Experimental Protocols

Scheme 336 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (103)

Suppliers (51)

Suppliers (54)

31-614-CAS-37728592

Steps: 1 Yield: 98%

Cross-coupling by a noncanonical mechanism involving the addition of aryl halide to Cu(II)

1.1 Reagents: 1-Fluoronaphthalene

By: Delaney, Connor P.; et al

Catalysts: Cuprous iodide, *N*¹,*N*²-Bis([1,1'-biphenyl]-2-yl)ethanediamide

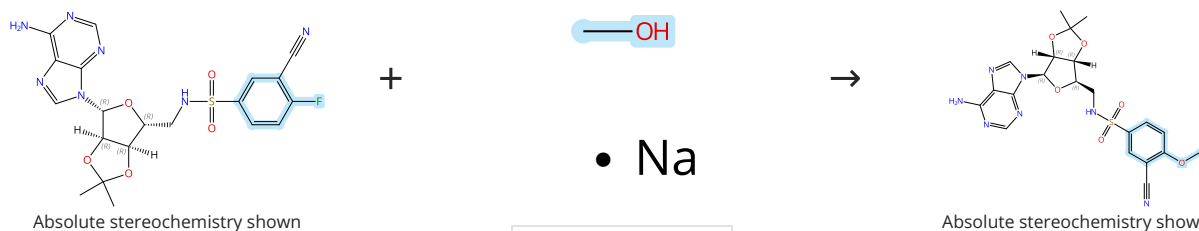
Solvents: Dimethyl sulfoxide; < 5 min, rt; 20 min, 100 °C

Science (Washington, DC, United States) (2023), 381(6662), 1079-1085.

Experimental Protocols

Scheme 337 (1 Reaction)

Steps: 1 Yield: 98%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (99)

31-614-CAS-31967633

Steps: 1 Yield: 98%

Potent Inhibition of SARS-CoV-2 nsp14 N7-Methyltransferase by Sulfonamide-Based Bisubstrate Analogues

1.1 Solvents: Methanol; 48 h, 50 °C

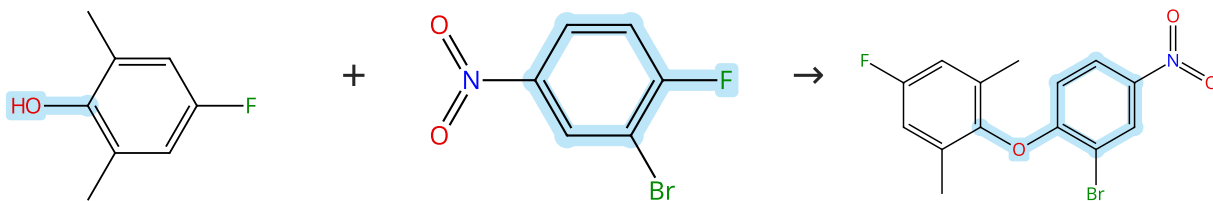
By: Ahmed-Belkacem, Rostom; et al

Experimental Protocols

Journal of Medicinal Chemistry (2022), 65(8), 6231-6249.

Scheme 338 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (53)

Suppliers (83)

Suppliers (6)

31-614-CAS-39886752

Steps: 1 Yield: 98%

1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 4 h, rt → 100 °C

Experimental Protocols

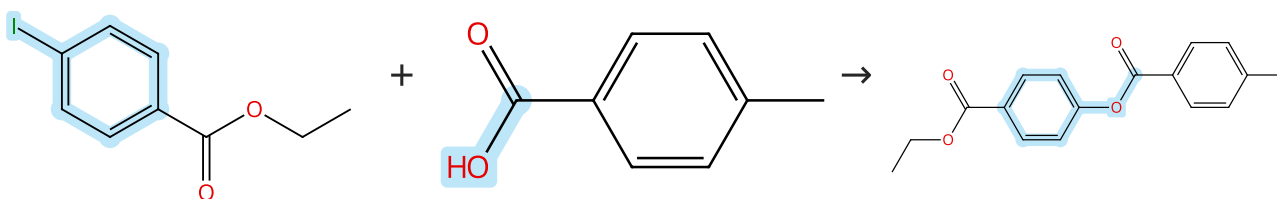
Structure-Based Discovery and Optimization of Furo[3,2-c]pyridin-4(5H)-one Derivatives as Potent and Second Bromodomain (BD2)-Selective Bromo and Extra Terminal Domain (BET) Inhibitors

By: Li, Junhua; et al

Journal of Medicinal Chemistry (2022), 65(7), 5760-5799.

Scheme 339 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (84)

Suppliers (99)

Supplier (1)

31-614-CAS-35219854

Steps: 1 Yield: 98%

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Tetrahydrofuran; 5 min, rt

1.2 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene; 36 h, 30 °C1.3 **Solvents:** Water

Experimental Protocols

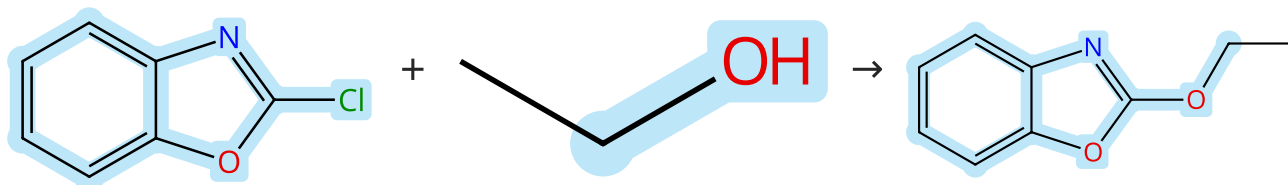
Photoinduced Nickel-Catalyzed Carbon-Heteroatom Coupling

By: Luo, Hang; et al

Chemistry - A European Journal (2023), 29(1), e202202385.

Scheme 340 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

Suppliers (618)

Suppliers (4)

31-614-CAS-31331042

Steps: 1 Yield: 98%

1.1 **Reagents:** Cesium carbonate; 16 h, 100 °C

Experimental Protocols

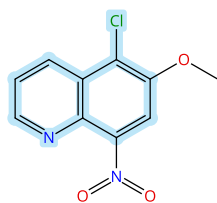
Cs₂CO₃-Promoted C-O Coupling Protocol Enables Solventless (Hetero)aryl Ether Synthesis under Air Atmosphere

By: Jiang, Bowen; et al

Chemistry - An Asian Journal (2022), 17(6), e202101370.

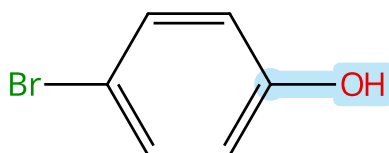
Scheme 341 (1 Reaction)

Steps: 1 Yield: 98%



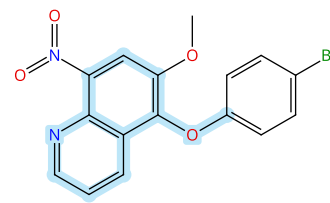
Suppliers (4)

+



Suppliers (82)

→



Supplier (1)

31-614-CAS-24716239

Steps: 1 Yield: 98%

5-Phenoxy Primaquine Analogs and the Tetraoxane Hybrid as Antimalarial Agents

1.1 Reagents: Lithium hydroxide
Solvents: Dimethyl sulfoxide; rt; 4 h, 100 °C

By: Jansongsaeng, Somrudee; et al

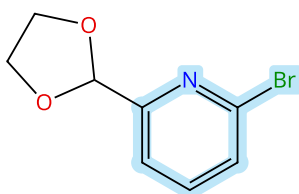
1.2 Reagents: Water

Molecules (2021), 26(13), 3991.

Experimental Protocols

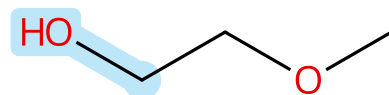
Scheme 342 (1 Reaction)

Steps: 1 Yield: 98%



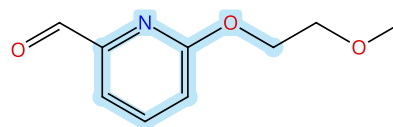
Suppliers (38)

+



Suppliers (132)

→



Supplier (1)

31-614-CAS-24021849

Steps: 1 Yield: 98%

Iminopyridyl ligands bearing polyethylene glycol unit for nickel catalyzed ethylene polymerization

1.1 Reagents: Sodium hydroxide; 100 °C; 15 h, 100 °C; 2 h, 140 °C

By: Yu, Fan; et al

1.2 Reagents: Hydrochloric acid
Solvents: Tetrahydrofuran, Water; 18 h, 100 °C; 100 °C → 0 °C

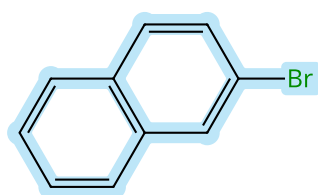
Polymer (2021), 229, 124023.

1.3 Reagents: Sodium bicarbonate
Solvents: Water; 0 °C

Experimental Protocols

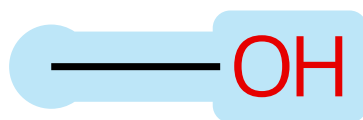
Scheme 343 (1 Reaction)

Steps: 1 Yield: 98%



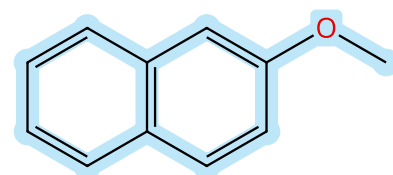
Suppliers (88)

+



Suppliers (444)

→



Suppliers (99)

31-614-CAS-38944059

Steps: 1 Yield: 98%

Bottom-Up Construction of Ni(II)-Incorporated Covalent Organic Framework for Metallaphotoredox Catalysis

1.1 Reagents: Triethylamine
Catalysts: 3026433-57-5
Solvents: Methanol, Acetonitrile; 24 h, rt

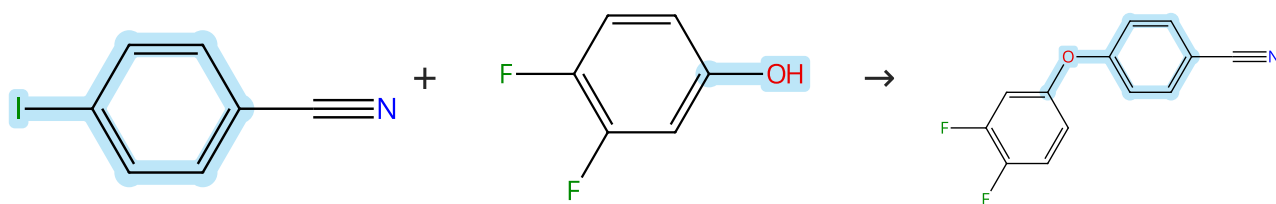
By: Hu, Hai-Chao; et al

Chemistry - A European Journal (2024), 30(10), e202303476.

Experimental Protocols

Scheme 344 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (81)

Suppliers (77)

Suppliers (36)

31-614-CAS-36007621

Steps: 1 Yield: 98%

Natural cellulose-based Cu(II) complex: An eco-friendly nanocatalyst for Ullmann condensations at room temperature

By: Rahman, Lutfor Md; et al

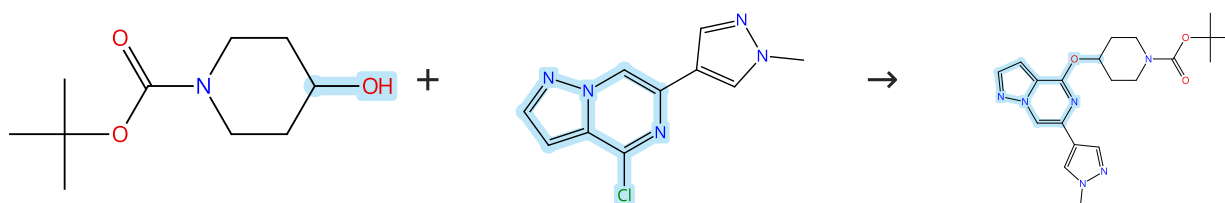
Journal of Cleaner Production (2023), 390, 136015.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Hydroxylamine hydrochloride, Copper, Cellulose, polymer with 2-propenenitrile, graft (reaction products with hydroxylamine hydrochloride, copper complexes)
Solvents: Acetonitrile; 4.5 h, 40 °C

Experimental Protocols

Scheme 345 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

Suppliers (33)

31-614-CAS-40236693

Steps: 1 Yield: 98%

Discovery and Preclinical Characterization of BIIB129, a Covalent, Selective, and Brain-Penetrant BTK Inhibitor for the Treatment of Multiple Sclerosis

By: Himmelbauer, Martin K.; et al

Journal of Medicinal Chemistry (2024), 67(10), 8122-8140.

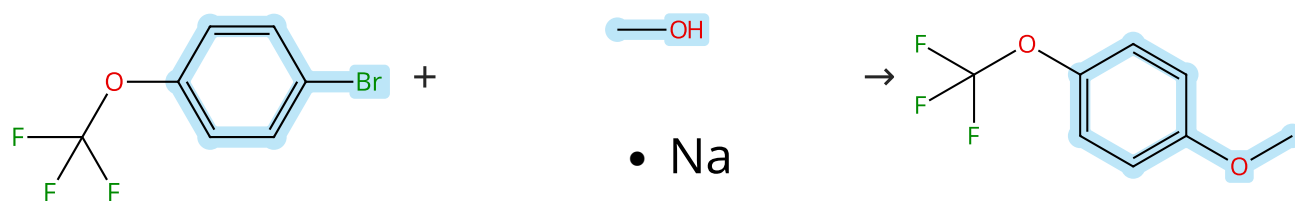
- 1.1 **Reagents:** Sodium hydride
Solvents: Dimethylformamide; 45 min, 0 °C

1.2 overnight, rt

Experimental Protocols

Scheme 346 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (75)

Suppliers (99)

Suppliers (72)

31-614-CAS-41520393

Steps: 1 Yield: 98%

Ligand-Boosted Copper Catalysis for Air-Tolerant Methoxylation of (Hetero)aryl Halides With Sodium Methoxide

By: Xu, Ke; et al

Applied Organometallic Chemistry (2024), 38(12), e7708.

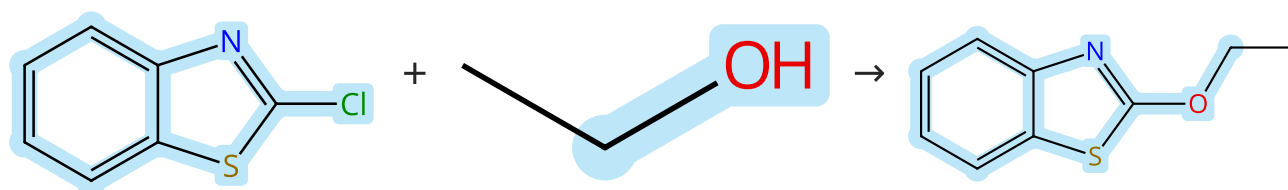
- 1.1 **Catalysts:** 2,2'-Bipyridine, Cupric chloride
Solvents: Methanol, Dimethylformamide; 2 h, 100 °C; 100 °C → rt

- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water; neutralized, rt

Experimental Protocols

Scheme 347 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (97)

Suppliers (618)

Suppliers (9)

31-614-CAS-31331048

Steps: 1 Yield: 98%

CS₂CO₃-Promoted C-O Coupling Protocol Enables Solventless (Hetero)aryl Ether Synthesis under Air Atmosphere

1.1 Reagents: Cesium carbonate; 16 h, 25 °C

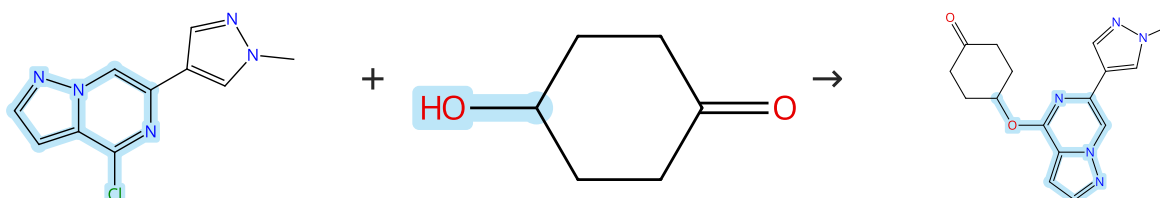
Experimental Protocols

By: Jiang, Bowen; et al

Chemistry - An Asian Journal (2022), 17(6), e202101370.

Scheme 348 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (33)

Suppliers (79)

31-614-CAS-40236720

Steps: 1 Yield: 98%

Discovery and Preclinical Characterization of BIB129, a Covalent, Selective, and Brain-Penetrant BTK Inhibitor for the Treatment of Multiple Sclerosis
1.1 Reagents: Potassium *tert*-butoxide
Solvents: Tetrahydrofuran; overnight, rt

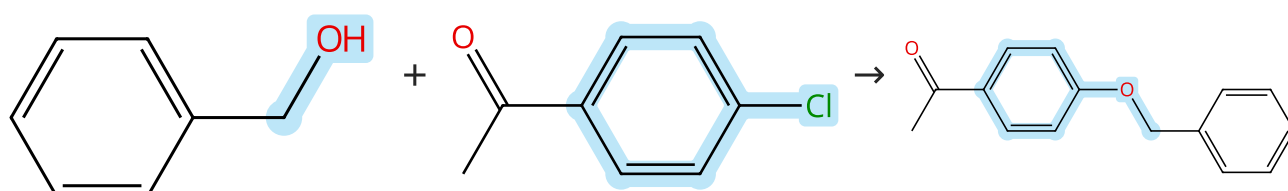
Experimental Protocols

By: Himmelbauer, Martin K.; et al

Journal of Medicinal Chemistry (2024), 67(10), 8122-8140.

Scheme 349 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (150)

Suppliers (68)

Suppliers (92)

31-614-CAS-40260587

Steps: 1 Yield: 98%

Thermal Nickel-Catalyzed Carbon-Oxygen Cross-Coupling of (Hetero)aryl Halides with Alcohols Enabled by the Use of a Silane Reductant Approach
1.1 Catalysts: Nickel bromide, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Toluene; 30 min, rt

1.2 Reagents: Phenylsilane, 1,8-Diazabicyclo[5.4.0]undec-7-ene; 24 h, 120 °C

Experimental Protocols

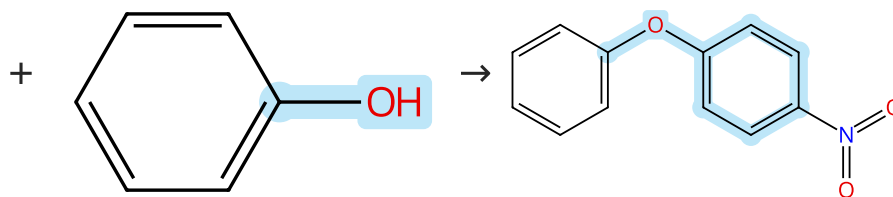
By: Yang, Liu; et al

ACS Catalysis (2024), 14(10), 7846-7852.

Scheme 350 (1 Reaction)

Steps: 1 Yield: 98%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (207)

Suppliers (68)

31-614-CAS-31534008

Steps: 1 Yield: 98%

Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates

By: Kikushima, Kotaro; et al

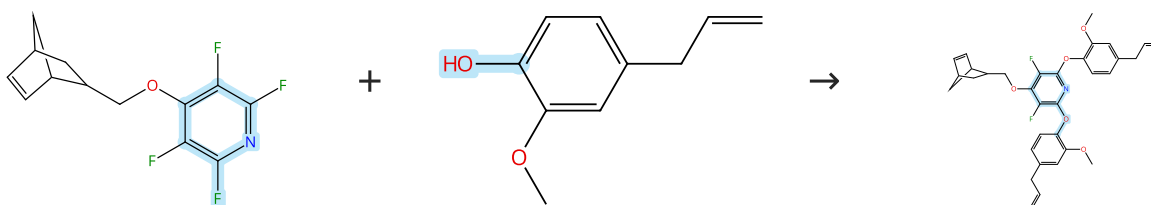
Organic Letters (2022), 24(10), 1924-1928.

- 1.1 **Reagents:** Sodium carbonate
Solvents: Dichloromethane, Water; 18 h, 40 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 351 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (120)

31-170-CAS-23809714

Steps: 1 Yield: 98%

Toward Taming the Chemical Reversibility of Perfluoro pyridine through Molecular Design with Applications to Pre- and Postmodifiable Polymer Architectures

By: Houck, Matthew B.; et al

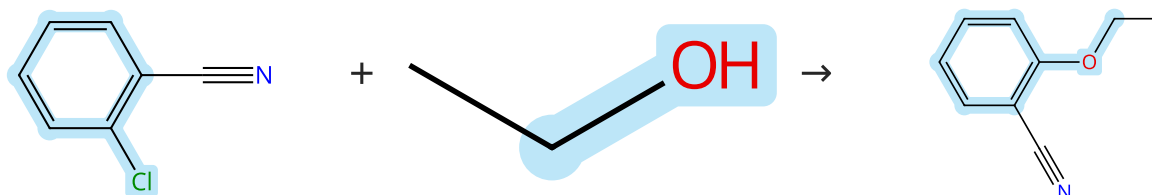
Macromolecules (Washington, DC, United States) (2021), 54(12), 5586-5594.

- 1.1 **Reagents:** Cesium carbonate
Solvents: Tetrahydrofuran; 18 h, reflux

Experimental Protocols

Scheme 352 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (69)

Suppliers (618)

Suppliers (75)

31-170-CAS-22584336

Steps: 1 Yield: 98%

Aryl Ether Syntheses via Aromatic Substitution Proceeding under Mild Conditions

By: Ando, Shin; et al

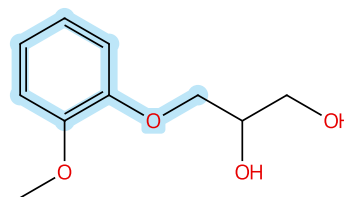
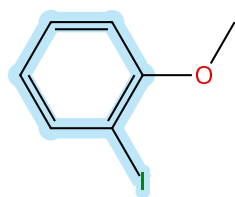
Journal of Organic Chemistry (2020), 85(17), 11181-11189.

- 1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 24 h, 25 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water; 0 °C

Scheme 353 (1 Reaction)

Steps: 1 Yield: 98%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (85)

Suppliers (95)

31-614-CAS-43785879

Steps: 1 Yield: 98%

**Ligand-Free Copper-Catalyzed Ullmann-Type C-O Bond
Formation in Non-Innocent Deep Eutectic Solvents under
Aerobic Conditions**

By: Quivelli, Andrea Francesca; et al

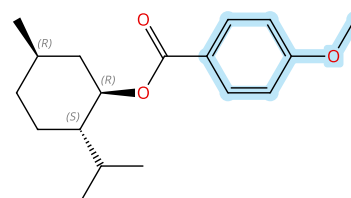
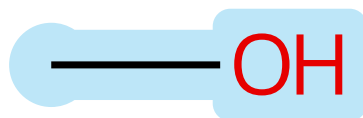
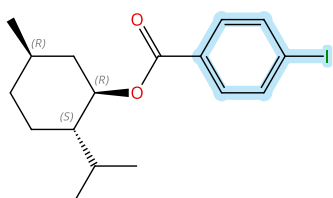
ChemSusChem (2022), 15(1), e202102211.

1.1 **Reagents:** Potassium carbonate
Catalysts: Cuprous iodide
Solvents: Ethanaminium, 2-hydroxy-*N,N,N*-trimethyl-, chloride
(1:1), mixt. with 1,2,3-propanetriol; 6 h, 80 °C

Experimental Protocols

Scheme 354 (1 Reaction)

Steps: 1 Yield: 98%



Absolute stereochemistry shown,
Rotation (-)

Absolute stereochemistry shown

Suppliers (444)

31-614-CAS-35849219

Steps: 1 Yield: 98%

**Ligand-Enabled Gold-Catalyzed C(sp²)-O Cross-Coupling
Reactions**

By: Das, Avishek; et al

ACS Catalysis (2023), 13(6), 3847-3853.

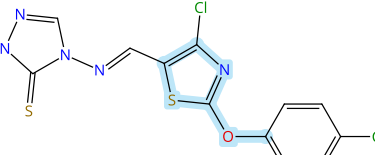
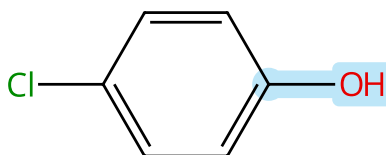
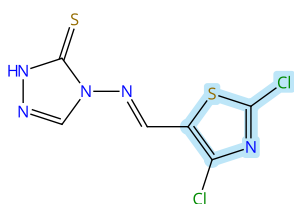
1.1 **Reagents:** Tripotassium phosphate
Catalysts: [2-[Bis(tricyclo[3.3.1.1^{3,7}]dec-1-yl)phosphino-κP]-*N,N*-
N-dimethylbenzenamine]chlorogold
Solvents: 1,2-Dichlorobenzene; 5 min, rt

1.2 **Reagents:** Silver hexafluoroantimonate; 24 h, 80 °C

Experimental Protocols

Scheme 355 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (90)

31-170-CAS-23032068

Steps: 1 Yield: 98%

**Synthesis, characterization and biological studies of novel
series of triazolyl schiff bases bearing thiazole moiety and
their s-derivatives**

By: Nkurunziza, Jean Baptiste; et al

Journal of Applicable Chemistry (Lumami, India) (2020), 9(4), 556-565.

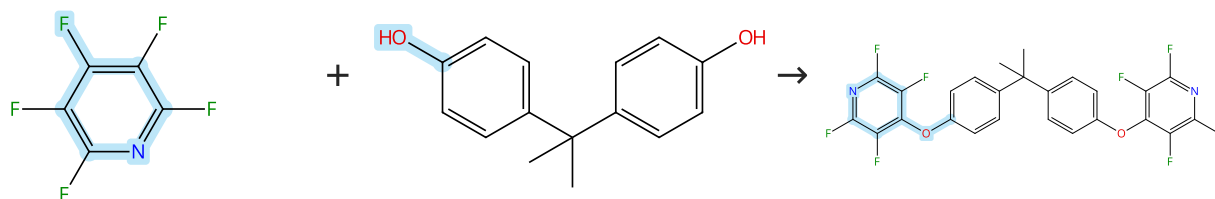
1.1 **Catalysts:** Potassium carbonate
Solvents: Acetonitrile; 10 min, rt

1.2 **Solvents:** Acetonitrile; 10 h, rt

Experimental Protocols

Scheme 356 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (71)

Suppliers (111)

31-614-CAS-41427247

Steps: 1 Yield: 98%

Fluorinated Pyridine Aryl Ether Polymers with Defined Microstructures, Diverse Architectures, and Tunable Thermal Properties

1.1 **Reagents:** Cesium carbonate
Solvents: Acetonitrile; 24 h, rt

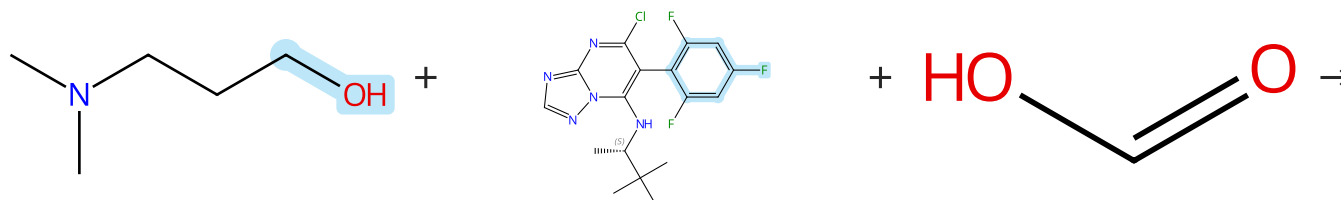
Experimental Protocols

By: Surina, Nicholas W.; et al

Macromolecules (Washington, DC, United States) (2024), 57(10), 5007-5017.

Scheme 357 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (58)

Absolute stereochemistry shown

Supplier (1)

Suppliers (176)

Multi-component
structure image
available in CAS
SciFinder

31-614-CAS-35335047

Steps: 1 Yield: 98%

Microtubule-Stabilizing 1,2,4-Triazolo[1,5-a]pyrimidines as Candidate Therapeutics for Neurodegenerative Disease: Matched Molecular Pair Analyses and Computational Studies Reveal New Structure-Activity Insights

1.1 **Reagents:** Sodium hydride
Solvents: Dimethyl sulfoxide; 1 h, rt → 60 °C
1.2 **Solvents:** Dimethyl sulfoxide; 3 h, 60 °C; 60 °C → rt
1.3 rt

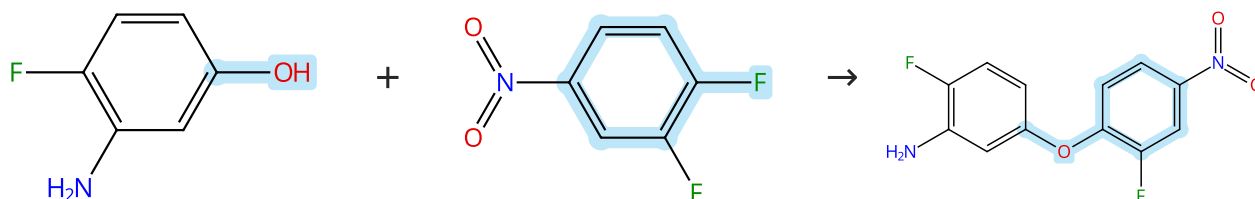
Experimental Protocols

By: Alle, Thibault; et al

Journal of Medicinal Chemistry (2023), 66(1), 435-459.

Scheme 358 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (76)

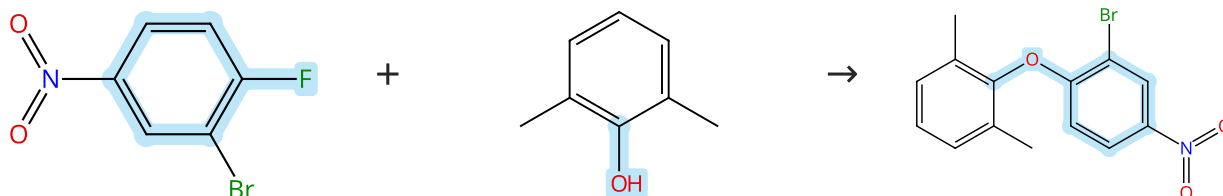
Suppliers (88)

Supplier (1)

31-614-CAS-36058197 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 2 h, 85 °C Experimental Protocols	Targeting Receptor-Interacting Protein Kinase 1 by Novel Benzothiazole Derivatives: Treatment of Acute Lung Injury through the Necroptosis Pathway By: Zhang, Xinqi; et al Journal of Medicinal Chemistry (2023), 66(7), 5261-5278.
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Scheme 359 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (83)

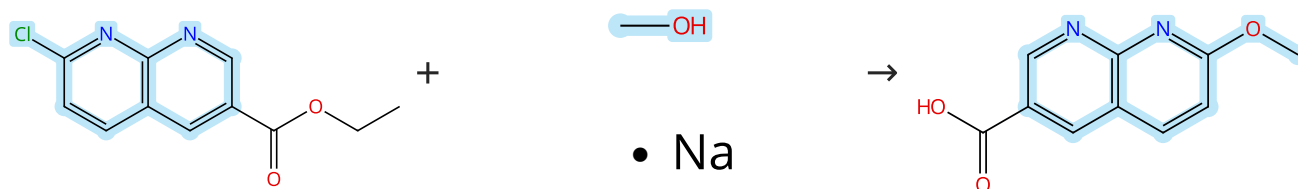
Suppliers (88)

Suppliers (10)

31-614-CAS-39886747 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 4 h, rt → 100 °C Experimental Protocols	Structure-Based Discovery and Optimization of Furo[3,2-c]pyridin-4(5H)-one Derivatives as Potent and Second Bromodomain (BD2)-Selective Bromo and Extra Terminal Domain (BET) Inhibitors By: Li, Junhua; et al Journal of Medicinal Chemistry (2022), 65(7), 5760-5799.
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Scheme 360 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (4)

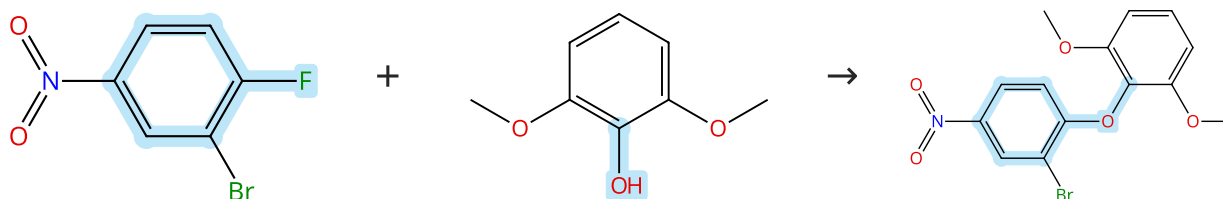
Suppliers (99)

Suppliers (3)

31-614-CAS-38094959 Steps: 1 Yield: 98% 1.1 Solvents: Methanol; rt; 2 h, 60 °C 1.2 Solvents: Water; 75 min, 60 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; pH 4 - 5 Experimental Protocols	The exploration of aza-quinolines as hematopoietic prostaglandin D synthase (H-PGDS) inhibitors with low brain exposure By: Cadilla, Rodolfo; et al Bioorganic & Medicinal Chemistry (2020), 28(23), 115791.
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Scheme 361 (1 Reaction)

Steps: 1 Yield: 98%



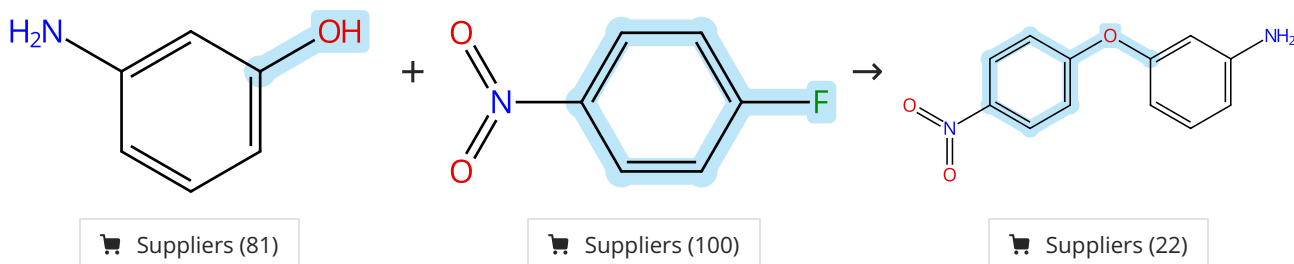
Suppliers (83)

Suppliers (99)

31-614-CAS-39886774	Steps: 1 Yield: 98%	Structure-Based Discovery and Optimization of Furo[3,2-c]pyridin-4(5H)-one Derivatives as Potent and Second Bromodomain (BD2)-Selective Bromo and Extra Terminal Domain (BET) Inhibitors By: Li, Junhua; et al Journal of Medicinal Chemistry (2022), 65(7), 5760-5799.
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 4 h, rt → 100 °C		
Experimental Protocols		

Scheme 362 (1 Reaction)

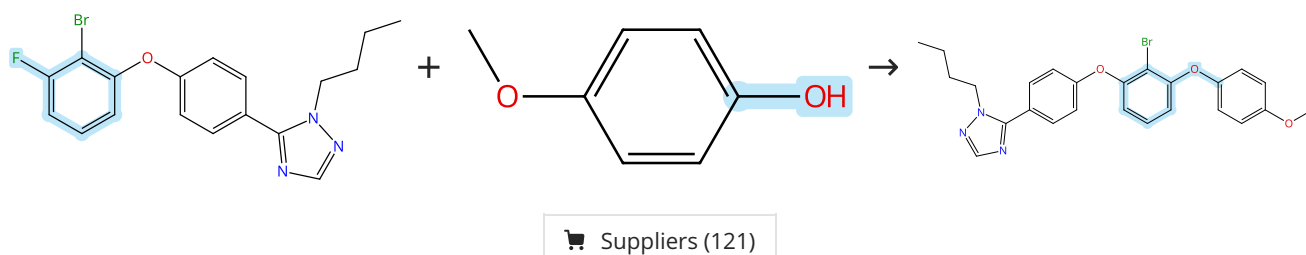
Steps: 1 Yield: 98%



31-614-CAS-40338183	Steps: 1 Yield: 98%	Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors By: Xu, Yue; et al Bioorganic Chemistry (2023), 131, 106339.
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C		
Experimental Protocols		

Scheme 363 (1 Reaction)

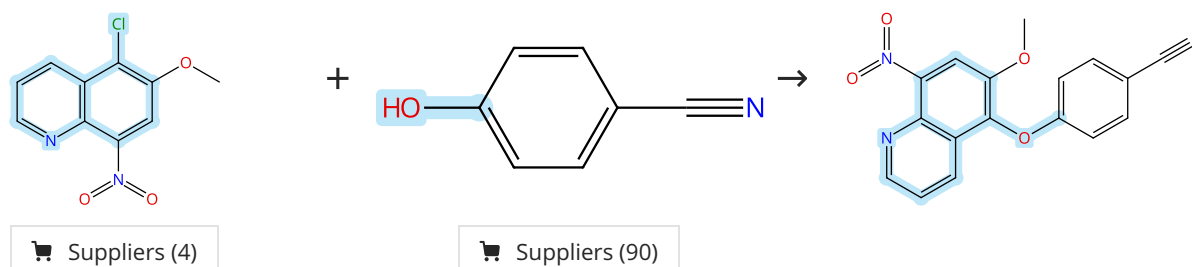
Steps: 1 Yield: 98%



31-170-CAS-22423904	Steps: 1 Yield: 98%	Triazole functionalized 5,9-dioxo-13b-boranaphtho[3,2,1-de]anthracene: a new family of multi-stimuli responsive materials By: Meng, Guoyun; et al Journal of Materials Chemistry C: Materials for Optical and Electronic Devices (2020), 8(23), 7749-7754.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; rt; 24 h, 150 °C		
Experimental Protocols		

Scheme 364 (1 Reaction)

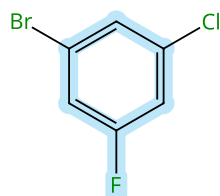
Steps: 1 Yield: 98%



31-614-CAS-24716234	Steps: 1 Yield: 98%	5-Phenoxy Primaquine Analogs and the Tetraoxane Hybrid as Antimalarial Agents
1.1 Reagents: Lithium hydroxide Solvents: Dimethyl sulfoxide; rt; 4 h, 100 °C		By: Jansongsaeng, Somrudee; et al
1.2 Reagents: Water		Molecules (2021), 26(13), 3991.
Experimental Protocols		

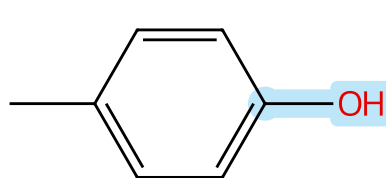
Scheme 365 (1 Reaction)

Steps: 1 Yield: 98%



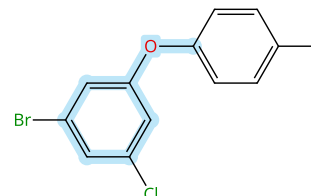
Suppliers (72)

+



Suppliers (89)

→

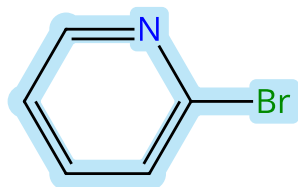


Suppliers (3)

31-170-CAS-22561195	Steps: 1 Yield: 98%	Solution-Processable Pure Green Thermally Activated Delayed Fluorescence Emitter Based on the Multiple Resonance Effect
1.1 Reagents: Potassium carbonate Solvents: <i>N</i> -Methyl-2-pyrrolidone; rt; 17 h, 150 °C		By: Ikeda, Naoya; et al
Experimental Protocols		Advanced Materials (Weinheim, Germany) (2020), 32(40), 2004072.

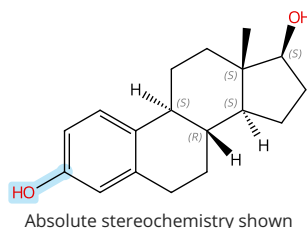
Scheme 366 (1 Reaction)

Steps: 1 Yield: 98%



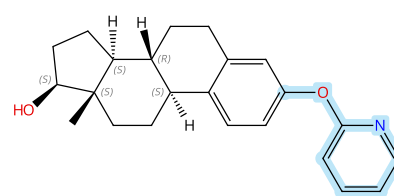
Suppliers (88)

+



Suppliers (132)

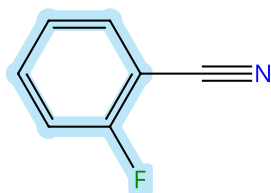
→



31-170-CAS-22502480	Steps: 1 Yield: 98%	Concise synthesis of 2-methoxyestradiol through C(sp ²)-H methoxylation
1.1 Reagents: Tripotassium phosphate Catalysts: Picolinic acid, Cuprous iodide Solvents: Dimethyl sulfoxide; 24 h, 90 °C		By: Sun, Mo-Ran; et al
Experimental Protocols		Steroids (2020), 162, 108697.

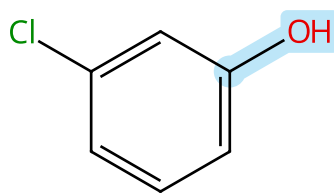
Scheme 367 (1 Reaction)

Steps: 1 Yield: 98%



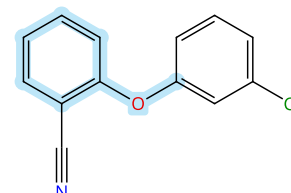
Suppliers (87)

+



Suppliers (85)

→

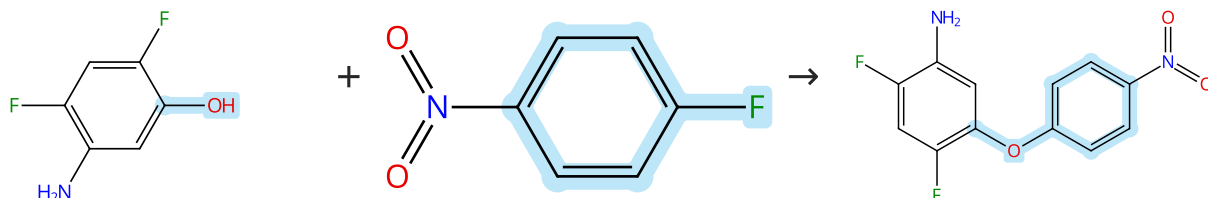


Suppliers (6)

31-614-CAS-32528358	Steps: 1 Yield: 98%	Access to ortho-Hydroxyphenyl Ketimines via Imine Anion-Mediated Smiles Rearrangement
1.1 Reagents: Potassium hydroxide Solvents: Dimethyl sulfoxide; 14.5 h, 90 °C; 90 °C → rt		By: Jinno, Shunki; et al
1.2 Reagents: Water; rt		Organic Letters (2022), 24(23), 4140-4144.
Experimental Protocols		

Scheme 368 (1 Reaction)

Steps: 1 Yield: 98%



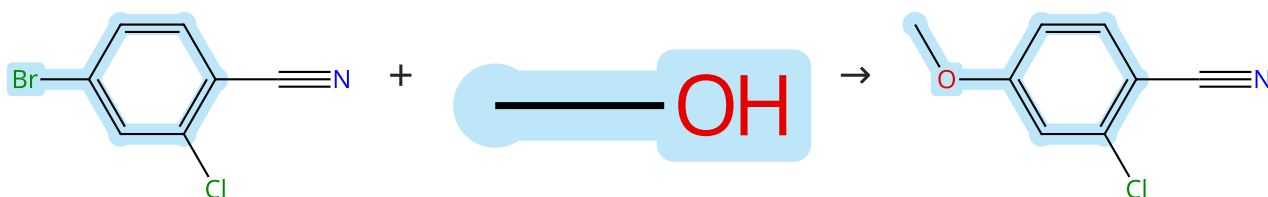
Suppliers (56)

Suppliers (100)

31-614-CAS-40338188	Steps: 1 Yield: 98%	Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C		By: Xu, Yue; et al
Experimental Protocols		Bioorganic Chemistry (2023), 131, 106339.

Scheme 369 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (78)

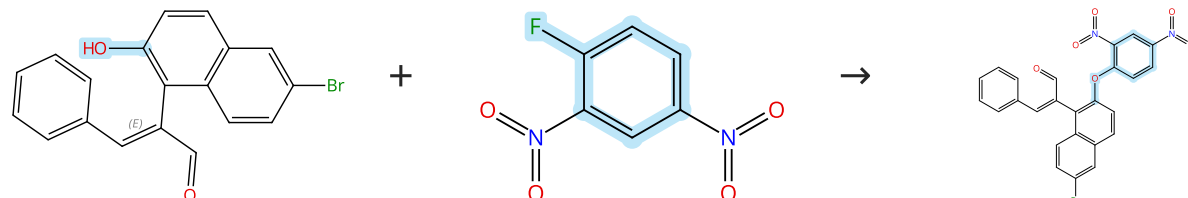
Suppliers (444)

Suppliers (60)

31-614-CAS-38944050	Steps: 1 Yield: 98%	Bottom-Up Construction of Ni(II)-Incorporated Covalent Organic Framework for Metallaphotoredox Catalysis
1.1 Reagents: Triethylamine Catalysts: 3026433-57-5 Solvents: Methanol, Acetonitrile; 24 h, rt		By: Hu, Hai-Chao; et al
Experimental Protocols		Chemistry - A European Journal (2024), 30(10), e202303476.

Scheme 370 (1 Reaction)

Steps: 1 Yield: 98%



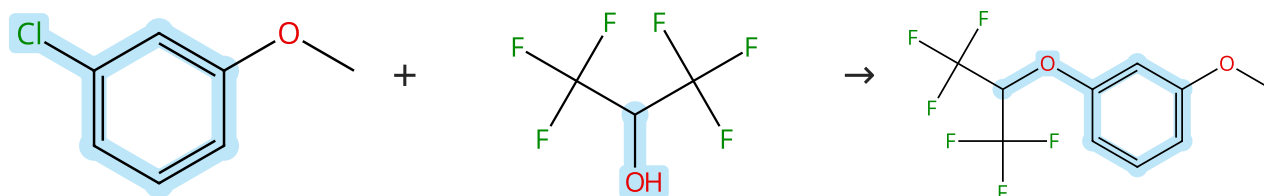
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560118 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
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Scheme 371 (1 Reaction)

Steps: 1 Yield: 98%



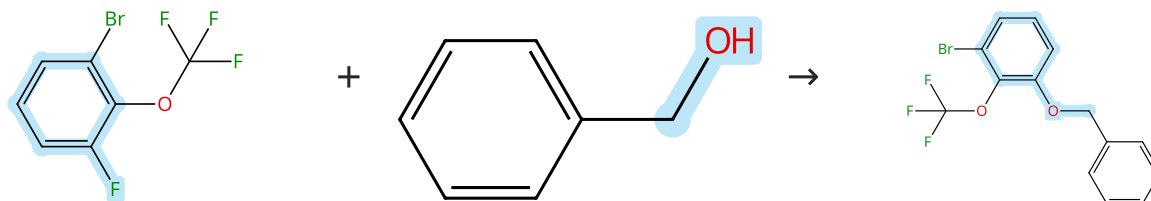
Suppliers (71)

Suppliers (154)

31-614-CAS-36552450 Steps: 1 Yield: 98% 1.1 Reagents: Silver carbonate Catalysts: Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium], [1,1,1-Trifluoro-<i>N</i>-[(trifluoromethyl)sulfonyl-κO]methanesulfonamido-κO]silver; 12 h, 80 °C Experimental Protocols	Catalytic S_NAr Hexafluoroisopropoxylation of Aryl Chlorides and Bromides By: Su, Jiang; et al Angewandte Chemie, International Edition (2023), 62(24), e202302908.
--	--

Scheme 372 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (47)

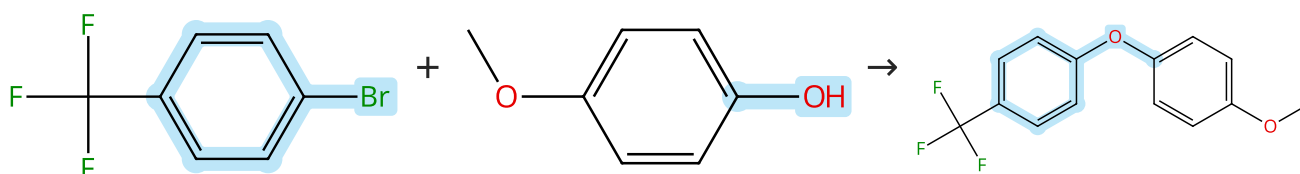
Suppliers (150)

Suppliers (3)

31-614-CAS-37086732 Steps: 1 Yield: 98% 1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 16 h, 0 °C → rt	Structure-property relationships of fluorinated carboxylic acid bioisosteres By: Alle, Thibault; et al Bioorganic & Medicinal Chemistry Letters (2023), 91, 129363.
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Scheme 373 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (79)

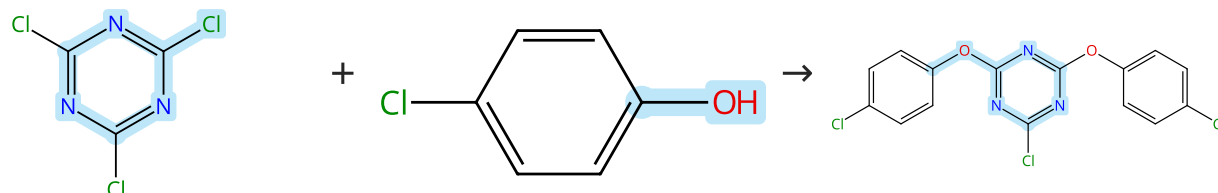
Suppliers (121)

Suppliers (3)

31-614-CAS-43873487	Steps: 1 Yield: 98%	Electromagnetic milling promoted mechanochemical copper-catalyzed C-O coupling reaction for diaryl ether formation
1.1 Reagents: Cesium carbonate Catalysts: Copper(II) acetylacetonate, <i>N</i> ¹ , <i>N</i> ² -Di-1-naphthale nylethanediamide Solvents: Dimethylformamide; 3 h, 79 °C		By: Zeng, Zhaoxuan; et al Organic Chemistry Frontiers (2025), 12(5), 1528-1536.
Experimental Protocols		

Scheme 374 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (71)

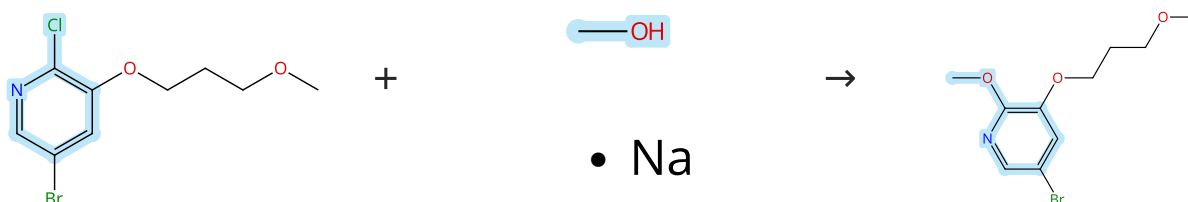
Suppliers (90)

Suppliers (4)

31-614-CAS-35047952	Steps: 1 Yield: 98%	Molecular duplexes featuring NH...N, CH...O and CH... π interactions in solid-state self-assembly of triazine based compounds
1.1 Reagents: Potassium carbonate Solvents: Tetrahydrofuran; 72 h, neutralized, rt		By: Asghar, Shazia; et al Royal Society Open Science (2022), 9(11), 220603.
Experimental Protocols		

Scheme 375 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (5)

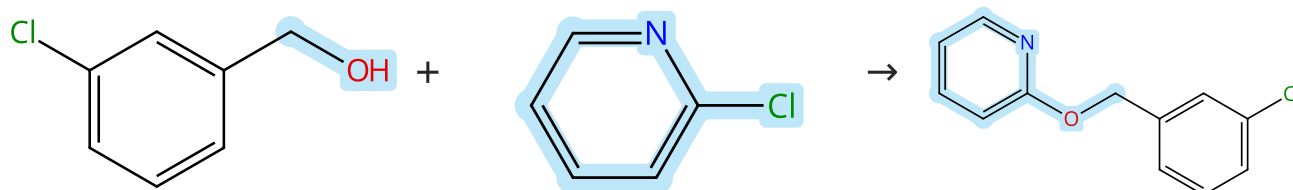
Suppliers (99)

Suppliers (24)

31-614-CAS-39001011	Steps: 1 Yield: 98%	Structure-Activity Relationships and Discovery of (S)-6-Isopropyl-2-methoxy-3-(3-methoxypropoxy)-10-oxo-5,10-dihydro-6H-pyrido[1,2-h][1,7]naphthyridine-9-carboxylic Acid (AB-452), a Novel Orally Available HBV RNA Destabilizer
1.1 Solvents: Methanol; rt → 80 °C; 3 h, 80 °C		By: Gotchev, Dimitar; et al Journal of Medicinal Chemistry (2024), 67(2), 1421-1446.
Experimental Protocols		

Scheme 376 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (70)

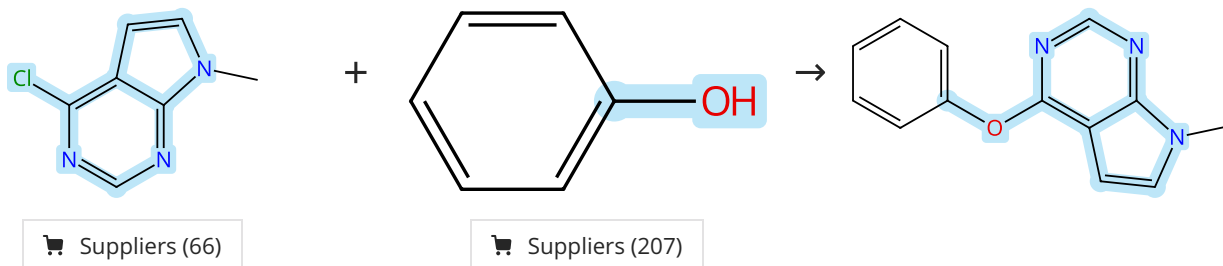
Suppliers (66)

Suppliers (7)

31-614-CAS-37711184	Steps: 1 Yield: 98%	Homocoupling of benzyl pyridyl ethers via visible light-mediated deoxygenation By: Chen, Jingtian; et al Chemical Communications (Cambridge, United Kingdom) (2023), 59(76), 11409-11412.
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: 1,4-Dioxane; rt → 98 °C		
Experimental Protocols		

Scheme 377 (1 Reaction)

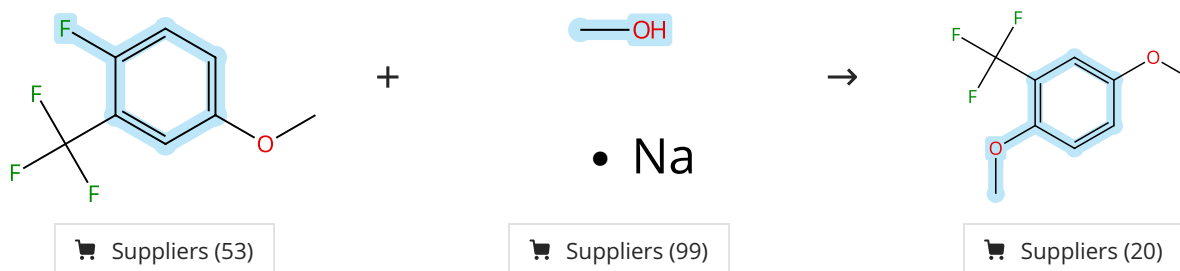
Steps: 1 Yield: 98%



31-170-CAS-22802428	Steps: 1 Yield: 98%	Diversification of Heteroaryl-Aryl Ether via Ligand-Free, Copper-Catalyzed O-Arylation Under Microwave Heating By: Choi, Sung Min; et al Bulletin of the Korean Chemical Society (2020), 41(8), 837-842.
1.1 Reagents: Tripotassium phosphate Catalysts: Cuprous iodide Solvents: Dimethylformamide; 1 h, 180 °C		
Experimental Protocols		

Scheme 378 (1 Reaction)

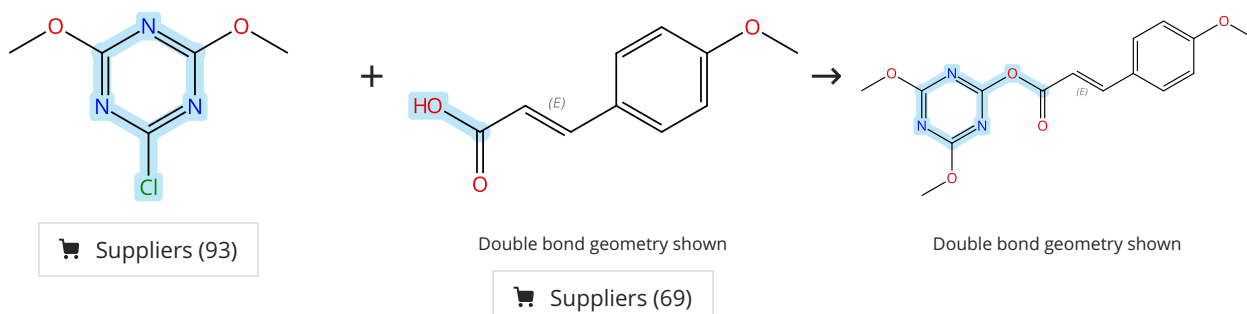
Steps: 1 Yield: 98%



31-614-CAS-40106006	Steps: 1 Yield: 98%	Discovery and Structure-Activity Relationships of 2,5-Dimethoxyphenylpiperidines as Selective Serotonin 5-HT_{2A} Receptor Agonists By: M. Rorsted, Emil; et al Journal of Medicinal Chemistry (2024), 67(9), 7224-7244.
1.1 Solvents: Dimethyl sulfoxide; 19 h, 120 °C 1.2 Reagents: Water; 0 °C		
Experimental Protocols		

Scheme 379 (1 Reaction)

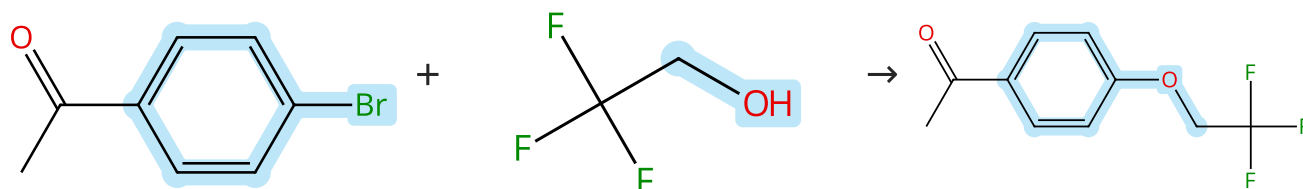
Steps: 1 Yield: 98%



31-614-CAS-36274447	Steps: 1 Yield: 98%	Palladium-catalysed Suzuki-Miyaura coupling of α,β-unsaturated superactive triazine esters
1.1 Reagents: 4-Methylmorpholine Solvents: Toluene; 1 h, rt		By: Lin, Dezhi; et al
Experimental Protocols		Chemical Communications (Cambridge, United Kingdom) (2023), 59(32), 4810-4813.

Scheme 380 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (85)

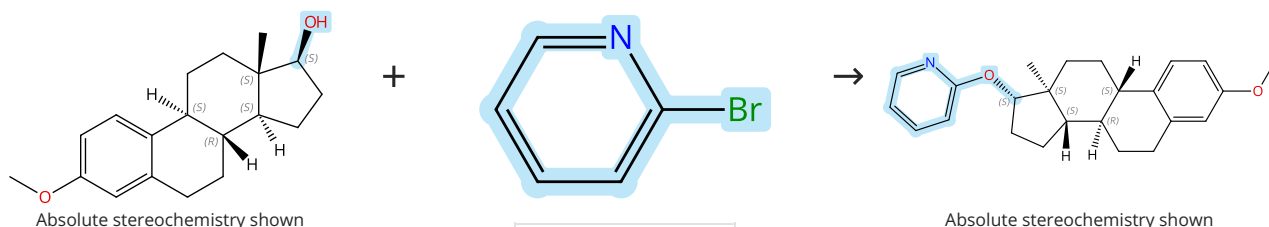
Suppliers (98)

Suppliers (45)

31-614-CAS-41430497	Steps: 1 Yield: 98%	Synthesis of Hexafluoroisopropyl Aryl Ethers via Dual Photoredox/Nickel-Catalyzed C-O Coupling
1.1 Catalysts: Methanesulfonic acid, 1,1,1-trifluoro-, nickel(2+) salt (2:1), [2,2'-Bipyridine]-4,4'-diamine, N^4,N^4,N^4,N^4 -tetramethyl- Solvents: Toluene; 1 h, 100 °C; 100 °C $\rightarrow$ rt		By: He, Xu; et al
1.2 Reagents: 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine Catalysts: Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine- $\kappa N^1,\kappa N^1$]bis[2-(2-pyridinyl- κM)phenyl- κC]-, (<i>OC</i> -6-33)-, hexafluorophosphate(1-) (1:1) Solvents: 1,1,2,2-Tetrafluoro-1-(2,2,2-trifluoroethoxy)ethane; 24 h, 60 °C		Organic Letters (2024), 26(31), 6782-6786.
Experimental Protocols		

Scheme 381 (1 Reaction)

Steps: 1 Yield: 98%



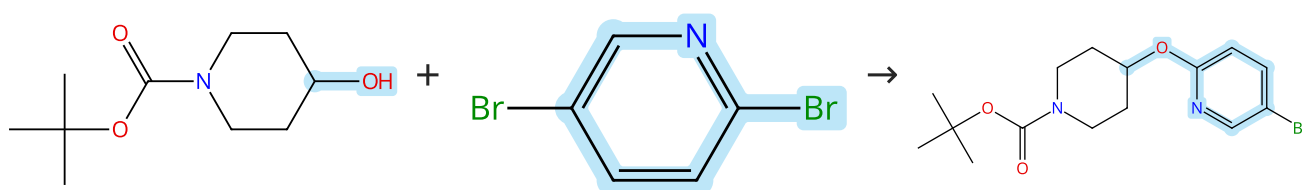
Suppliers (39)

Suppliers (88)

31-614-CAS-36348777	Steps: 1 Yield: 98%	Nucleophilic etherification of heteroaryl alkyl ethers, heteroaryl halides with (deuterated) alcohols
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: 1,4-Dioxane; rt $\rightarrow$ 100 °C; 16 h, 100 °C		By: Wang, Xia; et al
Experimental Protocols		Youji Huaxue (2021), 41(2), 795-805.

Scheme 382 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

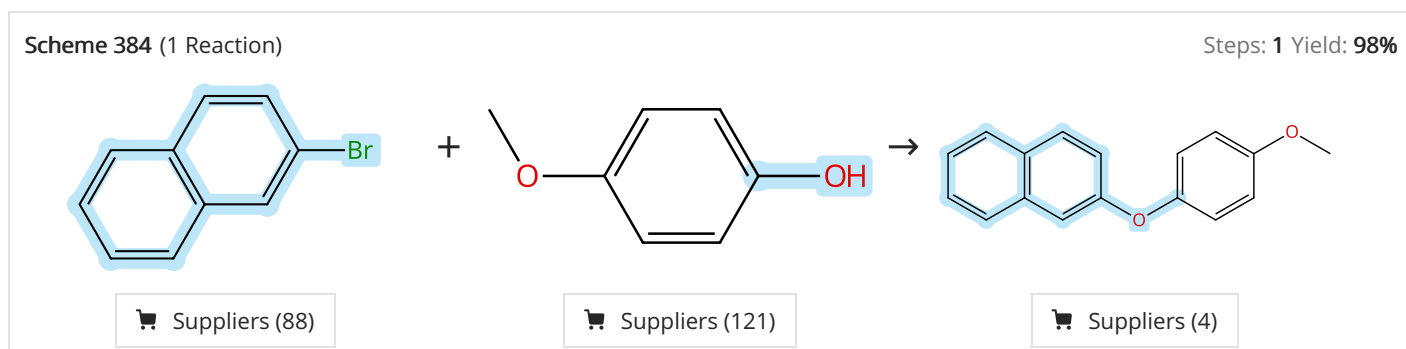
Suppliers (98)

Suppliers (55)

31-614-CAS-37081837 Steps: 1 Yield: 98% 1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; rt; 3 h, reflux	Synthesis, crystal structure and DFT study of tert-butyl 4-(5-bromopyridin-2-yloxy)piperidine-1-carboxylate By: Hou, Yuanyuan; et al Molecular Crystals and Liquid Crystals (2023), 767(1), 147-158.
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31-614-CAS-31864915 Steps: 1 Yield: 98% 1.1 Solvents: Methanol; 30 min, 0 °C; 1 h, rt 1.2 Reagents: Water; rt Experimental Protocols	Thieno[2,3-d]pyrimidine-Based Positive Allosteric Modulators of Human Mas-Related G Protein-Coupled Receptor X1 (MRGP RX1) By: Berhane, Ilyas; et al Journal of Medicinal Chemistry (2022), 65(4), 3218-3228.
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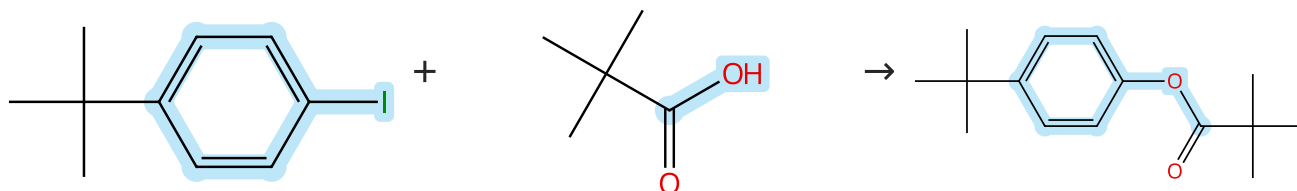
31-614-CAS-41355271 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Catalysts: Cuprous iodide, <i>N</i>¹,<i>N</i>²-Bis(2-methoxyethyl)ethanediamide Solvents: Dimethyl sulfoxide; 10 min, rt 1.2 24 h, 110 °C Experimental Protocols	Oxalamide ligands with additional coordinating groups for Cu-catalyzed arylation of alcohols and phenols By: Yao, Xiantong; et al Chemical Communications (Cambridge, United Kingdom) (2024), 60(69), 9210-9213.
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31-614-CAS-38873237	Steps: 1 Yield: 98%	Nickel-Coordinated Carbon Nitride as a Metallaphotoredox Platform for the Cross-Coupling of Aryl Halides with Alcohols
1.1 Reagents: Quinuclidine Catalysts: Imidazole, Nickel, Carbon nitride (C ₃ N ₄) Solvents: Dimethylacetamide; 8 h, 25 °C		By: Zhao, Xin; et al
Experimental Protocols		ACS Catalysis (2020), 10(24), 15178-15185.

Scheme 386 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (68)

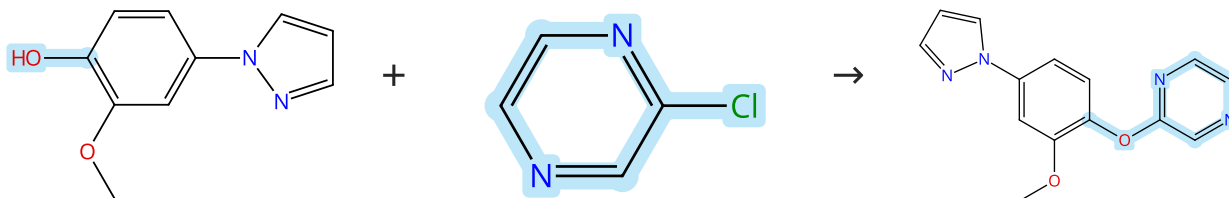
Suppliers (86)

Suppliers (10)

31-614-CAS-34709916	Steps: 1 Yield: 98%	Shining Visible Light on Reductive Elimination: Acridine-Pd-Catalyzed Cross-Coupling of Aryl Halides with Carboxylic Acids
1.1 Reagents: Cesium carbonate Catalysts: Bis(dibenzylideneacetone)palladium, 9-[3,5-Bis(trifluoromethyl)phenyl]-4-(diphenylphosphino)acridine Solvents: Dimethylacetamide; 24 h, 35 °C		By: Toriumi, Naoyuki; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		Journal of the American Chemical Society (2022), 144(42), 19592-19602.
Experimental Protocols		

Scheme 387 (1 Reaction)

Steps: 1 Yield: 98%



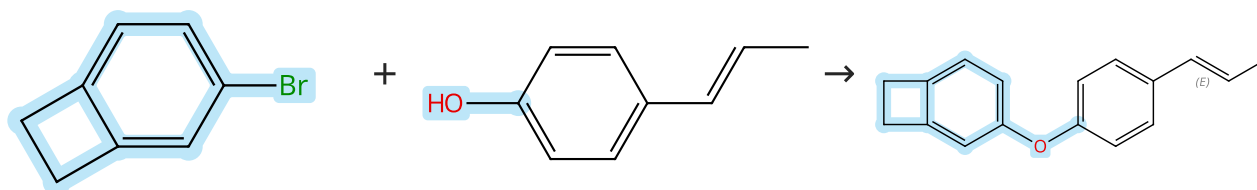
Suppliers (3)

Suppliers (90)

31-614-CAS-40984053	Steps: 1 Yield: 98%	Amination of Phenols and Halophenols via Pyridinium-Iridium Dual Photocatalysis
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt → 110 °C		By: Carson, Matthew C.; et al
Experimental Protocols		ACS Catalysis (2024), 14(16), 12173-12180.

Scheme 388 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (69)

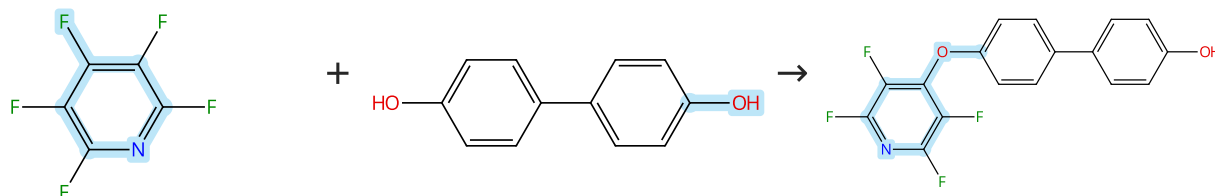
Suppliers (5)

Double bond geometry shown

31-614-CAS-36015012	Steps: 1 Yield: 98%	Constructing low-k polymers at high frequency from two propenyl-containing biomasses through the grubbs reaction
1.1 Reagents: Cesium carbonate Catalysts: Sodium ascorbate, Cuprous chloride, <i>N</i> ¹ , <i>N</i> ² -Bis(2-thienylmethyl)ethanediamide Solvents: 1,4-Dioxane; rt → 120 °C; 8 h, 120 °C		By: Shi, Manling; et al Polymer Chemistry (2023), 14(9), 999-1006.
Experimental Protocols		

Scheme 389 (1 Reaction)

Steps: 1 Yield: 98%



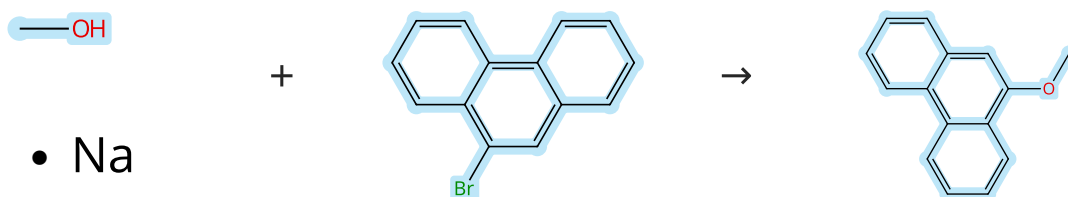
Suppliers (71)

Suppliers (100)

31-170-CAS-22000024	Steps: 1 Yield: 98%	Protecting Group-Controlled Remote Regioselective Electrophilic Aromatic Halogenation Reactions
1.1 Reagents: Potassium carbonate Solvents: Acetonitrile; 16 h, rt		By: Brittain, William D. G.; et al Journal of Organic Chemistry (2020), 85(11), 6862-6871.
Experimental Protocols		

Scheme 390 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (99)

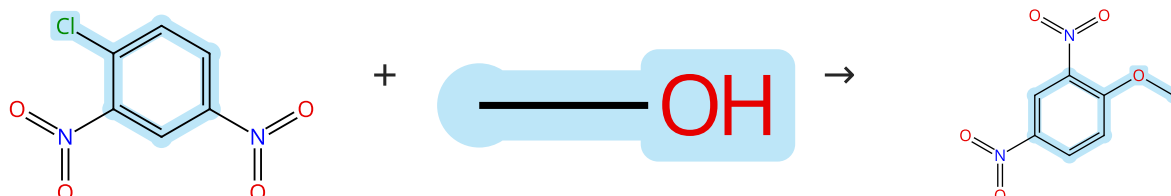
Suppliers (91)

Suppliers (19)

31-170-CAS-23756479	Steps: 1 Yield: 98%	A simple Hueckel model-driven strategy to overcome electronic barriers to retro-Brook silylation relevant to aryne and bisaryne precursor synthesis
1.1 Catalysts: Cuprous iodide Solvents: Methanol, Pyridine; rt → 80 °C 1.2 Solvents: Pyridine; 10 min, 80 °C; 42 h, 80 °C		By: Neal, Edward A.; et al Chemical Communications (Cambridge, United Kingdom) (2021), 57(13), 1663-1666.
Experimental Protocols		

Scheme 391 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (75)

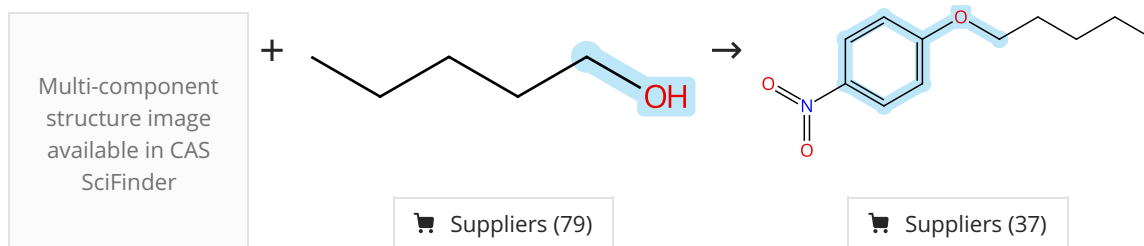
Suppliers (444)

Suppliers (28)

31-614-CAS-35190786 Steps: 1 Yield: 98% 1.1 Reagents: Sodium hydroxide Catalysts: Chloromethyl methyl ether, Triethylamine, hydrochloride (reaction products with PEEK), Poly(oxy-1,4-phenylene neoxy-1,4-phenylenecarbonyl-1,4-phenylene) (reaction products with triethylammonium chloride) Solvents: Methanol, Water; 2 h, 65 °C Experimental Protocols	Quaternary ammonium salt functionalized polyetheretherketone fiber: A novel fibrous material for phase-transfer catalysis By: Liu, Shuangshuang; et al Reactive & Functional Polymers (2023), 182, 105479.
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Scheme 392 (1 Reaction)

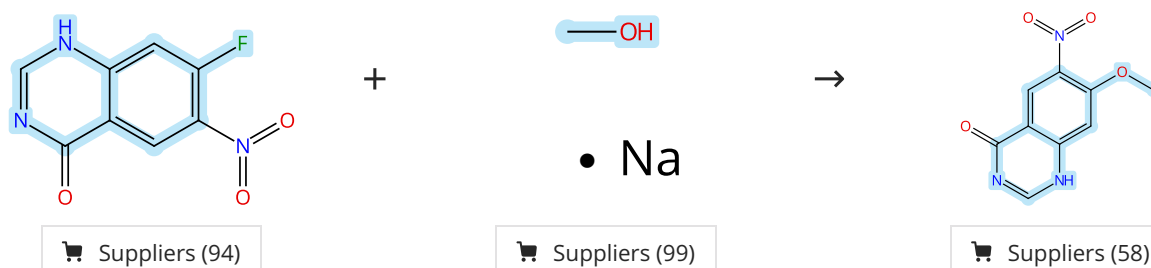
Steps: 1 Yield: 98%



31-614-CAS-38587463 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Solvents: Toluene; 2 h, 55 °C Experimental Protocols	Synthesis of Alkyl Aryl Ethers by O-Arylation of Alcohols with Diaryliodonium Salts: Scope, Limitations, and Mechanism By: Gallagher, Rory T.; et al Synlett (2024), 35(8), 889-894.
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Scheme 393 (1 Reaction)

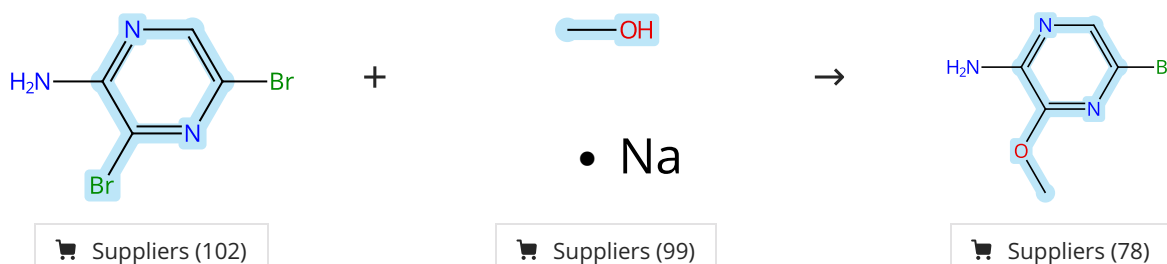
Steps: 1 Yield: 98%



31-170-CAS-23254079 Steps: 1 Yield: 98% 1.1 Solvents: Methanol; 30 °C; 30 °C → 65 °C; 2 h, 65 °C; 65 °C → rt 1.2 Reagents: Hydrochloric acid Solvents: Water; pH 5 - 6, rt 1.3 Solvents: Water; 30 min, rt Experimental Protocols	Design, synthesis and biological evaluation of new bivalent quinazoline analogues as IAP antagonists By: Bae, Inhwan; et al Bioorganic & Medicinal Chemistry Letters (2021), 34, 127676.
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Scheme 394 (1 Reaction)

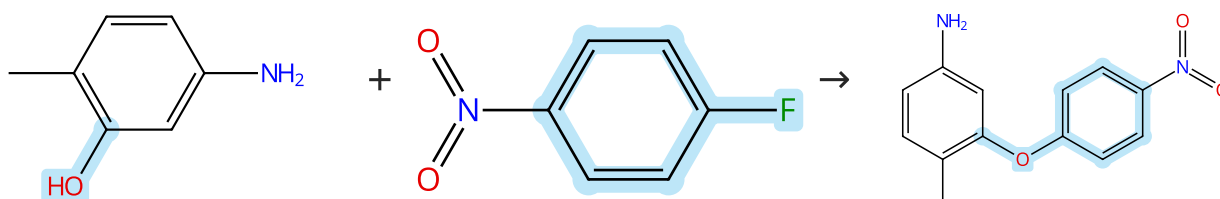
Steps: 1 Yield: 98%



31-170-CAS-22770372 Steps: 1 Yield: 98% 1.1 Solvents: 1,4-Dioxane; 18 h, rt → 100 °C; 100 °C → 0 °C 1.2 Reagents: Ammonium chloride Solvents: Water; 15 min, 0 °C Experimental Protocols	Design and synthesis of novel methoxypyridine-derived gamma-secretase modulators By: Ryneerson, Kevin D.; et al Bioorganic & Medicinal Chemistry (2020), 28(22), 115734.
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Scheme 395 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (95)

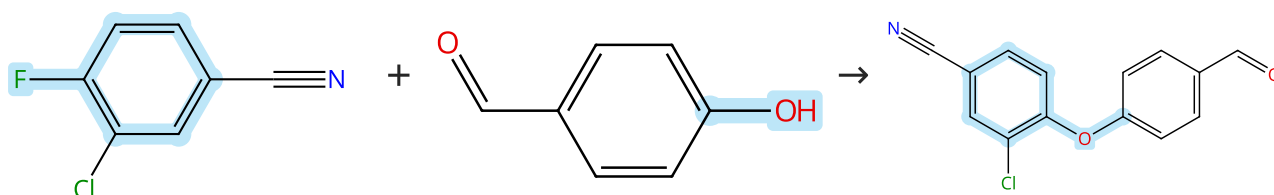
Suppliers (100)

Suppliers (2)

31-614-CAS-40338207 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C Experimental Protocols	Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors By: Xu, Yue; et al Bioorganic Chemistry (2023), 131, 106339.
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Scheme 396 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (78)

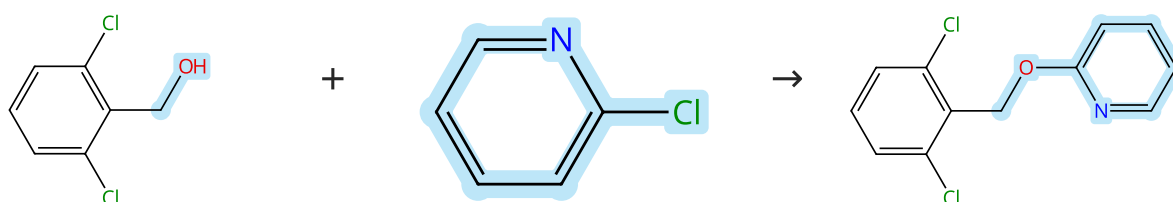
Suppliers (123)

Suppliers (24)

31-614-CAS-35482771 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 12 h, 100 °C	Strategies for the Production of [¹¹C]LY2795050 for Clinical Use By: Kaur, Tanpreet; et al Organic Process Research & Development (2023), 27(2), 373-381.
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Scheme 397 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (72)

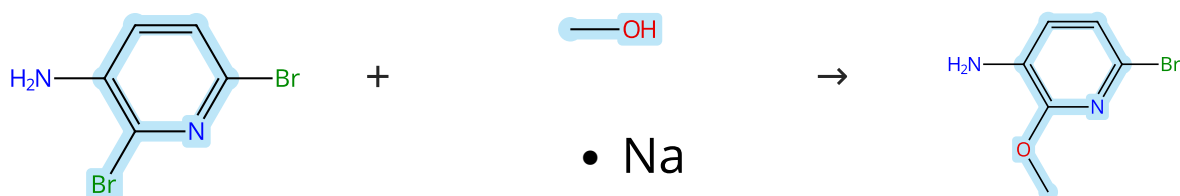
Suppliers (66)

Suppliers (7)

31-614-CAS-34387480	Steps: 1 Yield: 98%	Visible Light-Induced Deoxygenation and Allylation/Vinylation of Pyridyl Ethers
1.1 Reagents: Potassium hydroxide Catalysts: 18-Crown-6 Solvents: Toluene; rt → reflux		By: Wang, Fan; et al
Experimental Protocols		Organic Letters (2022), 24(40), 7309-7314.

Scheme 398 (1 Reaction)

Steps: 1 Yield: 98%



Chemical reaction scheme 398 shows the conversion of 2,5-dibromo-4-aminopyridine to 2,5-dibromo-4-aminopyridine-3-yl methyl ether. The reactant is 2,5-dibromo-4-aminopyridine, a pyridine ring with an amino group (H₂N) at position 4 and bromine atoms (Br) at positions 2 and 5. It reacts with methanol (CH₃OH) in the presence of sodium (Na). The product is 2,5-dibromo-4-aminopyridine-3-yl methyl ether, where the hydrogen at position 3 of the pyridine ring has been replaced by a methoxy group (-OCH₃).

 Suppliers (90)

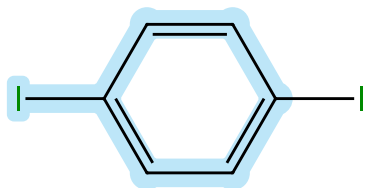
 Suppliers (99)

 Suppliers (77)

31-170-CAS-22770365	Steps: 1 Yield: 98%	Design and synthesis of novel methoxypyridine-derived gamma-secretase modulators
1.1 Solvents: 1,4-Dioxane; 18 h, rt → 100 °C; 100 °C → 0 °C		By: Rynearson, Kevin D.; et al
1.2 Reagents: Ammonium chloride Solvents: Water; 15 min, 0 °C		Bioorganic & Medicinal Chemistry (2020), 28(22), 115734.
Experimental Protocols		

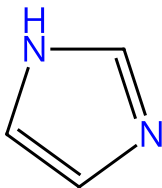
Scheme 399 (1 Reaction)

Steps: 1 Yield: 98%



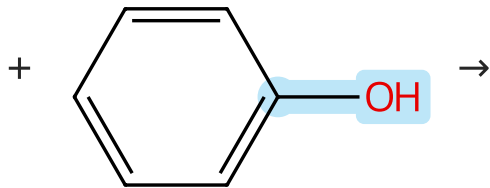
Chemical structure of 1,4-diiodobenzene (a benzene ring with iodine atoms at the para positions).

 Suppliers (103)



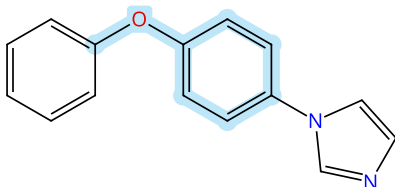
Chemical structure of Imidazole (a five-membered aromatic heterocycle with two nitrogen atoms).

 Suppliers (207)



Chemical structure of Phenol (a benzene ring with a hydroxyl group).

 Suppliers (207)



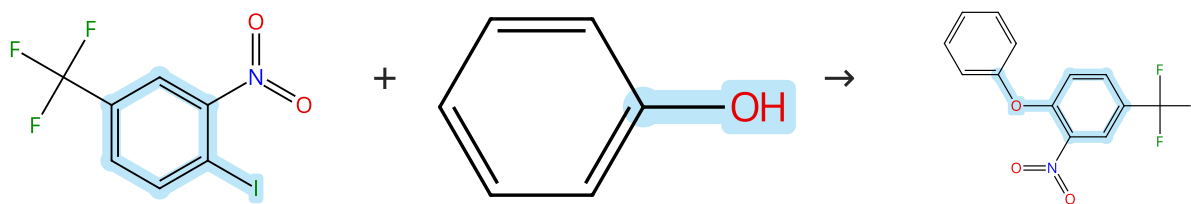
Chemical structure of the product, 1-(4-(benzyloxy)phenyl)imidazole, which is a benzimidazole derivative with a benzyloxy group.

 Suppliers (20)

31-170-CAS-23963685	Steps: 1 Yield: 98%	Bis(NHC)-Pd-catalyzed one-pot competitive C-C*C-C, C-C*C-O, C-C*C-N, and C-O*C-N cross-coupling reactions on an aryl dihalide catalyzed by a homogenous basic ionic liquid (TAlm[O H]) under base-free, ligand-free, and solvent-free conditions
1.1 Catalysts: Palladium diacetate, Water, 2623265-78-9; 3 min, reflux		By: Zhu, Yanfang; et al
Experimental Protocols		New Journal of Chemistry (2021), 45(26), 11662-11671.

Scheme 400 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (57)

Suppliers (207)

Suppliers (21)

31-614-CAS-42990177

Steps: 1 Yield: 98%

Selective Arene Photomnitration via Iron-Complex β -Homolysis

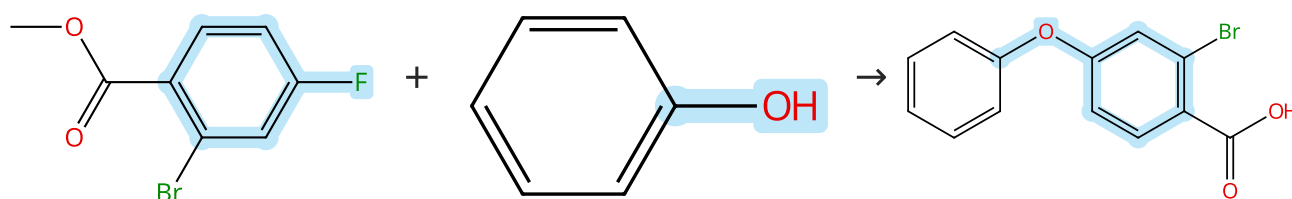
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 4 h, 100 °C

By: Liu, Shuyang; et al
 JACS Au (2024), 4(12), 4899-4909.

Experimental Protocols

Scheme 401 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (63)

Suppliers (207)

Suppliers (13)

31-614-CAS-31495577

Steps: 1 Yield: 98%

A Scalable Synthesis of Roxadustat (FG-4592)

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; rt; 18 - 20 h, 110 °C; 110 °C → 80 °C

By: Pisa, Ondrej; et al
 Organic Process Research & Development (2022), 26(3), 915-924.

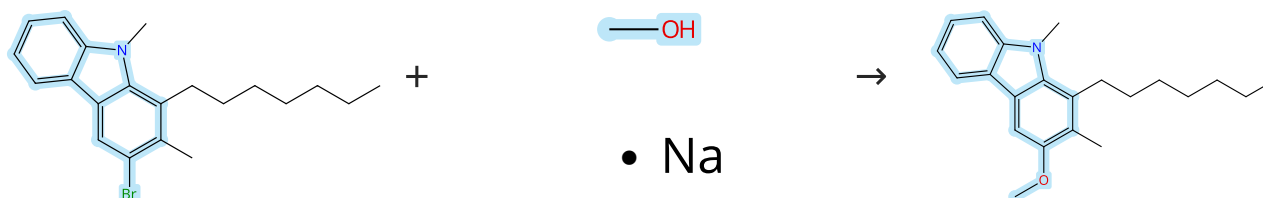
1.2 **Reagents:** Potassium hydroxide
Solvents: Water; < 30 min, 80 °C; 1 h, 80 °C; 80 °C → 40 °C

1.3 **Reagents:** Hydrochloric acid
Solvents: Water; pH 2; 1 h, rt

Experimental Protocols

Scheme 402 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (99)

Suppliers (3)

31-614-CAS-33898529

Steps: 1 Yield: 98%

An approach to functionalized carbazoles from Z-enoate propargylic alcohols. A unified total synthesis of N-Me-carazos tatin, N-Me-carbazquinocin C and N-Me-lipocarbazole A4

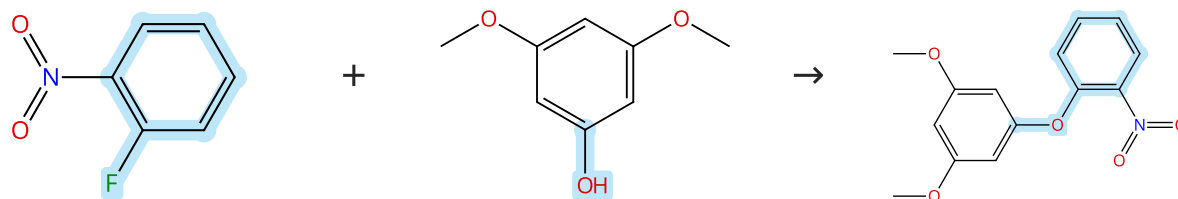
By: Roy, Debayan; et al
 Chemical Communications (Cambridge, United Kingdom) (2022), 58(73), 10210-10213.

1.1 **Reagents:** Cuprous iodide
Solvents: Dimethylformamide; 15 h, 115 °C

Experimental Protocols

Scheme 403 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (93)

Suppliers (92)

Supplier (1)

31-170-CAS-22969462

Steps: 1 Yield: 98%

Synthesis of Cycloheptatrienes, Oxepines, Thiepinines, and Silepinines: A Comparison between Brønsted Acid and Au-Catalysis

By: Sprenger, Kristin; et al

European Journal of Organic Chemistry (2020), 2020(39), 6245-6254.

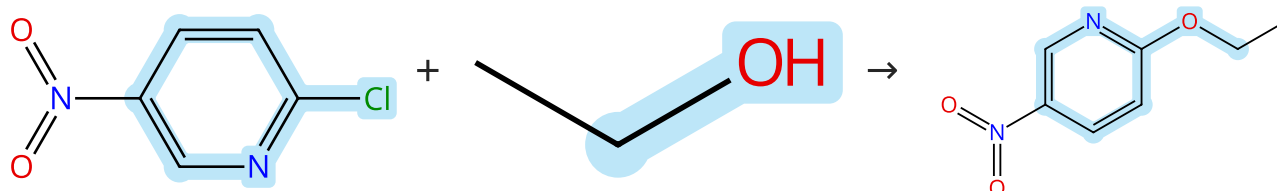
1.1 Reagents: Potassium carbonate

Solvents: Dimethyl sulfoxide; rt → 95 °C; 26 h, 95 °C

Experimental Protocols

Scheme 404 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (100)

Suppliers (618)

Suppliers (82)

31-614-CAS-43744137

Steps: 1 Yield: 98%

Reductive Synthesis of Azoxypyridines from Nitropyridines Using Hydroxides in Alcoholic Media

By: Caruana, Lorenzo; et al

Synthesis (2025), 57(4), 846-854.

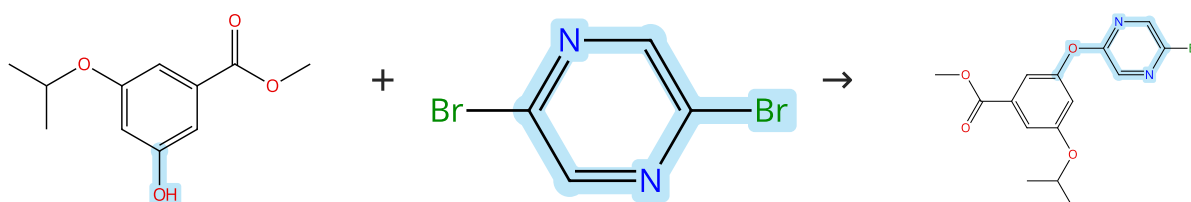
1.1 Reagents: Sodium ethoxide

Solvents: Ethanol; 1 h; overnight; 2 h, reflux

Experimental Protocols

Scheme 405 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (49)

Suppliers (80)

31-614-CAS-40832843

Steps: 1 Yield: 98%

Discovery of a Partial Glucokinase Activator Clinical Candidate: Diethyl ((3-(3-((5-(Azetidine-1-carbonyl)pyrazin-2-yl)oxy)-5-isopropoxybenzamido)-1H-pyrazol-1-yl)methyl)phosphonate (BMS-820132)

By: Shi, Yan; et al

Journal of Medicinal Chemistry (2022), 65(5), 4291-4317.

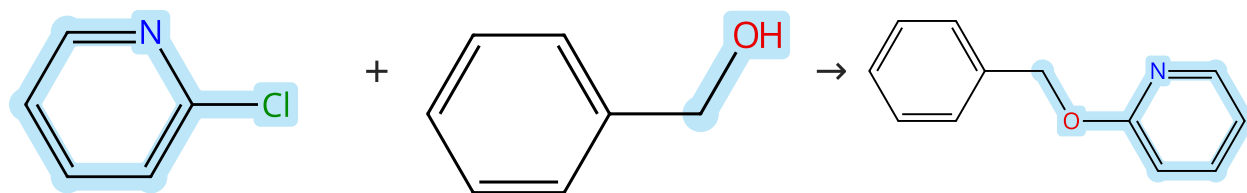
1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 18 h, 86 °C

Experimental Protocols

Scheme 406 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (66)

Suppliers (150)

Suppliers (52)

31-614-CAS-34387478

Steps: 1 Yield: 98%

Visible Light-Induced Deoxygenation and Allylation/Vinylation of Pyridyl Ethers

1.1 Reagents: Potassium hydroxide
Catalysts: 18-Crown-6
Solvents: Toluene; rt → reflux

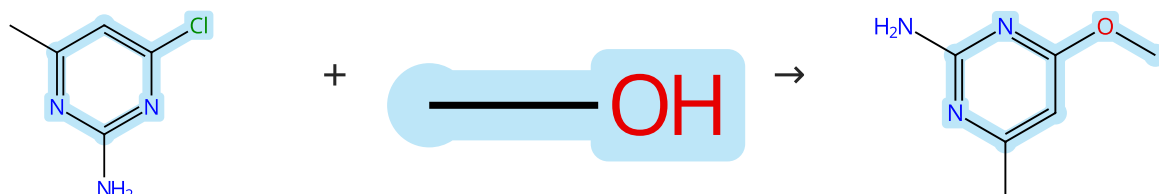
By: Wang, Fan; et al

Organic Letters (2022), 24(40), 7309-7314.

Experimental Protocols

Scheme 407 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (96)

Suppliers (444)

Suppliers (68)

31-614-CAS-44316138

Steps: 1 Yield: 98%

Design of Selective BRD4 Inhibitors for the Treatment of Autosomal Dominant Polycystic Kidney Disease

1.1 Reagents: Sodium methoxide
Solvents: Methanol; 6 h, rt → 65 °C

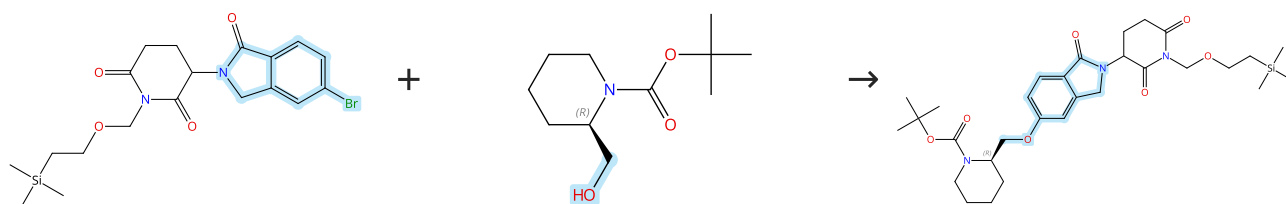
By: Yang, Yueyue; et al

Journal of Medicinal Chemistry (2025), 68(5), 5257-5274.

Experimental Protocols

Scheme 408 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (35)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (69)

31-614-CAS-47306789

Steps: 1 Yield: 98%

Discovery and Optimization of First-in-Class Molecular Glue Degradors of the WIZ Transcription Factor for Fetal Hemoglobin Induction to Treat Sickle Cell Disease

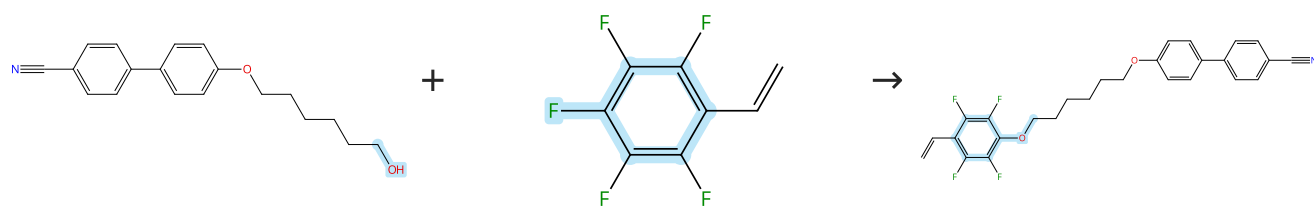
1.1 Reagents: 2,2,6,6-Tetramethylpiperidine
Catalysts: Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine, Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-κN¹,κN^{1'}][bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl-κN]phenyl-κC]-, (OC-6-33)-, hexafluorophosphate(1-) (1:1)
Solvents: Acetonitrile; 20 h, rt

By: Kerrigan, John Ryan; et al

Journal of Medicinal Chemistry (2024), 67(22), 20682-20694.

Scheme 409 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (8)

Suppliers (80)

31-614-CAS-25098791

Steps: 1 Yield: 98%

Influence of multifluorophenoxy terminus on the mesomorphism of the alkoxy and alkyl cyanobiphenyl compounds in search of new ambient nematic liquid crystals and mixtures

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; overnight, 40 - 50 °C

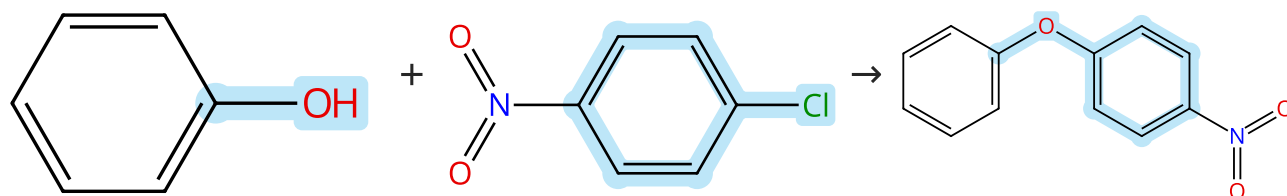
By: Wang, Kunlun; et al

Liquid Crystals (2021), 48(5), 672-688.

Experimental Protocols

Scheme 410 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (207)

Suppliers (65)

Suppliers (68)

31-614-CAS-41541876

Steps: 1 Yield: 98%

CuSO₄/N-(9H-carbazol-9-yl)picolinamide-Catalyzed C-O Coupling of (Hetero)Aryl Chlorides with Phenols on Water

1.1 Reagents: Tripotassium phosphate
Catalysts: Copper sulfate pentahydrate, 2-Pyridinecarboxamide, N-9H-carbazol-9-yl-
Solvents: Water; 24 h, 120 °C

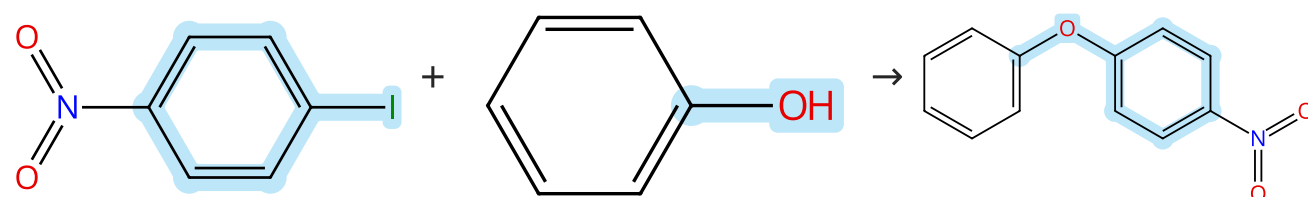
By: Hong, Peng; et al

Organic Letters (2024), 26(34), 7202-7206.

Experimental Protocols

Scheme 411 (2 Reactions)

Steps: 1 Yield: 97-98%



Suppliers (70)

Suppliers (207)

Suppliers (68)

31-614-CAS-30037264

Steps: 1 Yield: 98%

Synthesis and characterization of nano-cellulose immobilized phenanthroline-copper (I) complex as a recyclable and efficient catalyst for preparation of diaryl ethers, N-aryl amides and N-aryl heterocycles

1.1 Reagents: Potassium carbonate
Catalysts: 2758146-87-9 (nanocellulose supported)
Solvents: Toluene; 8 h, 110 °C

By: Hosseinzadeh, Rahman; et al

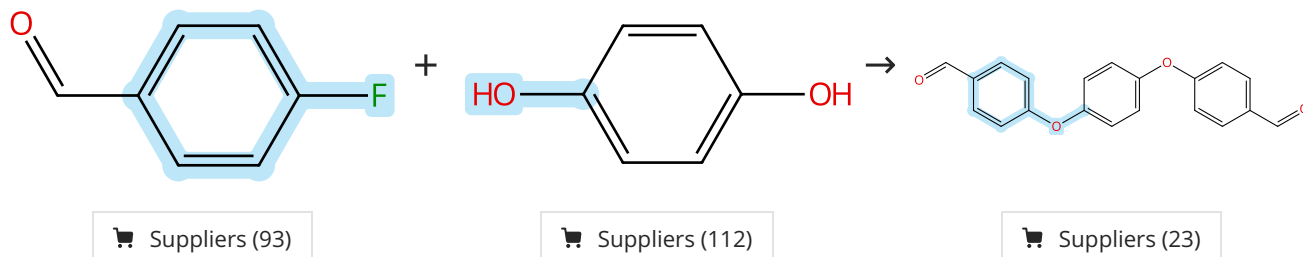
Polyhedron (2022), 213, 115631.

Experimental Protocols

31-170-CAS-23695356	Steps: 1 Yield: 97%	An efficient, cost-effective, and magnetically recoverable copper catalyst for O-arylation of phenols with aryl halides in choline chloride-based deep eutectic solvents By: Niakan, Mahsa; et al Colloids and Surfaces, A: Physicochemical and Engineering Aspects (2021), 620, 126603.
1.1 Reagents: Potassium carbonate Catalysts: 2642302-91-6 (supported on Fe ₃ O ₄ -SiO ₂) Solvents: Urea, Choline chloride; 6 h, 100 °C		
Experimental Protocols		

Scheme 412 (1 Reaction)

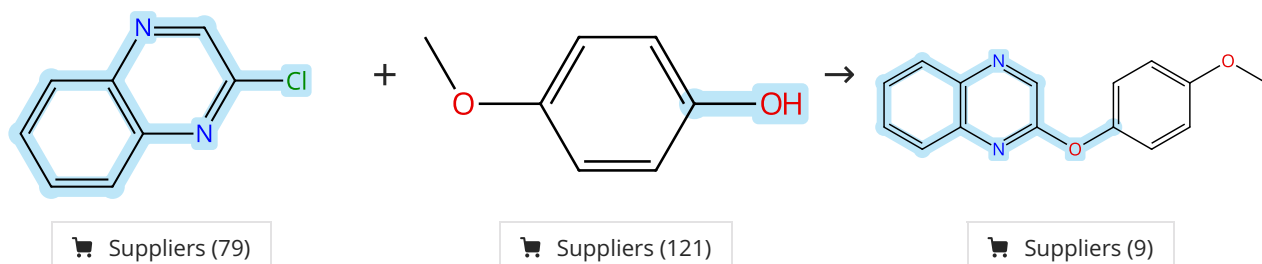
Steps: 1 Yield: 98%



31-614-CAS-32693620	Steps: 1 Yield: 98%	Synthesis of poly(pyrazolopyridine)s by Hantzsch multicomponent polymerization By: Saxer, Samantha; et al Polymer (2022), 252, 124935.
1.1 Reagents: Potassium carbonate Solvents: Dimethylacetamide; 22 h, rt → 150 °C		
Experimental Protocols		

Scheme 413 (1 Reaction)

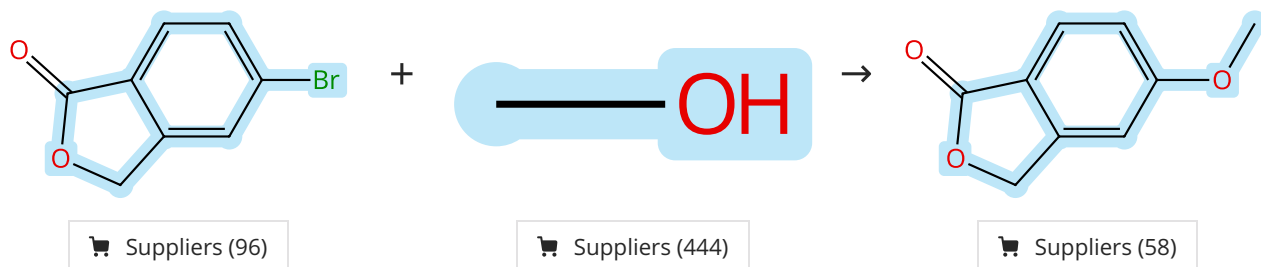
Steps: 1 Yield: 98%



31-614-CAS-37155773	Steps: 1 Yield: 98%	DMSO-Assisted K₃PO₄-Catalyzed Cooperative Metal-Free, Base-Free Etherification of Chloroheteroarenes at Low Temperature By: Khandagale, Deepali; et al Chemistry - An Asian Journal (2023), 18(16), e202300377.
1.1 Catalysts: Tripotassium phosphate Solvents: Dimethyl sulfoxide; 12 h, 30 °C		
Experimental Protocols		

Scheme 414 (1 Reaction)

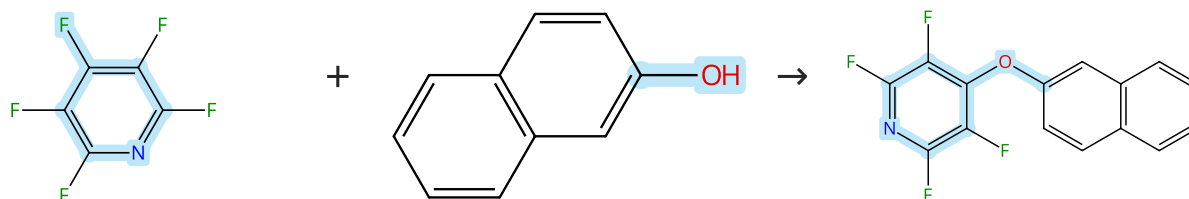
Steps: 1 Yield: 98%



31-614-CAS-38944061	Steps: 1 Yield: 98%	Bottom-Up Construction of Ni(II)-Incorporated Covalent Organic Framework for Metallaphotoredox Catalysis By: Hu, Hai-Chao; et al Chemistry - A European Journal (2024), 30(10), e202303476.
1.1 Reagents: Triethylamine Catalysts: 3026433-57-5 Solvents: Methanol, Acetonitrile; 24 h, rt		
Experimental Protocols		

Scheme 415 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (71)

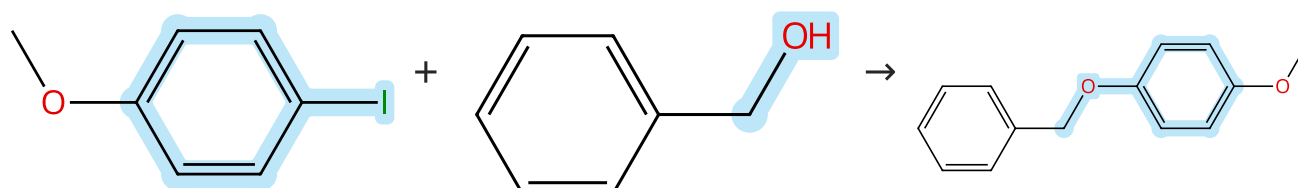
Suppliers (117)

Suppliers (4)

31-170-CAS-21983428	Steps: 1 Yield: 98%	Ionic hydrogenation of naphthyl tetrafluoropyridin-4-yl ethers as a new route to 5,6,7,8-tetrahydronaphthols By: Zhu, Zhongwei; et al Mendelev Communications (2020), 30(2), 190-191.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 1 h, 20 °C		
Experimental Protocols		

Scheme 416 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (89)

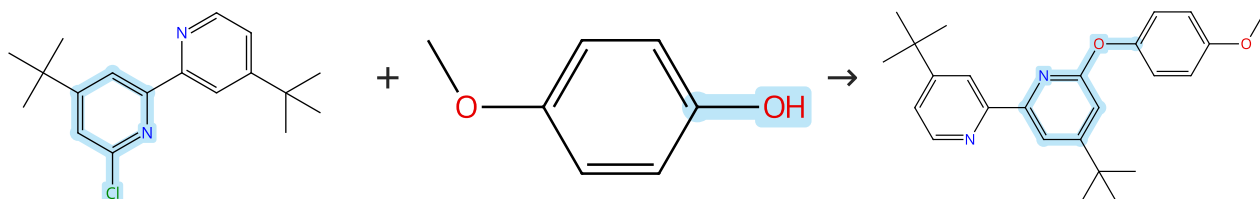
Suppliers (150)

Suppliers (45)

31-614-CAS-46386010	Steps: 1 Yield: 98%	Visible-Light-Promoted Copper-Catalyzed Etherification of Aryl Bromides via Energy Transfer By: Yao, Xiantong; et al Organic Letters (2025), 27(27), 7313-7318.
1.1 Reagents: Sodium <i>tert</i> -butoxide Catalysts: Cuprous iodide, Eosin, <i>N</i> ¹ , <i>N</i> ² -Bis(2-phenoxyethyl)ethanediamide Solvents: 1,4-Dioxane; 24 h, 32 - 36 °C		
1.2 Reagents: Hydrochloric acid Solvents: Water; neutralized Experimental Protocols		

Scheme 417 (2 Reactions)

Steps: 1 Yield: 98%



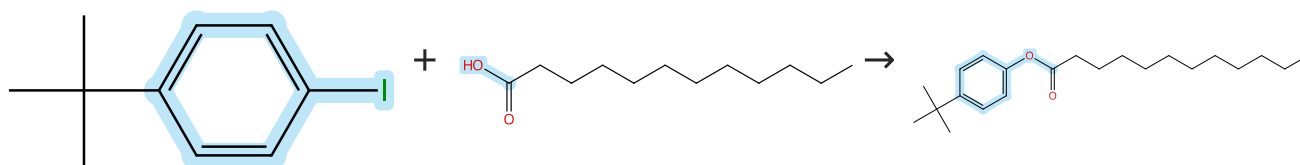
Suppliers (7)

Suppliers (121)

31-170-CAS-23866297	Steps: 1 Yield: 98%	Bipyridine-Type Bidentate Auxiliary-Enabled Copper-Mediated C-H/C-H Biaryl Coupling of Phenols and 1,3-Azoles
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 21 h, 160 °C		By: Kajiwara, Rikuo; et al
1.2 Solvents: Water		Organic Letters (2021), 23(14), 5405-5409.
Experimental Protocols		
31-614-CAS-24403438	Steps: 1 Yield: 98%	Copper-mediated Regioselective C-H Cyanation of Phenols with Assistance of Bipyridine-type Bidentate Auxiliary
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 21 h, 160 °C		By: Kajiwara, Rikuo; et al
1.2 Reagents: Water		Chemistry Letters (2021), 50(10), 1814-1817.
Experimental Protocols		

Scheme 418 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (68)

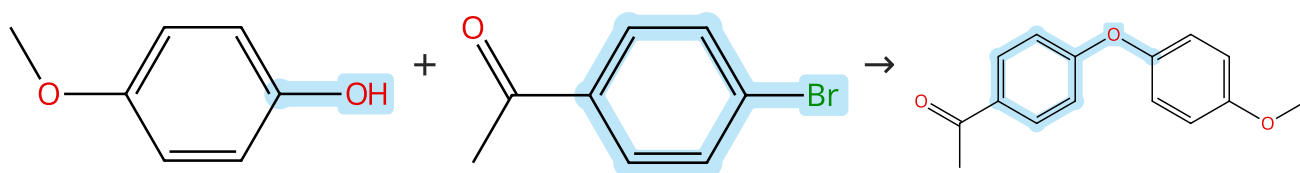
Suppliers (117)

Suppliers (2)

31-614-CAS-34709922	Steps: 1 Yield: 98%	Shining Visible Light on Reductive Elimination: Acridine-Pd-Catalyzed Cross-Coupling of Aryl Halides with Carboxylic Acids
1.1 Reagents: Cesium carbonate Catalysts: Bis(dibenzylideneacetone)palladium, 9-[3,5-Bis(trifluoromethyl)phenyl]-4-(diphenylphosphino)acridine Solvents: Dimethylacetamide; 24 h, 35 °C		By: Toriumi, Naoyuki; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		Journal of the American Chemical Society (2022), 144(42), 19592-19602.
Experimental Protocols		

Scheme 419 (2 Reactions)

Steps: 1 Yield: 98%



Suppliers (121)

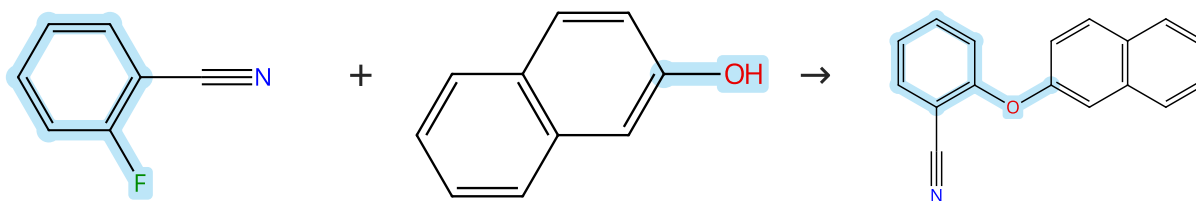
Suppliers (85)

Suppliers (62)

31-614-CAS-43873538	Steps: 1 Yield: 98%	Electromagnetic milling promoted mechanochemical copper-catalyzed C-O coupling reaction for diaryl ether formation
1.1 Reagents: Cesium carbonate Catalysts: Copper(II) acetylacetonate, <i>N</i> ¹ -[2-(1-Methylethyl)phenyl]- <i>N</i> ² -(phenylmethyl)ethanediamide Solvents: Dimethylformamide; 3 h, 79 °C		By: Zeng, Zhaoxuan; et al
Experimental Protocols		Organic Chemistry Frontiers (2025), 12(5), 1528-1536.
31-614-CAS-43873491	Steps: 1 Yield: 98%	Electromagnetic milling promoted mechanochemical copper-catalyzed C-O coupling reaction for diaryl ether formation
1.1 Reagents: Cesium carbonate Catalysts: Copper(II) acetylacetonate, <i>N</i> ¹ , <i>N</i> ² -Di-1-naphthylethanediamide Solvents: Dimethylformamide; 3 h, 79 °C		By: Zeng, Zhaoxuan; et al
Experimental Protocols		Organic Chemistry Frontiers (2025), 12(5), 1528-1536.

Scheme 420 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

Suppliers (117)

Suppliers (43)

31-614-CAS-38942839

Steps: 1 Yield: 98%

Update of the Imine-Anion-Mediated Smiles Rearrangement: Application to Migration of Electron-Neutral/Rich Aromatic Groups

By: Jinno, Shunki; et al

Synlett (2024), 35(13), 1565-1568.

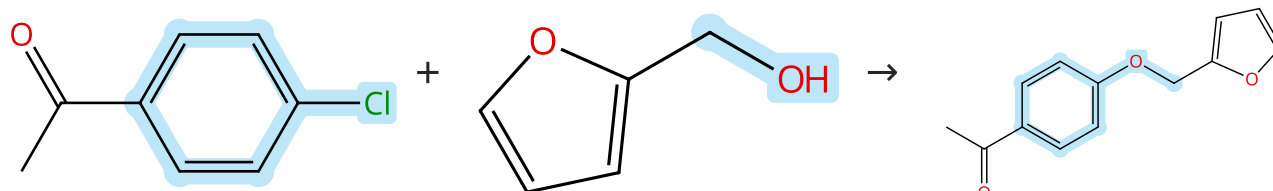
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; 15 h, 90 °C; 90 °C → rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 421 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (68)

Suppliers (71)

Suppliers (2)

31-614-CAS-40260646

Steps: 1 Yield: 98%

Thermal Nickel-Catalyzed Carbon-Oxygen Cross-Coupling of (Hetero)aryl Halides with Alcohols Enabled by the Use of a Silane Reductant Approach

By: Yang, Liu; et al

ACS Catalysis (2024), 14(10), 7846-7852.

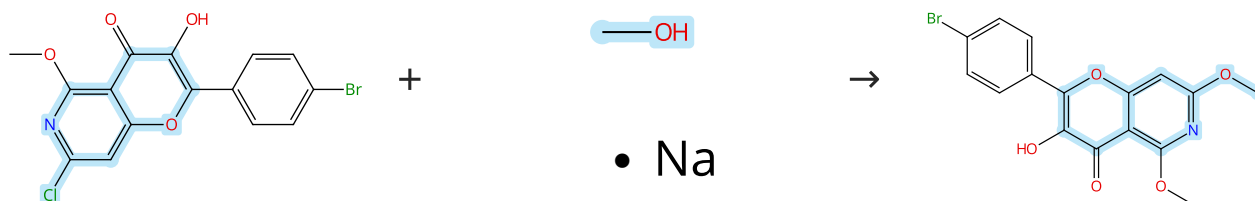
1.1 **Catalysts:** Nickel bromide, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Toluene; 30 min, rt

1.2 **Reagents:** Phenylsilane, 1,8-Diazabicyclo[5.4.0]undec-7-ene; 24 h, 120 °C

Experimental Protocols

Scheme 422 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (2)

Suppliers (99)

Suppliers (5)

31-170-CAS-22369285

Steps: 1 Yield: 98%

Design of Development Candidate eFT226, a First in Class Inhibitor of Eukaryotic Initiation Factor 4A RNA Helicase

By: Ernst, Justin T.; et al

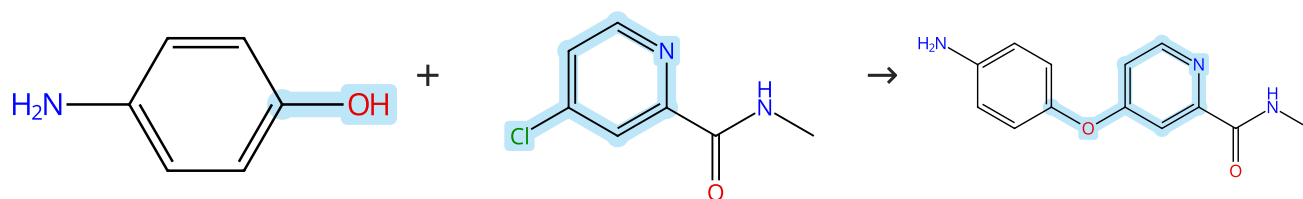
Journal of Medicinal Chemistry (2020), 63(11), 5879-5955.

1.1 **Solvents:** Methanol, Dimethylformamide; rt; 1 h, 80 °C

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; pH 6, cooled

Scheme 423 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (100)

Suppliers (86)

Suppliers (87)

31-614-CAS-37924151

Steps: 1 Yield: 98%

A Practical and Efficient Method for the Synthesis of Sorafenib and Regorafenib

By: Prachi, Ramteke; et al

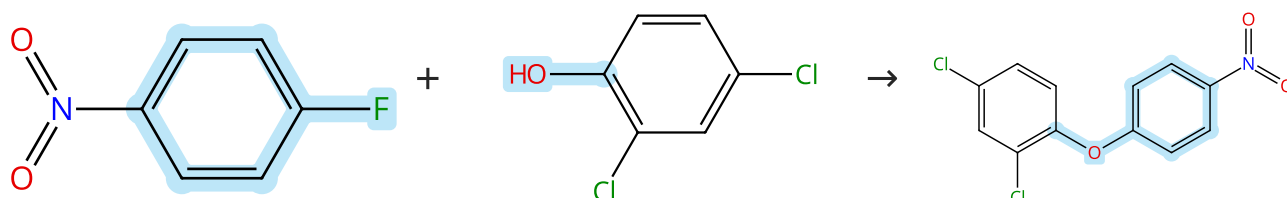
SynOpen (2023), 7(3), 422-429.

- 1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 10 min, rt
- 1.2 2 h, 110 °C; 110 °C → rt
- 1.3 **Solvents:** Water; rt

Experimental Protocols

Scheme 424 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (100)

Suppliers (102)

Suppliers (64)

31-614-CAS-37615377

Steps: 1 Yield: 98%

Discovery of N-Arylsulfonyl-Indole-2-Carboxamide Derivatives as Galectin-3 and Galectin-8 C-Terminal Domain Inhibitors

By: Zhang, Haoming; et al

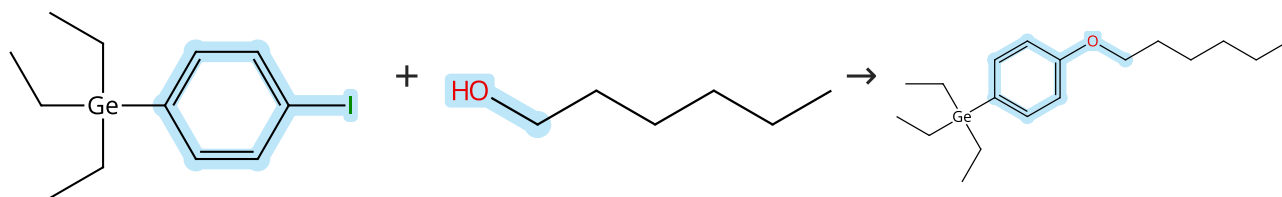
ACS Medicinal Chemistry Letters (2023), 14(9), 1257-1265.

- 1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 6 h, 50 °C

Experimental Protocols

Scheme 425 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (100)

31-614-CAS-36090858

Steps: 1 Yield: 97%

Orthogonal C-O Bond Construction with Organogermanes

By: Dahiya, Amit; et al

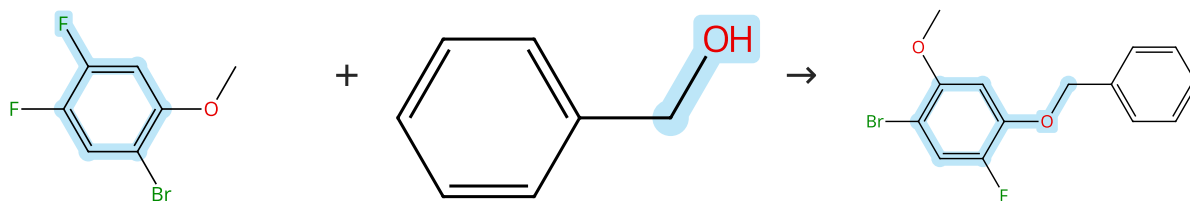
Journal of the American Chemical Society (2023), 145(14), 7729-7735.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: 3,4,7,8-Tetramethyl-1,10-phenanthroline, Cuprous iodide
Solvents: Toluene; 16 h, 110 °C

Experimental Protocols

Scheme 426 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (64)

Suppliers (150)

Suppliers (18)

31-614-CAS-45682279

Steps: 1 Yield: 97%

Discovery and Characterization of PVTX-321 as a Potent and Orally Bioavailable Estrogen Receptor Degradar for E R+/HER2-Breast Cancer

By: Xu, Guozhang; et al

Journal of Medicinal Chemistry (2025), 68(11), 11299-11321.

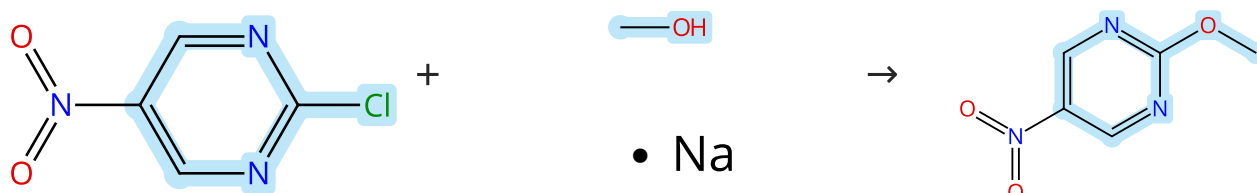
1.1 Reagents: Sodium *tert*-butoxide

Solvents: Dimethylformamide; 10 - 20 °C; 12 h, 20 °C

Experimental Protocols

Scheme 427 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (91)

Suppliers (99)

Suppliers (59)

31-614-CAS-39013782

Steps: 1 Yield: 97%

Parallel Optimization of Potency and Pharmacokinetics Leading to the Discovery of a Pyrrole Carboxamide ERK5 Kinase Domain Inhibitor

By: Miller, Duncan C.; et al

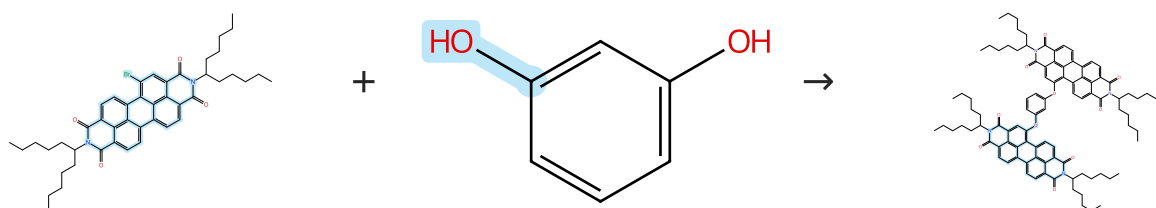
Journal of Medicinal Chemistry (2022), 65(9), 6513-6540.

1.1 Solvents: Methanol; 1 h, 65 °C

Experimental Protocols

Scheme 428 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (17)

Suppliers (109)

31-614-CAS-47037377

Steps: 1 Yield: 97%

Synthesis of ether-bonded perylene diimide oligomers without noble-metal catalysts for organic solar cells

By: Wang, Zhu; et al

Dyes and Pigments (2025), 241, 112892.

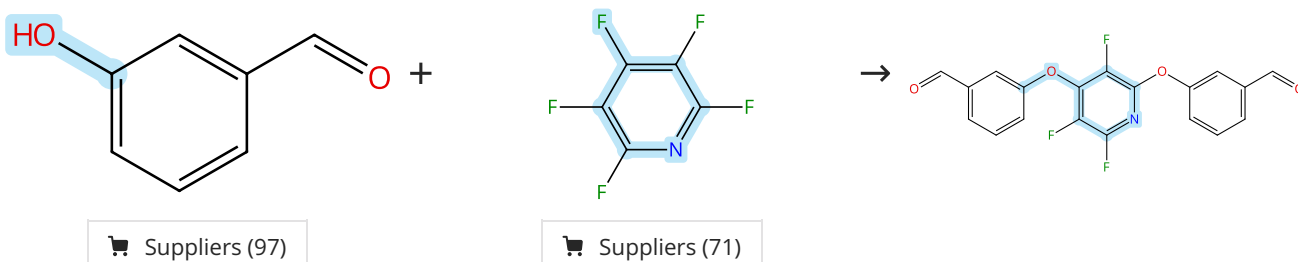
1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 12 h, 80 °C

Experimental Protocols

Scheme 429 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-36601614

Steps: 1 Yield: 97%

Site-selective nucleophilic substitution reactions of pentafluoropyridine with hydroxybenzaldehydes: synthesis of triarylmethanes comprising perfluoropyridine moieties

By: Vari, Zahra; et al

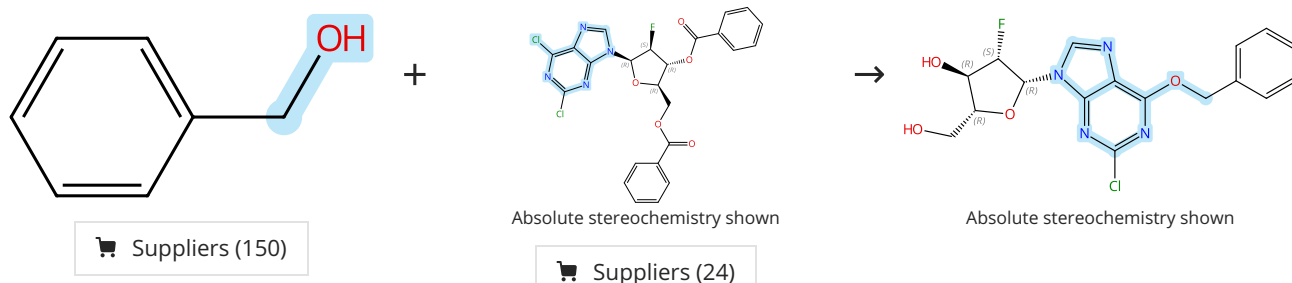
New Journal of Chemistry (2023), 47(22), 10645-10658.

- 1.1 Reagents: Potassium carbonate
Solvents: Acetonitrile; 30 min, rt
- 1.2 24 h, reflux

Experimental Protocols

Scheme 430 (1 Reaction)

Steps: 1 Yield: 97%



31-170-CAS-22799429

Steps: 1 Yield: 97%

Discovery of AB680: A Potent and Selective Inhibitor of C D73

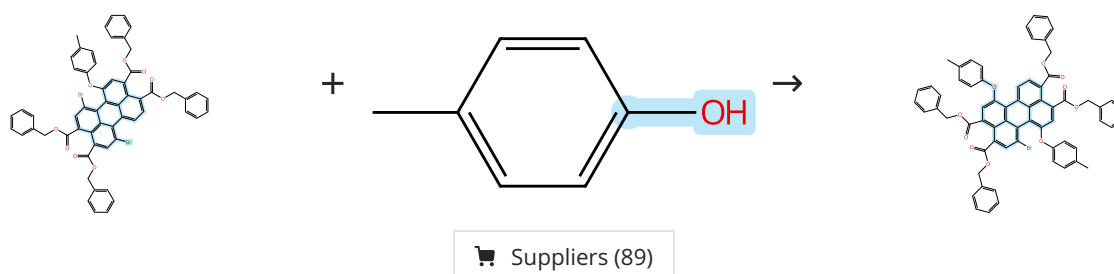
By: Lawson, Kenneth V.; et al

Journal of Medicinal Chemistry (2020), 63(20), 11448-11468.

- 1.1 Reagents: Sodium hydride; 15 min, rt
- 1.2 2 h, rt

Scheme 431 (1 Reaction)

Steps: 1 Yield: 97%



31-170-CAS-23443052

Steps: 1 Yield: 97%

Curved Perylene Diimides Fused with Seven- Membered Rings

By: Fujimoto, Keisuke; et al

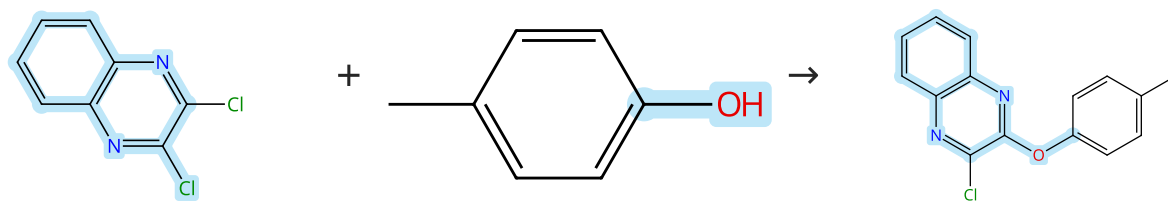
Chemistry - An Asian Journal (2021), 16(6), 690-695.

- 1.1 Reagents: Potassium carbonate, 18-Crown-6
Solvents: Dimethylformamide; rt; 12 h, 50 °C
- 1.2 Reagents: Water

Experimental Protocols

Scheme 432 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (92)

Suppliers (89)

Suppliers (2)

31-614-CAS-37054082

Steps: 1 Yield: 97%

Bioorthogonal Ligation and Cleavage by Reactions of Chloroquinoxalines with ortho-Dithiophenols

By: Li, Youshan; et al

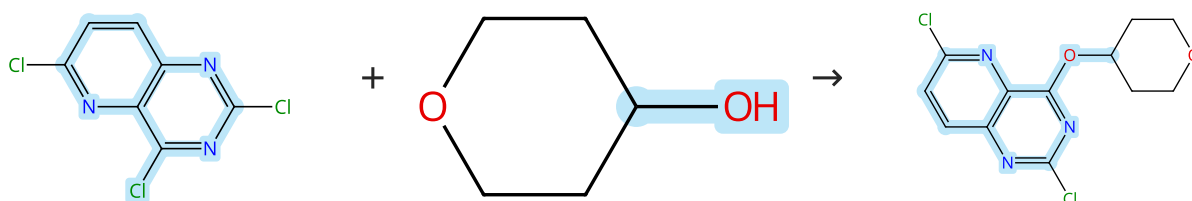
Angewandte Chemie, International Edition (2020), 59(9), 3671-3677.

 1.1 Reagents: Potassium carbonate
 Solvents: Dimethylformamide; 12 h, rt

Experimental Protocols

Scheme 433 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (61)

Suppliers (95)

31-614-CAS-24131840

Steps: 1 Yield: 97%

Improving metabolic stability and removing aldehyde oxidase liability in a 5-azaquinazoline series of IRAK4 inhibitors

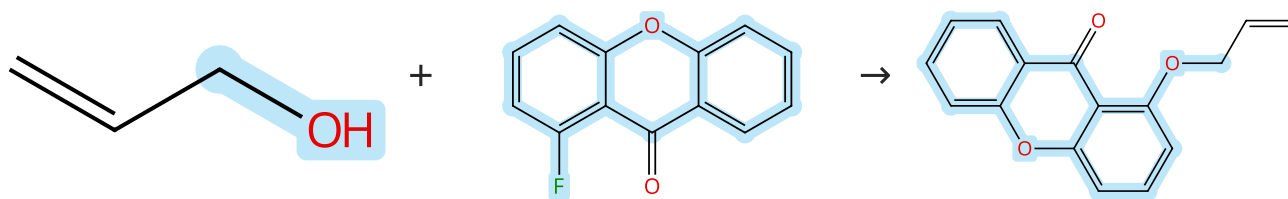
By: Degorce, Sebastien L.; et al

Bioorganic & Medicinal Chemistry (2020), 28(23), 115815.

 1.1 Reagents: Sodium hydride
 Solvents: Tetrahydrofuran; 16 h, rt

Scheme 434 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (47)

Supplier (1)

Supplier (1)

31-170-CAS-22503809

Steps: 1 Yield: 97%

An Efficient Two-Step Protocol for the Isoprenylation of Xanthone at the C2 Position Starting from 1-Fluoroxanthone Derivative

By: Fujimoto, Yuuki; et al

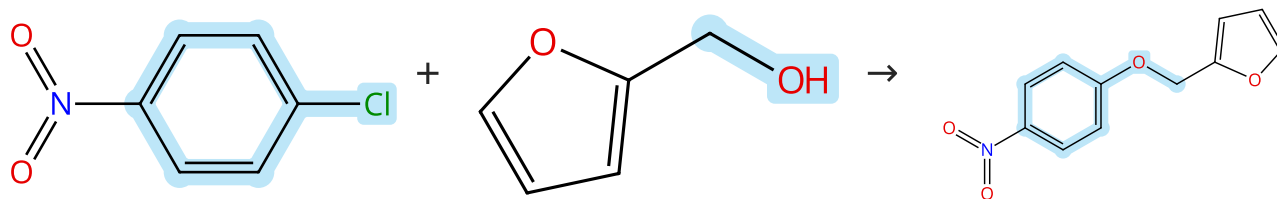
Synlett (2020), 31(14), 1423-1429.

 1.1 Reagents: Sodium hydride
 Solvents: Dimethylformamide; 0 °C; 30 min, 25 °C
 1.2 Solvents: Dimethylformamide; 0 °C; 15 min, 25 °C
 1.3 Solvents: Water; pH 7

Experimental Protocols

Scheme 435 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (65)

Suppliers (71)

Suppliers (20)

31-614-CAS-34564533

Steps: 1 Yield: 97%

Fe₃O₄@SiO₂Supported Pd (II)-selenoether N-heterocyclic carbene: A highly active and reusable heterogeneous catalyst for C-O cross-coupling of alcohols and chloro arenes

By: Khandaka, Himanshu; et al

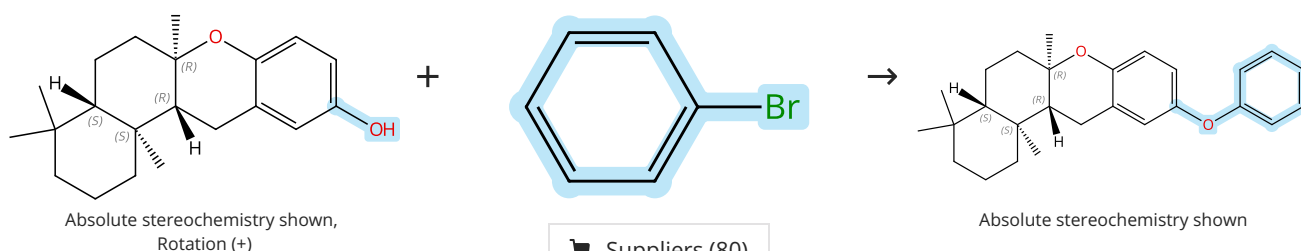
Tetrahedron Letters (2022), 111, 154163.

1.1 **Reagents:** Cesium carbonate
Catalysts: Iron oxide (Fe₃O₄), 2855128-96-8 (Fe₃O₄ supported)
Solvents: Toluene; 5 h, 80 °C

Experimental Protocols

Scheme 436 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (2)

Suppliers (80)

31-614-CAS-32422187

Steps: 1 Yield: 97%

Facile and divergent optimization of chromazonarol enabled the identification of simplified drimane meroter penoids as novel pharmaceutical leads

By: Wang, Xia; et al

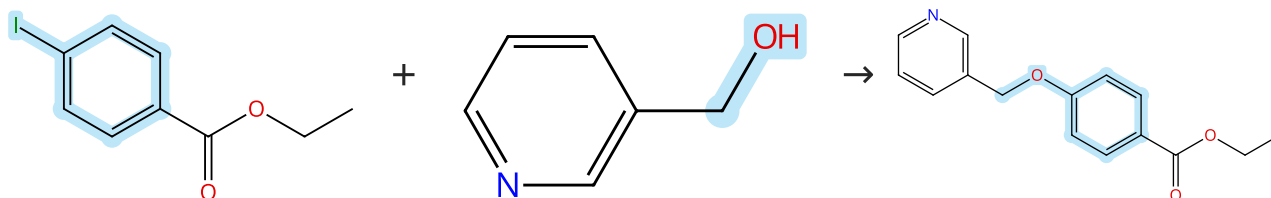
European Journal of Medicinal Chemistry (2022), 227, 113912.

1.1 **Reagents:** Cesium carbonate
Catalysts: *N,N*-Dimethylglycine, Cuprous iodide
Solvents: 1,4-Dioxane; 24 h, 105 °C

Experimental Protocols

Scheme 437 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (84)

Suppliers (88)

Suppliers (6)

31-614-CAS-35219842

Steps: 1 Yield: 97%

Photoinduced Nickel-Catalyzed Carbon-Heteroatom Coupling

By: Luo, Hang; et al

Chemistry - A European Journal (2023), 29(1), e202202385.

1.1 **Catalysts:** Nickel dichloride, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Tetrahydrofuran; 5 min, rt

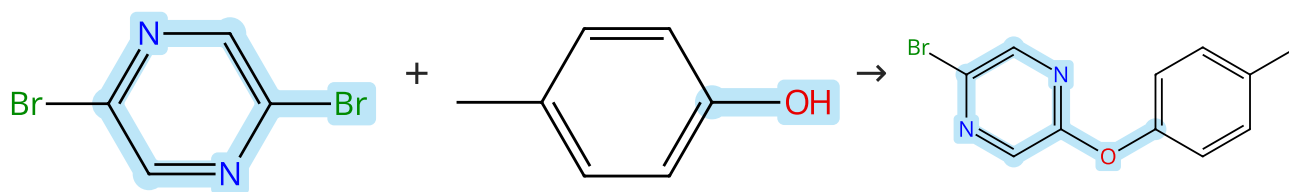
1.2 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene; 18 h, 30 °C

1.3 **Solvents:** Water

Experimental Protocols

Scheme 438 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (80)

Suppliers (89)

Suppliers (3)

31-614-CAS-31557603

Steps: 1 Yield: 97%

HTS-based discovery and optimization of novel positive allosteric modulators of the $\alpha 7$ nicotinic acetylcholine receptor

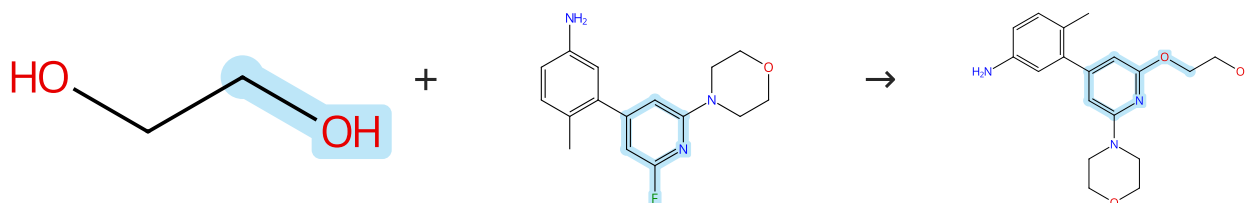
By: Ledneczki, Istvan; et al

European Journal of Medicinal Chemistry (2021), 222, 113560.

- 1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; overnight, reflux
- 1.2 Reagents: Water

Scheme 439 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (166)

Suppliers (11)

Suppliers (2)

31-170-CAS-21642307

Steps: 1 Yield: 97%

Design and Discovery of N-(3-(2-(2-Hydroxyethoxy)-6-morpholinopyridin-4-yl)-4-methylphenyl)-2-(trifluoromethyl)isonicotinamide, a Selective, Efficacious, and Well-Tolerated RAS AF Inhibitor Targeting RAS Mutant Cancers: The Path to the Clinic

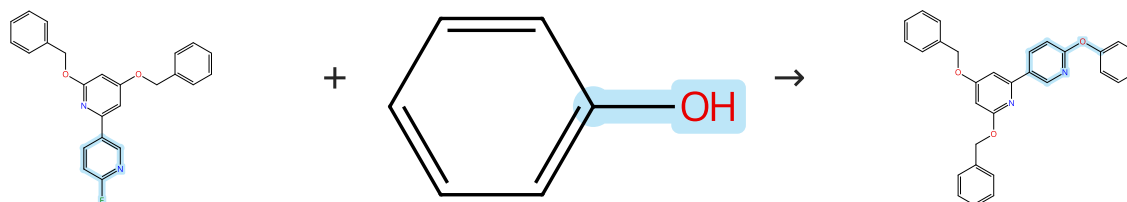
By: Ramurthy, Savithri; et al

Journal of Medicinal Chemistry (2020), 63(5), 2013-2027.

- 1.1 Reagents: Sodium hydride
Solvents: 1,4-Dioxane, *N*-Methyl-2-pyrrolidone; 20 min, rt
- 1.2 Solvents: 1,4-Dioxane; 18 h, 90 °C; 30 min, 150 °C
- 1.3 Reagents: Sodium bicarbonate
Solvents: Water

Scheme 440 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (207)

31-614-CAS-39486844

Steps: 1 Yield: 97%

Structure-Activity Relationship Studies and Optimization of 4-Hydroxypyridones as GPR84 Agonists

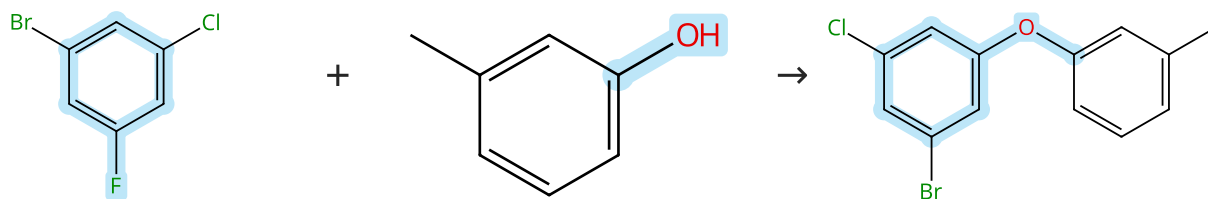
By: Ieremias, Loukas; et al

Journal of Medicinal Chemistry (2024), 67(5), 3542-3570.

- 1.1 Solvents: Dimethylformamide; 24 h, 100 °C
- Experimental Protocols

Scheme 441 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (72)

Suppliers (91)

Suppliers (3)

31-614-CAS-23926366

Steps: 1 Yield: 97%

Hypsochromic Shift of Multiple-Resonance-Induced Thermally Activated Delayed Fluorescence by Oxygen Atom Incorporation

By: Tanaka, Hiroyuki; et al

Angewandte Chemie, International Edition (2021), 60(33), 17910-17914.

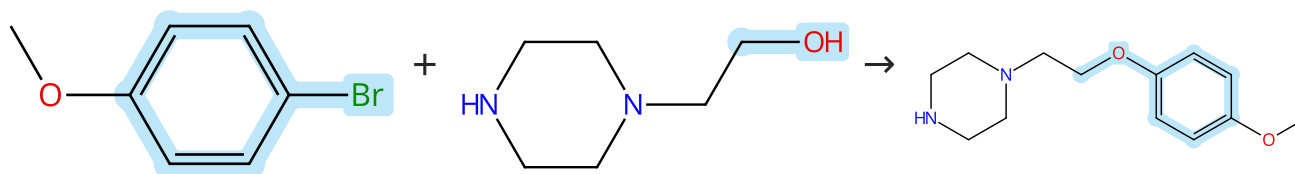
1.1 Reagents: Potassium carbonate

Solvents: *N*-Methyl-2-pyrrolidone; 22 h, 150 °C

Experimental Protocols

Scheme 442 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (71)

Suppliers (89)

Suppliers (55)

31-614-CAS-40801509

Steps: 1 Yield: 97%

Cu-Catalyzed Amination of Base-Sensitive Aryl Bromides and the Chemoselective N- and O-Arylation of Amino Alcohols

By: Strauss, Michael J.; et al

Journal of the American Chemical Society (2024), 146(27), 18616-18625.

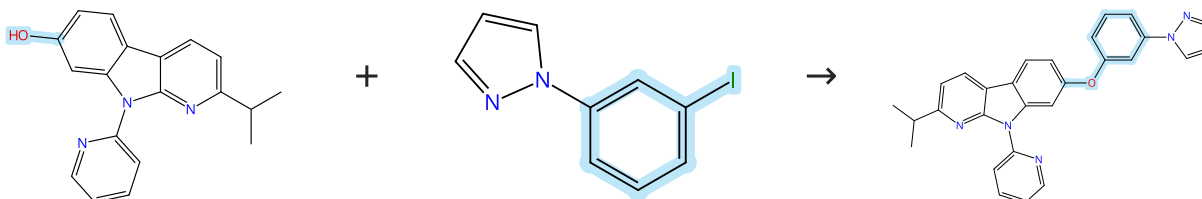
1.1 Reagents: Sodium *tert*-butoxideCatalysts: Cuprous iodide, 1,2-Benzenediamine, *N*¹,*N*²-bis(2-phenyl-1-naphthalenyl)-

Solvents: Dimethyl sulfoxide; 24 °C; 16 h, 24 °C

Experimental Protocols

Scheme 443 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (4)

31-614-CAS-37225961

Steps: 1 Yield: 97%

Blue-Phosphorescent Pt(II) Complexes of Tetradentate Pyridyl-Carbolinyl Ligands: Synthesis, Structure, Photophysics, and Electroluminescence

By: Yu, Feiling; et al

Inorganic Chemistry (2020), 59(19), 14493-14500.

1.1 Reagents: Cesium carbonate

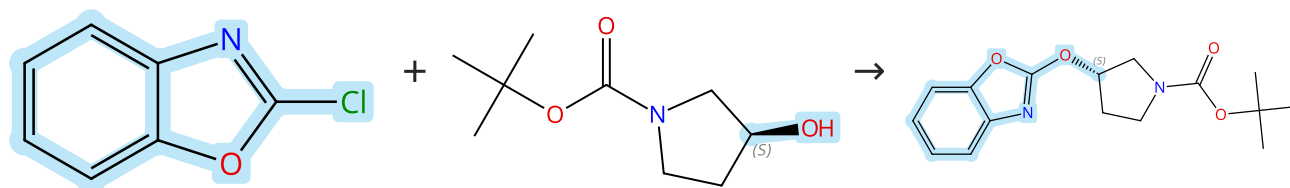
Catalysts: *N,N*-Dimethylglycine, Cuprous iodide

Solvents: Dimethyl sulfoxide; rt → 120 °C; 24 h, 120 °C

Experimental Protocols

Scheme 444 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (87)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (93)

Absolute stereochemistry shown

Suppliers (5)

31-170-CAS-21659781

Steps: 1 Yield: 97%

Palladium Catalyzed C-O Coupling of Amino Alcohols for the Synthesis of Aryl Ethers

By: Mikus, Malte S.; et al

Advanced Synthesis & Catalysis (2020), 362(2), 430-436.

1.1 Reagents: Cesium carbonate

Catalysts: Tris(dibenzylideneacetone)dipalladium, 5-[Bis(1,1-dimethylethyl)phosphino]-1',3',5'-triphenyl-1,4'-bi-1*H*-pyrazole

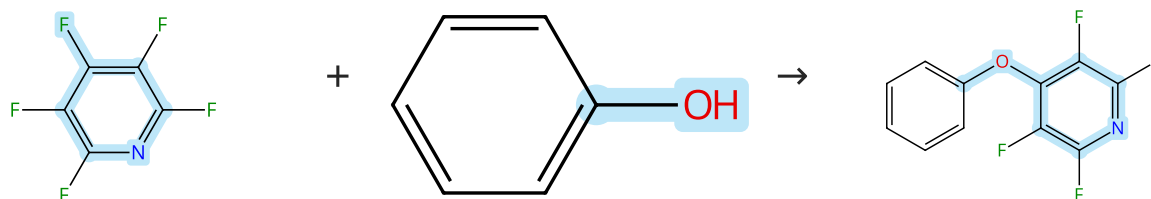
Solvents: Toluene; 10 - 15 min, 85 °C

1.2 Solvents: Toluene; 14 h, 85 °C

Experimental Protocols

Scheme 445 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (71)

Suppliers (207)

Suppliers (4)

31-170-CAS-22000016

Steps: 1 Yield: 97%

Protecting Group-Controlled Remote Regioselective Electrophilic Aromatic Halogenation Reactions

By: Brittain, William D. G.; et al

Journal of Organic Chemistry (2020), 85(11), 6862-6871.

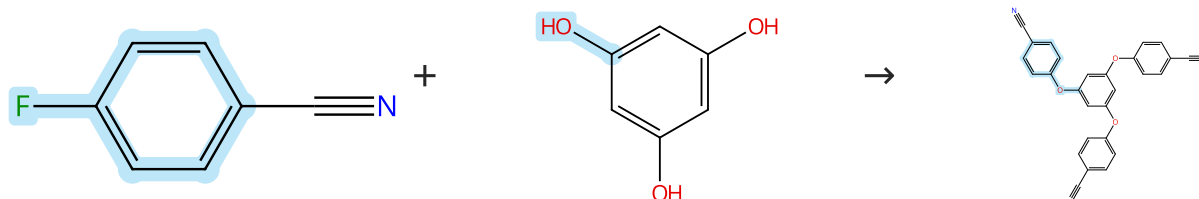
1.1 Reagents: Potassium carbonate

Solvents: Acetonitrile; 16 h, rt

Experimental Protocols

Scheme 446 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (92)

Suppliers (92)

Suppliers (44)

31-614-CAS-32405909

Steps: 1 Yield: 97%

Synthetic subnanochannels in porous aromatic frameworks accelerate selective water permeation in membrane desalination

By: Wang, Lei; et al

Science China Materials (2022), 65(7), 1920-1928.

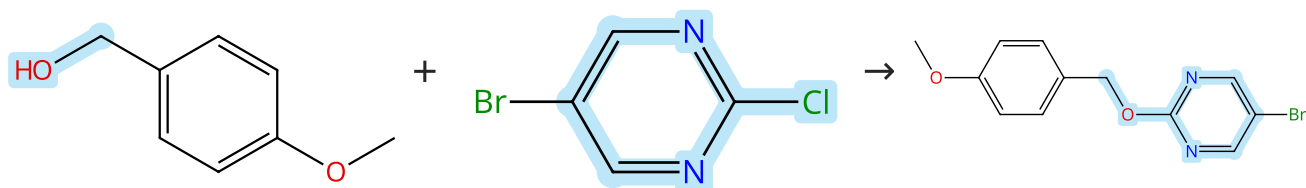
1.1 Reagents: Potassium carbonate

Solvents: Toluene, *N*-Methyl-2-pyrrolidone; 4 h, rt → 423 K; 423 K → 453 K; 16 h, 453 K

Experimental Protocols

Scheme 447 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (106)

Suppliers (107)

Suppliers (47)

31-170-CAS-22329393

Steps: 1 Yield: 97%

Synthesis and Anticancer Activity of Some New 2- Benzyloxy-5-alkyne Substituted Pyrimidines: An Application to Sonogashira Coupling

By: Reddy Onteddu, Surendranatha; et al

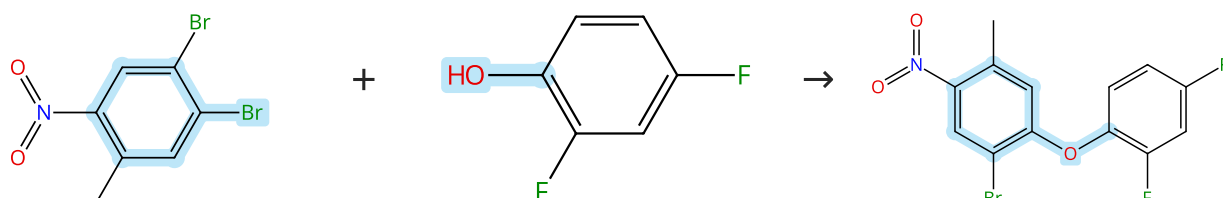
ChemistrySelect (2020), 5(27), 8194-8197.

- 1.1 Reagents: Cesium carbonate
Solvents: Dimethylformamide, Acetonitrile; rt; 10 min, rt
- 1.2 12 h, rt

Experimental Protocols

Scheme 448 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (47)

Suppliers (81)

Suppliers (8)

31-614-CAS-39824412

Steps: 1 Yield: 97%

Development and Scale-Up of a Key Copper-Catalyzed Biaryl Ether Formation for the Multikilogram Synthesis of Emprum apimod

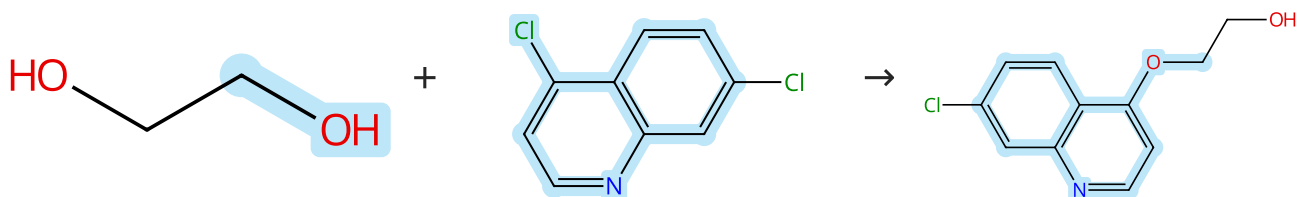
By: Brearley, Christopher; et al

Organic Process Research & Development (2024), 28(4), 1253-1259.

- 1.1 Reagents: Potassium carbonate
Solvents: Dimethylacetamide

Scheme 449 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (166)

Suppliers (105)

Suppliers (6)

31-170-CAS-23279200

Steps: 1 Yield: 97%

Morita-Baylis-Hillman reaction with 7-chloroquinoline derivatives-new compounds with potential anticancer activity

By: Oliveira, Joao Paulo G.; et al

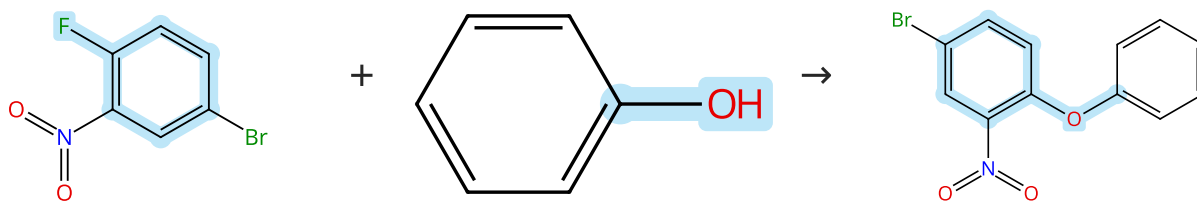
Journal of the Brazilian Chemical Society (2021), 32(2), 347-354.

- 1.1 Catalysts: Potassium *tert*-butoxide
Solvents: *tert*-Butanol; rt; 18 h, 80 °C
- 1.2 Reagents: Sodium bicarbonate
Solvents: Water

Experimental Protocols

Scheme 450 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (90)

Suppliers (207)

Suppliers (40)

31-614-CAS-34637560

Steps: 1 Yield: 97%

Synthesis of Square Planar Cu₄ Clusters

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 d, 80 °C

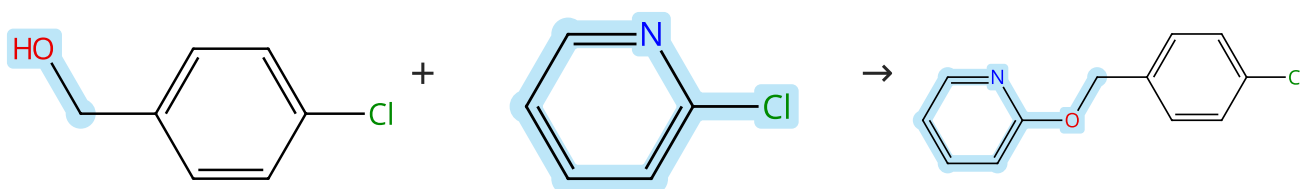
By: Osei, Manasseh Kusi; et al

Angewandte Chemie, International Edition (2022), 61(41), e202209529.

Experimental Protocols

Scheme 451 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (86)

Suppliers (66)

Suppliers (18)

31-614-CAS-34387483

Steps: 1 Yield: 97%

Visible Light-Induced Deoxygenation and Allylation/Vinylation of Pyridyl Ethers

1.1 Reagents: Potassium hydroxide
Catalysts: 18-Crown-6
Solvents: Toluene; rt → reflux

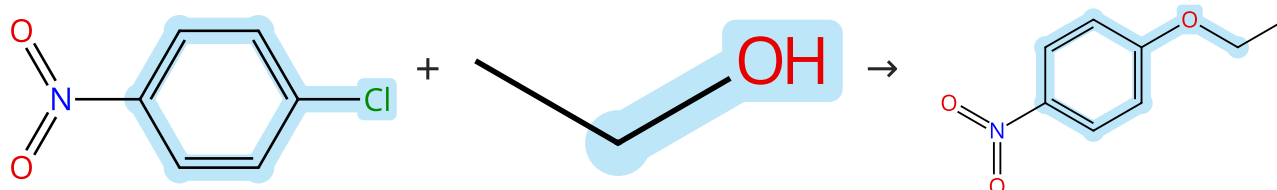
By: Wang, Fan; et al

Organic Letters (2022), 24(40), 7309-7314.

Experimental Protocols

Scheme 452 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (65)

Suppliers (618)

Suppliers (37)

31-170-CAS-22584329

Steps: 1 Yield: 97%

Aryl Ether Syntheses via Aromatic Substitution Proceeding under Mild Conditions

1.1 Reagents: Potassium *tert*-butoxide
Solvents: Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 10 min, 0 °C

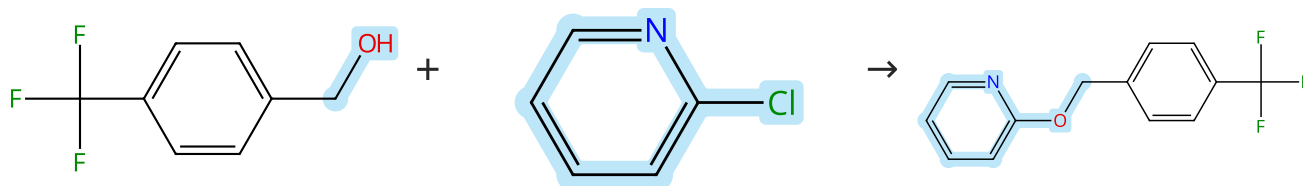
By: Ando, Shin; et al

Journal of Organic Chemistry (2020), 85(17), 11181-11189.

1.2 Reagents: Ammonium chloride
Solvents: Water; 0 °C

Scheme 453 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (79)

Suppliers (66)

Suppliers (7)

31-614-CAS-34387481

Steps: 1 Yield: 97%

Visible Light-Induced Deoxygenation and Allylation/Vinylation of Pyridyl Ethers

1.1 Reagents: Potassium hydroxide

Catalysts: 18-Crown-6

Solvents: Toluene; rt → reflux

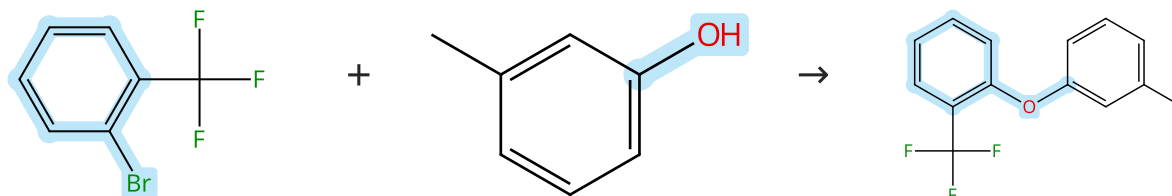
By: Wang, Fan; et al

Organic Letters (2022), 24(40), 7309-7314.

Experimental Protocols

Scheme 454 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (78)

Suppliers (91)

31-614-CAS-36696309

Steps: 1 Yield: 97%

Synthesis of Fluorenones and Xanthenes through Intramolecular C-F Arylation

1.1 Reagents: Potassium carbonate

Catalysts: Copper, Potassium iodide

Solvents: Pyridine; 24 h, reflux

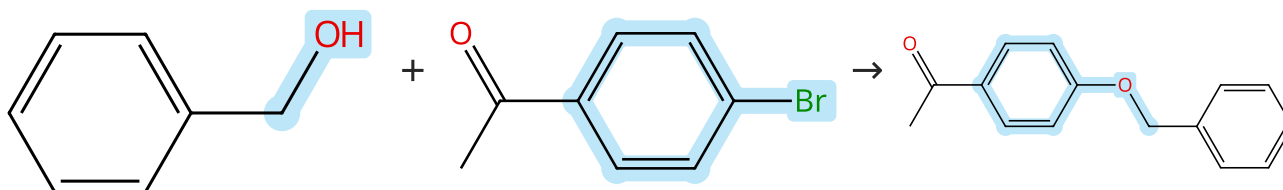
By: Hamada, Shusuke; et al

Bulletin of the Chemical Society of Japan (2023), 96(5), 401-405.

Experimental Protocols

Scheme 455 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (150)

Suppliers (85)

Suppliers (92)

31-614-CAS-40260586

Steps: 1 Yield: 97%

Thermal Nickel-Catalyzed Carbon-Oxygen Cross-Coupling of (Hetero)aryl Halides with Alcohols Enabled by the Use of a Silane Reductant Approach

1.1 Catalysts: Nickel bromide, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine

Solvents: Toluene; 30 min, rt

1.2 Reagents: Phenylsilane, 1,8-Diazabicyclo[5.4.0]undec-7-ene; 24 h, 120 °C

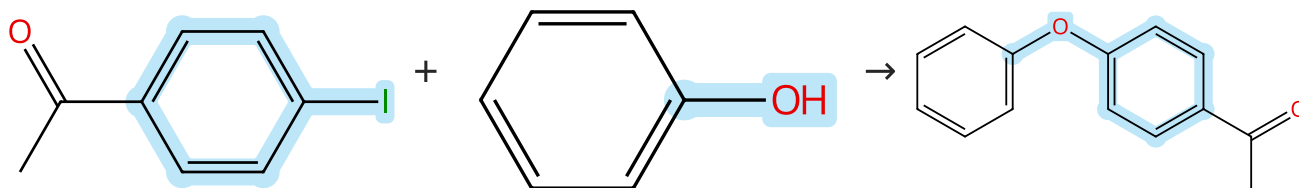
By: Yang, Liu; et al

ACS Catalysis (2024), 14(10), 7846-7852.

Experimental Protocols

Scheme 456 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (88)

Suppliers (207)

Suppliers (86)

31-614-CAS-33446666

Steps: 1 Yield: 97%

Highly Active and Stable Bis Imidazolium-Based Copper N-heterocyclic Carbene Modified Graphene Oxide for O-arylation and N-arylation Reactions in Water

By: Imanpour, Morteza; et al

Catalysis Letters (2023), 153(5), 1390-1409.

1.1 Catalysts: 2803960-69-0

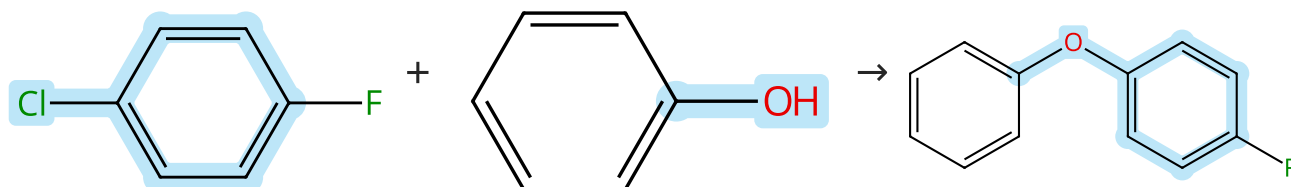
Solvents: Water; 10 min, 85 °C

1.2 Reagents: Potassium carbonate; 5 h, 80 °C

Experimental Protocols

Scheme 457 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (60)

Suppliers (207)

Suppliers (54)

31-614-CAS-41541874

Steps: 1 Yield: 97%

CuSO₄/N-(9H-carbazol-9-yl)picolinamide-Catalyzed C-O Coupling of (Hetero)Aryl Chlorides with Phenols on Water

By: Hong, Peng; et al

Organic Letters (2024), 26(34), 7202-7206.

1.1 Reagents: Tripotassium phosphate

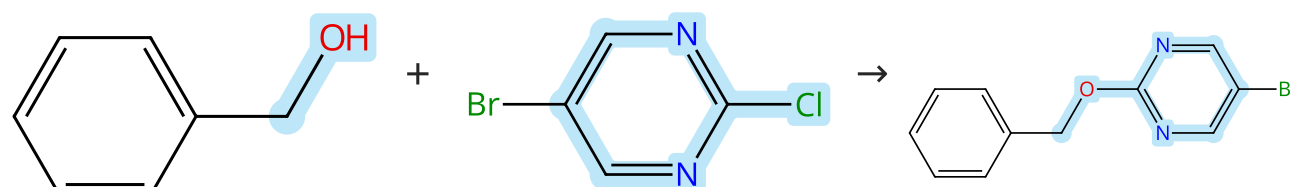
Catalysts: Copper sulfate pentahydrate, 2-Pyridinecarboxamide, N-9H-carbazol-9-yl-

Solvents: Water; 24 h, 120 °C

Experimental Protocols

Scheme 458 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (150)

Suppliers (107)

Suppliers (56)

31-170-CAS-22329392

Steps: 1 Yield: 97%

Synthesis and Anticancer Activity of Some New 2- Benzyloxy-5-alkyne Substituted Pyrimidines: An Application to Sonogashira Coupling

By: Reddy Onteddu, Surendranatha; et al

ChemistrySelect (2020), 5(27), 8194-8197.

1.1 Reagents: Cesium carbonate

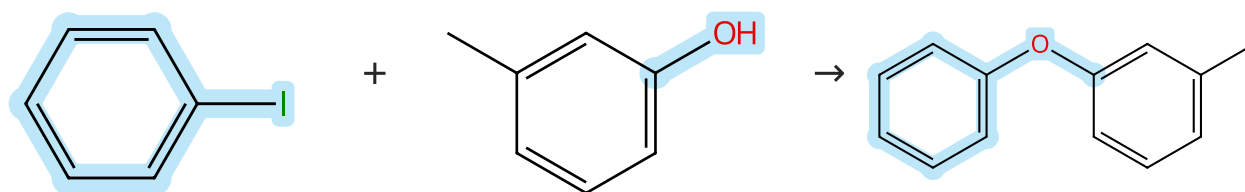
Solvents: Dimethylformamide, Acetonitrile; rt; 10 min, rt

1.2 12 h, rt

Experimental Protocols

Scheme 459 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (96)

Suppliers (91)

Suppliers (71)

31-170-CAS-22063629

Steps: 1 Yield: 97%

Pd based on 2-Aminopyrimidine and 1 H-benzo[d]imidazol-2-amine functionalized Fe₃O₄ nanoparticles as novel recyclable magnetic nanocatalysts for Ullmann coupling reaction

By: Chen, Jin; et al

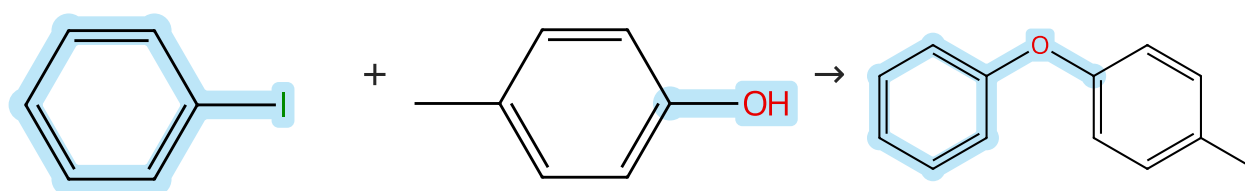
Applied Organometallic Chemistry (2020), 34(8), e5708.

1.1 **Reagents:** Potassium carbonate
Catalysts: *N*-[3-(Trimethoxysilyl)propyl]-2-pyrimidinamine (Fe₃O₄ magnetic nanoparticles; post-treated with Pd(OAc)₂)
Solvents: Dimethylformamide; 1.5 h, 70 °C

Experimental Protocols

Scheme 460 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (96)

Suppliers (89)

Suppliers (62)

31-170-CAS-24171139

Steps: 1 Yield: 97%

An advancement in the synthesis of nano Pd@magnetic amine-Functionalized UiO-66-NH₂ catalyst for cyanation and O-arylation reactions

By: Moghaddam, Firouz Matloubi; et al

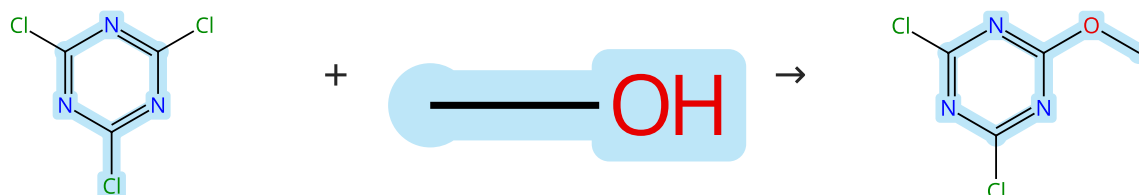
Scientific Reports (2021), 11(1), 11387.

1.1 **Reagents:** Potassium hydroxide
Catalysts: Palladium (complex with magnetite-supported and triazine- and tetrazole-functiona...), Hexakis[μ-[2-amino-1,4-benzenedicarboxylato(2-)-κO¹:κO¹]]tetra-μ₃-hydroxytetra-μ₃-oxohexazirconium (magnetite nanoparticle-bound, reaction products with trichlorotriazine...)
Solvents: *N*-Methyl-2-pyrrolidone; rt → 80 °C; 3 h, 80 °C

Experimental Protocols

Scheme 461 (2 Reactions)

Steps: 1 Yield: 97%



Suppliers (71)

Suppliers (444)

Suppliers (72)

31-614-CAS-24281882

Steps: 1 Yield: 97%

Synthesis and Antimicrobial Evaluation of Bis-morpholine Triazine Quaternary Ammonium Salts

By: Morandini, Andrea; et al

ChemMedChem (2021), 16(20), 3172-3176.

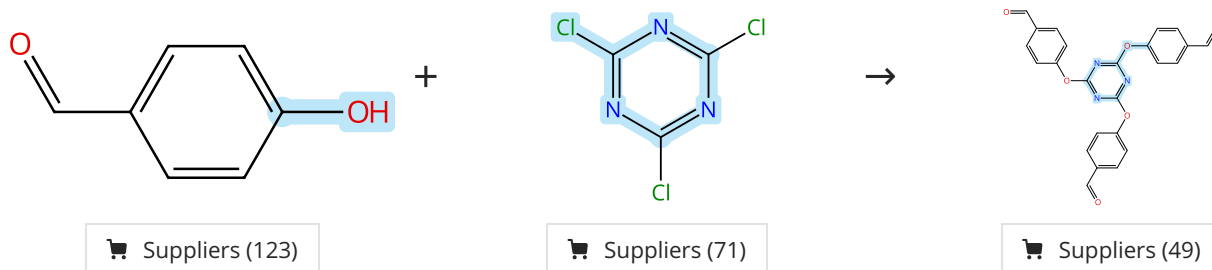
1.1 **Reagents:** Sodium bicarbonate
Solvents: Methanol; 0 °C; 1 h, 0 °C

Experimental Protocols

31-170-CAS-23512045	Steps: 1 Yield: 97%	Synthesis of Amidation Agents and Their Reactivity in Condensation Reactions
1.1 Reagents: Sodium bicarbonate Solvents: Methanol; 0 °C; 1 h, 0 °C		By: Sole, Roberto; et al
Experimental Protocols		Synthesis (2021), 53(9), 1672-1682.

Scheme 462 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-39399377	Steps: 1 Yield: 97%	Design and synthesis of a new tripod-chromogenic sensor based on a s-triazine and thiazolidine-2,4-dione ring (TCST) for naked-eye detection of Li⁺
1.1 Reagents: Sodium hydroxide Solvents: Acetone, Water; 5 h, 0 - 5 °C		By: Yazdani Nyaki, Hadiseh; et al
Experimental Protocols		Canadian Journal of Chemistry (2021), 99(5), 482-490.